CHAPTER

26

FIRE PROTECTION

CHAPTER 26
FIRE PROTECTION

Subject/Page	Date	COC
26-EFFECTIVE PAGES		
1 thru 7	Sep 15/2026	
8	BLANK	
26-CONTENTS		
1	May 15/2024	
2	May 15/2024	
3	Jan 15/2025	
O 4	Sep 15/2026	
O 5	Sep 15/2026	
6	Jan 15/2025	
7	May 15/2024	
8	May 15/2024	
9	May 15/2024	
10	May 15/2024	
11	Sep 15/2025	
12	May 15/2026	
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14	May 15/2026	
15	May 15/2026	
16	BLANK	
26-00-00		
901	May 15/2024	
902	May 15/2024	
903	May 15/2024	
904	May 15/2024	
905	May 15/2024	
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907	May 15/2024	
908	May 15/2024	
909	May 15/2024	
910	Jan 15/2025	
911	Jan 15/2025	
912	Jan 15/2025	
913	Jan 15/2025	
914	Jan 15/2025	
915	May 15/2024	
916	May 15/2024	
26-00-00 (cont)		
917	May 15/2024	
918	May 15/2024	
919	May 15/2024	
920	May 15/2024	
921	May 15/2024	
922	May 15/2024	
923	May 15/2024	
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926	Jan 15/2025	
927	Jan 15/2025	
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409	May 15/2026	
410	May 15/2026	
411	May 15/2024	
412	May 15/2026	
413	May 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
26-00-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
26-10-00		
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504	May 15/2024	
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R 517	Sep 15/2026	
R 518	Sep 15/2026	
R 519	Sep 15/2026	
R 520	Sep 15/2026	
R 521	Sep 15/2026	
R 522	Sep 15/2026	
O 523	Sep 15/2026	
R 524	Sep 15/2026	
R 525	Sep 15/2026	
R 526	Sep 15/2026	
O 527	Sep 15/2026	
O 528	Sep 15/2026	
O 529	Sep 15/2026	
O 530	Sep 15/2026	
A 531	Sep 15/2026	
A 532	Sep 15/2026	
26-10-01		
201	May 15/2024	
202	May 15/2024	
203	May 15/2025	

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Subject/Page	Date	COC
26-10-01 (cont)		
204	May 15/2024	
26-10-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2025	
404	May 15/2024	
26-11-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
26-11-00		
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506	Sep 15/2025	
507	May 15/2024	
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510	Sep 15/2024	
511	May 15/2024	
512	May 15/2024	
513	May 15/2024	
514	May 15/2024	
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516	BLANK	
26-11-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	

Subject/Page	Date	COC
26-11-01 (cont)		
409	May 15/2024	
410	May 15/2024	
411	May 15/2026	
412	May 15/2024	
413	May 15/2024	
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421	May 15/2026	
422	May 15/2026	
423	May 15/2024	
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26-11-01		
R 601	Sep 15/2026	
R 602	Sep 15/2026	
A 603	Sep 15/2026	
A 604	Sep 15/2026	
A 605	Sep 15/2026	
A 606	Sep 15/2026	
A 607	Sep 15/2026	
A 608	Sep 15/2026	
A 609	Sep 15/2026	
A 610	BLANK	
26-11-02		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
O 403	Sep 15/2026	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	

Subject/Page	Date	COC
26-11-02 (cont)		
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
R 418	Sep 15/2026	
R 419	Sep 15/2026	
R 420	Sep 15/2026	
R 421	Sep 15/2026	
R 422	Sep 15/2026	
R 423	Sep 15/2026	
R 424	Sep 15/2026	
R 425	Sep 15/2026	
R 426	Sep 15/2026	
R 427	Sep 15/2026	
R 428	Sep 15/2026	
R 429	Sep 15/2026	
R 430	Sep 15/2026	
R 431	Sep 15/2026	
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R 433	Sep 15/2026	
R 434	Sep 15/2026	
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R 436	Sep 15/2026	
R 437	Sep 15/2026	
R 438	Sep 15/2026	
R 439	Sep 15/2026	
R 440	Sep 15/2026	
A 441	Sep 15/2026	
A 442	Sep 15/2026	
A 443	Sep 15/2026	
A 444	Sep 15/2026	
A 445	Sep 15/2026	

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FIRE PROTECTION

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26-11-02 (cont)		
A 446	Sep 15/2026	
A 447	Sep 15/2026	
A 448	Sep 15/2026	
A 449	Sep 15/2026	
A 450	Sep 15/2026	
A 451	Sep 15/2026	
A 452	Sep 15/2026	
A 453	Sep 15/2026	
A 454	Sep 15/2026	
A 455	Sep 15/2026	
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26-15-00		
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26-15-00		
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516	Sep 15/2025	
517	May 15/2024	
518	May 15/2024	
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26-15-01		
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402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
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407	May 15/2026	
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410	Jan 15/2025	
411	May 15/2026	
412	Jan 15/2025	
413	Jan 15/2025	
414	BLANK	
26-15-01		
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602	BLANK	
26-15-02		
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402	May 15/2024	
403	Jan 15/2026	
404	May 15/2026	
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406	Jan 15/2025	
26-16-00		
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26-16-01		
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26-16-01		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
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26-16-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	

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405	May 15/2024	
406	May 15/2024	
26-18-00		
201	May 15/2024	
202	May 15/2024	
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26-18-00		
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26-18-01		
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510	May 15/2024	
26-18-01 (cont)		
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26-18-02		
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402	May 15/2026	
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404	May 15/2026	
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26-18-02		
501	Jan 15/2026	
26-18-02 (cont)		
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503	Jan 15/2026	
504	Jan 15/2026	
505	May 15/2025	
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507	May 15/2024	
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26-19-00		
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504	May 15/2026	
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26-19-01		
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404	May 15/2025	
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26-19-02		
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402	May 15/2024	
403	May 15/2024	
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405	May 15/2025	
406	May 15/2025	
26-20-00		
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602	May 15/2024	
603	May 15/2024	
604	May 15/2024	

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FIRE PROTECTION

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605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
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26-21-00		
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202	May 15/2024	
203	May 15/2024	
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205	May 15/2024	
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26-21-00		
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R 502	Sep 15/2026	
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R 505	Sep 15/2026	
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509	May 15/2024	
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R 511	Sep 15/2026	
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517	Sep 15/2024	
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520	May 15/2025	
521	Sep 15/2025	
522	Sep 15/2025	
523	Sep 15/2025	

Subject/Page	Date	COC
26-21-00 (cont)		
524	Sep 15/2025	
525	Sep 15/2024	
526	Sep 15/2024	
26-21-01		
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R 402	Sep 15/2026	
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R 405	Sep 15/2026	
R 406	Sep 15/2026	
407	May 15/2024	
408	May 15/2026	
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R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
A 415	Sep 15/2026	
A 416	BLANK	
26-21-01		
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602	May 15/2024	
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605	May 15/2024	
606	May 15/2024	
26-21-02		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	Sep 15/2025	
406	Sep 15/2025	
407	May 15/2026	
408	Sep 15/2025	

Subject/Page	Date	COC
26-22-00		
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502	Sep 15/2025	
R 503	Sep 15/2026	
504	May 15/2025	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
507	May 15/2024	
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509	May 15/2024	
510	Jan 15/2026	
511	Jan 15/2026	
512	Jan 15/2026	
513	Jan 15/2026	
514	May 15/2024	
26-22-01		
401	May 15/2024	
R 402	Sep 15/2026	
403	Jan 15/2025	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
406	May 15/2026	
407	May 15/2024	
R 408	Sep 15/2026	
409	Jan 15/2026	
R 410	Sep 15/2026	
411	May 15/2026	
412	Jan 15/2025	
26-22-01		
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602	May 15/2024	
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26-22-02		
401	May 15/2024	
402	Sep 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	

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406	May 15/2026	
407	Sep 15/2025	
408	Sep 15/2025	
26-22-03		
401	May 15/2024	
402	May 15/2024	
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R 506	Sep 15/2026	
R 507	Sep 15/2026	
R 508	Sep 15/2026	
R 509	Sep 15/2026	
510	Jan 15/2025	
R 511	Sep 15/2026	
R 512	Sep 15/2026	
513	May 15/2026	
R 514	Sep 15/2026	
R 515	Sep 15/2026	

Subject/Page	Date	COC
26-23-00 (cont)		
R 516	Sep 15/2026	
O 517	Sep 15/2026	
R 518	Sep 15/2026	
R 519	Sep 15/2026	
R 520	Sep 15/2026	
R 521	Sep 15/2026	
R 522	Sep 15/2026	
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R 524	Sep 15/2026	
R 525	Sep 15/2026	
R 526	Sep 15/2026	
R 527	Sep 15/2026	
R 528	Sep 15/2026	
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26-23-01		
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403	Sep 15/2025	
404	Jan 15/2026	
405	May 15/2024	
406	May 15/2024	
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414	May 15/2026	
415	May 15/2026	
416	May 15/2026	
417	May 15/2026	
418	May 15/2026	
R 419	Sep 15/2026	
420	Jan 15/2026	

Subject/Page	Date	COC
26-23-01 (cont)		
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R 422	Sep 15/2026	
423	Jan 15/2026	
424	May 15/2026	
425	Jan 15/2026	
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26-23-01		
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603	May 15/2026	
R 604	Sep 15/2026	
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R 702	Sep 15/2026	
R 703	Sep 15/2026	
R 704	Sep 15/2026	
705	May 15/2024	
706	BLANK	
26-23-02		
401	Jan 15/2026	
402	Jan 15/2026	

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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2026	
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410	May 15/2024	
26-23-04		
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403	May 15/2024	
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205	Jan 15/2026	
206	BLANK	
26-24-01		
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26-26-01		
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26-26-01		
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26-26-02		
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R 603	Sep 15/2026	
604	May 15/2024	
605	May 15/2024	
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607	May 15/2026	
608	May 15/2026	
609	May 15/2026	
610	BLANK	

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FIRE PROTECTION

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FIRE PROTECTION - DDG MAINTENANCE PROCEDURES	26-00-00		901	PTF ALL
DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Preparation TASK 26-00-00-040-808			902	PTF ALL
DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Restoration TASK 26-00-00-440-806			902	PTF ALL
DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Preparation TASK 26-00-00-040-809			902	PTF ALL
DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Restoration TASK 26-00-00-440-807			903	PTF ALL
DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Preparation TASK 26-00-00-040-802			903	PTF ALL
DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Restoration TASK 26-00-00-440-802			905	PTF ALL
DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative Or Not Installed - Preparation TASK 26-00-00-040-803			906	PTF ALL
DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative Or Not Installed - Restoration TASK 26-00-00-440-803			907	PTF ALL
DDG 26-18-01-01 Wheel Well Fire Detection Loops Inoperative - Preparation TASK 26-00-00-040-804			909	PTF ALL
DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Preparation TASK 26-00-00-040-810			910	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Restoration TASK 26-00-00-440-808			911	PTF ALL
DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Preparation TASK 26-00-00-040-819			911	PTF ALL
DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Restoration TASK 26-00-00-440-817			912	PTF ALL
DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Preparation TASK 26-00-00-040-820			913	PTF ALL
DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Restoration TASK 26-00-00-440-819			913	PTF ALL
DDG 26-20-01-02 APU Bottle Discharged Light Inoperative - Preparation TASK 26-00-00-040-821			914	PTF ALL
DDG 26-20-01-02 APU Bottle Discharged Light Inoperative - Restoration TASK 26-00-00-440-820			916	PTF ALL
DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Preparation TASK 26-00-00-040-805			916	PTF ALL
DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Restoration TASK 26-00-00-440-818			918	PTF ALL
DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Preparation TASK 26-00-00-040-811			918	PTF ALL
DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Restoration TASK 26-00-00-440-809			920	PTF ALL
DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Preparation TASK 26-00-00-040-812			921	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Restoration TASK 26-00-00-440-810			922	PTF ALL
DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Preparation TASK 26-00-00-040-814			923	PTF ALL
DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Restoration TASK 26-00-00-440-812			925	PTF ALL
DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Preparation TASK 26-00-00-040-807			925	PTF ALL
DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Restoration TASK 26-00-00-440-805			926	PTF ALL
DDG 26-26-01 Portable Fire Extinguishers Inoperative (EASA) - Preparation TASK 26-00-00-040-801			926	PTF ALL
DDG 26-26-01 Portable Fire Extinguishers Inoperative (EASA) - Restoration TASK 26-00-00-440-801			927	PTF ALL
ENGINE AND APU FIRE CONTROL PANEL - REMOVAL/INSTALLATION	26-00-01		401	PTF ALL
Engine and APU Fire Control Panel Removal TASK 26-00-01-000-801			401	PTF ALL
Engine and APU Fire Control Panel Installation TASK 26-00-01-400-801			405	PTF ALL
Engine and APU Fire Shutoff Switch Assembly Removal TASK 26-00-01-000-802			408	PTF ALL
Engine and APU Fire Shutoff Switch Assembly Installation TASK 26-00-01-400-802			413	PTF ALL
CARGO FIRE CONTROL PANEL - REMOVAL/INSTALLATION	26-00-02		401	PTF ALL
Cargo Fire Control Panel Removal TASK 26-00-02-000-801			401	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Cargo Fire Control Panel Installation TASK 26-00-02-400-801			404	PTF ALL
DETECTION - ADJUSTMENT/TEST	26-10-00		501	PTF ALL
Fire and Overheat Detection System - Operational Test TASK 26-10-00-710-801			501	PTF ALL
Overheat Detection Test Using Handheld LCRZ Meter TASK 26-10-00-700-802			516	PTF ALL
Overheat Detection Test Using UN-ATE Tool by Technologies Harness Scanner TASK 26-10-00-700-803			526	PTF ALL
ENGINE AND APU FIRE DETECTION CONTROL MODULE, M279 - MAINTENANCE PRACTICES	26-10-01		201	PTF ALL
Engine and APU Fire Detection Control Module - Deactivation TASK 26-10-01-040-801			201	PTF ALL
Engine and APU Fire Detection Control Module - Activation TASK 26-10-01-440-801			204	PTF ALL
ENGINE AND APU FIRE DETECTION CONTROL MODULE, M279 - REMOVAL/INSTALLATION	26-10-01		401	PTF ALL
Engine and APU Fire Detection Module Removal TASK 26-10-01-000-801			401	PTF ALL
Engine and APU Fire Detection Module Installation TASK 26-10-01-400-801			404	PTF ALL
ENGINE FIRE DETECTION - MAINTENANCE PRACTICES	26-11-00		201	PTF ALL
Engine Fire Detection - Deactivation TASK 26-11-00-040-801			201	PTF ALL
Engine Fire Detection - Activation TASK 26-11-00-440-801			204	PTF ALL
ENGINE FIRE DETECTION - ADJUSTMENT/TEST	26-11-00		501	PTF ALL
Engine Fire Detection - Operational Test TASK 26-11-00-710-801			501	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Engine Fire Detection Loop Resistance - System Test TASK 26-11-00-730-801			506	PTF ALL
Engine Fire Detection Circuit - System Test TASK 26-11-00-730-802			510	PTF ALL
ENGINE FIRE DETECTORS - REMOVAL/INSTALLATION	26-11-01		401	PTF ALL
Engine Fire Detector Element Removal TASK 26-11-01-000-801			401	PTF ALL
Engine Fire Detector Element Installation TASK 26-11-01-400-801			411	PTF ALL
Engine Fire Detector Assembly Removal TASK 26-11-01-000-802			413	PTF ALL
Engine Fire Detector Assembly Installation TASK 26-11-01-400-802			415	PTF ALL
Combustor Burn Through Fire Detector Assembly Removal TASK 26-11-01-000-803			417	PTF ALL
Combustor Burn Through Fire Detector Assembly Installation TASK 26-11-01-400-803			421	PTF ALL
ENGINE FIRE DETECTORS - INSPECTION/CHECK	26-11-01		601	PTF ALL
Engine Fire Detector Inspection TASK 26-11-01-211-801			601	PTF ALL
ENGINE FIRE DETECTOR HARNESS - REMOVAL/INSTALLATION	26-11-02		401	PTF ALL
Engine Fire Detection Harness Removal TASK 26-11-02-000-801			401	PTF ALL
Engine Fire Detection Harness Installation TASK 26-11-02-400-801			446	PTF ALL
LAVATORY SMOKE DETECTION - ADJUSTMENT/TEST	26-14-00		501	PTF ALL
Lavatory Smoke Detection - Self Test TASK 26-14-00-710-801			501	PTF ALL
Lavatory Smoke Detection - Smoke Test TASK 26-14-00-730-801			501	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LAVATORY SMOKE DETECTOR - REMOVAL/INSTALLATION	26-14-01		401	PTF ALL
Lavatory Smoke Detector Removal TASK 26-14-01-000-801			401	PTF ALL
Lavatory Smoke Detector Installation TASK 26-14-01-400-801			403	PTF ALL
LAVATORY SMOKE DETECTOR - CLEANING/PAINTING	26-14-01		701	PTF ALL
Lavatory Smoke Detector Cleaning TASK 26-14-01-100-801			701	PTF ALL
APU FIRE DETECTION - MAINTENANCE PRACTICES	26-15-00		201	PTF ALL
APU Fire Detection - Deactivation TASK 26-15-00-040-801			201	PTF ALL
APU Fire Detection - Activation TASK 26-15-00-440-801			205	PTF ALL
APU FIRE DETECTION - ADJUSTMENT/TEST	26-15-00		501	PTF ALL
APU Fire Detection - Operational Test TASK 26-15-00-710-801			501	PTF ALL
APU Fire Detection Loop Resistance - System Test TASK 26-15-00-730-801			507	PTF ALL
APU Fire Detection Circuit - System Test TASK 26-15-00-730-802			511	PTF ALL
APU OVERHEAT DETECTORS - REMOVAL/INSTALLATION	26-15-01		401	PTF ALL
APU Overheat Detector Element Removal TASK 26-15-01-000-801			401	PTF ALL
APU Overheat Detector Element Installation TASK 26-15-01-400-801			407	PTF ALL
APU Overheat Detector Assembly Removal TASK 26-15-01-000-802			410	PTF ALL
APU Overheat Detector Assembly Installation TASK 26-15-01-400-802			411	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
APU OVERHEAT DETECTORS - INSPECTION/CHECK	26-15-01		601	PTF ALL
APU Overheat Detectors Inspection TASK 26-15-01-210-801			601	PTF ALL
APU FIRE WARNING HORN - REMOVAL/INSTALLATION	26-15-02		401	PTF ALL
APU Fire Warning Horn Removal TASK 26-15-02-000-801			401	PTF ALL
APU Fire Warning Horn Installation TASK 26-15-02-400-801			404	PTF ALL
CARGO BAY SMOKE DETECTION - ADJUSTMENT/TEST	26-16-00		501	PTF ALL
Cargo Bay Smoke Detection - Operational Test TASK 26-16-00-710-801			501	PTF ALL
Cargo Bay Smoke Detection - Smoke Test TASK 26-16-00-730-801			507	PTF ALL
CARGO BAY SMOKE DETECTOR - REMOVAL/INSTALLATION	26-16-01		401	PTF ALL
Cargo Bay Smoke Detector Removal TASK 26-16-01-000-801			401	PTF ALL
Cargo Bay Smoke Detector Installation TASK 26-16-01-400-801			405	PTF ALL
CARGO BAY SMOKE DETECTOR - CLEANING/PAINTING	26-16-01		701	PTF ALL
Clean the Cargo Bay Smoke Detectors TASK 26-16-01-100-801			701	PTF ALL
CARGO ELECTRONIC UNIT - REMOVAL/INSTALLATION	26-16-02		401	PTF ALL
Cargo Electronic Unit Removal TASK 26-16-02-000-801			401	PTF ALL
Cargo Electronic Unit Installation TASK 26-16-02-400-801			405	PTF ALL
Cargo Electronic Unit Installation Test TASK 26-16-02-710-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WHEEL WELL, WING AND LOWER AFT BODY OVERHEAT DETECTION SYSTEM - MAINTENANCE PRACTICES	26-18-00		201	PTF ALL
Wheel Well, Wing and Lower Aft Body Overheat Detection System - Deactivation TASK 26-18-00-040-801			201	PTF ALL
Wheel Well, Wing and Lower Aft Body Overheat Detection System - Activation TASK 26-18-00-440-801			204	PTF ALL
WHEEL WELL, WING AND LOWER AFT BODY OVERHEAT DETECTION SYSTEM - ADJUSTMENT/TEST	26-18-00		501	PTF ALL
Wheel Well and Lower Aft Body Overheat Detection System - Operational Test TASK 26-18-00-710-801			501	PTF ALL
Wing and Lower Aft Body Overheat Detection System - Operational Test TASK 26-18-00-710-802			508	PTF ALL
COMPARTMENT OVERHEAT DETECTION CONTROL UNIT, M237 - REMOVAL/INSTALLATION	26-18-01		401	PTF ALL
Compartment Overheat Detection Control Unit Removal TASK 26-18-01-000-801			401	PTF ALL
Compartment Overheat Detection Control Unit Installation TASK 26-18-01-400-801			404	PTF ALL
COMPARTMENT OVERHEAT DETECTION CONTROL UNIT, M237 - ADJUSTMENT/TEST	26-18-01		501	PTF ALL
Compartment Overheat Detection Control Unit, M237 - BITE Test TASK 26-18-01-740-801			501	PTF ALL
Compartment Overheat System Automated Test with WiN MS Aero FLi Test Kit TASK 26-18-01-700-801			507	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WING, WHEEL WELL, AND LOWER AFT BODY OVERHEAT DETECTION SENSING ELEMENT - REMOVAL/INSTALLATION	26-18-02		401	PTF ALL
Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Removal TASK 26-18-02-000-801			401	PTF ALL
Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Installation TASK 26-18-02-400-801			423	PTF ALL
WING, WHEEL WELL, AND LOWER AFT BODY OVERHEAT DETECTION SENSING ELEMENT - ADJUSTMENT/TEST	26-18-02		501	PTF ALL
Wing, Wheel Well and Lower Aft Body Overheat Sensing Element Test TASK 26-18-02-720-801			501	PTF ALL
E/E COOLING SMOKE DETECTION - ADJUSTMENT/TEST	26-19-00		501	PTF ALL
E/E Cooling Smoke Detection - Operational Test TASK 26-19-00-710-801			501	PTF ALL
E/E Cooling Smoke Detector - Operational Test TASK 26-19-00-710-802			508	PTF ALL
Equipment Supply Smoke Detector - Removal/Installation	26-19-01		401	PTF ALL
Equipment Supply Smoke Detector Removal TASK 26-19-01-000-801			401	PTF ALL
Equipment Supply Smoke Detector Installation TASK 26-19-01-400-801			404	PTF ALL
Equipment Exhaust Smoke Detector - Removal/Installation	26-19-02		401	PTF ALL
Equipment Exhaust Smoke Detector Removal TASK 26-19-02-000-801			401	PTF ALL
Equipment Exhaust Smoke Detector Installation TASK 26-19-02-400-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FIRE EXTINGUISHING BOTTLE PRESSURE GAUGE - INSPECTION/CHECK	26-20-00		601	PTF ALL
Engine Fire Extinguishing Bottle Check TASK 26-20-00-210-801			601	PTF ALL
APU Fire Extinguishing Bottle Check TASK 26-20-00-210-802			606	PTF ALL
ENGINE FIRE EXTINGUISHING - MAINTENANCE PRACTICES	26-21-00		201	PTF ALL
Engine Fire Extinguishing System - Deactivation TASK 26-21-00-040-801			201	PTF ALL
Engine Fire Extinguishing System - Activation TASK 26-21-00-440-801			204	PTF ALL
ENGINE FIRE EXTINGUISHING - ADJUSTMENT/TEST	26-21-00		501	PTF ALL
Engine Fire Extinguishing Bottle Squib Circuit Test TASK 26-21-00-730-801			501	PTF ALL
Engine Fire Extinguishing Bottle Pressure Switch Circuit Test TASK 26-21-00-730-802			509	PTF ALL
Engine Fire Extinguishing Discharge Line Flow Test TASK 26-21-00-720-801			509	PTF ALL
Engine Fire Extinguishing Discharge Line Pressure Test TASK 26-21-00-730-803			516	PTF ALL
Engine Fire Switch System Shutdown Test TASK 26-21-00-720-802			520	PTF ALL
ENGINE FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION	26-21-01		401	PTF ALL
Engine Fire Extinguishing Bottle Removal TASK 26-21-01-000-801			401	PTF ALL
Engine Fire Extinguishing Bottle Installation TASK 26-21-01-400-801			408	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE FIRE EXTINGUISHING BOTTLE - INSPECTION/CHECK	26-21-01		601	PTF ALL
Engine Fire Extinguisher Distribution System Inspection TASK 26-21-01-210-801			601	PTF ALL
ENGINE FIRE EXTINGUISHER BOTTLE SQUIB - REMOVAL/INSTALLATION	26-21-02		401	PTF ALL
Engine Fire Extinguisher Bottle Squib Removal TASK 26-21-02-000-801			401	PTF ALL
Engine Fire Extinguisher Bottle Squib Installation TASK 26-21-02-400-801			405	PTF ALL
APU FIRE EXTINGUISHING - ADJUSTMENT/TEST	26-22-00		501	PTF ALL
APU Fire Extinguishing Bottle Squib Circuit Test TASK 26-22-00-730-801			501	PTF ALL
APU Fire Extinguishing Bottle Pressure Switch Circuit Test TASK 26-22-00-730-802			510	PTF ALL
APU Fire Switch System Shutdown Test TASK 26-22-00-720-801			510	PTF ALL
APU FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION	26-22-01		401	PTF ALL
APU Fire Extinguishing Bottle Removal TASK 26-22-01-000-801			401	PTF ALL
APU Fire Extinguishing Bottle Installation TASK 26-22-01-400-801			406	PTF ALL
APU FIRE EXTINGUISHING BOTTLE - INSPECTION/CHECK	26-22-01		601	PTF ALL
APU Fire Bottle Inspection TASK 26-22-01-210-801			601	PTF ALL
APU FIRE EXTINGUISHING BOTTLE SQUIB - REMOVAL/INSTALLATION	26-22-02		401	PTF ALL
APU Fire Extinguishing Bottle Squib Removal TASK 26-22-02-000-801			401	PTF ALL
APU Fire Extinguishing Bottle Squib Installation TASK 26-22-02-400-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
APU FIRE CONTROL PANEL - REMOVAL/INSTALLATION	26-22-03		401	PTF ALL
APU Fire Control Panel Removal TASK 26-22-03-000-801			401	PTF ALL
APU Fire Control Panel Installation TASK 26-22-03-400-801			405	PTF ALL
APU FIRE CONTROL PANEL - INSPECTION/CHECK	26-22-03		601	PTF ALL
APU Fire Control Panel Inspection TASK 26-22-03-210-801			601	PTF ALL
CARGO FIRE EXTINGUISHING - MAINTENANCE PRACTICES	26-23-00		201	PTF ALL
Cargo Compartment Fire Extinguishers and Squibs - Deactivation TASK 26-23-00-040-801			201	PTF ALL
Cargo Compartment Fire Extinguishers and Squibs - Activation TASK 26-23-00-440-801			203	PTF ALL
CARGO FIRE EXTINGUISHING - ADJUSTMENT/TEST	26-23-00		501	PTF ALL
Cargo Fire Extinguishing Bottle Squib - Circuit Test TASK 26-23-00-730-801			501	PTF ALL
Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test TASK 26-23-00-730-802			513	PTF ALL
Cargo Fire Extinguishing Bottle Low Rate Discharge on Ground Inhibit Test TASK 26-23-00-730-804			514	PTF ALL POST SB 737-26-1173
Cargo Fire Extinguishing Discharge Line Flow Test TASK 26-23-00-720-801			520	PTF ALL
Cargo Fire Extinguishing Discharge Line Pressure Test TASK 26-23-00-730-803			524	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - REMOVAL/INSTALLATION	26-23-01		401	PTF ALL
Cargo Fire Extinguisher Bottle Removal TASK 26-23-01-000-801			401	PTF ALL
Cargo Fire Extinguisher Bottle Installation TASK 26-23-01-400-801			408	PTF ALL
Cargo Fire Extinguisher Bottle Removal - With Tool TASK 26-23-01-000-802			415	PTF ALL
Cargo Fire Extinguisher Bottle Installation - With Tool TASK 26-23-01-400-802			418	PTF ALL
CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - INSPECTION/CHECK	26-23-01		601	PTF ALL
Discharge Head Leak Check (Method 1) TASK 26-23-01-200-801			601	PTF ALL
Discharge Head Leak Check (Method 2) TASK 26-23-01-200-802			614	PTF ALL
CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE SYSTEM - CLEANING/PAINTING	26-23-01		701	PTF ALL
Cargo Fire Extinguishing System Cleaning TASK 26-23-01-100-801			701	PTF ALL
CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE SQUIB - REMOVAL/INSTALLATION	26-23-02		401	PTF ALL
Cargo Fire Extinguisher Bottle Squib Removal TASK 26-23-02-000-801			401	PTF ALL
Cargo Fire Extinguisher Bottle Squib Installation TASK 26-23-02-400-801			407	PTF ALL
CARGO FIRE EXTINGUISHING FILTER/DRYER - REMOVAL/INSTALLATION	26-23-04		401	PTF ALL
Cargo Fire Extinguishing Filter/Dryer Removal TASK 26-23-04-000-801			401	PTF ALL
Cargo Fire Extinguishing Filter/Dryer Installation TASK 26-23-04-400-801			404	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LAVATORY WASTE COMPARTMENT FIRE EXTINGUISHING - MAINTENANCE PRACTICES	26-24-01		201	PTF ALL
Lavatory Waste Compartment Fire Extinguishing Bottle Removal/Installation TASK 26-24-01-900-801			201	PTF ALL
Lavatory Waste Compartment Fire Extinguishing Bottle Inspection/Check TASK 26-24-01-200-801			205	PTF ALL
LAVATORY WASTE COMPARTMENT - INSPECTION/CHECK	26-24-01		601	PTF ALL
Lavatory Waste Compartment Temperature Indicator Strip Inspection TASK 26-24-01-210-801			601	PTF ALL
HALON FIRE EXTINGUISHERS - REMOVAL/INSTALLATION	26-26-01		401	PTF ALL
Halon Fire Extinguisher Removal TASK 26-26-01-020-801			401	PTF ALL
Halon Fire Extinguisher - Installation TASK 26-26-01-430-801			403	PTF ALL
HALON FIRE EXTINGUISHER - INSPECTION/CHECK	26-26-01		601	PTF ALL
Halon Fire Extinguisher - Inspection/Check TASK 26-26-01-200-801			601	PTF ALL
Halon Fire Extinguisher - Check TASK 26-26-01-200-802			604	PTF ALL
WATER FIRE EXTINGUISHERS- REMOVAL/INSTALLATION	26-26-02		401	PTF ALL
Water Fire Extinguisher Removal TASK 26-26-02-000-801			401	PTF ALL
Water Fire Extinguisher Installation TASK 26-26-02-400-801			403	PTF ALL
WATER FIRE EXTINGUISHERS - INSPECTION/CHECK	26-26-02		601	PTF ALL
Water Fire Extinguishers - Inspection TASK 26-26-02-200-801			601	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Water Fire Extinguishers - Check TASK 26-26-02-200-802			606	PTF ALL

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FIRE PROTECTION - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks are used to prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the fire protection system:

(1) DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Preparation

(2) DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Restorataion

(3) DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Preparation

(4) DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Restoration

(5) DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Preparation

(6) DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Restoration

(7) DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative or Not Installed - Preparation

(8) DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative or Not Installed - Restoration

(9) DDG 26-18-01-01 Wheel Well Fire Detection Loops Inoperative - Preparation

(10) DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Preparation

(11) DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Restoration

(12) DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Preparation

(13) DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Restoration

(14) DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Preparation

(15) DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Restoration

(16) DDG 26-20–01–02 APU Bottle Discharged light Inoperative - Preparation

(17) DDG 26-20–01–02 APU Bottle Discharged light Inoperative - Restoration

(18) DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Preparation

(19) DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Restoration

(20) DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Preparation

(21) DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Restoration

(22) DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Preparation

(23) DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Restoration

(24) DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Preparation

(25) DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Restoration

(26) DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Preparation

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(27) DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Restoration

(28) DDG 26-26-01 Portable Fire Extinguishers Inoperative - Preparation

(29) DDG 26-26-01 Portable Fire Extinguishers Inoperative - Restoration.

TASK 26-00-00-040-808

2. DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the lavatory smoke detection system inoperative.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Lavatory Deactivation

SUBTASK 26-00-00-040-014

(1) Do the steps that follow:

(a) Empty the lavatory waste receptacle.

(b) Close and lock the lavatory door.

(c) Install an INOPERATIVE - DO NOT ENTER placard on the lavatory door.

NOTE: Lavatory use or inspections by crew members is allowed.

——— END OF TASK ———

TASK 26-00-00-440-806

3. DDG 26-14-01 Lavatory Smoke Detection System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the lavatory smoke detection system inoperative.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Lavatory Smoke Detection System Activation

SUBTASK 26-00-00-040-015

(1) Remove the INOP placard from the lavatory door.

SUBTASK 26-00-00-800-001

(2) Unlock the lavatory door.

——— END OF TASK ———

TASK 26-00-00-040-809

4. DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the lavatory smoke detector self test switch inoperative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
26-14-00-730-801	Lavatory Smoke Detection - Smoke Test (P/B 501)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Lavatory Smoke Detector Deactivation

SUBTASK 26-00-00-040-016

(1) For each smoke detector, do the lavatory smoke detection system test (TASK 26-14-00-730-801).

(a) The test switch may be inoperative provided that each lavatory smoke detector is verified to pass the smoke test.

——— END OF TASK ———

TASK 26-00-00-440-807

5. DDG 26-14-01-01 Lavatory Smoke Detector SELF TEST Switch Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with a lavatory smoke detector SELF TEST switch inoperative.

B. References

Reference	Title
26-14-01 P/B 401	LAVATORY SMOKE DETECTOR - REMOVAL/INSTALLATION

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Lavatory Smoke Detector Activation

SUBTASK 26-00-00-400-003

(1) Replace the lavatory smoke detector (PAGEBLOCK 26-14-01/401).

——— END OF TASK ———

TASK 26-00-00-040-802

6. DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the wheel well fire detection system inoperative and the brake temperature monitoring system (BTMS) operates normally.

B. References

Reference	Title
22-11-33-000-801	Flight Control Computer Removal (P/B 401)
22-11-33-400-801	Flight Control Computer Installation (P/B 401)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Wheel Well Fire Detection System Deactivation

SUBTASK 26-00-00-010-001

(1) Get access to the wheel well fire detection system compartment.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-00-00-740-001

(2) Deactivate the wheel well fire detection system, use one of the two methods.

SUBTASK 26-00-00-040-002

(3) Method 1: deactivate WHEEL WELL light and wheel well fire detection system input to fire warning system.

(a) Remove the two light bulbs from the WHEEL WELL light on the P8-1 panel.

1) Install an INOP - FIRE BELL DISCONNECTED placard on the light switch.

(b) Install a WHEEL WELL FIRE DETECTION SYSTEM INOP placard on the OVHT/FIRE TEST switch, on the P8-1 overheat and fire protection panel.

(c) Disconnect the fire bell:

1) Remove Flight Control Computer (FCC) B to get access to the wiring for the fire bell (TASK 22-11-33-000-801).

2) Disconnect the wires from D742, pin 16 and pin 25 on the E1-4 shelf.

3) Cap and stow the wires.

4) Install the FCC B (TASK 22-11-33-400-801).

(d) Install a WHEEL WELL WARNING LIGHT AND BELL INOP placard on the captain's and first officer's FIRE WARN lights, on the P7 panel.

SUBTASK 26-00-00-040-003

(4) Method 2: deactivate the detection elements of the wheel well fire detection system.

NOTE: NOTE: When in dual loop (normal mode) operations, both MWW LOOP A and MWW LOOP B sensors are required to be in a state other than STANDBY in order annunciate a wheel well fire. If only one sensor is faulted (MWW LOOP A or MWW LOOP B), it is recommended to put the CODM to the single loop configuration (TASK 26-00-00-040-804). This is to maintain wheel well fire detection system coverage rather than disabling the wheel well fire detection system entirely.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the terminal lugs A and B on both sides of the M270 and M3068 detection elements.

(b) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

1) Make sure that all tests pass except for the wheel well fire function with the disabled element.

(c) Stow the wiring and terminal lugs on either side of M270 and M3068 detection elements.

SUBTASK 26-00-00-410-011

(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 26-00-00-440-802

7. DDG 26-18-01A Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Operates Normally - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the wheel well fire detection systems inoperative and the brake temperature monitoring system (BTMS) operates normally.

B. References

Reference	Title
22-11-33-000-801	Flight Control Computer Removal (P/B 401)
22-11-33-400-801	Flight Control Computer Installation (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Wheel Well Fire Detection Systems Activation

SUBTASK 26-00-00-040-004

(1) Install the two light bulbs in the WHEEL WELL light on the P8-1 panel.

(a) Remove an INOP - FIRE BELL DISCONNECTED placard from the switch.

SUBTASK 26-00-00-040-025

(2) Remove a WHEEL WELL FIRE DETECTION SYSTEM INOP placard from the OVHT/FIRE TEST switch, on the P8-1 overheat and fire protection panel.

SUBTASK 26-00-00-010-002

(3) Remove Flight Control Computer (FCC) B to get access to the wiring for the fire bell (TASK 22-11-33-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-00-930-001

(4) Install the wire on D742, pin 16 and pin 25 on the E1-4 shelf.

(a) Remove the WHEEL WELL WARNING LIGHT AND BELL INOP placard from the captain's and first officer's FIRE WARN lights, on the P7 panel.

SUBTASK 26-00-00-410-001

(5) Install FCC B (TASK 22-11-33-400-801).

SUBTASK 26-00-00-420-006

(6) If it is necessary, install the terminal lugs A and B on both sides of the M270 and M3068 detection elements.

——— END OF TASK ———

TASK 26-00-00-040-803

8. DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative Or Not Installed - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the wheel well fire detection system inoperative and the brake temperature monitoring system (BTMS) operates inoperative or not installed.

B. References

Reference	Title
22-11-33-000-801	Flight Control Computer Removal (P/B 401)
22-11-33-400-801	Flight Control Computer Installation (P/B 401)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Wheel Well Fire Detection System Deactivation

SUBTASK 26-00-00-010-003

(1) Get access to the wheel well fire detection system compartment.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-00-00-740-002

(2) Deactivate the wheel well fire detection system, use one of the two methods.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-00-040-005

(3) Method 1: deactivate WHEEL WELL light and wheel well fire detection system input to fire warning system.

(a) Remove the two light bulbs from the WHEEL WELL light on the P8-1 panel.

1) Install an INOP - FIRE BELL DISCONNECTED placard on the light switch.

(b) Install a WHEEL WELL FIRE DETECTION SYSTEM INOP placard on the OVHT/FIRE TEST switch, on the P8-1 overheat and fire protection panel.

(c) Disconnect the fire bell.

1) Remove Flight Control Computer (FCC) B to get access to the wiring for the fire bell (TASK 22-11-33-000-801).

2) Disconnect the wires from D742, pin 16 and pin 25 on the E1-4 shelf.

3) Cap and stow the wires.

4) Install the FCC B (TASK 22-11-33-400-801).

(d) Install a WHEEL WELL WARNING LIGHT AND BELL INOP placard on the captain's and first officer's FIRE WARN lights, on the P7 panel.

SUBTASK 26-00-00-040-006

(4) Method 2: deactivate the detection elements of the wheel well fire detection system.

NOTE: When in dual loop (normal mode) operations, both MWW LOOP A and MWW LOOP B sensors are required to be in a state other than STANDBY in order annunciate a wheel well fire. If only one sensor is faulted (MWW LOOP A or MWW LOOP B), it is recommended to put the CODM to the single loop configuration (TASK 26-00-00-040-804). This is to maintain wheel well fire detection system coverage rather than disabling the wheel well fire detection system entirely.

(a) Disconnect the terminal lugs A and B on both sides of the M270 and M3068 detection elements.

(b) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

1) Make sure that all tests pass except for the wheel well fire function with the disabled element.

(c) Stow the wiring and terminal lugs on either side of M270 and M3068 detection elements.

SUBTASK 26-00-00-410-012

(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 26-00-00-440-803

9. DDG 26-18-01B Wheel Well Fire Detection System Inoperative - Brake Temperature Monitoring System Inoperative Or Not Installed - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the wheel well fire detection systems inoperative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
22-11-33-000-801	Flight Control Computer Removal (P/B 401)
22-11-33-400-801	Flight Control Computer Installation (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Wheel Well Fire Detection Systems Activation

SUBTASK 26-00-00-010-013

(1) Get access to the wheel well fire detection system compartment.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-00-00-040-007

(2) Install the two light bulbs in the WHEEL WELL light on the P8-1 panel.

(a) Remove an INOP - FIRE BELL DISCONNECTED placard from the switch.

SUBTASK 26-00-00-040-026

(3) Remove a WHEEL WELL FIRE DETECTION SYSTEM INOP placard from the OVHT/FIRE TEST switch, on the P8-1 overheat and fire protection panel.

SUBTASK 26-00-00-010-004

(4) Remove Flight Control Computer (FCC) B to get access to the wiring for the fire bell (TASK 22-11-33-000-801).

SUBTASK 26-00-00-930-002

(5) Install the wire on D742, pin 16 and pin 25 on the E1-4 shelf.

(a) Remove the WHEEL WELL WARNING LIGHT AND BELL INOP placard from the captain's and first officer's FIRE WARN lights, on the P7 panel.

SUBTASK 26-00-00-410-002

(6) Install FCC B (TASK 22-11-33-400-801).

SUBTASK 26-00-00-420-007

(7) If it is necessary, install the terminal lugs A and B on both sides of the M270 and M3068 detection elements.

SUBTASK 26-00-00-410-014

(8) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

——— END OF TASK ———

TASK 26-00-00-040-804

10. DDG 26-18-01-01 Wheel Well Fire Detection Loops Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the wheel well fire detection loops inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Wheel Well Fire Detection System Deactivation

SUBTASK 26-00-00-010-005

(1) Get access to the wheel well fire detection system Compartment Overheat Detection Module (CODM).

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-00-00-040-008

(2) On the E1-4 rack, do a CODM local test, LOC TEST and record the indication of a fault for the main wheel well, MWW, sensor loops.

SUBTASK 26-00-00-040-009

(3) Make sure that the CODM is in the single loop configuration.

(a) Make sure that the "SINGLE LOOP OPS" appears on the CODM front panel display.

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(c) On the P8 overheat and fire protection panel, hold the TEST switch in the OVHT/FIRE position.

(d) Make sure that these indications occur:

1) The fire warning bell comes on.

2) The master FIRE WARN lights come on.

3) The WHEEL WELL light comes on.

(4) Release the TEST switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(5) Make sure that these indications occur:

(a) The fire warning bell stops.

(b) The master FIRE WARN lights go off.

(c) The WHEEL WELL light goes off.

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-00-00-410-013

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 26-00-00-040-810

11. DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Wing Body Overheat Test System inoperative.

B. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Check the Wing-Body Overheat Test System

SUBTASK 26-00-00-040-017

(1) For each operative detector loop, apply heat from an applicable heat source for a period of approximately 30 seconds.

NOTE: Apply at any point along the single loop sensing elements for the left and right wing-body overheat detector.

NOTE: The test heat device must not produce heat more than 450°F (232°C). This temperature limit should be followed in the event that fuel vapors exist in the area.

SUBTASK 26-00-00-040-018

(2) When a detector element in the left system is heated, make sure that these lights come on:

(a) Left WING-BODY OVERHEAT light on the Air Conditioning Panel (P5).

(b) The AIR COND light on the glareshield P7.

(c) The MASTER CAUTION lights.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-00-040-019

(3) When a detector element in the right system is heated, make sure that these lights come on:

(a) Right WING-BODY OVERHEAT light on the Air Conditioning Panel (P5).

(b) The AIR COND light on the glareshield P7.

(c) The MASTER CAUTION lights.

SUBTASK 26-00-00-420-008

(4) Install the TEST SWITCH INOP placard on the OVHT TEST switch on the Air Conditioning panel, P5-10.

——— END OF TASK ———

TASK 26-00-00-440-808

12. DDG 26-18-02-03 Wing-Body Overheat Test System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Wing/Body Overheat Test Systems Inoperative.

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Wing/Body Overheat Test Systems Activation

SUBTASK 26-00-00-040-020

(1) Remove the TEST SWITCH INOP placard from the OVHT TEST switch on the Air Conditioning Panel, P5-10.

——— END OF TASK ———

TASK 26-00-00-040-819

13. DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Electronic Equipment (EE) cooling smoke detectors inoperative.

B. References

Reference	Title
26-19-00-710-802	E/E Cooling Smoke Detector - Operational Test (P/B 501)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Smoke Detector Deactivation

SUBTASK 26-00-00-740-003

(1) Do these steps to make sure that one EE cooling smoke detector operates correctly (TASK 26-19-00-710-802):

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) On the forward panel of J23 box, push and release the EQUIP COOL SMK DET TEST button (S1204).

(c) Make sure that the green EQUIP COOL SMK DET SPLY light (L1293) or the green EQUIP COOL SMK DET EXH light (L1294) comes on.

SUBTASK 26-00-00-860-010

(2) If the EQUIP COOL SMK DET SPLY light (L1293) does not come on, then do this step:

(a) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

SUBTASK 26-00-00-860-011

(3) If the EQUIP COOL SMK DET EXH light (L1294) does not come on, then do this step:

(a) Open this circuit breaker and install safety lock:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

SUBTASK 26-00-00-410-010

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 26-00-00-440-817

14. DDG 26-19-01-01 Equipment Cooling Smoke Detector Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the equipment cooling smoke detector inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Equipment Cooling Smoke Detector Activation

SUBTASK 26-00-00-860-012

(1) Remove the applicable safety locks and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

——— END OF TASK ———

TASK 26-00-00-040-820

15. DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the equipment cooling EQUIP SMOKE light inoperative.

B. References

Reference	Title
26-19-00-710-802	E/E Cooling Smoke Detector - Operational Test (P/B 501)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Equipment Cooling Smoke Detection Test

SUBTASK 26-00-00-740-004

(1) Make sure that the equipment cooling smoke detectors are operating correctly (TASK 26-19-00-710-802).

SUBTASK 26-00-00-420-009

(2) Install an INDICATOR INOP placard on the EQUIP SMOKE light on the overhead panel, P5.

——— END OF TASK ———

TASK 26-00-00-440-819

16. DDG 26-19-01-02 Equipment Cooling Smoke Detection EQUIP SMOKE Light Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the equipment cooling smoke detection EQUIP SMOKE light inoperative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Equipment Cooling SMOKE LIGHT Activation

SUBTASK 26-00-00-020-010

(1) Remove the INDICATOR INOP placard from the EQUIP SMOKE light on the overhead panel, P5.

——— END OF TASK ———

TASK 26-00-00-040-821

17. DDG 26-20-01-02 APU Bottle Discharged Light Inoperative - Preparation

(Figure 901)

A. General

(1) This task prepares the airplane for flight with the APU bottle discharged light inoperative.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left

C. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

D. APU Bottle Discharged Light Deactivation

SUBTASK 26-00-00-040-024

(1) Do these steps to make sure that the APU bottle pressure is acceptable for dispatch:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

(b) Make sure that the pressure on the gauge is greater than 400 psig (2758 kPa).

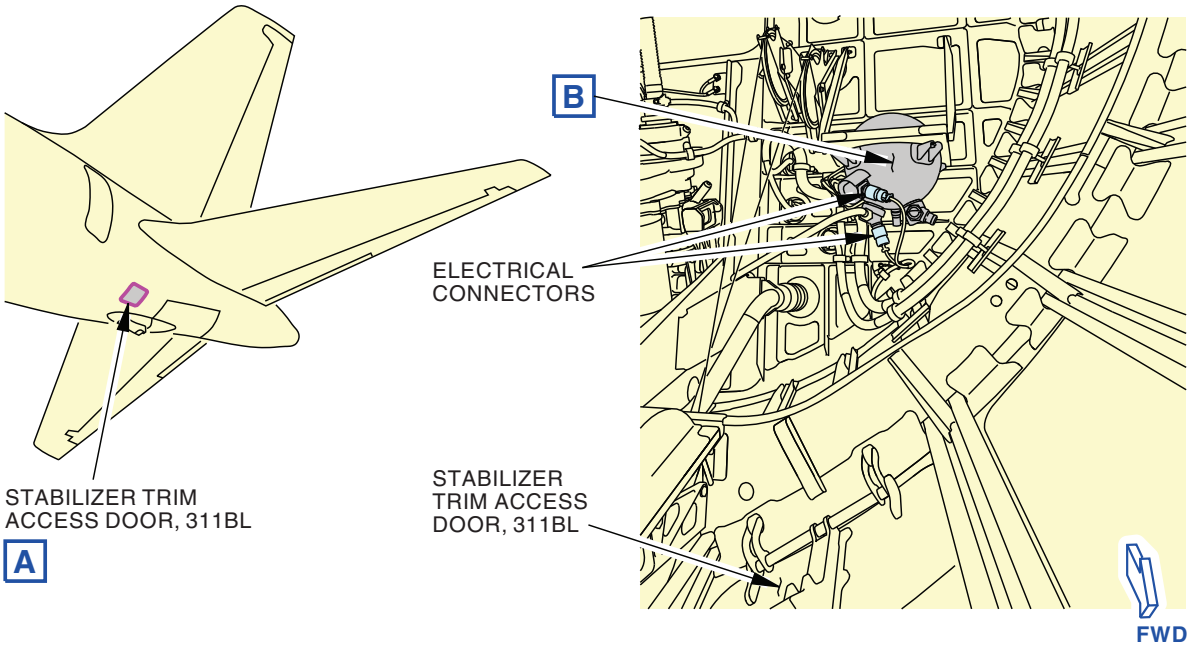
(c) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

(d) Install an INDICATOR INOP placard on the APU BOTTLE DISCHARGED light on P8-1 panel.

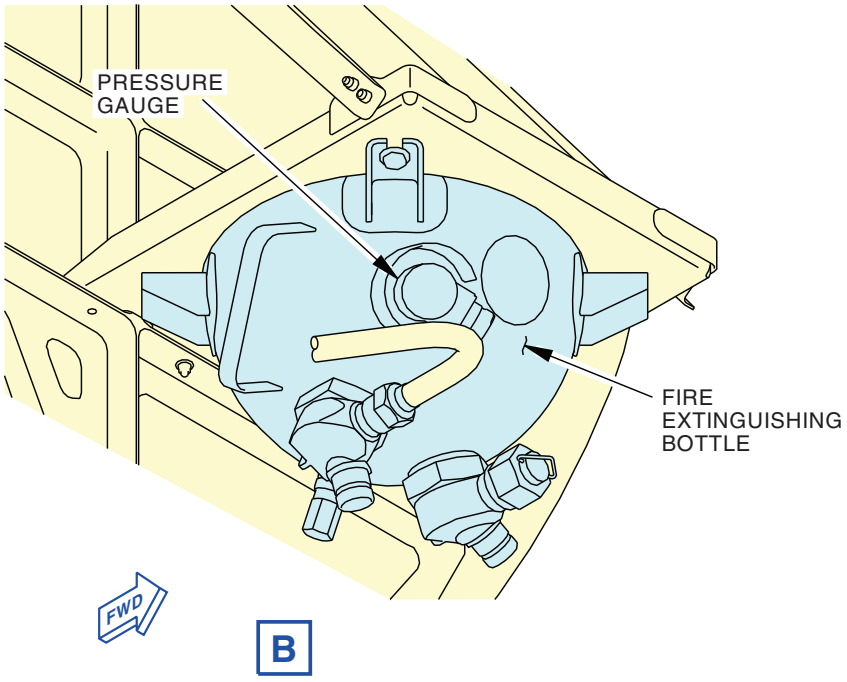
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACCESS DOOR, 311BL

A



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APU Fire Extinguishing Bottle Inspection
Figure 901/26-00-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-00-440-820

18. DDG 26-20-01-02 APU Bottle Discharged Light Inoperative - Restoration

A. General

(1) This task restores the airplane to its usual condition.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left

C. APU Bottle Discharged Light Activation

SUBTASK 26-00-00-440-005

(1) Remove the INDICATOR INOP placard from the APU BOTTLE DISCHARGED light on the P8-1 fire protection panel.

——— END OF TASK ———

TASK 26-00-00-040-805

19. DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Engine/APU Fire Extinguisher Test System Inoperative.

(2) For each inoperative squib test circuit, make sure that the failure is in the squib test circuit by making sure that the squib circuit operates normally once each flight day.

B. References

Reference	Title
26-21-00-730-801	Engine Fire Extinguishing Bottle Squib Circuit Test (P/B 501)
26-22-00-730-801	APU Fire Extinguishing Bottle Squib Circuit Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-5826	Lamp - 28V, 40 MA
STD-6286	Jumper Wire
STD-15135	Cable - Jumper (Make from 0.5 inch length 20 AWG wire, crimp or solder both end to pin contact size 16 P/N M39029/31-228)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Engine/APU Extinguisher System Deactivation

SUBTASK 26-00-00-700-003

(1) Method 1: Make sure that the squib circuit operates correctly, for each squib test circuit that does not operate:

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WARNING USE THE CORRECT LAMP WHEN YOU MAKE THE SQUIB TESTER. IF YOU USE AN INCORRECT LAMP, THE HIGH ELECTRICAL CURRENT CAN CAUSE THE SQUIB TO FIRE. THIS WILL RELEASE THE EXTINGUISHING AGENT. INJURIES TO PERSONNEL CAN OCCUR.

(a) Make a squib test lamp with pin contacts on the end of the pigtail leads.

1) Use a lamp, COM-5826, in an applicable holder and Jumper Wire, STD-6286, or jumper cable, STD-15135.

(b) Open these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

(c) Pull the engine 1, engine 2, and APU fire handles, on the P8-1 panel.

WARNING BE CAREFUL WHEN YOU OPEN THE P6-2 CIRCUIT BREAKER PANEL. THE P6-2 CIRCUIT BREAKER PANEL CONTAINS HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Loosen the screws to lower the P6-2 circuit breaker panel.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE ENGINES AND APU. ACCIDENTAL DISCHARGE OF THE FIRE EXTINGUISHING BOTTLE CAN CAUSE INJURY TO PERSONS.

(e) Connect the test lamp across the circuit breaker shown in the table.

(f) Turn the applicable fire handle in the direction as shown in the table.

1) Make sure that the lamp comes on for each position.

NOTE: If the lamp comes on, the squib is satisfactory.

Table 901/26-00-00-993-802

CIRCUIT BREAKER	FIRE HANDLE	DIRECTION
C296 FIRE PROTECTION EXTINGUISHERS LEFT	ENG 1 ENG 2	LEFT LEFT
C297 FIRE PROTECTION EXTINGUISHERS RIGHT	ENG 1 ENG 2	RIGHT RIGHT
C452 FIRE PROTECTION EXTINGUISHERS APU	APU APU	LEFT RIGHT
C1021 FIRE PROTECTION EXTINGUISHERS ALTN LEFT	ENG 1 ENG 2	LEFT LEFT
C1022 FIRE PROTECTION EXTINGUISHERS ALTN RIGHT	ENG 1 ENG 2	RIGHT RIGHT

(g) Remove the test lamp after you do the check for all of the applicable circuit breakers.

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(h) Push the engine 1, engine 2, and APU fire handles, on the P8-1 panel, to the normal position.

(i) Close the P6-2 panel and install the screws.

(j) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-00-00-040-010

(2) Method 2: Make sure that the Engine/APU fire extinguisher squib circuit operates correctly, do the applicable task:

- Engine Fire Extinguishing Bottle Squib Circuit Test, TASK 26-21-00-730-801
- APU Fire Extinguishing Bottle Squib Circuit Test, TASK 26-22-00-730-801.

SUBTASK 26-00-00-930-003

(3) Install a TEST INOP placard on the EXT TEST switch, on the P8-1 panel.

——— END OF TASK ———

TASK 26-00-00-440-818

20. DDG 26-20-02 Engine/APU Fire Extinguisher Test System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Engine/APU Squib Test Inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Engine/APU Extinguisher System Activation

SUBTASK 26-00-00-930-004

(1) Remove the TEST INOP placard from the EXT TEST switch, on the P8-1 panel.

——— END OF TASK ———

TASK 26-00-00-040-811

21. DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the lower cargo compartment DISCH light inoperative.

B. References

Reference	Title
26-00-02-000-801	Cargo Fire Control Panel Removal (P/B 401)
26-00-02-400-801	Cargo Fire Control Panel Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Lower Cargo Compartment DISCH Light Deactivation

SUBTASK 26-00-00-000-001

(1) Do this task: Cargo Fire Control Panel Removal, TASK 26-00-02-000-801.

SUBTASK 26-00-00-020-011

(2) Do the steps that follow for an inoperative DISCH light circuit to make sure that the fire extinguisher bottles have an adequate charge:

NOTE: It is necessary to use a 28VDC circuit test lamp with pigtail leads that terminate in pin contacts in an applicable holder. To make the lamp, lamp P/N 116-565-1042-001 (28 volt, 40 milliamps) or the equivalent is suggested, but you can use an applicable lamp or a digital multimeter.

(a) For bottle 1, do these steps:

1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	17	C01526	CARGO FIRE EXT 1

2) Connect the test lamp or digital multimeter between the airplane side of connector D12760, pin 48 (the +28 volt source for bottle 1) and pin 28 (pressure switch ground).

3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	17	C01526	CARGO FIRE EXT 1

4) If the test lamp stays off or digital multimeter shows an OPEN circuit, the fire extinguisher bottle has an adequate charge.

5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	17	C01526	CARGO FIRE EXT 1

6) Remove the test lamp.

7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	17	C01526	CARGO FIRE EXT 1

(b) For bottle 2, do these steps:

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1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2

2) Connect the test lamp or digital multimeter between the airplane side of connector D12760, pin 45 (the +28 volt source for bottle 2) and pin 28 (pressure switch ground).

3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2

4) If the test lamp stays off or digital multimeter shows an OPEN circuit, the fire extinguisher bottle has an adequate charge.

5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2

6) Remove the test lamp.

7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2

SUBTASK 26-00-00-400-004

(3) Do this task: Cargo Fire Control Panel Installation, TASK 26-00-02-400-801.

——— END OF TASK ———

TASK 26-00-00-440-809

22. DDG 26-23-01-01 Lower Cargo Compartment DISCH Light Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the lower cargo compartment DISCH light inoperative.

B. References

Reference	Title
26-23-00-730-802	Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left

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(Continued)

Zone	Area
212	Flight Compartment - Right

D. Lower Cargo Compartment DISCH Light Activation

SUBTASK 26-00-00-810-013

(1) Correct the fault.

(a) Find the fault code or description of the fault that occurred.

(b) Refer to the IFIM and do the applicable procedure(s).

SUBTASK 26-00-00-730-001

(2) Do this task: Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test, TASK 26-23-00-730-802.

——— END OF TASK ———

TASK 26-00-00-040-812

23. DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Preparation

A. General

(1) This task gives maintenance steps that prepare the airplane for flight with the lower cargo compartment extinguisher bottle pressure switch inoperative.

B. References

Reference	Title
26-23-01-000-801	Cargo Fire Extinguisher Bottle Removal (P/B 401)
26-23-01-400-801	Cargo Fire Extinguisher Bottle Installation (P/B 401)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Lower Cargo Compartment Extinguisher Bottle Pressure Switch Deactivation

SUBTASK 26-00-00-020-004

WARNING DO NOT TOUCH THE PINS ON THE SQUIB. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do the steps that follow to deactivate the applicable extinguisher bottle pressure switch:

(a) Do this task: Cargo Fire Extinguisher Bottle Removal, TASK 26-23-01-000-801.

(b) Disconnect the electrical connectors from the bottle.

(c) Cap and stow the bottle pressure switch electrical connector (D16576 or D16582).

SUBTASK 26-00-00-280-001

(2) Weigh the applicable extinguisher bottle on a scale that is accurate to within +/-0.1 lb (0.0 kg).

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(a) For P/N 473957-4 and 475976-1, make sure that the bottle weight is within 0.25 lb (0.11 kg) of the service weight marked on the bottle name plate.

(b) For P/N 475975-1, make sure that the bottle weight is within 0.13 lb (0.06 kg) of the service weight marked on the bottle name plate.

SUBTASK 26-00-00-420-002

(3) Do this task: Cargo Fire Extinguisher Bottle Installation, TASK 26-23-01-400-801.

SUBTASK 26-00-00-930-005

(4) Install the applicable extinguisher bottle pressure switch INOP Placard on the P8 cargo fire panel.

——— END OF TASK ———

TASK 26-00-00-440-810

24. DDG 26-23-01-02 Lower Cargo Compartment Extinguisher Bottle Pressure Switch Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the lower cargo compartment extinguisher bottle pressure switch inoperative.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Lower Cargo Compartment Extinguisher Bottle Pressure Switch Activation

SUBTASK 26-00-00-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-00-010-007

(2) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-00-00-010-012

(3) Remove the lining at the aft end of the forward cargo compartment to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-00-00-440-002

(4) Do the steps that follow to activate the applicable extinguisher bottle pressure switch:

(a) Remove the cap from the pressure switch electrical connector.

WARNING DO NOT TOUCH THE PINS ON THE SQUIB. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Connect the applicable extinguisher bottle pressure switch electrical connector to the pressure switch.

SUBTASK 26-00-00-410-004

(5) Install the liner at the aft end of the forward cargo compartment to the air conditioning distribution bay (TASK 50-11-02-400-801).

SUBTASK 26-00-00-410-005

(6) Close this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-00-00-930-006

(7) Remove the applicable Extinguisher Bottle pressure switch INOP placard on the Cargo Fire panel (P8).

SUBTASK 26-00-00-860-002

(8) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

END OF TASK

TASK 26-00-00-040-814

25. DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the lower cargo compartment EXT light inoperative.

B. References

Reference	Title
26-00-02-000-801	Cargo Fire Control Panel Removal (P/B 401)

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(Continued)

Reference	Title
26-00-02-400-801	Cargo Fire Control Panel Installation (P/B 401)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Lower Cargo Compartment EXT Light Deactivation

SUBTASK 26-00-00-000-002

(1) Do this task: Cargo Fire Control Panel Removal, TASK 26-00-02-000-801.

SUBTASK 26-00-00-750-002

WARNING KEEP PERSONNEL AWAY FROM THE EXTINGUISHER BOTTLE AREA DURING THE SQUIB CIRCUIT TEST. USE A MAXIMUM CURRENT OF 40 MILLIAMPS DURING THE TEST. USE ONLY TEST LAMP P/N 116-565-1042-001 (28 VOLT, 40 MILLIAMPS) OR ITS EQUIVALENT. DO NOT USE A MULTIMETER. IF THE EXTINGUISHER BOTTLE RELEASES THE EXTINGUISHING MATERIAL, IT CAN CAUSE INJURIES TO PERSONNEL.

(2) Do the steps that follow to make sure that the EXT light SQUIB circuit operates correctly:

NOTE: If the test lamp comes on, the squib circuit is good.

(a) Connect the test lamp between D12760 pin 48 (the +28 volt source for bottle 1) and pin 26 (EXT 1 FWD squib).

1) Make sure that the test lamp comes on.

(b) Connect the test lamp between D12760 pin 48 (the +28 volt source for bottle 1) and pin 24 (EXT 1 AFT squib).

1) Make sure that the test lamp comes on.

(c) Connect the test lamp between D12760 pin 45 (the +28 volt source for bottle 2) and pin 12 (EXT 2 FWD squib).

1) Make sure that the test lamp comes on.

(d) Connect the test lamp between D12760 pin 45 (the +28 volt source for bottle 2) and pin 23 (EXT 2 AFT squib).

1) Make sure that the test lamp comes on.

(e) Remove the test lamp.

SUBTASK 26-00-00-400-005

(3) Do this task: Cargo Fire Control Panel Installation, TASK 26-00-02-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-00-440-812

26. DDG 26-23-01-03 Lower Cargo Compartment EXT Light Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the lower cargo compartment EXT light inoperative.

B. References

Reference	Title
26-16-00-710-801	Cargo Bay Smoke Detection - Operational Test (P/B 501)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Lower Cargo Compartment EXT Light Activation

SUBTASK 26-00-00-810-014

(1) Correct the fault.

(a) Find the fault code or description of the fault that occurred.

(b) Refer to the IFIM and do the applicable procedure(s).

SUBTASK 26-00-00-710-011

(2) Do this task: Cargo Bay Smoke Detection - Operational Test, TASK 26-16-00-710-801.

——— END OF TASK ———

TASK 26-00-00-040-807

27. DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the lavatory fire extinguisher system inoperative.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Lavatory Deactivation

SUBTASK 26-00-00-040-012

(1) If the lavatory fire extinguisher system does not operate, do these steps.

(a) Empty the lavatory waste receptacle.

(b) Close and lock the lavatory door.

(c) Install an INOPERATIVE - DO NOT ENTER placard on the lavatory door.

NOTE: Lavatory use or inspections by crew members is allowed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-00-440-805

28. DDG 26-24-01B Lavatory Fire Extinguisher System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Lavatory Fire Extinguisher System Inoperative.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Lavatory Fire Extinguisher System Activation

SUBTASK 26-00-00-800-002

(1) Unlock the lavatory door.

SUBTASK 26-00-00-040-013

(2) Remove the INOP placard from the lavatory door.

——— END OF TASK ———

TASK 26-00-00-040-801

29. DDG 26-26-01 Portable Fire Extinguishers Inoperative (EASA) - Preparation

A. General

(1) This procedure gives the maintenance steps which prepare the airplane for flight with the inoperative portable fire extinguishers.

(2) This procedure is only for aircraft using the EASA MMEL. DDG Maintenance procedures with no regulatory authority identification in the procedure title apply to both FAA and EASA MMELs

B. References

Reference	Title
26-26-01-020-801	Halon Fire Extinguisher Removal (P/B 401)
26-26-02-000-801	Water Fire Extinguisher Removal (P/B 401)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Portable Fire Extinguisher Deactivation

SUBTASK 26-00-00-020-001

(1) Do the applicable task to remove the inoperative portable fire extinguisher:

- Halon Fire Extinguisher Removal, TASK 26-26-01-020-801
- Water Fire Extinguisher Removal, TASK 26-26-02-000-801.

SUBTASK 26-00-00-040-001

(2) Stow the extinguisher.

(a) Put the extinguisher in a bag.

(b) Make sure that the extinguisher can not be used by mistake.

NOTE: The extinguisher will be removed from the airplane at the next available maintenance base.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-00-610-001

(3) Make sure that the necessary supply of serviceable extinguishers is available throughout the airplane.

——— END OF TASK ———

TASK 26-00-00-440-801

30. DDG 26-26-01 Portable Fire Extinguishers Inoperative (EASA) - Restoration

A. General

(1) This procedure puts the airplane back to its usual condition after operation with the inoperative Portable Fire Extinguishers.

(2) This procedure is only for aircraft using the EASA MMEL. DDG Maintenance procedures with no regulatory authority identification in the procedure title apply to both FAA and EASA MMELs

B. References

Reference	Title
26-26-01-430-801	Halon Fire Extinguisher - Installation (P/B 401)
26-26-02-400-801	Water Fire Extinguisher Installation (P/B 401)

C. Reactivation of the Inoperative Portable Fire Extinguishers

SUBTASK 26-00-00-420-001

(1) Do the applicable task to install the portable fire extinguisher:

- Halon Fire Extinguisher - Installation, TASK 26-26-01-430-801
- Water Fire Extinguisher Installation, TASK 26-26-02-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE AND APU FIRE CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Engine and APU Fire Control Panel Removal

(2) Engine and APU Fire Control Panel Installation

(3) Engine and APU Fire Shutoff Switch Assembly Removal

(4) Engine and APU Fire Shutoff Switch Assembly Installation.

B. The Engine and APU Fire Control Panel is installed on the P8 aisle control stand in the flight compartment.

TASK 26-00-01-000-801

2. Engine and APU Fire Control Panel Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Engine and Auxiliary Power Unit (APU) Fire Control Panel.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 26-00-01-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
B	19	C01344	APU FIRE SW POWER
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND
C	10	C00284	PANEL & INSTR ELEX PANEL
C	12	C01276	MASTER CAUTION ANNUNCIATOR CONT 2
D	8	C00701	EMER PANEL LTG
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	14	C00033	AUX POWER UNIT CONT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
F	12	C01285	GENERATOR APU GEN CONT UNIT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	9	C01326	APU GEN CONT UNIT

SUBTASK 26-00-01-860-002

(2) If it is necessary, set the engine start levers to IDLE to get access to remove the engine and APU fire control panel [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Engine and APU Fire Control Panel Removal

SUBTASK 26-00-01-860-009

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Before you touch the engine and APU fire control panel [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 26-00-01-020-001

(2) Loosen the four quarter turn fasteners that attach the engine and APU fire control panel [1] to the frame.

SUBTASK 26-00-01-020-002

(3) Slide the engine and APU fire control panel [1] from the frame.

SUBTASK 26-00-01-020-003

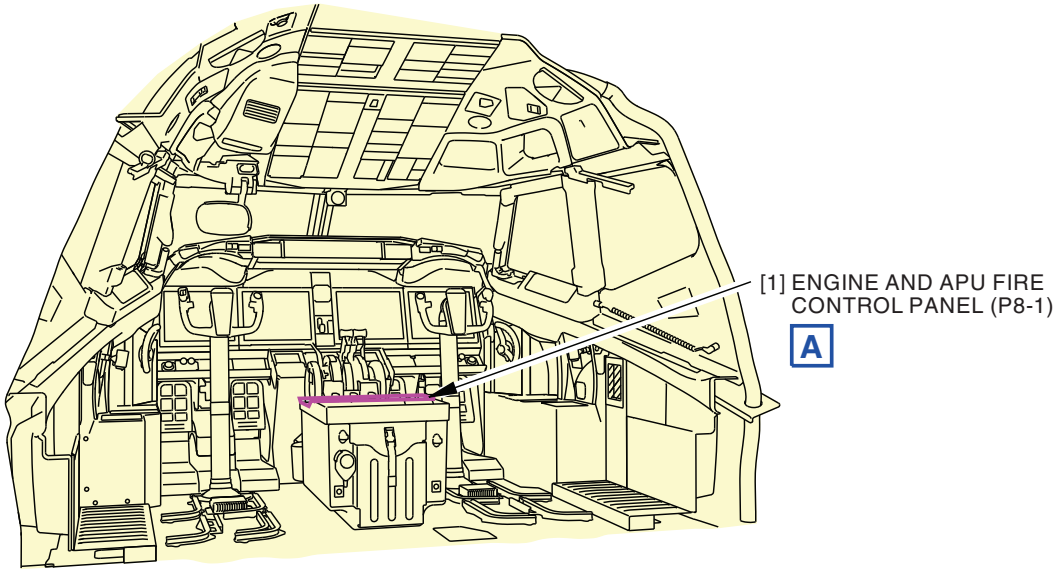
(4) Tag and remove the five electrical connectors.

SUBTASK 26-00-01-860-003

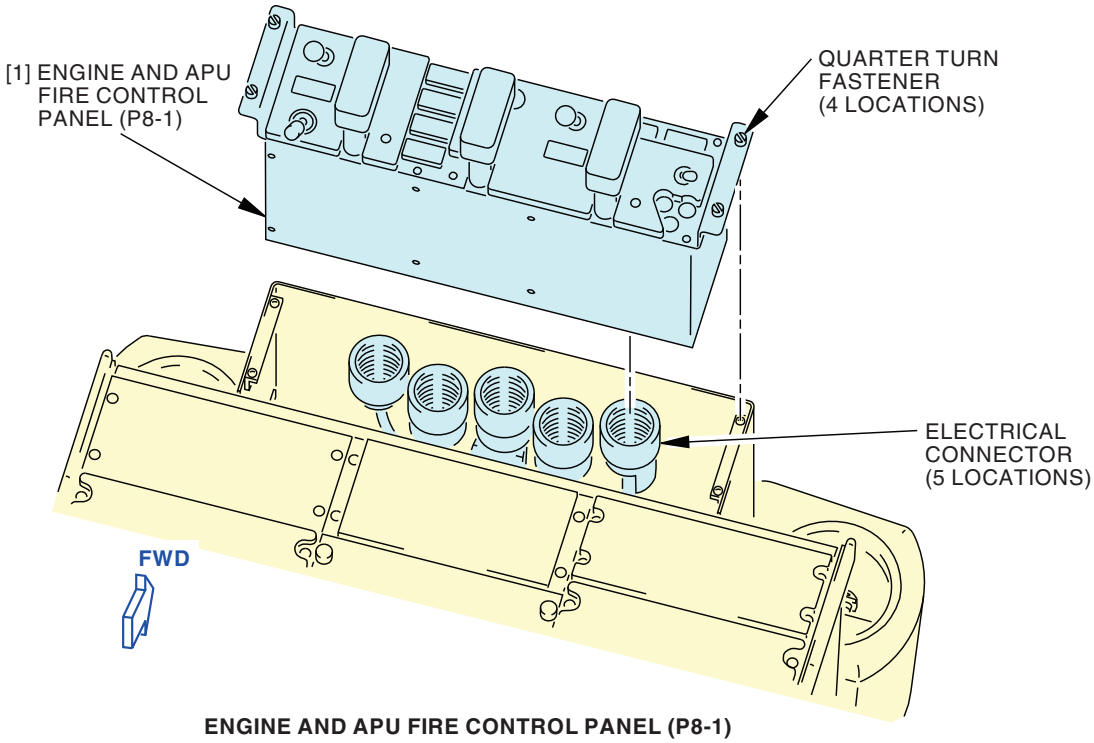
(5) Make sure that the engine start levers switch to CUTOFF.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



ENGINE AND APU FIRE CONTROL PANEL (P8-1)

A

2412058 S00061529827_V2

Fire Control Panel Installation
Figure 401/26-00-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-01-400-801

3. Engine and APU Fire Control Panel Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Engine and Auxiliary Power Unit (APU) Fire Control Panel.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-10-00-710-801	Fire and Overheat Detection System - Operational Test (P/B 501)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)
26-21-00-720-802	Engine Fire Switch System Shutdown Test (P/B 501)
26-21-00-730-801	Engine Fire Extinguishing Bottle Squib Circuit Test (P/B 501)
26-21-00-730-802	Engine Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)
26-22-00-720-801	APU Fire Switch System Shutdown Test (P/B 501)
26-22-00-730-801	APU Fire Extinguishing Bottle Squib Circuit Test (P/B 501)
26-22-00-730-802	APU Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Engine and APU fire control panel	26-00-01-01-010	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Installation

SUBTASK 26-00-01-860-005

(1) If it is necessary, set the engine start levers to IDLE to get access to install the engine and APU fire control panel [1].

F. Engine and APU Fire Control Panel Installation

SUBTASK 26-00-01-860-010

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Before you touch the engine and APU fire control panel [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-00-01-200-001

(2) Make sure that the pins are not bent or damaged and the connector is not damaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-01-420-001

(3) Install the five connectors in accordance with tag information and remove the tags.

SUBTASK 26-00-01-020-004

(4) Slide the engine and APU fire control panel [1] into the frame and secure with the four quarter turn fasteners.

SUBTASK 26-00-01-860-006

(5) Make sure that the engine start levers switch to CUTOFF.

G. Engine and APU Fire Detection Control Unit Installation Test

SUBTASK 26-00-01-720-001

(1) Do this task: Fire and Overheat Detection System - Operational Test, TASK 26-10-00-710-801.

SUBTASK 26-00-01-720-002

(2) Do this task: Engine Fire Switch System Shutdown Test, TASK 26-21-00-720-802.

SUBTASK 26-00-01-710-001

(3) Do this task: Engine Fire Extinguishing Bottle Squib Circuit Test, TASK 26-21-00-730-801.

SUBTASK 26-00-01-720-003

(4) Do this task: Engine Fire Extinguishing Bottle Pressure Switch Circuit Test, TASK 26-21-00-730-802.

SUBTASK 26-00-01-720-004

(5) Do this task: APU Fire Switch System Shutdown Test, TASK 26-22-00-720-801.

SUBTASK 26-00-01-720-005

(6) Do this task: APU Fire Extinguishing Bottle Squib Circuit Test, TASK 26-22-00-730-801.

SUBTASK 26-00-01-720-006

(7) Do this task: APU Fire Extinguishing Bottle Pressure Switch Circuit Test, TASK 26-22-00-730-802.

SUBTASK 26-00-01-710-002

(8) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-00-01-860-007

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
B	19	C01344	APU FIRE SW POWER
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND
C	10	C00284	PANEL & INSTR ELEX PANEL
C	12	C01276	MASTER CAUTION ANNUNCIATOR CONT 2
D	8	C00701	EMER PANEL LTG
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
E	11	C00313	INDICATOR MASTER DIM SECT 1
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	14	C00033	AUX POWER UNIT CONT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
F	12	C01285	GENERATOR APU GEN CONT UNIT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	9	C01326	APU GEN CONT UNIT

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-01-000-802

4. Engine and APU Fire Shutoff Switch Assembly Removal

(Figure 402)

A. General

(1) This task gives instructions to remove the Engine 1, Engine 2, or Auxiliary Power Unit (APU) Fire Shutoff Switch Assembly.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 26-00-01-865-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
B	19	C01344	APU FIRE SW POWER
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	5	C00540	FUEL SPAR VALVE IND
C	12	C01276	MASTER CAUTION ANNUNCIATOR CONT 2
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	14	C00033	AUX POWER UNIT CONT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
F	12	C01285	GENERATOR APU GEN CONT UNIT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	9	C01326	APU GEN CONT UNIT

E. Engine and APU Fire Shutoff Switch Removal

SUBTASK 26-00-01-860-011

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Before you touch the engine and APU fire control panel [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 26-00-01-030-001

(2) Loosen the quarter turn fasteners that attach the engine and APU fire control panel [1] to the frame.

SUBTASK 26-00-01-030-002

(3) Carefully pull the engine and APU fire control panel [1] away from the frame insert.

(a) Make sure that you can get sufficient access to the connectors that are connected to the fire shutoff switches.

SUBTASK 26-00-01-030-003

(4) Disconnect the electrical connector from the applicable fire shutoff switch.

SUBTASK 26-00-01-030-004

(5) Put a protective cover on the connector of the applicable fire shutoff switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-00-01-480-001

(6) Put a protective cover on the airplane electrical connector disconnected from the applicable fire shutoff switch.

SUBTASK 26-00-01-030-005

(7) Remove the cover [14].

(a) Remove the screws [15] that attach the cover [14] to the engine and APU fire control panel [1].

(b) Carefully remove the sealant between the cover [14] and the baseplate.

SUBTASK 26-00-01-020-005

(8) If necessary, remove the lightplate [4].

(a) Remove the screws [5].

SUBTASK 26-00-01-020-006

(9) If necessary, remove the lightplate [6].

(a) Remove the bell cutout handle [2].

1) Remove the screws [3].

(b) Remove the screws [7].

SUBTASK 26-00-01-020-007

(10) Remove the applicable fire shutoff switch:

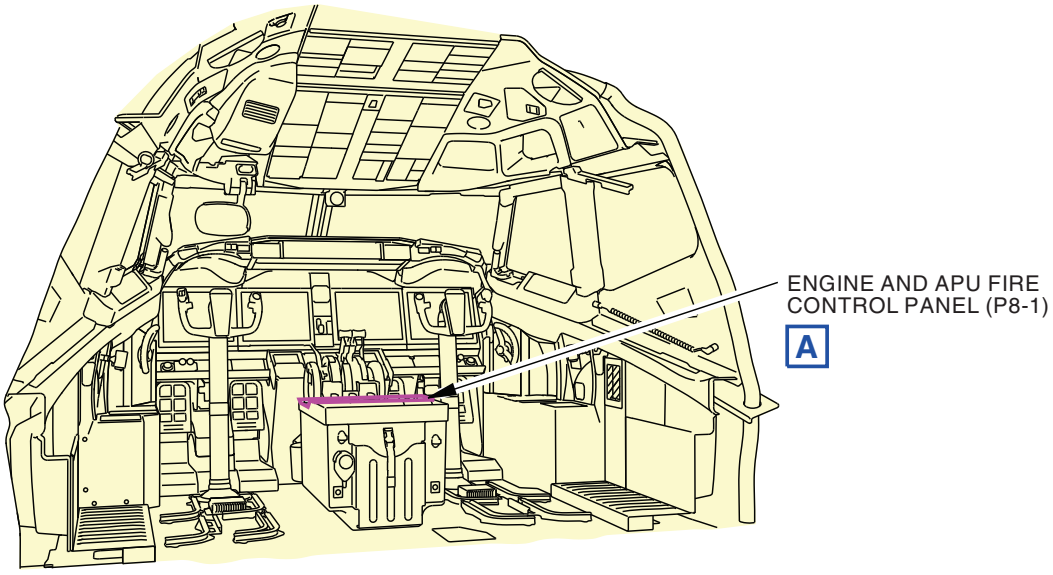
(a) For the engine 1 fire shutoff switch [9], remove the screws [8].

(b) For the engine 2 fire shutoff switch [11], remove the screws [10].

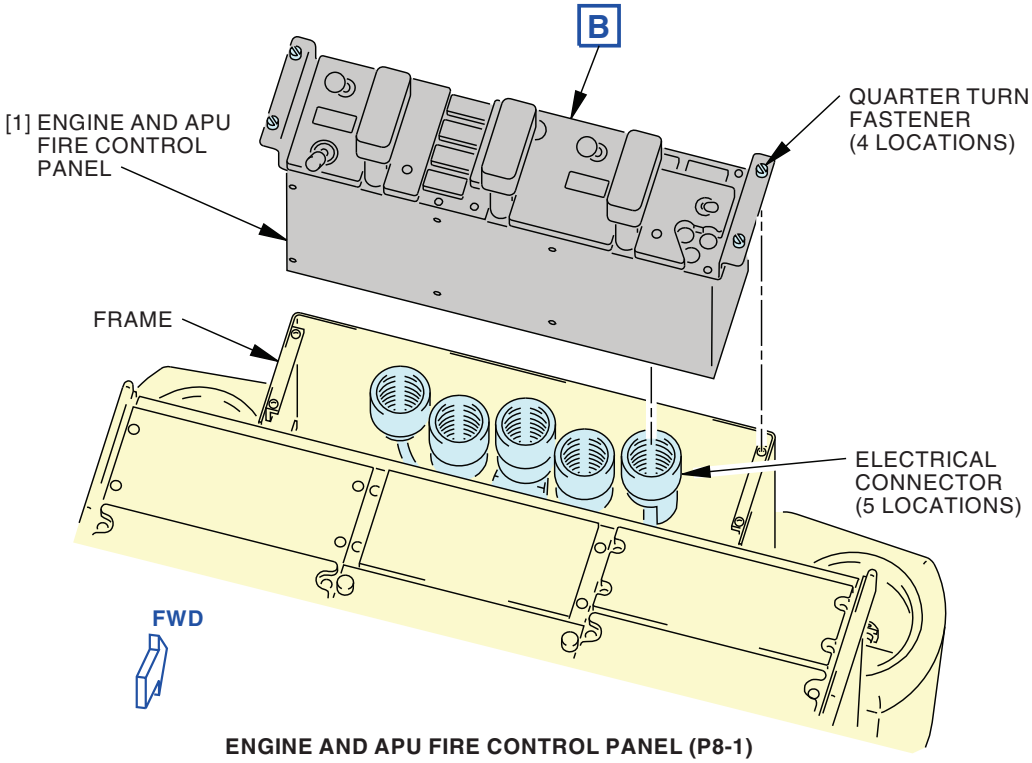
(c) For the APU fire shutoff switch [13], remove the screws [12].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



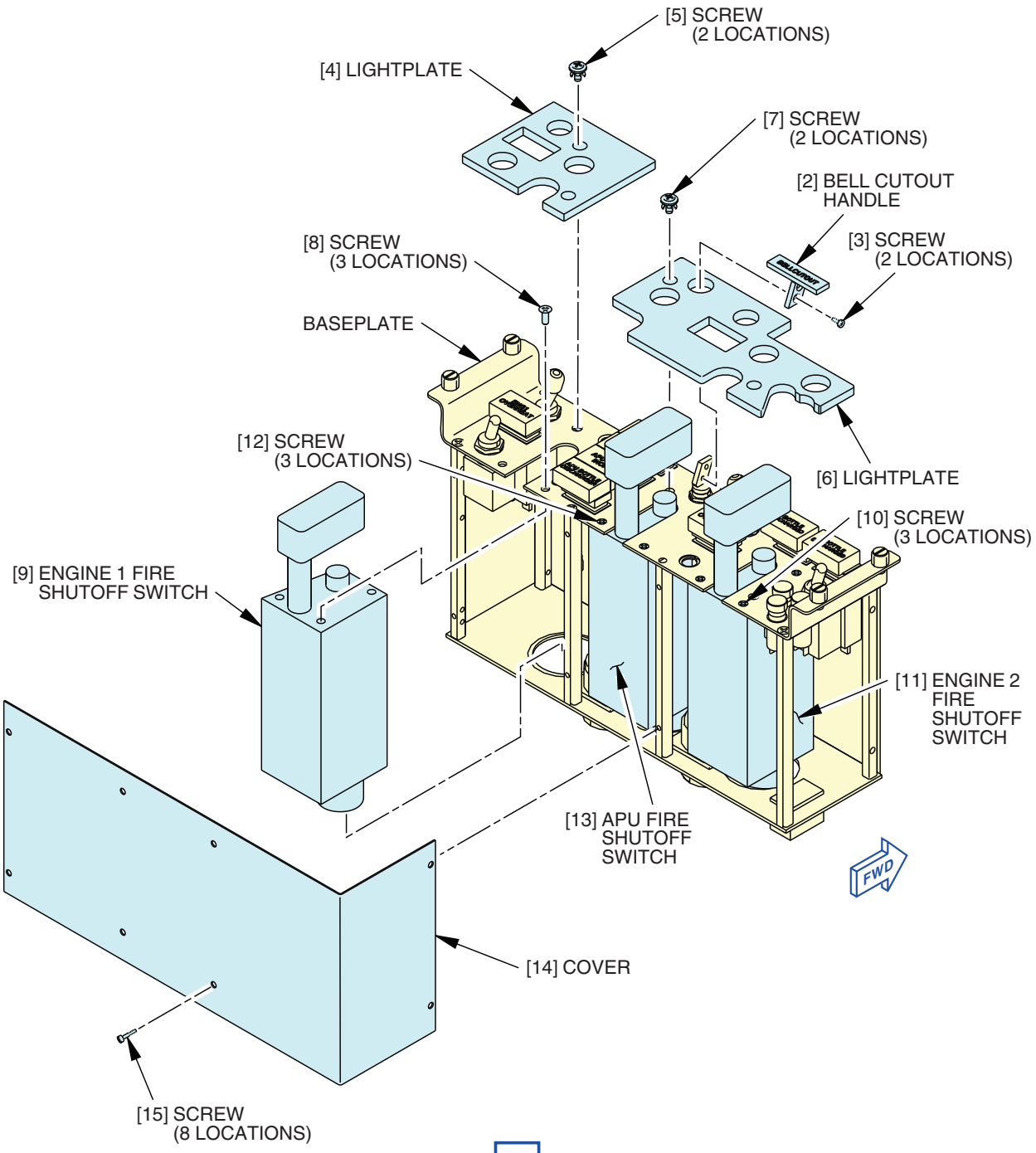
ENGINE AND APU FIRE CONTROL PANEL (P8-1)

A

2412059 S00061529830_V3

Engine and APU Fire Shutoff Switch Assembly Installation
Figure 402/26-00-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine and APU Fire Shutoff Switch Assembly Installation
Figure 402/26-00-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-01-400-802

5. Engine and APU Fire Shutoff Switch Assembly Installation

(Figure 402)

A. General

(1) This task gives instructions to install the Engine 1, Engine 2, or Auxiliary Power Unit (APU) Fire Shutoff Switch Assembly.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-21-00-720-802	Engine Fire Switch System Shutdown Test (P/B 501)
26-22-00-720-801	APU Fire Switch System Shutdown Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A00142	Sealant - Temperature Resistant, Fuel Pressure, And Weather Sealant	BMS5-44
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Engine 1 fire shutoff switch	26-00-01-01-020	PTF ALL
11	Engine 2 fire shutoff switch	26-00-01-01-025	PTF ALL
13	APU fire shutoff switch	26-00-01-01-030	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Engine and APU Fire Shutoff Switch Installation

SUBTASK 26-00-01-860-012

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Before you touch the engine and APU fire control panel [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-00-01-420-002

(2) Make sure that you have sufficient access to install the applicable electrical connector for the fire shutoff switch.

(a) If you do not have access to the connector, carefully pull the engine and APU fire control panel [1] away from the frame.

SUBTASK 26-00-01-420-003

(3) Put the applicable fire shutoff switch in its position in the engine and APU fire control panel [1].

NOTE: There are indexing pins on the switch.

SUBTASK 26-00-01-420-004

(4) Install the applicable fire shutoff switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) For the engine 1 fire shutoff switch [9], install the screws [8].

(b) For the engine 2 fire shutoff switch [11], install the screws [10].

(c) For the APU fire shutoff switch [13], install the screws [12].

SUBTASK 26-00-01-420-008

(5) If necessary, install the lightplate [6]:

(a) Put the lightplate [6] in its position on the baseplate.

(b) Install the screws [7].

(c) Install the bell cutout handle [2].

1) Install the screws [3].

SUBTASK 26-00-01-420-009

(6) If necessary, install the lightplate [4]:

(a) Put the lightplate [4] in its position on the baseplate.

(b) Install the screws [5].

SUBTASK 26-00-01-420-010

(7) Install the cover [14], do these steps:

(a) If necessary, remove the old sealant from the cover [14].

(b) Put the cover [14] in its position.

(c) Install the screws [15].

(d) Apply sealant, A00142, or sealant, A50110, between the cover [14] and baseplate.

1) Make sure that there is a good seal around all the cover edges and the baseplate.

NOTE: This is to give a moisture-tight seal to the engine and APU fire control panel.

SUBTASK 26-00-01-420-005

(8) Remove the protective cover from the electrical connector of the applicable fire shutoff switch.

SUBTASK 26-00-01-080-001

(9) Remove the protective covers from the applicable airplane electrical connector.

SUBTASK 26-00-01-420-006

(10) Examine the electrical connector for bent, broken pins, dirt or damage.

(a) If it is necessary, clean or repair the electrical connector.

SUBTASK 26-00-01-430-001

(11) Connect the electrical connector to the applicable fire shutoff switch.

SUBTASK 26-00-01-420-007

(12) Install the engine and APU fire control panel [1].

(a) Secure with the quarter-turn fasteners.

SUBTASK 26-00-01-860-013

(13) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	2	C01317	ENGINE 1 START LEVER CHAN B
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1
B	6	C01318	ENGINE 2 START LEVER CHAN A
B	7	C01319	ENGINE 2 START LEVER CHAN B
B	19	C01344	APU FIRE SW POWER
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L
C	7	C00277	ENGINE 2 THRUST REVERSER CONT
C	18	C00348	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 2
C	19	C00347	HYDRAULIC SYSTEM SHUTOFF VALVE ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1
B	5	C00540	FUEL SPAR VALVE IND
C	12	C01276	MASTER CAUTION ANNUNCIATOR CONT 2
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	14	C00033	AUX POWER UNIT CONT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
F	12	C01285	GENERATOR APU GEN CONT UNIT

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	9	C01326	APU GEN CONT UNIT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Electrical System Installation Test

SUBTASK 26-00-01-700-001

(1) Do this task, as applicable: Engine Fire Switch System Shutdown Test, TASK 26-21-00-720-802 or APU Fire Switch System Shutdown Test, TASK 26-22-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO FIRE CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cargo fire control panel

(2) An installation of the cargo fire control panel.

B. The cargo fire control panel is installed on the P8 aisle control stand in the flight compartment.

TASK 26-00-02-000-801

2. Cargo Fire Control Panel Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the cargo fire control panel.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 26-00-02-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	8	C00701	EMER PANEL LTG
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
F	13	C01179	INDICATOR MASTER DIM SECT 7

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Cargo Fire Control Panel Removal

SUBTASK 26-00-02-910-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before the cargo fire control panel [1] is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 26-00-02-020-001

(2) Loosen the quarter turn fasteners that attach the cargo fire control panel [1] to the frame.

SUBTASK 26-00-02-020-002

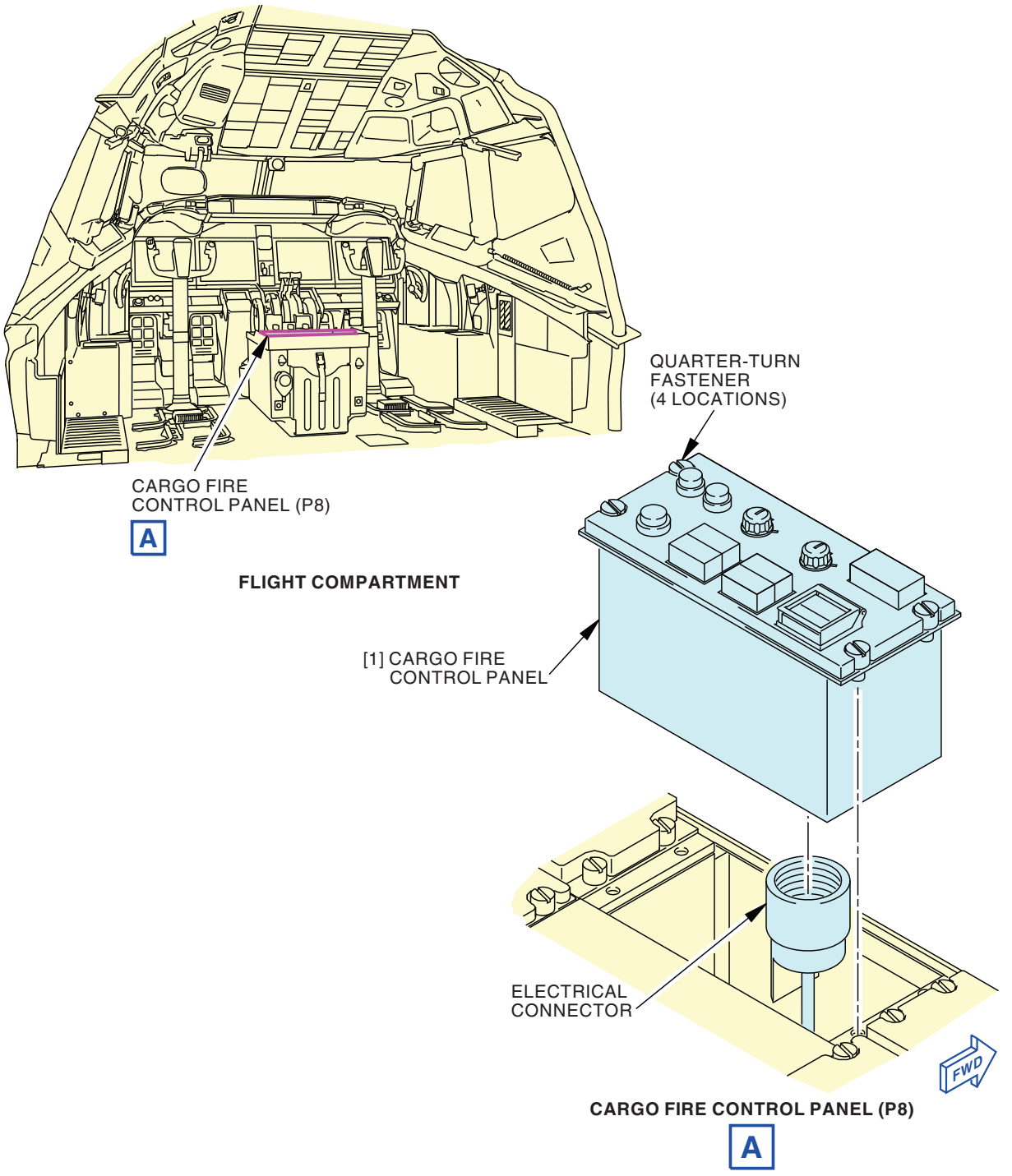
(3) Slide the cargo fire control panel [1] from the frame.

SUBTASK 26-00-02-020-003

(4) Tag and remove the electrical connectors.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Fire Control Panel Installation
Figure 401/26-00-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-00-02-400-801

3. Cargo Fire Control Panel Installation

(Figure 401)

A. General

(1) This task gives instructions to install the cargo fire control panel.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-23-00-730-802	Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cargo fire control panel	26-00-02-01-010	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Installation

SUBTASK 26-00-02-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	8	C00701	EMER PANEL LTG
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
F	13	C01179	INDICATOR MASTER DIM SECT 7

F. Cargo Fire Control Panel Installation

SUBTASK 26-00-02-910-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before the cargo fire control panel [1] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-00-02-420-001

(2) Install the connectors on the cargo fire control panel [1] and remove the tags.

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SUBTASK 26-00-02-020-004

(3) Slide the cargo fire control panel [1] into the frame and attach with the quarter turn fasteners.

SUBTASK 26-00-02-860-003

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	8	C00701	EMER PANEL LTG
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
F	13	C01179	INDICATOR MASTER DIM SECT 7

G. Cargo Fire Control Panel Installation Test

SUBTASK 26-00-02-720-001

(1) Push the TEST switch on the cargo fire control panel.

(a) Make sure that the DETECTOR FAULT light stays off.

(b) Make sure that the red FWD and AFT ARM lights come on.

(c) Make sure that the FWD and AFT EXT lights come on.

(d) Make sure that the DISCHARGE light comes on.

SUBTASK 26-00-02-720-002

(2) Do this task: Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test, TASK 26-23-00-730-802.

——— END OF TASK ———

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DETECTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the fire and overheat detection system

(2) An overheat detection test using handheld LCRZ meter

(3) An overheat detection test using UN-ATE tool by Technologies Harness Scanner.

B. The operational test uses the test switches on the Engine and APU Fire Control Panel, P8-1, in the flight compartment. It also uses the test switch on the Fire Detection Control Unit, M279, in the Electrical and Electronics Compartment.

C. The fire and overheat detection system consists of the following systems:

(1) The engine fire detection system

(2) The APU fire detection system

(3) The wheel well, wing, and aft lower body (compartment) overheat detection system.

TASK 26-10-00-710-801

2. Fire and Overheat Detection System - Operational Test

(Figure 501 or Figure 502)

A. General

(1) The operational test makes sure that the engine, Auxiliary Power Unit (APU), and compartment fire/overheat detection systems operate correctly. The operational test uses only equipment that is installed on the airplane and it does not disturb the airplane systems. The operational test consists of these tests:

(a) The FIRE/OVHT test for loops A and B.

(b) The FIRE/OVHT test for loop A.

(c) The FIRE/OVHT test for loop B.

(d) The FAULT/INOP test for loops A and B.

(e) The test for an Engine 1 Detection Fault.

(f) The test for an Engine 2 Detection Fault.

(g) The test for an APU Detection Fault.

(h) The Air Mode Test for the APU Fire Control Panel, P28.

(2) This test will do a check of all the systems affected by the detection test switches on the Engine and APU Fire Control Panel. In other words, when you do the detection tests, you will look for every indication that is affected by the test.

(3) A brief system description is included at the end of the procedure, refer to Table 501, Table 502, Table 503.

B. References

Reference	Title
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

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(Continued)

Zone	Area
118	Electrical and Electronics Compartment - Right
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Operational Test

SUBTASK 26-10-00-860-002

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 26-10-00-860-003

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

SUBTASK 26-10-00-860-004

(3) Make sure the MASTER CAUTION lights and the OVHT/FIRE lights on the Pilots Glareshield, P7, are not on.

E. Do the FIRE/OVHT Test for Loops A and B

SUBTASK 26-10-00-710-001

(1) Make sure the two OVHT DET switches on the Engine and APU Fire Control Panel, P8-1, are in the NORMAL positions.

SUBTASK 26-10-00-710-002

(2) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

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(a) On the Engine and APU Fire Control Panel, make sure these lights come on:
   1) Engine 1 Fire Handle light (Red).
   2) Engine 2 Fire Handle light (Red).
   3) APU Fire Handle light (Red).
   4) ENG 1 OVERHEAT light (Amber).
   5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights come on:
   1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
   2) The Captain's and First Officer's FIRE WARN lights (Red).
   3) The OVHT/DET light (Amber).

(c) Make sure you hear the bell in the Aural Warning Module, M315, in the flight compartment.

(d) Make sure the APU fire warning horn, M277, in the right main wheel well comes on.

(e) Make sure the APU Remote Fire red light, on the APU fire control panel, P28, in the right main wheel well flashes on and off at approximately one cycle per second.

(f) Push the BELL CUTOUT switch on the Engine and APU Fire Control Panel, P8-1.

NOTE: If you push the HORN CUTOUT switch on the P28 panel, or either of the FIRE WARN lights on the P7 panel, the same conditions will occur. The BELL CUTOUT switch, the HORN CUTOUT switch, and the FIRE WARN switch cause the same resets but from different locations.

   1) Make sure you do not hear the bell in the Aural Warning Module, M315, in the flight compartment.
   2) Make sure the two FIRE WARN lights on the Pilot's Glareshield, P7, go off.
   3) Make sure the APU fire warning horn in the right main wheel well stops.
   4) Make sure the APU Remote Fire red light on the P28 panel in the right main wheel well comes on and does not flash.

SUBTASK 26-10-00-710-003

(3) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) On the Engine and APU Fire Control Panel, make sure these lights go off:
   1) Engine 1 Fire Handle light (Red).
   2) Engine 2 Fire Handle light (Red).
   3) APU Fire Handle light (Red).
   4) ENG 1 OVERHEAT light (Amber).
   5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights go off:
   1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
   2) The Captain's and First Officer's FIRE WARN lights (Red).
   3) The OVHT/DET light (Amber).

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F. Do the FIRE/OVHT Test for Loop A

SUBTASK 26-10-00-710-004

(1) Move the two OVHT DET switches on the Engine and APU Fire Control Panel, P8-1, to the A positions.

SUBTASK 26-10-00-710-005

(2) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

(a) On the Engine and APU Fire Control Panel, make sure these lights come on:

1) Engine 1 Fire Handle light (Red).
2) Engine 2 Fire Handle light (Red).
3) APU Fire Handle light (Red).
4) ENG 1 OVERHEAT light (Amber).
5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
2) The Captain's and First Officer's FIRE WARN lights (Red).
3) The OVHT/DET light (Amber).

(c) Make sure the bell in the Aural Warning Module, M315, comes on.

(d) Make sure the APU fire warning horn, M277, in the right main wheel well comes on.

(e) Make sure the APU Remote Fire red light on the APU fire control panel, P28, in the right main wheel well flashes on and off at approximately one cycle per second.

SUBTASK 26-10-00-710-006

(3) Release the TEST switch on the Engine and APU Fire Control Panel.

G. Do the FIRE/OVHT Test for Loop B

SUBTASK 26-10-00-710-007

(1) Move the two OVHT DET switches on the Engine and APU Fire Control Panel, P8-1, to the B positions.

SUBTASK 26-10-00-710-008

(2) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

(a) On the Engine and APU Fire Control Panel, make sure these lights come on:

1) Engine 1 Fire Handle light (Red).
2) Engine 2 Fire Handle light (Red).
3) APU Fire Handle light (Red).
4) ENG 1 OVERHEAT light (Amber).
5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
2) The Captain's and First Officer's FIRE WARN lights (Red).
3) The OVHT/DET light (Amber).

(c) Make sure the bell in the Aural Warning Module, M315, comes on.

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(d) Make sure the APU fire warning horn, M277, in the right main wheel well comes on.

(e) Make sure the APU Remote Fire red light on the APU fire control panel, P28, in the right main wheel well flashes on and off at approximately one cycle per second.

SUBTASK 26-10-00-710-009

(3) Release the TEST switch on the Engine and APU Fire Control Panel.

H. Do the FAULT/INOP Test for Loops A and B

SUBTASK 26-10-00-710-010

(1) Move the two OVHT DET switches on the Engine and APU Fire Control Panel, P8-1, to the NORMAL positions.

SUBTASK 26-10-00-710-011

(2) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the FAULT/INOP position.

(a) On the Engine and APU Fire Control Panel, make sure these lights come on:

1) FAULT light (Amber).

2) APU DET INOP light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

(c) On the Engine and APU Fire Detection Control Unit, M279, on the E2-2 shelf in the Electrical and Electronics Compartment, make sure these lights come on:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) ENGINE 2 LOOP A (Amber).

4) ENGINE 2 LOOP B (Amber).

5) APU (Amber).

6) The four FAULT CODE lights are ON-ON-ON-ON.

SUBTASK 26-10-00-710-012

(3) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) On the Engine and APU Fire Control Panel, make sure these lights are off:

1) FAULT light (Amber).

2) APU DET INOP light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights are off:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

(c) On the Engine and APU Fire Detection Control Unit, M279, in the Electrical and Electronics Compartment, make sure these lights are off:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) ENGINE 2 LOOP A (Amber).

4) ENGINE 2 LOOP B (Amber).

5) APU (Amber).

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6) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

I. Do the Test for an Engine 1 Detection Fault

SUBTASK 26-10-00-860-005

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(a) On the Engine and APU Fire Control Panel, P8-1, make sure the FAULT light (Amber) is on.

(b) On the Fire Detection Control Module, M279, in the Electrical and Electronics Compartment, make sure these lights come on:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) The four FAULT CODE lights are ON-OFF-ON-OFF.

SUBTASK 26-10-00-860-006

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(a) On the Engine and APU Fire Control Module, make sure the FAULT light (Amber) is off.

(b) On the Fire Detection Control Module, in the Electrical and Electronics Compartment, make sure these lights go off:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

J. Do the Test for an Engine 2 Detection Fault

SUBTASK 26-10-00-860-007

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2

(a) On the Engine and APU Fire Control Panel, P8-1, make sure the FAULT light (Amber) is on.

(b) On the Fire Detection Control Module, M279, in the Electrical and Electronics Compartment, make sure these lights come on:

1) ENGINE 2 LOOP A (Amber).

2) ENGINE 2 LOOP B (Amber).

3) The four FAULT CODE lights are ON-OFF-ON-OFF.

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SUBTASK 26-10-00-860-008

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2

(a) On the Engine and APU Fire Control Panel, make sure the FAULT light (Amber) is off.

(b) On the Fire Detection Control Module, in the Electrical and Electronics Compartment, make sure these lights go off (Figure 501 or Figure 502).

1) ENGINE 2 LOOP A (Amber).

2) ENGINE 2 LOOP B (Amber).

3) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

K. Do the Test for an APU Detection Fault

SUBTASK 26-10-00-860-009

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU

(a) On the Pilot's Glareshield, P7, make sure the Captain's and First Officer's MASTER CAUTION lights (Amber) are on.

(b) On the Pilot's Glareshield, make sure the OVHT/DET light (Amber) is on.

(c) On the Engine and APU Fire Control Panel, P8-1, make sure the APU DET INOP light (Amber) is on.

(d) On the Fire Detection Control Module, M279, in the Electrical and Electronics Compartment, make sure the APU FAULT AREA light (Amber) comes on.

(e) On the Fire Detection Control Module, make sure the four FAULT CODE lights are ON-OFF-ON-OFF.

SUBTASK 26-10-00-860-010

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU

(a) On the Pilot's Glareshield, make sure the Captain's and First Officer's MASTER CAUTION lights (Amber) are off.

(b) On the Pilot's Glareshield, make sure the OVHT/DET light (Amber) is off.

(c) On the Engine and APU Fire Control Panel, make sure the APU DET INOP light (Amber) is off.

(d) On the Fire Detection Control Module, in the Electrical and Electronics Compartment, make sure the APU FAULT AREA light (Amber) is off.

(e) On the Fire Detection Control Module, make sure the four FAULT CODE lights are OFF-OFF-OFF-OFF.

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L. Do the Air Mode Test for the APU Fire Control Panel, P28

SUBTASK 26-10-00-860-011

(1) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 26-10-00-710-013

(2) Move and hold the TEST switch on the Engine and APU Fire Control Panel, P8-1, to the OVHT/FIRE position.

NOTE: Other indications will occur. Ignore all the indications that are not listed in the steps that follow.

(a) Make sure the APU fire warning horn in the right main wheel well does not come on.

(b) Make sure the red light on the APU fire control panel, P28, in the right main wheel well flashes on and off at approximately one cycle per second.

(c) Push the HORN CUTOUT switch on the APU fire control panel.

(d) Make sure the red light on the APU fire control panel stays on and does not flash.

SUBTASK 26-10-00-710-014

(3) Release the TEST switch on the Engine and APU Fire Control Panel.

SUBTASK 26-10-00-860-012

(4) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

M. Put the Airplane Back to its Usual Condition

SUBTASK 26-10-00-860-013

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

N. System Description

SUBTASK 26-10-00-800-001

(1) The Engine and APU Fire Detection Systems are monitored and controlled by the Fire Detection Control Module, M279, in the Electrical and Electronics Compartment. The Compartment Overheat System is monitored and controlled by the Compartment Overheat Detection Controller, M237, in the Electrical and Electronics Compartment. The Module and the Controller send fire, overheat, and fault signals to the Engine and APU Fire Control Panel, P8-1, and the Pilot's Glareshield, P7, in the flight compartment. The module also sends an APU fire signal to the APU Fire Control Panel, P28, in the right main wheel well.

(a) The Engine and APU Fire, Overheat, and Fault Warning Indications are as follows:

Table 501/26-10-00-993-801

System	Condition	Indication	Location
ENGINE 1	FIRE	Engine 1 Fire Handle Light (red)	P8-1
		MASTER CAUTION Lights (amber)	P7
		FIRE WARN Lights (red)	P7
		Flight Compartment Fire Bell	M315
ENGINE 2	FIRE	Engine 2 Fire Handle Light (red)	P8-1

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Table 501/26-10-00-993-801 (Continued)

System	Condition	Indication	Location
		MASTER CAUTION Lights (amber)	P7
		FIRE WARN Lights (red)	P7
————————	————————	Flight Compartment Fire Bell	M315
ENGINE 1	OVER-HEAT	ENG 1 OVERHEAT Light (amber)	P8-1
		MASTER CAUTION Lights (amber)	P7
		OVHT/DET Light (amber)	P7
ENGINE 2	OVER-HEAT	ENG 2 OVERHEAT Light (amber)	P8-1
		MASTER CAUTION Lights (amber)	P7
		OVHT/DET Light (amber)	P7
ENGINE 1	FAULT	FAULT Light (amber)	P8-1
		ENGINE NO. 1 LOOP A Light (amber)	M279
		ENGINE NO. 1 LOOP B Light (amber)	M279
ENGINE 2	FAULT	FAULT Light (amber)	P8-1
		ENGINE NO. 2 LOOP A Light (amber)	M279
		ENGINE NO. 2 LOOP B Light (amber)	M279
APU	FIRE	APU Fire Handle Light (red)	P8-1
		MASTER CAUTION Lights (amber)	P7
		FIRE WARN Lights (red)	P7
		Flight Compartment Fire Bell	M315
		Right Wheel Well APU Fire Warning Horn	M277
		APU Remote Fire Light (red)	P28
APU	OVER-HEAT	MASTER CAUTION Light (amber)	P7
		OVHT/DET Light (amber)	P7
APU	FAULT	APU DET INOP Light (amber)	P8-1
		APU FAULT AREA Light (amber)	M279

(b) The Engine Fire Detection system has dual sensors in each loop and operates in "AND" logic. Therefore, both loops must sense a fire or a fault before the fire or fault indication is shown. The following table is a summary of the indications that will be given for the various conditions of the sensing loops:

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Table 502/26-10-00-993-802

LOOP 1	LOOP 2	INDICATIONS
NORMAL	NORMAL	NONE
NORMAL	INOP	NONE
NORMAL	FAULT	NONE
NORMAL	FIRE	NONE
INOP	NORMAL	NONE
INOP	INOP	NONE
INOP	FAULT	FAULT
INOP	FIRE	FIRE
FAULT	NORMAL	NONE
FAULT	INOP	FAULT
FAULT	FAULT	FAULT
FAULT	FIRE	FIRE
FIRE	NORMAL	NONE
FIRE	INOP	FIRE
FIRE	FAULT	FIRE
FIRE	FIRE	FIRE

(c) The following table shows all the fire, overheat, and fault indications in the flight compartment. These are the indications that will occur when the OVHT DET switches are in the normal position and you set the TEST switch to the FIRE/OVHT position, then to the NORMAL position, then to the FAULT INOP position, and then to the NORMAL position:

Table 503/26-10-00-993-803

INDICATOR	LOC	FIRE/ OVHT	NORMAL	FAULT/ INOP	NORMAL
Engine "1" Light	P8-1	ON	OFF	OFF	OFF
Engine "2" Light	P8-1	ON	OFF	OFF	OFF
"APU" Light	P8-1	ON	OFF	OFF	OFF
"ENG 1 OVHT" Light	P8-1	ON	OFF	OFF	OFF
"ENG 2 OVHT" Light	P8-1	ON	OFF	OFF	OFF
"WHEEL WELL" Light	P8-1	ON	OFF	OFF	OFF
"APU DET INOP" Light	P8-1	OFF	OFF	ON	OFF
"FAULT" Light	P8-1	OFF	OFF	ON	OFF
"FIRE WARN" Light *[1]	P7	ON	OFF	OFF	OFF
"MASTER CAUTION" Light	P7	ON	*[2]	ON	*[2]
"OVHT/DET" Light	P7	ON	*[2]	ON	*[2]

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Table 503/26-10-00-993-803 (Continued)

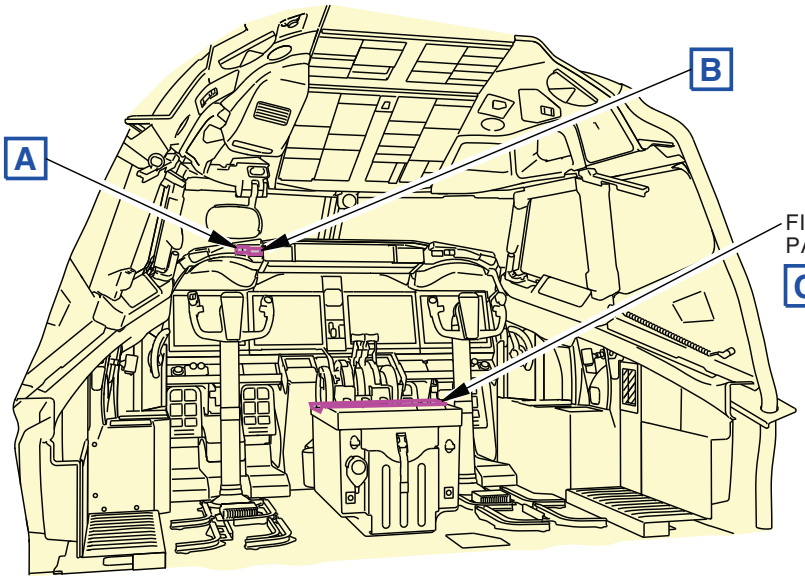
INDICATOR	LOC	FIRE/ OVHT	NORMAL	FAULT/ INOP	NORMAL
Flight Compt Bell *[1]	M315	ON	OFF	OFF	OFF
Wheel Well Bell *[1]	M277	ON	OFF	OFF	OFF
Remote Fire Light *[1]	P28	ON	OFF	OFF	OFF

*[1] Reset by pulling the "BELL CUTOUT" switch or by pushing the "FIRE/WARN" light, or by pushing the "HORN CUTOUT" switch on the P28 panel.

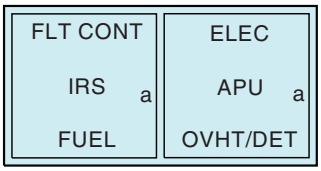
*[2] Reset by pushing the "MASTER/CAUTION" light.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

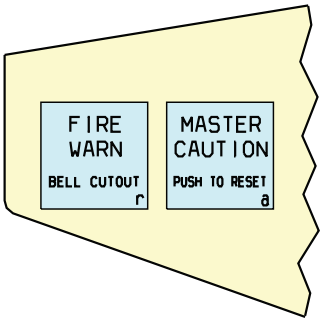


FLIGHT COMPARTMENT



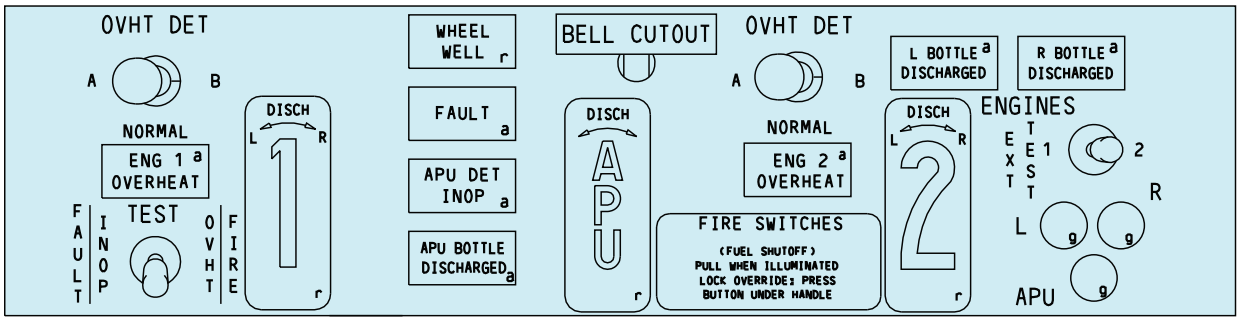
CAPTAIN'S WARNING
LIGHTS (P7)

B



CAPTAIN'S FIRE
WARNING SWITCH (P7)
(FIRST OFFICER'S IS OPPOSITE)

A



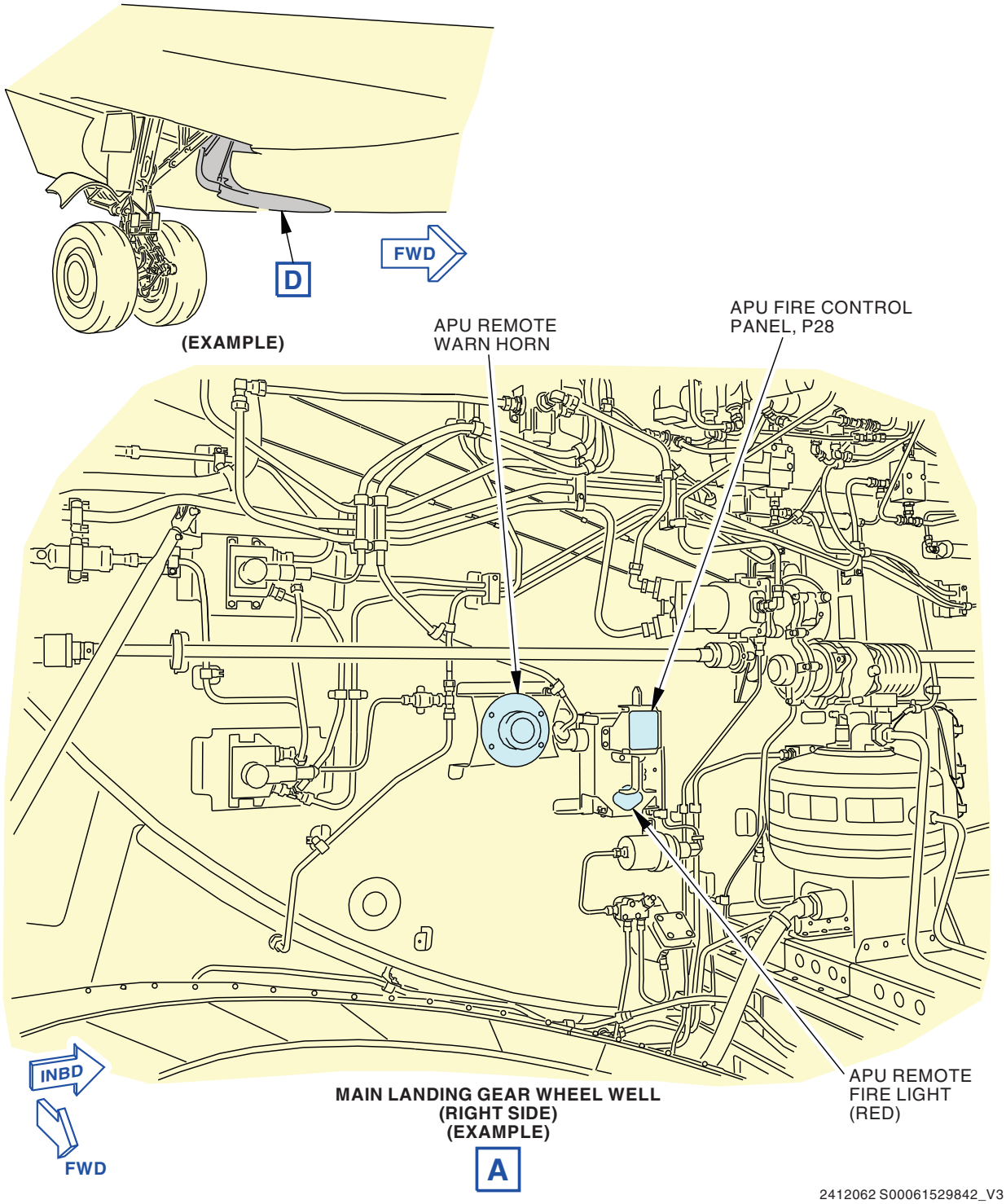
FIRE CONTROL PANEL (P8-1)

C

2412061 S00061529841_V2

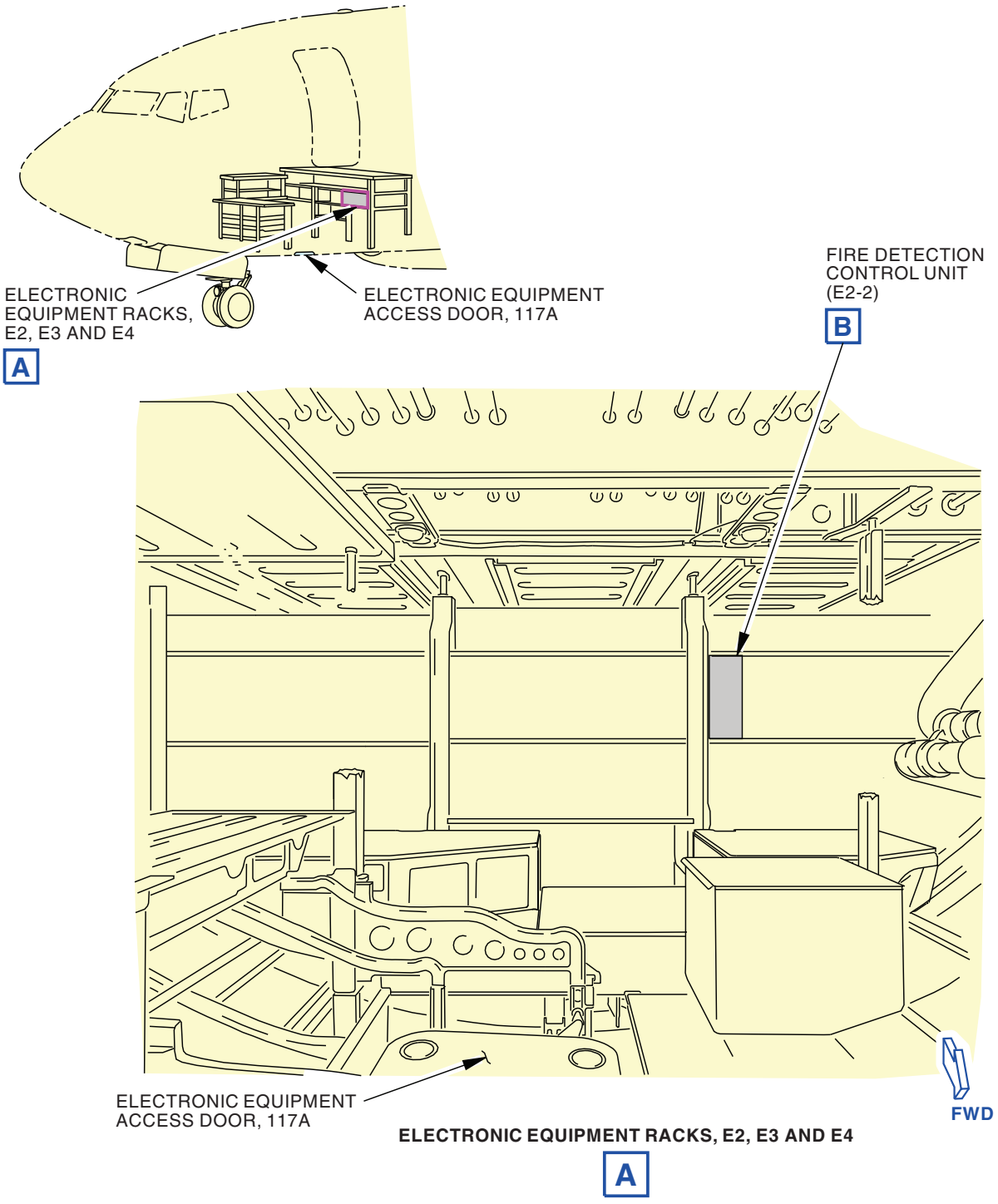
Auxiliary Power Unit (APU) Fire Detection Test
Figure 501/26-10-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



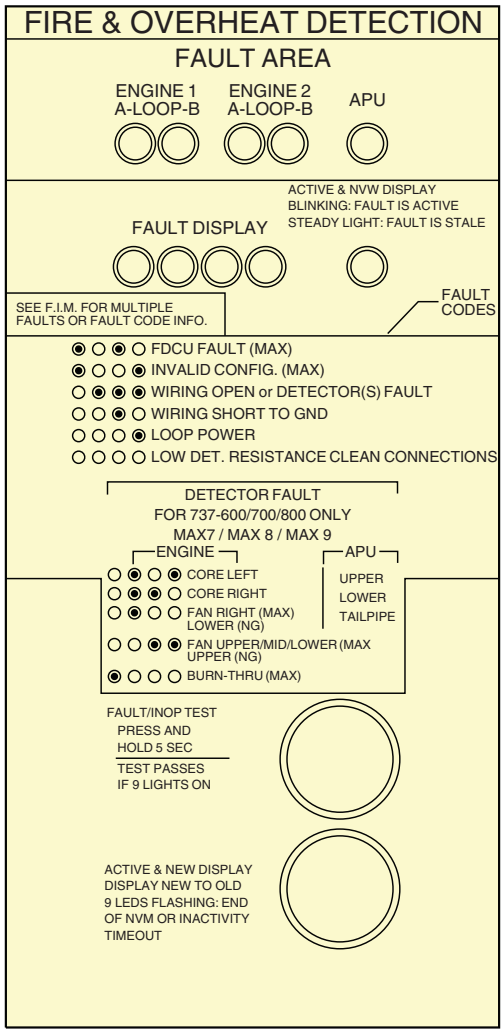
Auxiliary Power Unit (APU) Fire Detection Test
Figure 501/26-10-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine and APU Fire Detection Control Unit Installation
Figure 502/26-10-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIRE DETECTION CONTROL UNIT

B

2412064 S00061529844_V3

Engine and APU Fire Detection Control Unit Installation
Figure 502/26-10-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-10-00-700-802

3. Overheat Detection Test Using Handheld LCRZ Meter

(Figure 503)

A. General

(1) This task consists of the following procedures using a 1v/1kHz AC LCRZ impedance meter:

(a) Instructions for the health check of the compartment overheat loop detectors and wiring.

1) This procedure is used to detect shorts, open, and degraded overheat loop detectors in the wing and body overheat detection system.

(b) Instructions for locating faulty detector or wiring for compartment overheat loop that reads less than nominal range.

1) This procedure is used to isolate an entire loop to a degraded or failed detector, or to a wiring problem.

B. References

Reference	Title
22-11-33-400-801	Flight Control Computer Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
26-18-01-000-801	Compartment Overheat Detection Control Unit Removal (P/B 401)
26-18-01-400-801	Compartment Overheat Detection Control Unit Installation (P/B 401)
26-18-02-000-801	Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Removal (P/B 401)
26-18-02-400-801	Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Installation (P/B 401)
WDM 26-12-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-17158	Controller - Loop Opt Part #: 9240 SI Supplier: FBCN9
COM-17417	Controller - Loop Part #: 20549 SI Supplier: FBCN9
COM-20443	Meter - LCR and Z (Inductance, Capacitance, Resistance and Impedance) Part #: 880 Supplier: 08098
SPL-12921	Test Equipment - Compartment Overheat Detection Module Part #: C26007-98 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735

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E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

F. Prepare for Overheat Detection Loop Health Check and Troubleshooting

SUBTASK 26-10-00-860-016

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

NOTE: Electrical power must be off to prevent incorrect LCRZ meter readings because of Electromagnetic Interference (EI).

SUBTASK 26-10-00-860-017

(2) Make sure that the battery switch is in the OFF position on the P5 overhead panel.

SUBTASK 26-10-00-860-018

(3) Make sure that all ground power is disconnected.

SUBTASK 26-10-00-020-003

(4) Remove the M237 compartment overheat detection control unit (TASK 26-18-01-000-801).

SUBTASK 26-10-00-480-013

(5) Install the extender box assembly of the overheat detection module test equipment, SPL-12921, on the compartment overheat detection control unit mount.

(a) Do not connect the test box assembly to the extended box assembly with adapter cable assembly.

SUBTASK 26-10-00-700-040

(6) Prepare the loop controller, COM-17158, or Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443 (preferred), or loop controller, COM-17417, for the test.

CAUTION

DO NOT USE A NON-EQUIVALENT METER, OR A DC METER, FOR THE LOOP ELEMENTS. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(a) If the loop controller, COM-17158, or Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443 (preferred), is used, do the following steps:

1) Connect the black test lead to the black receptacle port.

2) Connect the red test lead to the red receptacle port.

CAUTION

DO NOT USE A NON-EQUIVALENT METER, OR A DC METER, FOR THE LOOP ELEMENTS. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(b) If the loop controller, COM-17417, is used, do the following steps:

1) Connect the black test lead to the black receptacle port.

2) Connect the red test lead to the red receptacle port.

3) Connect the blue test lead to the blue receptacle port.

G. Overheat Detection Loop Health Check with COM-17158

SUBTASK 26-10-00-480-016

(1) Connect the black lead to aircraft ground.

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SUBTASK 26-10-00-480-017

(2) Connect the red test lead to breakout jack of the extender box assembly in Table 504 (View C, Figure 503).

Table 504/26-10-00-993-806 Overheat Detection Loop Health Check

Compartment Overheat Detection Controller Receptacle D742 Pin*[1]	Expected Reading (Impedance)
2	24k ohms +/- 5k
4	20k ohms +/- 5k
8	24k ohms +/- 5k
9	28k ohms +/- 5k
10	20k ohms +/- 5k
11	20k ohms +/- 5k
12	20k ohms +/- 5k
13	14k ohms +/- 5k
21	30k ohms +/- 5k
24	30k ohms +/- 5k
27	14k ohms +/- 5k
29	28k ohms +/- 5k
30	28k ohms +/- 5k
31	30k ohms +/- 5k
32	37k ohms +/- 5k
33	37k ohms +/- 5k
34	30k ohms +/- 5k
35	28k ohms +/- 5k
36	17k ohms +/- 5k
37	17k ohms +/- 5k
39	28k ohms +/- 5k
40	28k ohms +/- 5k
41	24k ohms +/- 5k
42	24k ohms +/- 5k

*[1] Refer to WDM 26-12-11 .

SUBTASK 26-10-00-700-020

(3) Prepare the loop controller, COM-17158.

(a) For each measurement, do these steps:

1) Allow reading to stabilize and record reading.

2) Set the switch to TEST then back to ISOL to reset unit.

3) Allow reading to stabilize and record reading.

4) Make sure that both readings are similar to provide accurate results.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Check for a short or degraded detector by taking the measurements per Table 504.

1) If a reading is greater than the expected reading, check the wiring between the D742 connector and the first detector in the loop for an open circuit condition.

2) If a reading is between 5k ohms and the nominal range, a scheduled repeat test for degradation or replacement of the detector is recommended at the operator's convenience.

NOTE: This indicates that one of the detectors is degraded. The degraded detector may continue to degrade resulting in intermittent failures until it completely fails, that is typically within 180 days.

a) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring For Overheat Detection Loop”.

3) If a reading is < 5k ohms, do the following steps to isolate the problem:

NOTE: This indicates that either one of the detectors is shorted between the center conductor and ground, or that there is a problem in the wiring.

a) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring For Overheat Detection Loop”.

SUBTASK 26-10-00-080-007

(4) Disconnect the black test lead from ground.

SUBTASK 26-10-00-080-008

(5) Disconnect the black test lead from the loop controller, COM-17158, and connect to the green receptacle port.

SUBTASK 26-10-00-480-018

(6) Connect the red test lead and green test lead to breakout jack of the extender box assembly in Table 505 (View C, Figure 503).

Table 505/26-10-00-993-807 Overheat Loop Detection Open Check

Compartment Overheat Detection Controller Receptacle D742 Pin*[1]	Expected Reading
2 to 8	< 15 ohms
4 to 12	< 15 ohms
9 to 29	< 25 ohms
10 to 11	< 15 ohms
13 to 27	< 25 ohms
39 to 40	< 15 ohms
30 to 35	< 15 ohms
31 to 34	< 15 ohms
21 to 24	< 15 ohms
36 to 37	< 25 ohms
41 to 42	< 15 ohms
32 to 33	< 15 ohms

*[1] Refer to WDM 26-12-11 .

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-10-00-700-021

(7) Set the loop controller, COM-17158, switch to the CONT position.

(a) For each measurement, do these steps:

1) Allow reading to stabilize and record reading.

2) Set the switch to the TEST position then back to the CONT position to reset unit.

3) Allow reading to stabilize and record reading.

4) Make sure that both readings are similar to provide accurate results.

SUBTASK 26-10-00-700-022

(8) Check for an open detector or circuit by taking the measurements per Table 505.

(a) If a reading is > 15 ohms, do continuity checks of the applicable loop segments to isolate to the failed loop or wiring.

NOTE: If a reading is greater than 15 ohms, it is possible that the detector connectors and/or the mating electrical lead connectors are contaminated or there is corrosion on the electrical leads. Contaminated connectors can be cleaned using alcohol, B00130 and blown dry with low pressure air or nitrogen.

(b) Replace the faulty detector, or repair wiring, as applicable.

H. Overheat Detection Loop Health Check with COM-20443

SUBTASK 26-10-00-480-019

(1) Connect the black lead to aircraft ground.

SUBTASK 26-10-00-480-020

(2) Connect the red test lead to breakout jack of the extender box assembly in Table 504 (View C, Figure 503).

SUBTASK 26-10-00-700-023

(3) Prepare the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443.

(a) Press the power button for one second to turn on meter.

(b) Press PRI/AUTO button until “Z” is displayed (for impedance measurement).

(c) Press FREQ/REC button until 1 kHz is displayed.

(d) Press and hold the SEC/LEV button for 2 seconds to access the test voltage level options.

1) Press the SEC/LEV button momentarily to toggle setting to 1V.

(e) Press the RATE/P-S button momentarily until “SLOW” is displayed for a more stable reading.

(f) Press and hold RATE/P-S button for 2 seconds to toggle to parallel mode.

1) Press the RATE/P-S button momentarily to toggle to PAL (parallel) mode.

(g) For each measurement allow reading to stabilize and record reading.

SUBTASK 26-10-00-700-024

(4) Check for a short or degraded detector by taking the measurements per Table 504.

(a) If a reading is greater than the expected reading, check the wiring between the D742 connector and the first detector in the loop for an open circuit condition.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If a reading is between 5k ohms and the nominal range, a scheduled repeat test for degradation or replacement of the detector is recommended at the operator's convenience.

NOTE: This indicates that one of the detectors is degraded. The degraded detector may continue to degrade resulting in intermittent failures until it completely fails, that is typically within 180 days.

1) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring for Overheat Detection Loop”.

(c) If a reading is < 5k ohms, do the following steps to isolate the problem:

NOTE: This indicates that either one of the detectors is shorted between the center conductor and ground, or that there is a problem in the wiring.

1) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring for Overheat Detection Loop”.

SUBTASK 26-10-00-080-009

(5) Disconnect the black test lead from ground.

SUBTASK 26-10-00-480-021

(6) Connect the red test lead and black test lead to breakout jack of the extender box assembly in Table 505 (View C, Figure 503).

SUBTASK 26-10-00-700-025

(7) Set the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, to measure continuity via DCR (DC Resistance).

(a) If it is necessary, press the power button for two seconds to turn on meter.

(b) Press the PRI/AUTO button until DCR is displayed to select direct current resistance measurement.

(c) For each measurement allow reading to stabilize and record reading.

SUBTASK 26-10-00-700-026

(8) Check for an open detector or circuit by taking the measurements per Table 505.

(a) If a reading is > 15 ohms, do continuity checks of the applicable loop segments to isolate to the failed loop or wiring.

NOTE: If a reading is greater than 15 ohms, it is possible that the detector connectors and/or the mating electrical lead connectors are contaminated or there is corrosion on the electrical leads. Contaminated connectors can be cleaned using alcohol, B00130 and blown dry with low pressure air or nitrogen.

(b) Replace the faulty detector, or repair wiring, as applicable.

I. Overheat Detection Loop Health Check with COM-17417

SUBTASK 26-10-00-480-022

(1) Connect the black lead to aircraft ground.

SUBTASK 26-10-00-480-023

(2) Connect the red test lead to breakout jack of the extender box assembly in Table 504 (View C, Figure 503).

SUBTASK 26-10-00-700-027

(3) Press the INSULATION button on loop controller, COM-17417.

(a) For each measurement, allow reading to stabilize and record reading.

(b) Check for a short or degraded detector by taking the measurements per Table 504.

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1) If a reading is greater than the expected reading, check the wiring between the D742 connector and the first detector in the loop for an open circuit condition.

2) If a reading is between 5k ohms and the nominal range, a scheduled repeat test for degradation or replacement of the detector is recommended at the operator's convenience.

NOTE: This indicates that one of the detectors is degraded. The degraded detector may continue to degrade resulting in intermittent failures until it completely fails, that is typically within 180 days.

a) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring For Overheat Detection Loop”.

3) If a reading is < 5k ohms, do the following steps to isolate the problem:

NOTE: This indicates that either one of the detectors is shorted between the center conductor and ground, or that there is a problem in the wiring.

a) To isolate to the degraded detector, proceed to section “Locating Faulty Detector or Wiring For Overheat Detection Loop”.

4) If all of the readings are within tolerance, continue to the next jack per Table 504.

SUBTASK 26-10-00-080-010

(4) Disconnect the black test lead from ground.

SUBTASK 26-10-00-080-011

(5) Disconnect the black test lead from the loop controller, COM-17417, and connect to the blue receptacle port.

SUBTASK 26-10-00-480-024

(6) Connect the red test lead and blue test lead to breakout jack of the extender box assembly in Table 505 (View C, Figure 503).

SUBTASK 26-10-00-700-028

(7) Press the CONTINUITY push-button on the loop controller, COM-17417.

(a) For each measurement, allow reading to stabilize and record reading.

SUBTASK 26-10-00-700-029

(8) Check for an open detector or circuit by taking the measurements per Table 505.

(a) If a reading is > 15 ohms, do continuity checks of the applicable loop segments to isolate to the failed loop or wiring.

NOTE: If a reading is greater than 15 ohms, it is possible that the detector connectors and/or the mating electrical lead connectors are contaminated or there is corrosion on the electrical leads. Contaminated connectors can be cleaned using alcohol, B00130 and blown dry with low pressure air or nitrogen.

(b) Replace the faulty detector, or repair wiring, as applicable.

J. Locating Faulty Detector or Wiring For Overheat Detection Loop

SUBTASK 26-10-00-700-030

(1) For the applicable loop section, disconnect the loop at the approximate half way point (WDM 26-12-11).

NOTE: This will allow you to determine that side is faulty. For example, for the loop that connects to pins 4 and 12 of connector D742, disconnect between detectors M1761 and M3057.

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SUBTASK 26-10-00-700-031

(2) Check the readings again at each pin at the rack connector.

(a) The side that shows the lower reading is the faulty side.

(b) The resistance values should roughly double from the original expected value.

NOTE: For example, if the original expected reading was 21k ohms ± 5k ohms and you measured 5k ohms, when you disconnect the loops at the halfway point, the resistance should be +42k ohms on each side on a good system. If one side reads 42k ohms, and the other side reads 5k ohms, the faulty side is the one reading 5k ohms.

1) Note the faulty side and reconnect the loop segments that were disconnected in the previous step.

2) Disconnect the loop at the approximate half way point of the faulty side.

3) Check the readings again at each pin at the connector rack.

a) The side that has the lower reading is the faulty side.

4) If it is necessary, continue to check to isolate to the faulty loop section.

5) Replace the faulty detector (TASK 26-18-02-000-801 and TASK 26-18-02-400-801), or repair wiring (Standard Wiring Practices Manual (SWPM)), as applicable.

NOTE: Value for individual loops is 100k ohms, but this number varies by length. A very long section, 200+ inches, could read 100k ohms or below.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-10-00-080-012

(1) Remove the overheat detection module test equipment, SPL-12921, from the compartment overheat detection control unit mount.

SUBTASK 26-10-00-420-002

(2) If the flight control computer B was removed, install the flight control computer B (TASK 22-11-33-400-801).

SUBTASK 26-10-00-410-002

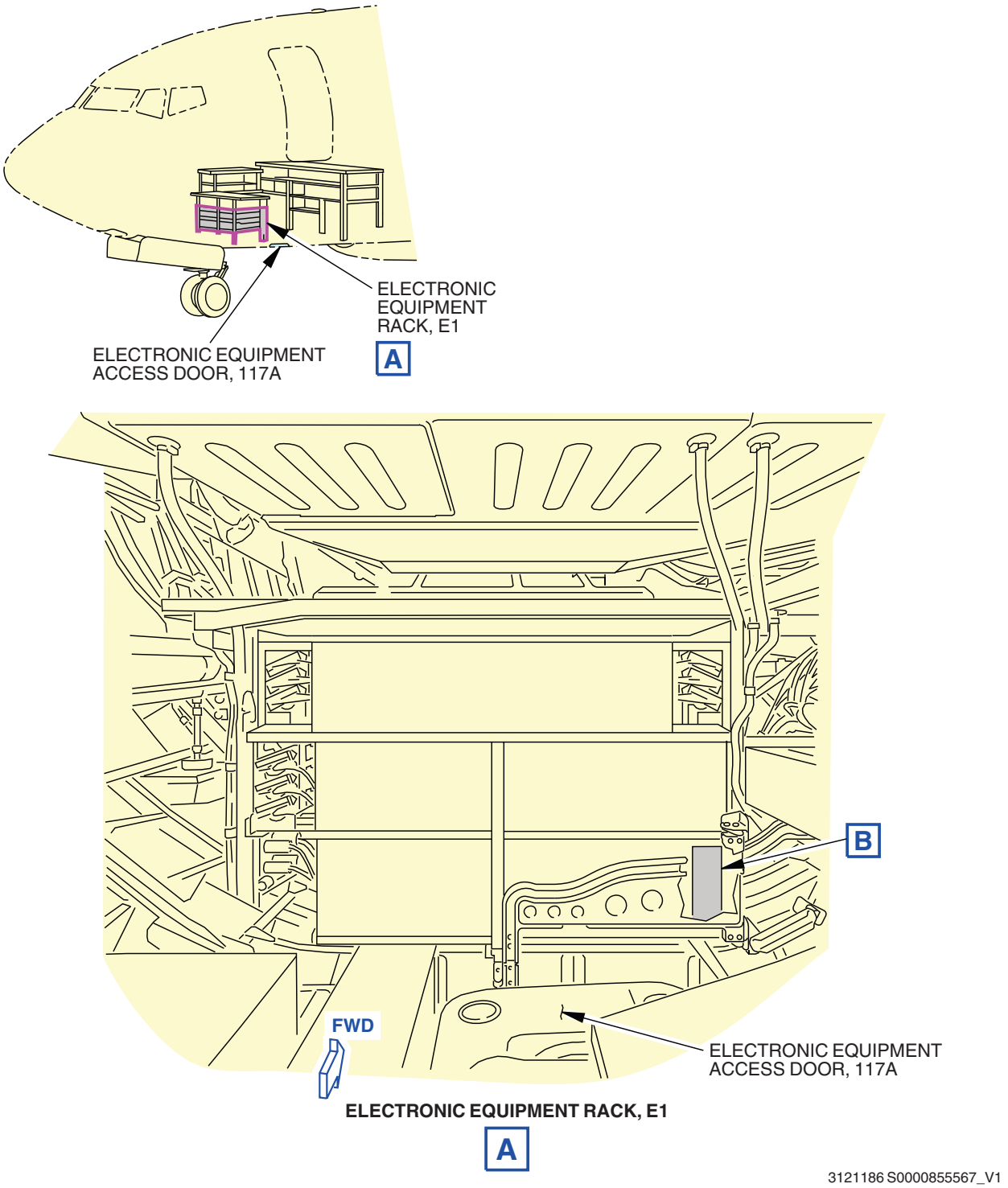
(3) Install the compartment overheat detection control unit (TASK 26-18-01-400-801).

SUBTASK 26-10-00-860-015

(4) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

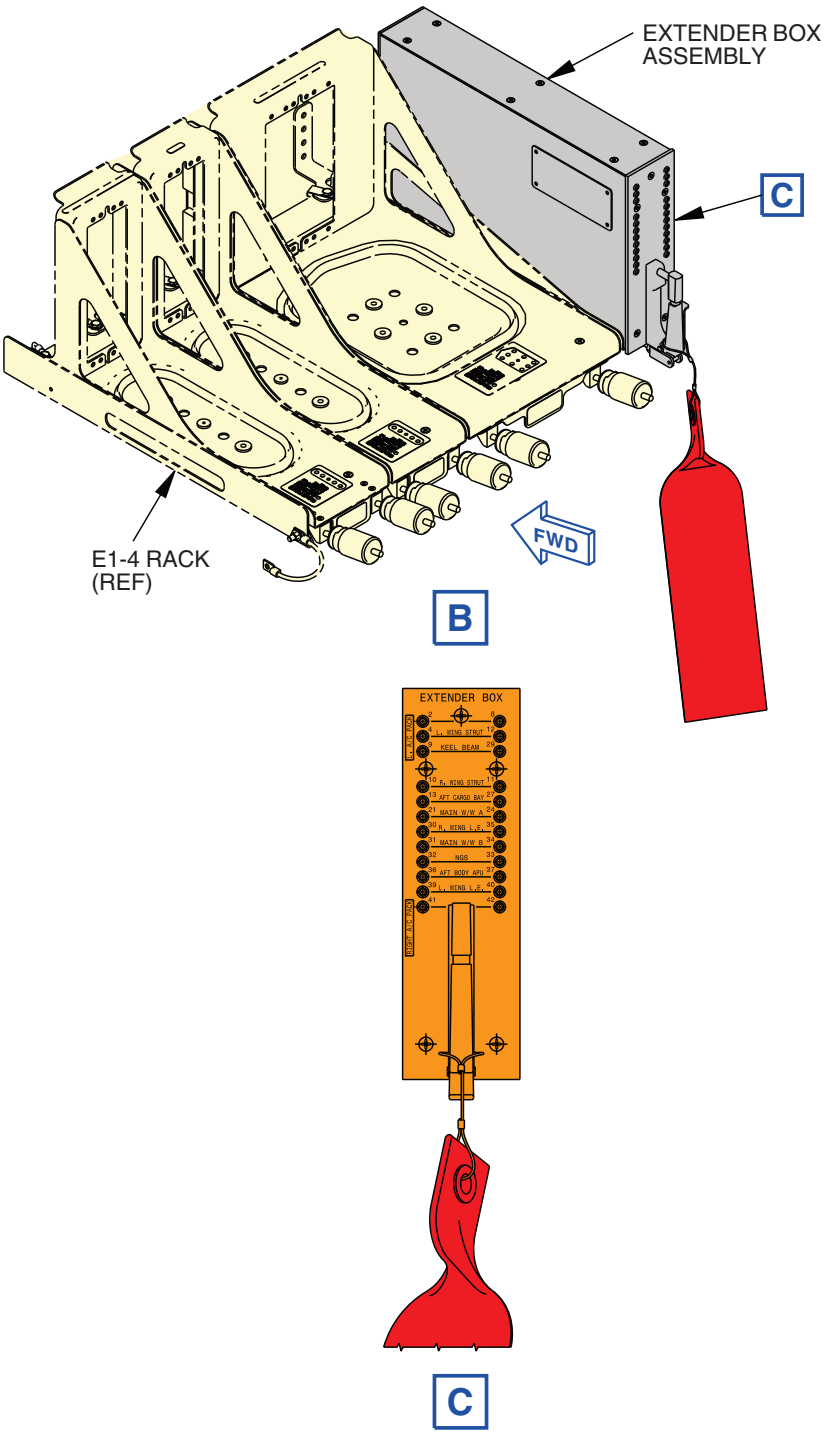
——— END OF TASK ———

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Overheat Detection Test Using Handheld LCRZ Meter
Figure 503/26-10-00-990-804 (Sheet 1 of 2)

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Overheat Detection Test Using Handheld LCRZ Meter
Figure 503/26-10-00-990-804 (Sheet 2 of 2)

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TASK 26-10-00-700-803

4. Overheat Detection Test Using UN-ATE Tool by Technologies Harness Scanner

(Figure 504)

A. General

(1) This task gives the instructions to do a health check of the compartment overheat loop detectors and wiring using the UN-ATE tool.

(a) This procedure is used to detect shorts, open, and degraded overheat loop detectors in the wing and body overheat detection system.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
26-18-01-000-801	Compartment Overheat Detection Control Unit Removal (P/B 401)
26-18-01-400-801	Compartment Overheat Detection Control Unit Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-17989	Wiring Tester - Automated Wireless Part #: UN-ATE BASIC Supplier: 87Z60
SPL-17990	Interface Cable - UN-ATE Aircraft Bleed OH Loop Part #: A42 Supplier: 87Z60
SPL-17991	Test Kit - Bleed Overheat Loop Solution Part #: SS-B737MX-BOHL Supplier: 87Z60

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Prepare for the Test

SUBTASK 26-10-00-020-005

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

NOTE: Electrical power must be off to prevent incorrect fault indications or fault messages from occurring.

SUBTASK 26-10-00-860-014

(2) Make sure that the battery switch is in the OFF position on the P-5 overhead panel.

SUBTASK 26-10-00-760-001

(3) Make sure that all ground power is disconnected.

SUBTASK 26-10-00-020-006

(4) Remove the M237 compartment overheat detection control unit (TASK 26-18-01-000-801).

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SUBTASK 26-10-00-480-025

(5) Insert the interface cable, SPL-17990, to the M237 compartment overheat detection control unit's connector on the airplane, with the optional electronic dust cap to verify if there is any power on the connector.

(a) If the electronic dust cap has power, remove the power.

SUBTASK 26-10-00-020-007

(6) If no power is confirmed, remove the electronic dust cap and connect the wiring tester, COM-17989, to the interface cable, SPL-17990.

(a) Do not ground the test box unit for the bleed overheat loop test.

NOTE: This may introduce error on the overheat loop readings.

SUBTASK 26-10-00-480-026

(7) Log into to the tablet to provide testing and traceability.

(a) Start up the test kit, SPL-17991, and login with your user ID.

(b) Identify the location of the aircraft being maintained.

(c) Select the aircraft model.

(d) Select the aircraft subsystem (example: bleed OH loop).

(e) Identify the reason for your test (training/diagnostic/preventative).

(f) The software will now guide you into the set-up with pictograph.

F. Overheat Detection Test with UN-ATE

NOTE: Do not ground the test box unit for the bleed overheat loop test.

SUBTASK 26-10-00-700-032

(1) UN-ATE will make a scan to provide an overview of all OH loops.

(a) Select “Scan” on the UN-ATE test kit, SPL-17991.

1) The wiring tester, COM-17989, will automatically make impedance measurements on each loop circuit and present results as per wiring diagram.

NOTE: Each loop circuit will have three impedance measurements such as the Table 506 below.

Table 506/26-10-00-993-808 Overheat Loop Impedance Measurements

Measurement type	Impedance result	
Inner to inner core	Resistance (ohm)	Capacitance (nF)
One side inner to ground	Resistance (ohm)	Capacitance (nF)
Other side inner to ground	Resistance (ohm)	Capacitance (nF)

2) The wiring tester, COM-17989, will analyze the three measurement results of each loop circuit and highlight in color the problematic results based on the following condition scenarios in Table 507 below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 507/26-10-00-993-809 Overheat Detection Loop Health Check

Wiring Example	Condition Scenarios	Resistance (ohm)	Capacitance (nF)	Loop health status	Reference procedure step
Loop Inner to Inner core	1	< 12 ohms*[1]	-	Good inner core	-
	2	> 12 ohms*[1]	-	Outside tolerance	Step (3)
	3	Any reading	1 to 100 nF	Opened inner core	Step (4)
Loop Inner to ground on both sides	4	1 M ohms-	1 to 100 nF	Excellent loop	Step (2)
	5	< 15 k ohms	1 to 100 nF	Shortened bad loop	Step (5)

*[1] The resistance values provided here are only arbitrary. The prescribed maximum tolerance values for each circuit will be populated into the baseline signature reference.

SUBTASK 26-10-00-700-033

(2) Healthy loop - All measurements are within their tolerance.

(a) Once the scan completed, wiring tester, COM-17989, shows NO HIGHLIGHT on any loop circuit indicating all loop circuit are HEALTHY.

(b) Go to Step (7).

SUBTASK 26-10-00-700-034

(3) Loop inner core measurement outside of tolerance.

(a) The wiring tester, COM-17989, flags an OUT OF TOLERANCE inner to inner core loop circuit.

1) This problem may be an indication of the degrading wire/loop connection (corrosion, pin insert, etc).

2) The procedure is to validate with conventional procedures and visual inspection.

SUBTASK 26-10-00-700-035

(4) Opened inner core loop fault location procedure.

(a) The UN-ATE flags an OPENED or discontinuity along the loop inner core.

NOTE: The wiring tester, COM-17989's fault detection feature is used to locate the discontinuity point.

1) Select the problematic loop and select FIRELOOP DIAGNOSTIC and then IDENTITY BAD LOOP to locate the closest point to the open fault location.

NOTE: The wiring tester, COM-17989's diagnostic feature may take up to ten minutes to process. Once the process is completed, the closest overheat fireloop segment causing the fault is highlighted.

2) Confirm the damage part (wire or fireloop segment) that cause the problem, do these steps:

a) Go to the fireloop segment located by the wiring tester, COM-17989, and do a check for any visible damage that may cause the discontinuity in the loop.

b) Manually disconnect one side of the fireloop segment closest to the wiring tester, COM-17989, located point.

c) Connect the INNER core lead at both sides of the connector to the ground (aircraft chassis) using two short cables with alligator clips and lead adaptors.

NOTE: This location is the manually grounded point.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) In the test kit, SPL-17991, select REFRESH IMPEDANCE to re-measure the three impedance measurements of the selected loop (inner to inner core and inner to ground x 2).

e) Do the following steps to confirm if the fault is located at the RIGHT or LEFT side of the previously grounded point according to the measurement results in Table 508 below:

Table 508/26-10-00-993-810 Opened Inner Core Loop Check

Scenario	Measurement	Resistance (ohm)	Capacitance (nF)	Fault Location
-	Inner to inner	-	-	-
1	1st Inner to ground	XX ohm	-	Fault at the RIGHT side of the grounded point
	2nd Inner to ground	-	XX nF	
2	1st Inner to ground	-	XX nF	Fault at the LEFT side of the grounded point
	2nd Inner to ground	XX ohm	-	

<1> Go to the previously grounded point undo the grounding and reconnect the disconnected fireloop segment.

<2> Move to the next wire/fireloop segment (RIGHT or LEFT according to the previous deduction) and disconnect one side of the that segment and connect the inner core leads to the ground.

NOTE: This location is the second grounding point.

<3> In the test kit, SPL-17991, redo the measurement by selecting REFRESH IMPEDANCE.

<4> Confirm the side (RIGHT or LEFT) of the faulty segment.

NOTE: After this step, the faulty part may be precisely located. In some cases, other disconnecting and grounding processes may be required, especially in cases where multiple wire segments are inserted between overheat fireloop segments.

f) Once the faulty part located, confirmed its condition using conventional Boeing or operator procedure.

g) Replace or repair the part.

h) Do the complete scan in Step (1) again to confirm the repair.

SUBTASK 26-10-00-700-036

(5) Shorted bad loop fault location procedure.

(a) The wiring tester, COM-17989, flags a SHORTED loop or alteration of the loop insulation quality.

NOTE: The wiring tester, COM-17989's fault detection feature is used to locate the short point.

(b) Select the problematic loop and select FIRELOOP DIAGNOSTIC and then IDENTIFY BAD LOOP to locate the closest point to the short fault location.

NOTE: The wiring tester, COM-17989's diagnostic feature may take up to ten minutes to process. Once the process is completed, the closest wire/fireloop segment causing the fault is highlighted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Confirm the damage part (wire or fireloop segment) that cause the problem, do these steps:

1) Go to the wire/fireloop segment located by the wiring tester, COM-17989, and check for any visible damage that may cause the short in the loop.

2) Manually disconnect one side of the fireloop segment closest to the wiring tester, COM-17989, located point.

NOTE: This location is the floating point.

3) In the test kit, SPL-17991, select REFRESH IMPEDANCE to measure the three impedance measurements of the selected loop (inner to inner core and inner to ground x 2).

4) Do the following steps to confirm if the fault is located at the RIGHT or LEFT side of the previously floating point according to the measurement results in Table 509 below:

Table 509/26-10-00-993-811 Shorted Bad Loop Check

Scenario	Measurement	Resistance (ohm)	Capacitance (nF)	Fault Location
-	Inner to inner	-	-	-
1	1st Inner to ground	Short Resistance ohm	-	Fault at the LEFT side of the floating point
	2nd Inner to ground	-	XX nF	
2	1st Inner to ground	-	XX nF	Fault at the LEFT side of the floating point
	2nd Inner to ground	Short Resistance ohm		

a) Go to the previously floating point and reconnect the disconnected wire/fireloop segment.

b) Move to the next wire/fireloop segment (RIGHT or LEFT according to the previous deduction) and disconnect one side of that segment.

NOTE: This location is the second floating point.

c) In the test kit, SPL-17991, redo the measurement by selecting REFRESH IMPEDANCE.

d) Confirm the side (RIGHT or LEFT) of the faulty segment.

NOTE: After this step, the faulty part may be precisely located. In some cases, other disconnecting processes may be required, especially if the short resistance is between 10 k ohms to 15 k ohms.

5) Once the faulty part located, confirmed its condition using conventional Boeing/Airline procedures.

6) Replace or repair part.

7) Do the complete scan in Step (1) again to confirm the repair.

SUBTASK 26-10-00-700-038

(6) Analysis and documentation.

(a) You have now completed the wiring tester, COM-17989's diagnostic and should document your work.

1) Insert your detailed comments about the repairs performed in the wiring tester, COM-17989's test notes (optional).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Save the test results.

a) Click on the proper tab that best describes the diagnostic tests results.

NOTE: These comments are preset and will be correlated to the reason of your test in the Prepare for the Test Section, Step (6) (not relevant, no issue found, found my problem, potential degradation).

b) The wiring tester, COM-17989, will save it using traceability data (aircraft type, sub-system, serial number, date).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 26-10-00-080-013

(1) Disconnect the wiring tester, COM-17989, from the interface cable, SPL-17990.

SUBTASK 26-10-00-080-014

(2) Disconnect the interface cable, SPL-17990, from the M237 compartment overheat detection control unit connector on the airplane.

SUBTASK 26-10-00-420-003

(3) Install the M237 compartment overheat detection control unit (TASK 26-18-01-400-801).

SUBTASK 26-10-00-840-001

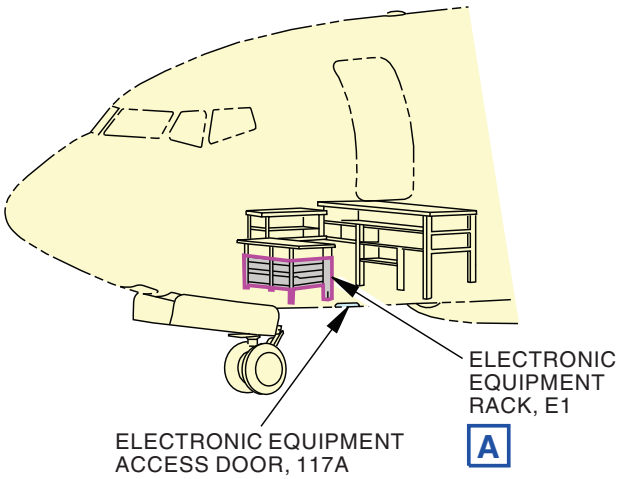
(4) Make sure that the battery switch is in the ON position on the P-5 overhead panel.

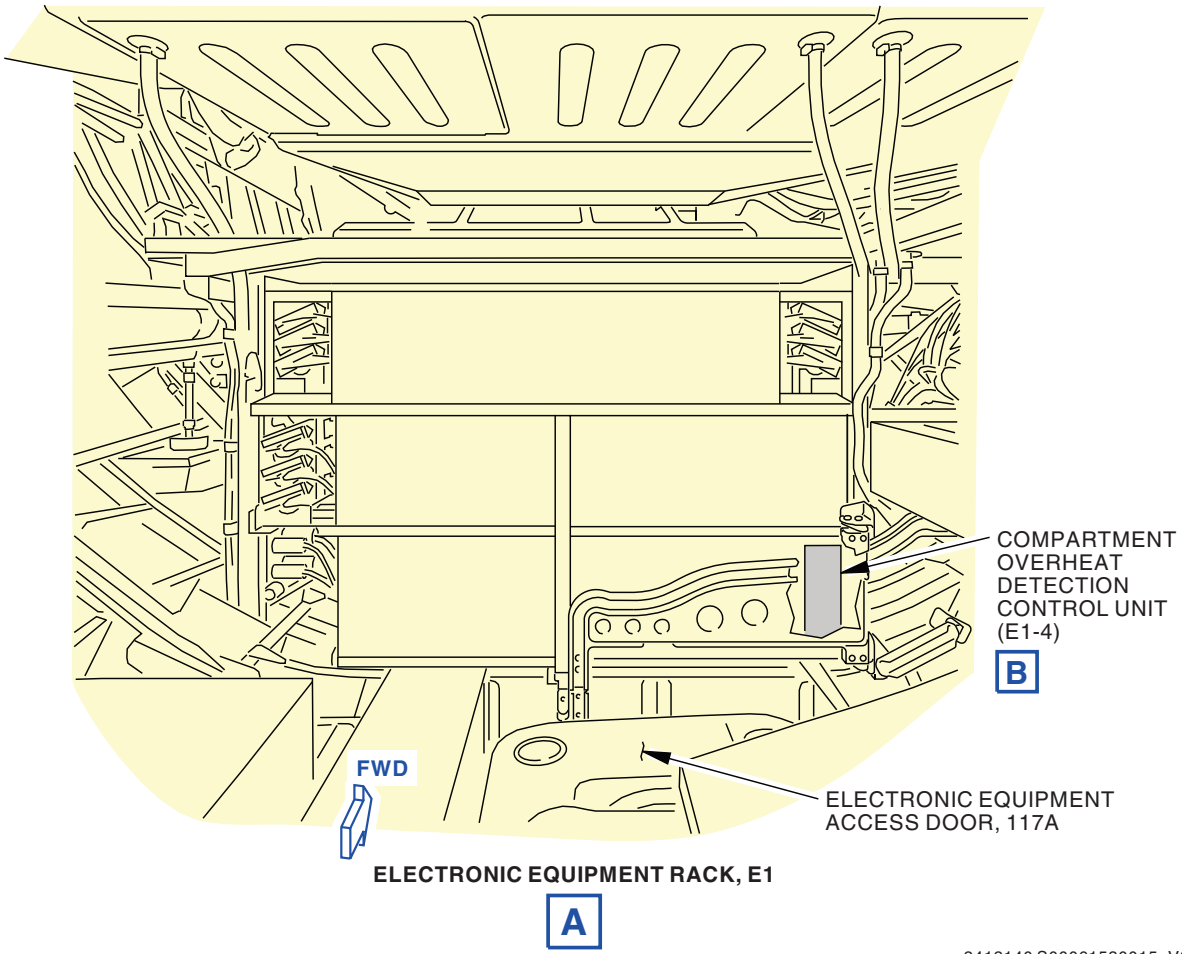
SUBTASK 26-10-00-840-002

(5) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2412140 S00061530015_V2

Compartment Overheat Detection Control Unit Location
Figure 504/26-10-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE AND APU FIRE DETECTION CONTROL MODULE, M279 - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Engine and APU fire detection control module - deactivation.

(2) Engine and APU fire detection control module - activation.

TASK 26-10-01-040-801

2. Engine and APU Fire Detection Control Module - Deactivation

A. General

(1) This task will deactivate the engine and APU fire detection control module, M279.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Engine and APU Fire Detection Control Module Deactivation

SUBTASK 26-10-01-020-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

D. Engine and APU Fire Detection Control Module - Tryout

NOTE: This tryout is to make sure that the engine and APU fire detection control module is in a zero energy state.

SUBTASK 26-10-01-210-001

(1) On the Engine and APU Fire Control Panel, P8-1, make sure that the FAULT light (Amber) is on.

SUBTASK 26-10-01-210-002

(2) On the Fire Detection Control Module, M279, in the Electrical and Electronics Compartment, make sure that these lights come on:

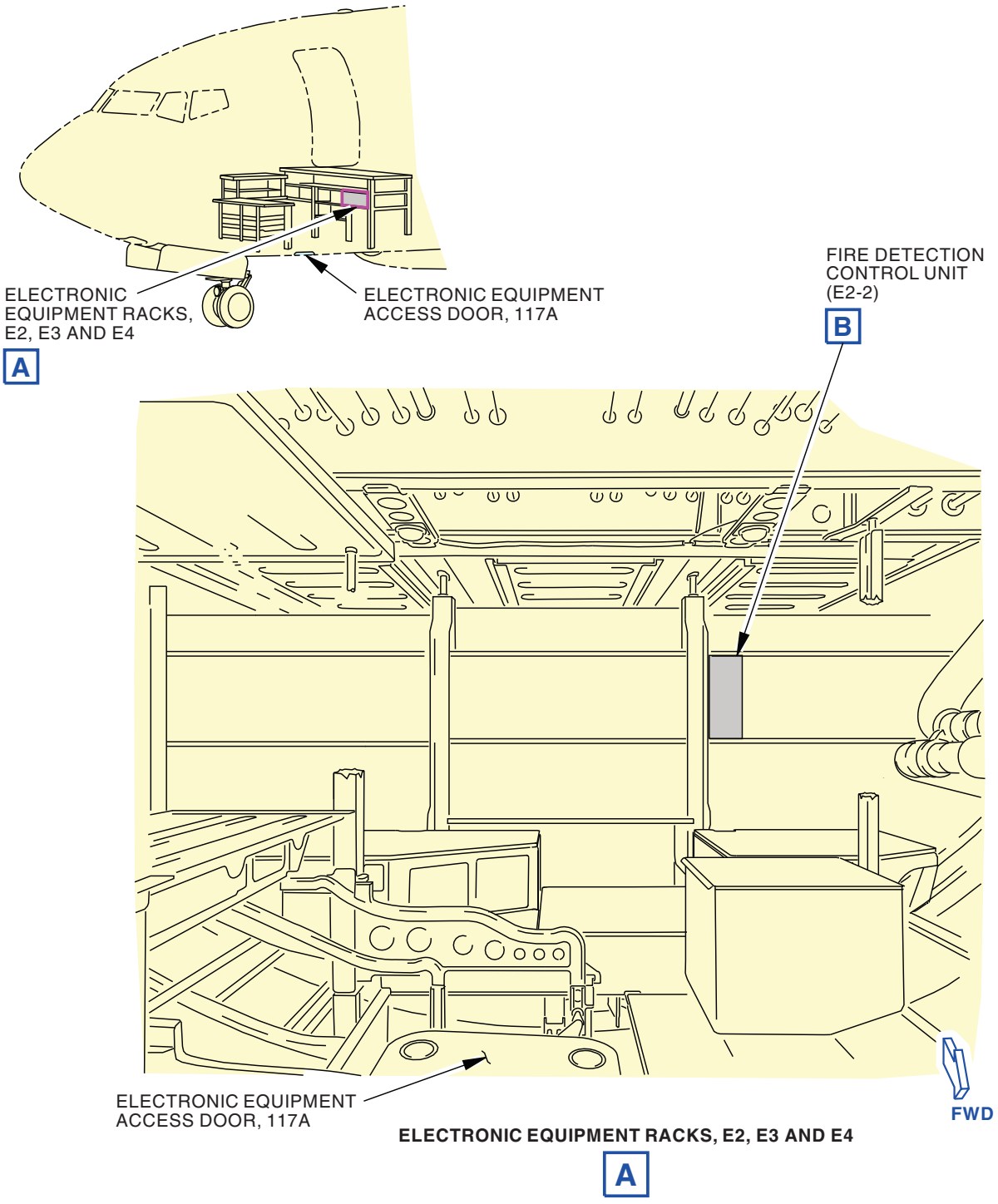
(a) ENGINE 1 LOOP A (Amber).

(b) ENGINE 1 LOOP B (Amber).

(c) The four FAULT CODE lights are OFF-OFF-ON-OFF.

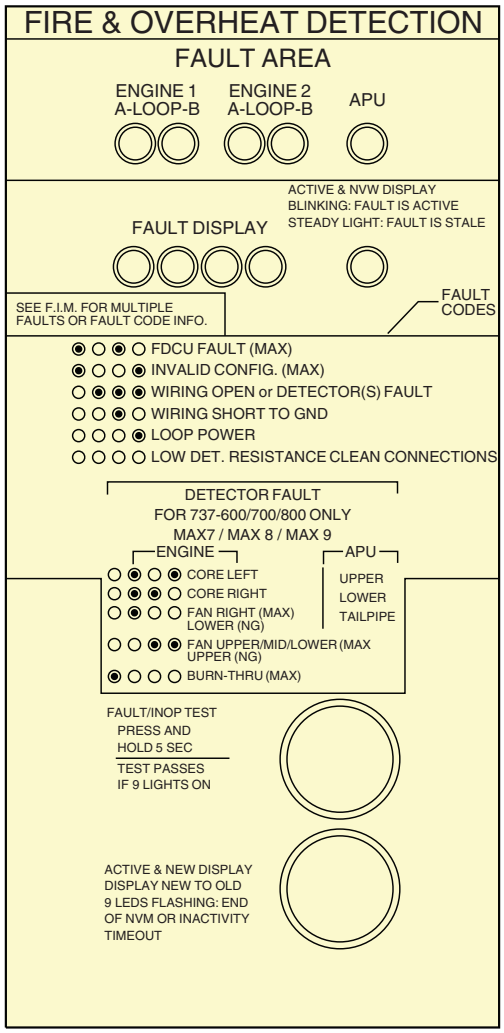
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine and APU Fire Detection Control Module
Figure 201/26-10-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIRE DETECTION CONTROL UNIT

B

2412064 S00061529844_V3

Engine and APU Fire Detection Control Module
Figure 201/26-10-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-10-01-440-801

3. Engine and APU Fire Detection Control Module - Activation

(Figure 201)

A. General

(1) This task will activate the engine and APU fire detection control module.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Engine and APU Fire Detection Control Module Activation

SUBTASK 26-10-01-420-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE AND APU FIRE DETECTION CONTROL MODULE, M279 - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Engine and APU Fire Detection Control Module (M279).

(2) An installation of the Engine and APU Fire Detection Control Module.

B. The Engine and APU Fire Detection Control Module is installed on the E2-2 shelf in the electronic equipment compartment.

TASK 26-10-01-000-801

2. Engine and APU Fire Detection Module Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Engine and APU Fire Detection Module.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 26-10-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-10-01-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

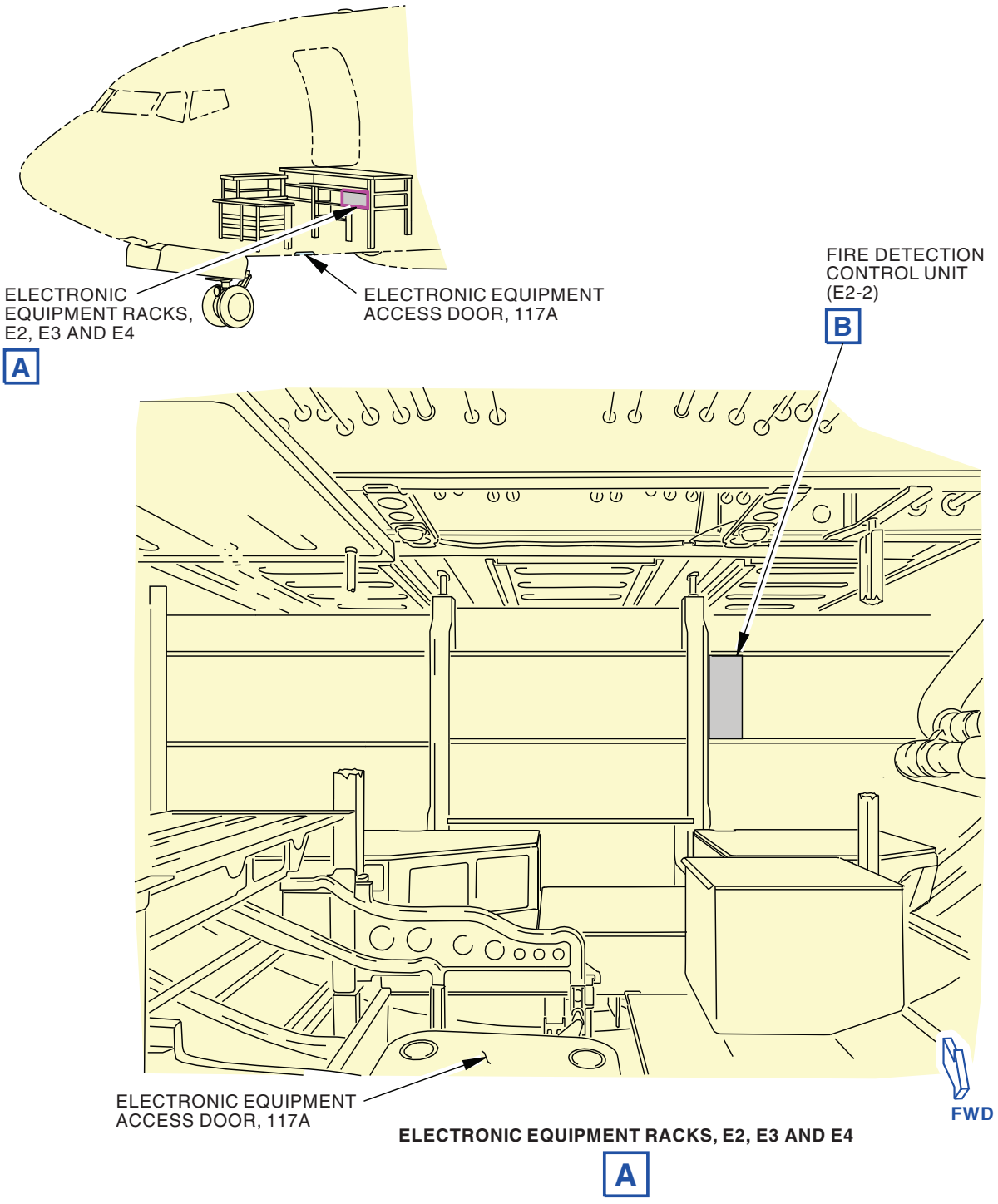
F. Engine and APU Fire Detection Module Removal.

SUBTASK 26-10-01-020-002

(1) To remove the fire detection control unit [1], do this task: E/E Box Removal, TASK 20-10-07-000-801.

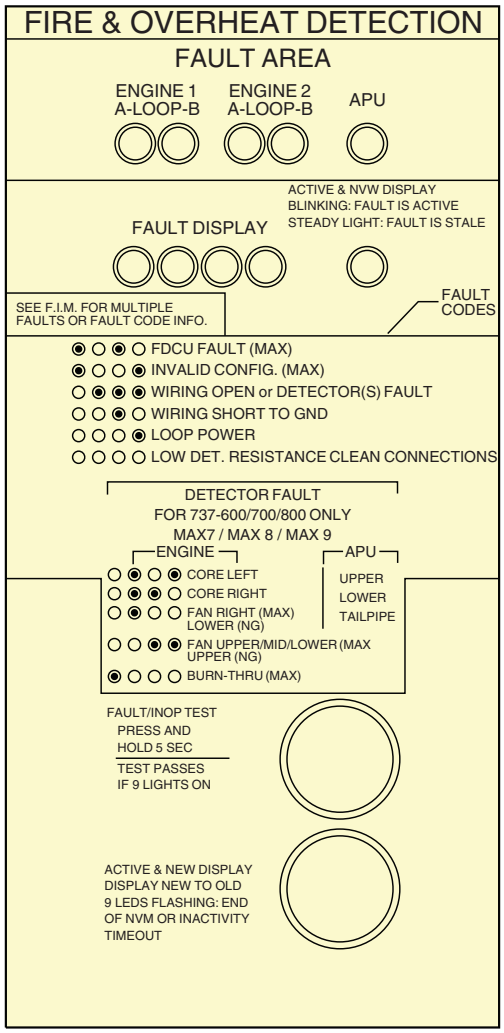
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine and APU Fire Detection Control Unit Installation
Figure 401/26-10-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIRE DETECTION CONTROL UNIT

B

2412064 S00061529844_V3

Engine and APU Fire Detection Control Unit Installation
Figure 401/26-10-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-10-01-400-801

3. Engine and APU Fire Detection Module Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Engine and APU Fire Detection Module.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
26-11-00-710-801	Engine Fire Detection - Operational Test (P/B 501)
26-15-00-710-801	APU Fire Detection - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fire detection control unit	26-10-01-02-010	PTF ALL

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Engine and APU Fire Detection Control Module Installation

SUBTASK 26-10-01-420-002

(1) To install the fire detection control unit [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

G. Engine and APU Fire Detection Control Unit Installation Test

SUBTASK 26-10-01-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-10-01-710-001

(2) Do this task: Engine Fire Detection - Operational Test, TASK 26-11-00-710-801.

SUBTASK 26-10-01-710-002

(3) Do this task: APU Fire Detection - Operational Test, TASK 26-15-00-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-10-01-010-002

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE DETECTION - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Engine fire detection - deactivation.

(2) Engine fire detection - activation.

TASK 26-11-00-040-801

2. Engine Fire Detection - Deactivation

A. General

(1) This task will deactivate the engine fire detection system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-11-00-020-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

D. Engine Fire Detection - Tryout

NOTE: This tryout is to make sure that the engine fire detection system is in a zero energy state.

SUBTASK 26-11-00-210-001

(1) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

(a) On the Engine and APU Fire Control Panel, make sure that none of these lights come on:

1) Engine 1 Fire Handling light.

2) Engine 2 Fire Handling light.

3) ENG 1 OVERHEAT light.

4) ENG 2 OVERHEAT light.

(b) On the Pilot's Glareshield, P7, make sure that none of these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights.

2) The Captain's and First Officer's FIRE WARN lights.

3) The OVHT/DET light.

(c) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, does not come on.

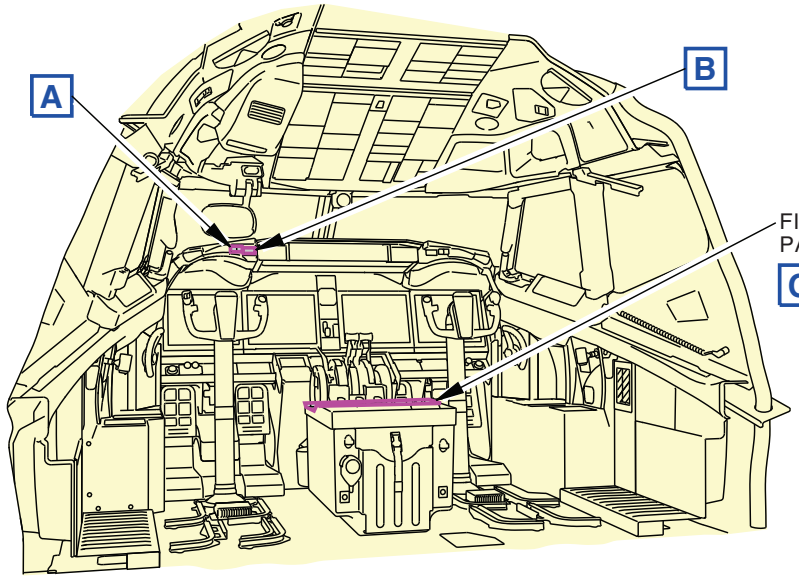
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-11-00-210-002

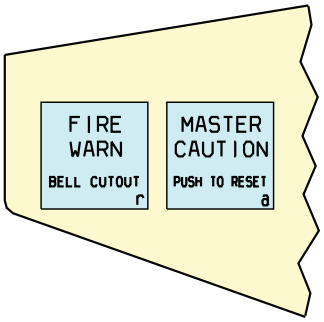
(2) Release the TEST switch on the Engine and APU Fire Control Panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

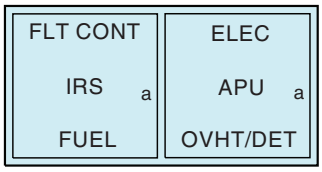


FLIGHT COMPARTMENT



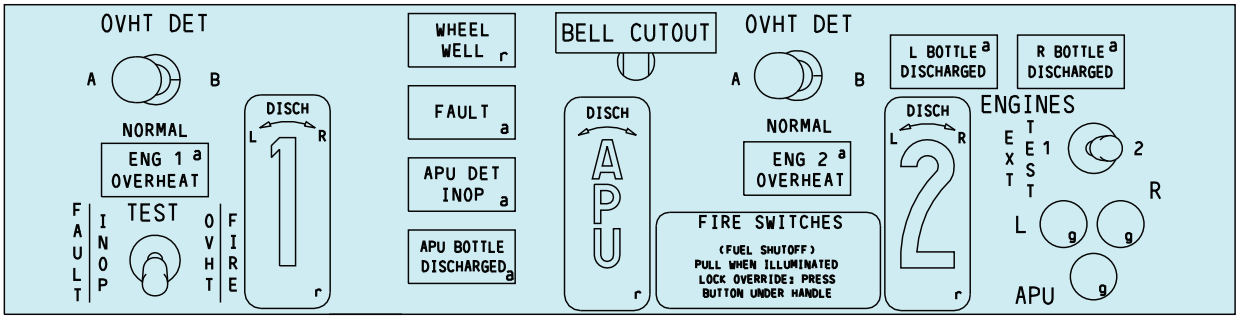
CAPTAIN'S FIRE
WARNING SWITCH (P7)
(FIRST OFFICER'S IS
OPPOSITE)

A



CAPTAIN'S WARNING
LIGHTS (P7)

B



FIRE CONTROL PANEL (P8-1)

C

2412065 S00061529856_V1

Engine Fire Detection System
Figure 201/26-11-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-00-440-801

3. Engine Fire Detection - Activation

(Figure 201)

A. General

(1) This task will activate the engine fire detection system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-11-00-420-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE DETECTION - ADJUSTMENT/TEST

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure consists of these tasks:

(1) Engine Fire Detection - Operational Test.

(2) Engine Fire Detection Loop Resistance - System Test.

(3) Engine Fire Detection Circuit - System Test.

TASK 26-11-00-710-801

2. Engine Fire Detection - Operational Test

(Figure 501, Figure 502)

A. General

(1) The Operational Test does a check of the Engine Fire Detection System without disturbing the detection system and it only uses the equipment that is on the airplane. This test will verify the circuit from the Engine and APU Fire Control Panel, P8-1, to the Engine and APU Fire Detection Control Module, M279. The test will verify that the Engine and APU Fire Detection Control Module, which continuously monitors the detection loops, operates correctly.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 26-11-00-860-003

(1) Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

SUBTASK 26-11-00-710-001

(2) Make sure the two OVHT DET switches on the Engine and APU Fire Control Panel, P8-1, in the flight compartment are in the NORMAL positions.

D. Do the FIRE/OVHT Test

(Figure 501)

SUBTASK 26-11-00-710-002

(1) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

(a) On the Engine and APU Fire Control Panel, make sure these lights come on:

1) Engine 1 Fire Handle light (Red).

2) Engine 2 Fire Handle light (Red).

3) APU Fire Handle light (Red)

4) ENG 1 OVERHEAT light (Amber).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The Captain's and First Officer's FIRE WARN lights (Red).

3) The OVHT/DET light (Amber).

(c) Make sure the lights on the left and right Engine Start Levers, on the Engine Start Lever Panel, P10, come on.

(d) Make sure the flight compartment fire bell in the Aural Warning Module, M315, comes on.

SUBTASK 26-11-00-710-003

(2) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) On the Engine and APU Fire Control Panel, make sure these lights go off:

1) Engine 1 Fire Handle light (Red).

2) Engine 2 Fire Handle light (Red).

3) APU Fire Handle light (Red)

4) ENG 1 OVERHEAT light (Amber).

5) ENG 2 OVERHEAT light (Amber).

(b) On the Pilot's Glareshield, make sure these lights go off:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The Captain's and First Officer's FIRE WARN lights (Red).

3) The OVHT/DET light (Amber).

(c) Make sure the lights on the left and right Engine Start Levers, on the Engine Start Lever Panel, P10, go off.

(d) Make sure that the fire bell in the flight compartment on the Aural Warning Module, M315, stops.

NOTE: Fire Bell may take up to 2 seconds to stop after TEST switch is released.

E. Do the FAULT/INOP Test

(Figure 501)

SUBTASK 26-11-00-710-004

(1) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the FAULT/INOP position.

(a) On the Engine and APU Fire Control Panel, make sure these lights come on:

1) FAULT light (Amber).

2) APU DET INOP (Amber)

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

SUBTASK 26-11-00-710-005

(2) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) On the Engine and APU Fire Control Panel, make sure these lights are off:

1) FAULT light (Amber).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) APU DET INOP (Amber)

(b) On the Pilot's Glareshield, P7, make sure these lights are off:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

F. Do the FAULT/INOP Test from the Engine and APU Fire Detection Control Module, M279

(Figure 502)

SUBTASK 26-11-00-710-006

(1) Push and hold the FAULT/INOP TEST switch on the Engine and APU Fire Detection Control Module, in the Electrical and Electronics Compartment.

(a) On the Engine and APU Fire Detection Control Module, make sure these lights come on:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) ENGINE 2 LOOP A (Amber).

4) ENGINE 2 LOOP B (Amber).

5) APU (Amber)

6) The four FAULT CODE lights are ON-ON-ON-ON.

SUBTASK 26-11-00-710-007

(2) Release the FAULT/INOP TEST switch on the Engine and APU Fire Detection Control Module.

(a) On the Engine and APU Fire Detection Control Module, make sure these lights are off:

1) ENGINE 1 LOOP A (Amber).

2) ENGINE 1 LOOP B (Amber).

3) ENGINE 2 LOOP A (Amber).

4) ENGINE 2 LOOP B (Amber).

5) APU (Amber)

6) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

G. Put the Airplane Back to its Usual Condition

SUBTASK 26-11-00-860-004

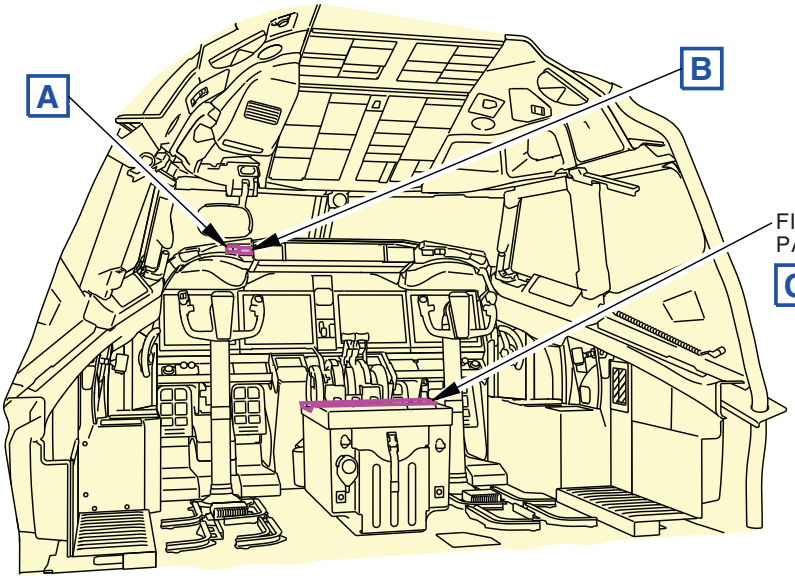
(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

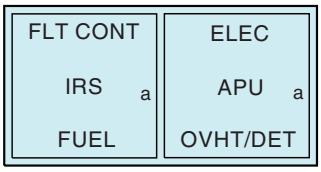
Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

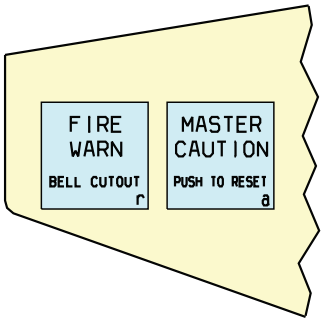


FLIGHT COMPARTMENT



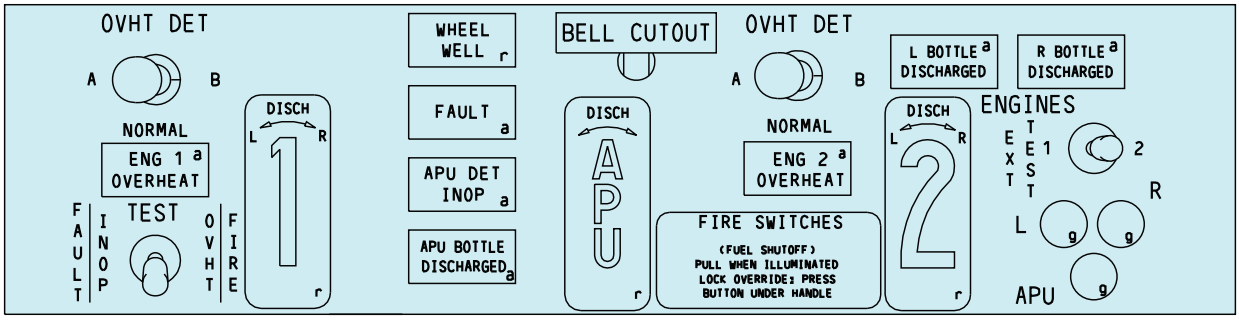
CAPTAIN'S WARNING
LIGHTS (P7)

B



CAPTAIN'S FIRE
WARNING SWITCH (P7)
(FIRST OFFICER'S IS
OPPOSITE)

A



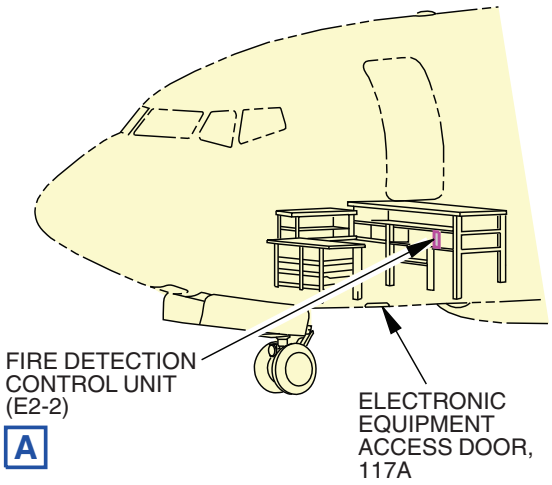
FIRE CONTROL PANEL (P8-1)

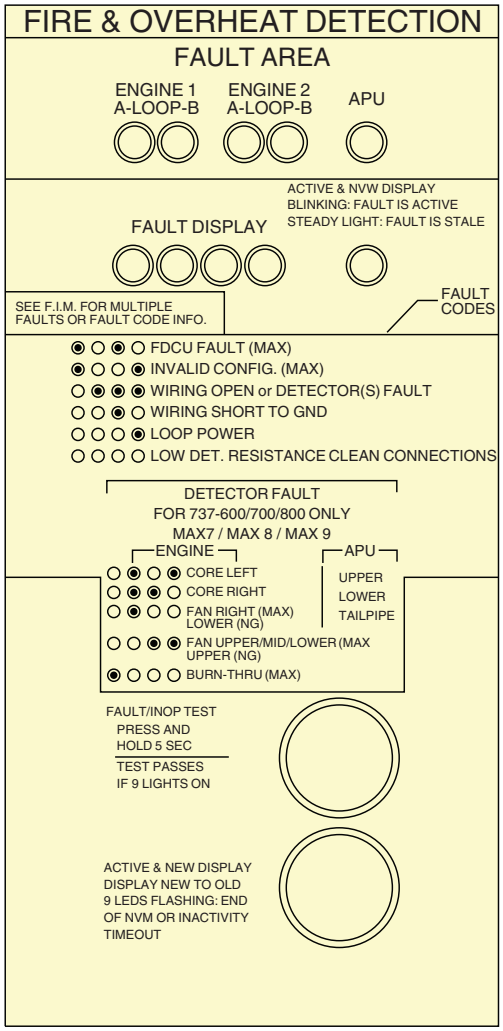
C

2412065 S00061529856_V1

Engine Fire Detection Operational Test
Figure 501/26-11-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details





FIRE DETECTION CONTROL UNIT

A

2412066 S00061529861_V2

Engine and APU Fire Detection Control Module, M279
Figure 502/26-11-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-00-730-801

3. Engine Fire Detection Loop Resistance - System Test

(Figure 503)

A. General

(1) The Loop Resistance Test does a check of the Engine Fire Detection Loop Circuit. It disturbs the engine fire detection system by removing the Engine and APU Fire Detection Control Module, M279. Some test equipment will also be necessary.

B. References

Reference	Title
26-10-01-000-801	Engine and APU Fire Detection Module Removal (P/B 401)
26-10-01-400-801	Engine and APU Fire Detection Module Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Loop Resistance Test

SUBTASK 26-11-00-860-006

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-11-00-010-001

(2) Get access to the Electronic Equipment Bay:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

G. Engine Fire Detection Loop Resistance Test

SUBTASK 26-11-00-020-002

(1) Do this task: Engine and APU Fire Detection Module Removal, TASK 26-10-01-000-801.

SUBTASK 26-11-00-730-001

(2) Use a digital/analog multimeter, COM-1793 to measure the resistance across pin 25 of connector D1002 and airplane ground.

(a) Make sure that the resistance is 881 ±40 ohms.

SUBTASK 26-11-00-730-002

(3) Use a digital/analog multimeter, COM-1793 to measure the resistance across pin 12 of connector D1002 and airplane ground.

(a) Make sure that the resistance is 881 ±40 ohms.

SUBTASK 26-11-00-730-003

(4) Use a digital/analog multimeter, COM-1793 to measure the resistance across pin 25 of connector D998 and airplane ground.

(a) Make sure that the resistance is 881 ±40 ohms.

SUBTASK 26-11-00-730-004

(5) Use a digital/analog multimeter, COM-1793 to measure the resistance across pin 12 of connector D998 and airplane ground.

(a) Make sure that the resistance is 881 ±40 ohms.

SUBTASK 26-11-00-420-002

(6) Do this task: Engine and APU Fire Detection Module Installation, TASK 26-10-01-400-801.

H. Put the Airplane Back to its Usual Condition

SUBTASK 26-11-00-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-11-00-860-007

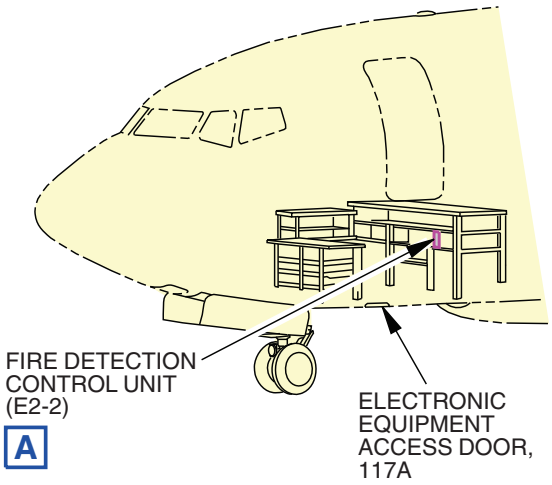
(2) Close these circuit breakers:

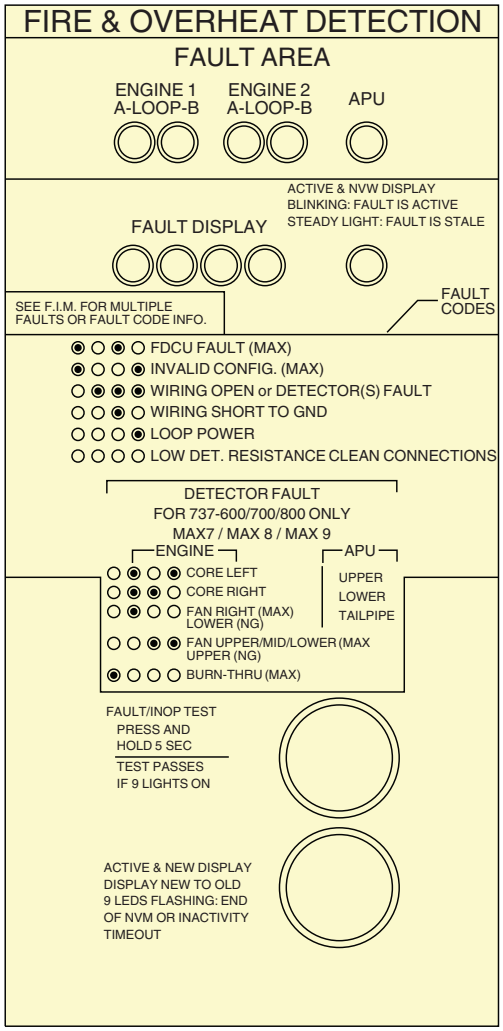
F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





FIRE DETECTION CONTROL UNIT

A

2412066 S00061529861_V2

Engine Fire Detection Resistance Test
Figure 503/26-11-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-00-730-802

4. Engine Fire Detection Circuit - System Test

(Figure 501, Figure 504)

A. General

(1) The Circuit Test does a check of all the Engine Fire Detection System. It does not disturb the engine fire detection system but test equipment will be necessary.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1655	Test Equipment - APU/Engine Fire Detection System Part #: C26005-1 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Circuit Test

SUBTASK 26-11-00-860-009

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
D	15	C01401	LANDING GEAR AIR/GND RELAY
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 26-11-00-860-010

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

SUBTASK 26-11-00-010-002

(3) Get access to the Electronic Equipment Bay:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-11-00-010-003

(4) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

SUBTASK 26-11-00-730-006

(5) Make sure that the two OVHT DET switches, on the Engine and APU Fire Control Panel, P8-1, are in the NORMAL positions.

SUBTASK 26-11-00-480-001

(6) On Engine 1 (2), connect the test box equipment, SPL-1655 to the two terminal lugs (A/1 and B/2) of the FAN LOWER Fire Detector, M3061, and airplane ground.

NOTE: This test is written for engine 1. The test is the same for engine 2, except as noted by the data in parenthesis. Use the data in parenthesis to do the test for engine 2.

(a) Make sure that all the toggle switches on the test box are in the NORMAL positions.

(b) Connect one pair of alligator clips to each loop:

1) Connect one alligator clip to the responder bracket or to a grounding lug.

2) Connect the other alligator clip to the responder lug.

G. Do a test of the fault indication circuit

SUBTASK 26-11-00-730-007

(1) On the test box, move the ENGINE LOOP A/1 GND FAULT switch to the GND FAULT position.

(a) On the Engine and APU Fire Detection Control Module, M279, make sure that these conditions occur:

1) The ENGINE 1 (2) LOOP A FAULT AREA light comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) The four FAULT CODE lights are OFF-OFF-ON-OFF.

SUBTASK 26-11-00-730-008

(2) On the test box, move the ENGINE LOOP B/2 GND FAULT switch to the GND FAULT position.

(a) Make sure that the FAULT light on the Engine and APU Fire Control Panel, P8-1, comes on.

(b) On the Engine and APU Fire Detection Control Module, M279, make sure that these conditions occur:

1) The ENGINE 1 (2) LOOP A FAULT AREA light comes on.

2) The ENGINE 1 (2) LOOP B FAULT AREA light comes on.

3) The four FAULT CODE lights are OFF-OFF-ON-OFF.

SUBTASK 26-11-00-730-009

(3) On the test box, move the ENGINE LOOP A/1 AND B/2 GND FAULT switches to the NORMAL position.

(a) Make sure that the FAULT light on the Engine and APU Fire Control Panel, P8-1, goes off.

(b) On the Engine and APU Fire Detection Control Module, M279, make sure that these conditions occur:

1) The ENGINE 1 (2) LOOP A FAULT AREA light goes off.

2) The ENGINE 1 (2) LOOP B FAULT AREA light goes off.

3) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

H. Do a test of the overheat indication circuit

SUBTASK 26-11-00-730-010

(1) On the test box, move the ENGINE LOOP A/1 AND B/2 OVHT switches to the OVHT position.

(a) Make sure that the ENG 1 (2) OVERHEAT light on the Engine and APU Fire Control Panel, P8-1, comes on.

(b) On the Pilot's Glareshield, P7, make sure that these conditions occur:

1) The Captain and the First Officer's MASTER CAUTION light come on.

2) The OVHT DET light comes on.

SUBTASK 26-11-00-730-011

(2) On the test box, move the ENGINE LOOP A/1 AND B/2 OVHT switches to the NORMAL position.

(a) Make sure the ENG 1 (2) OVERHEAT light on the Engine and APU Fire Control Panel, P8-1, goes off.

(b) On the Pilot's Glareshield, P7, make sure that these conditions occur:

1) The Captain and First Officer's MASTER CAUTION light goes off.

2) The OVHT DET light goes off.

I. Do a test of the fire indication circuit

SUBTASK 26-11-00-730-012

(1) On the test box, move the ENGINE LOOP A/1 AND B/2 FIRE switches to the FIRE position.

(a) Make sure that the engine 1 (2) Fire Handle light, on the Engine and APU Fire Control Panel, P8-1, comes on.

(b) On the Pilot's Glareshield, P7, make sure that these conditions occur:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) The Captain and the First Officer's MASTER CAUTION light comes on.

2) The Captain and First Officer's FIRE WARN light comes on.

(c) Make sure that the flight compartment fire bell, on the Aural Warning Module, M315, comes on.

SUBTASK 26-11-00-730-013

(2) On the test box, move the ENGINE LOOP A/1 AND B/2 FIRE switches to the NORMAL position.

(a) Make sure that the engine 1 (2) Fire Handle light, on the Engine and APU Fire Control Panel, P8-1, goes off.

(b) On the Pilot's Glareshield, P7, make sure that these conditions occur:

1) The Captain and First Officer's MASTER CAUTION light goes off.

2) The Captain and First Officer's FIRE WARN light goes off.

(c) Make sure that the flight compartment fire bell, on the Aural Warning Module, M315, stops.

SUBTASK 26-11-00-480-002

(3) On Engine 1 (2), disconnect the test box equipment, SPL-1655 from the two terminal lugs (A/1 and B/2) of the FAN LOWER Fire Detector, M3061, and airplane ground.

SUBTASK 26-11-00-730-014

(4) Make sure that you do the circuit test for engine 1 and engine 2.

J. Put the airplane back to its usual condition.

SUBTASK 26-11-00-410-002

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 26-11-00-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 26-11-00-860-011

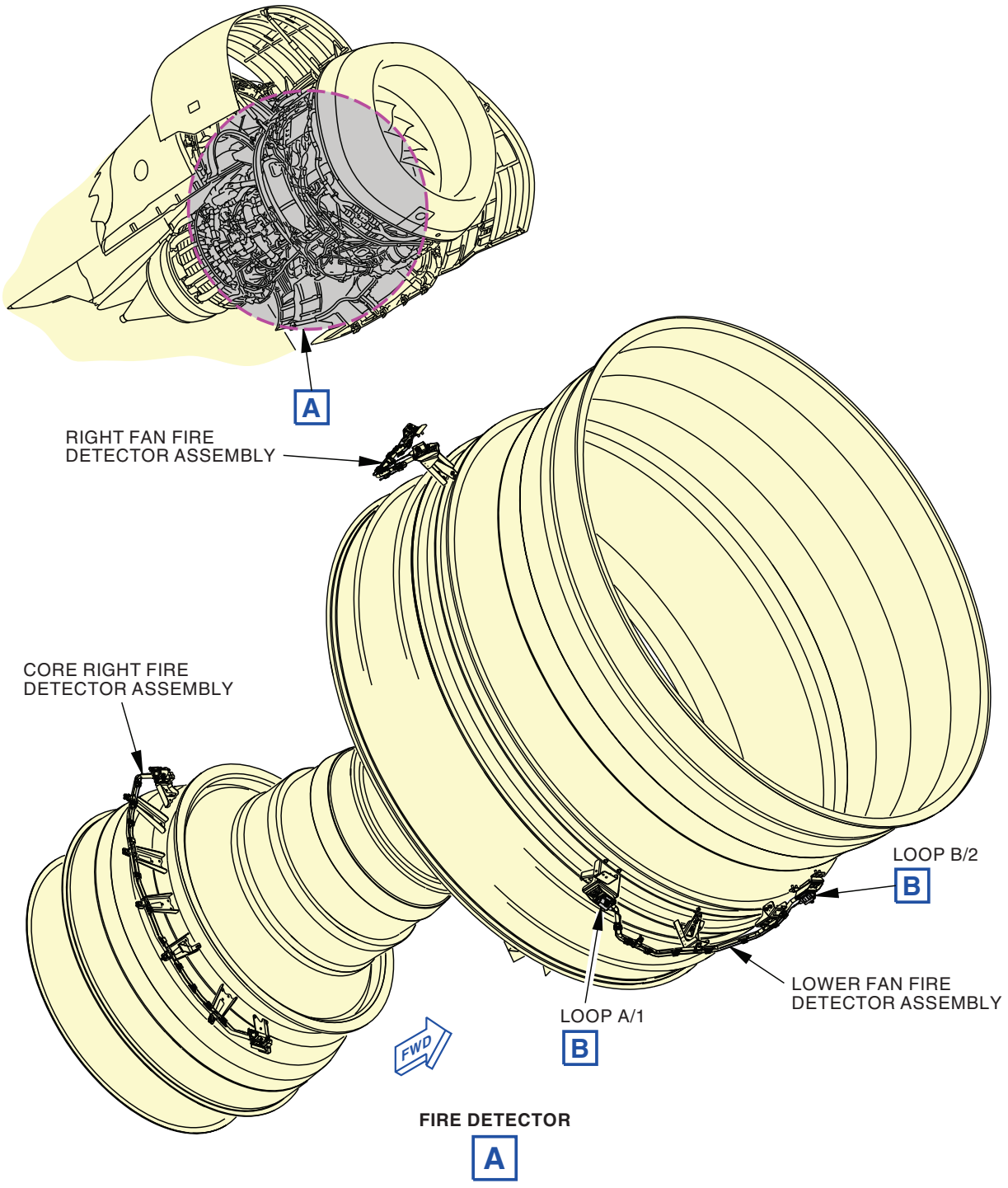
(3) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

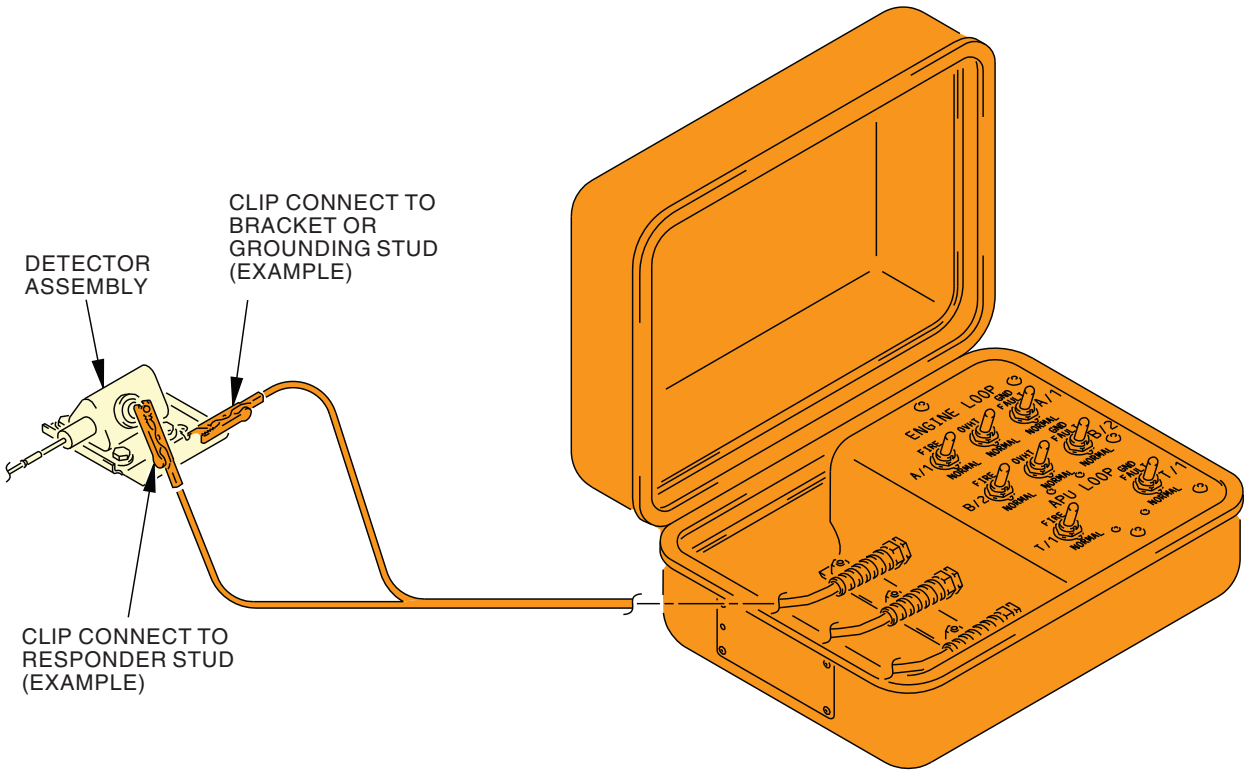
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detection Circuit Test
Figure 504/26-11-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

B

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Engine Fire Detection Circuit Test
Figure 504/26-11-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE DETECTORS - REMOVAL/INSTALLATION

1. General

A. This procedure has the following tasks:

(1) An engine fire detector element removal

(2) An engine fire detector element installation

(3) An engine fire detector assembly removal

(4) An engine fire detector assembly installation

(5) A combustor burn through fire detector assembly removal

(6) A combustor burn through fire detector assembly installation.

B. All of the engine fire detectors are dual loop detectors. Each fire detector assembly has two elements which can be removed separately.

C. There are seven different fire detectors on each engine:

(1) One is installed in the upper fan area.

(2) One is installed in the left mid fan area.

(3) One is installed in the lower fan area.

(4) One is installed in the right fan area.

(5) One is installed in the left engine core.

(6) One is installed in the right engine core.

(7) A combustor burn through detector is installed on the engine pylon lower area.

TASK 26-11-01-000-801

2. Engine Fire Detector Element Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the engine fire detector element.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 26-11-01-010-001

(1) To access the fan fire detector assemblies, do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-11-01-010-002

WARNING DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To access the core fire detector assemblies, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-11-01-860-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

E. Engine Fire Detector Element Removal

SUBTASK 26-11-01-020-001

(1) Do these steps to remove a fire detector element [7]:

CAUTION WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Remove the nut [2] and washer [3] from the responder stud.

CAUTION WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Remove the nut [4] and washer [5] from the ground stud.

(c) Disconnect the leads [6] from the studs.

(d) Loosen the screw [15] on each double loop clamp [13] and single loop clamp [14].

(e) Remove the fire detector element [7] from each double loop clamp [13] and single loop clamp [14].

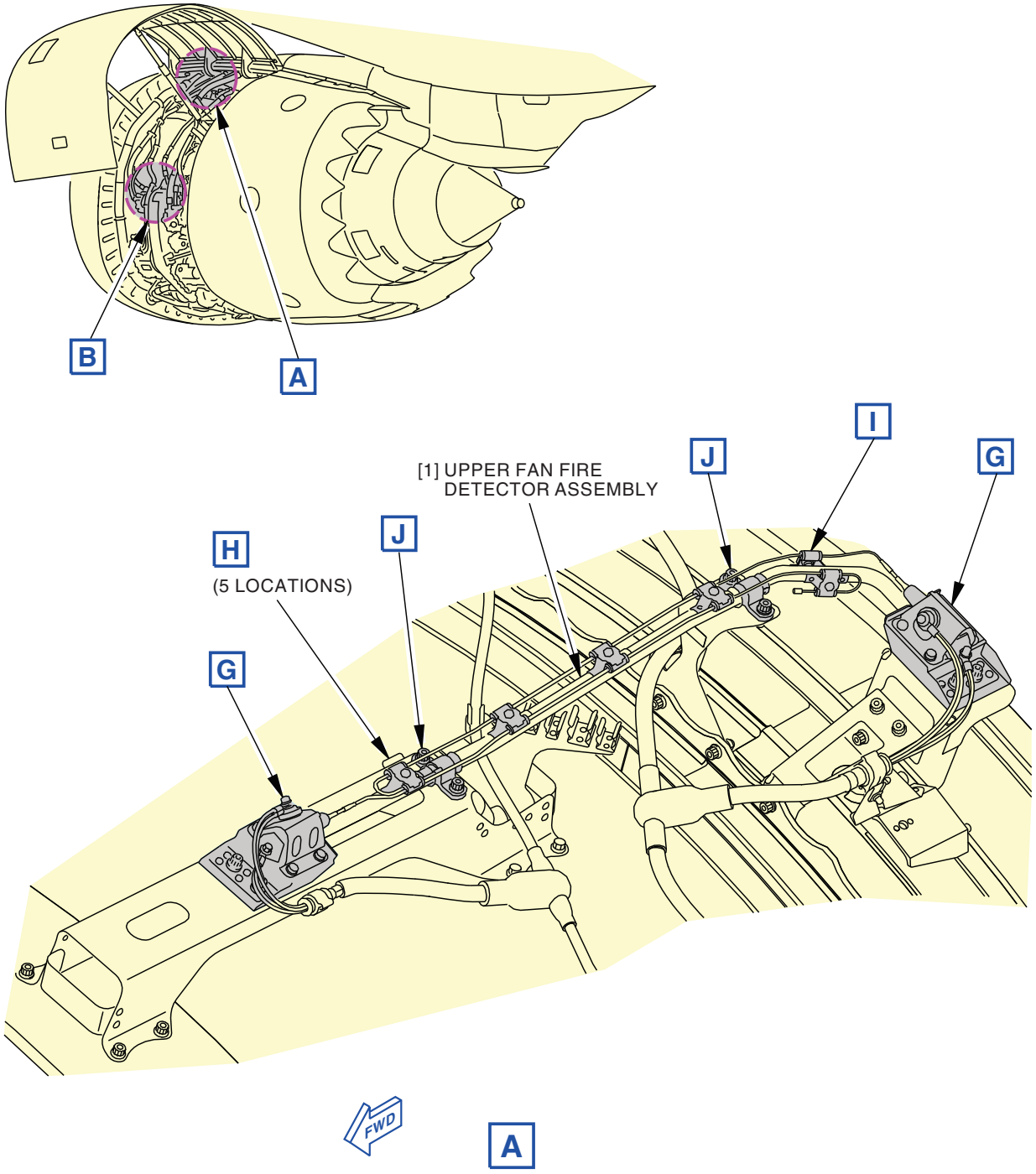
(f) Remove the grommets [16] from the fire detector element [7] to use on the new element.

(g) Remove the four screws [9], washers [10], and nuts [11] that attach the fire detector housing [8] to the mounting bracket.

(h) Remove the fire detector element [7].

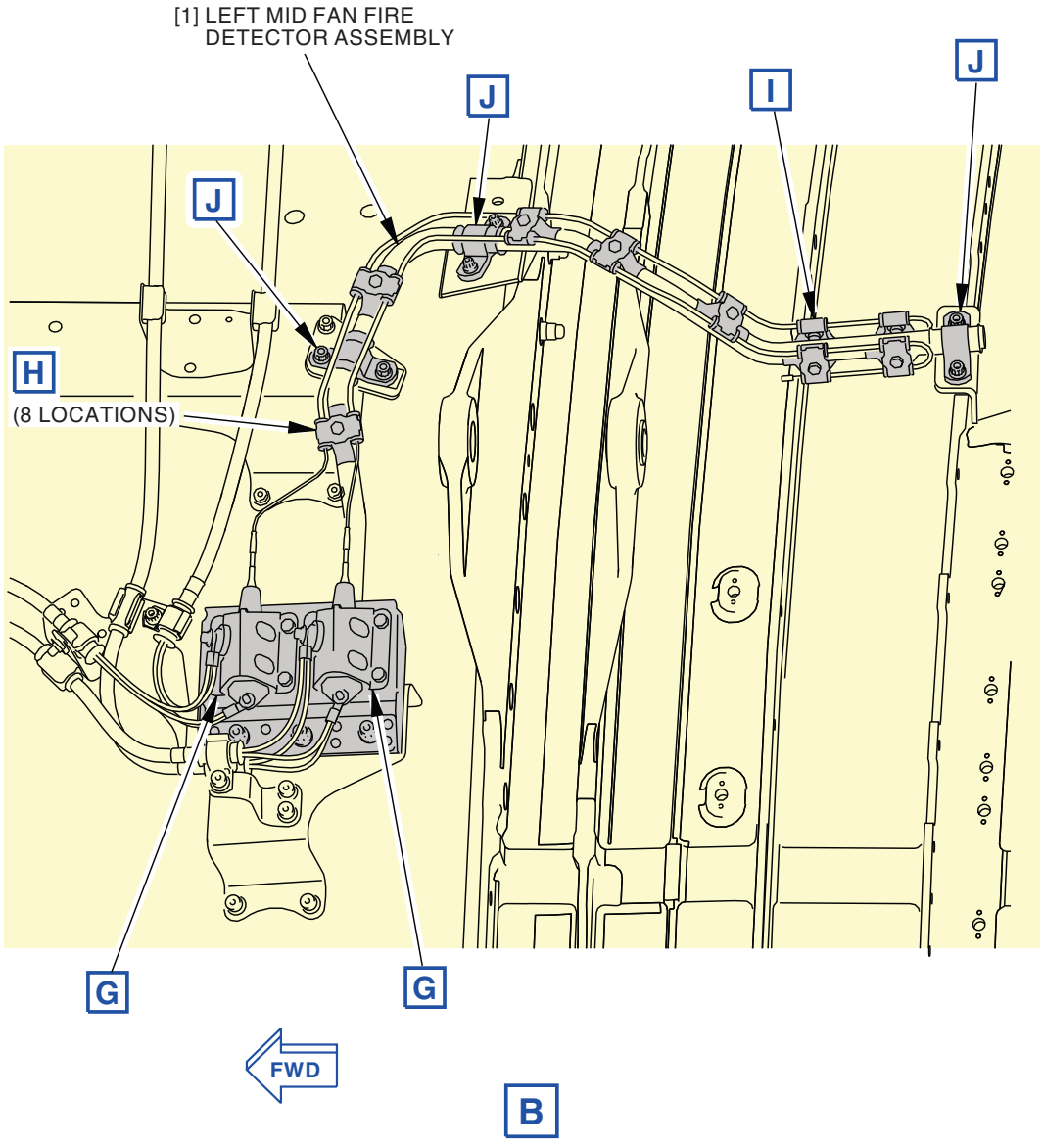
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 1 of 8)

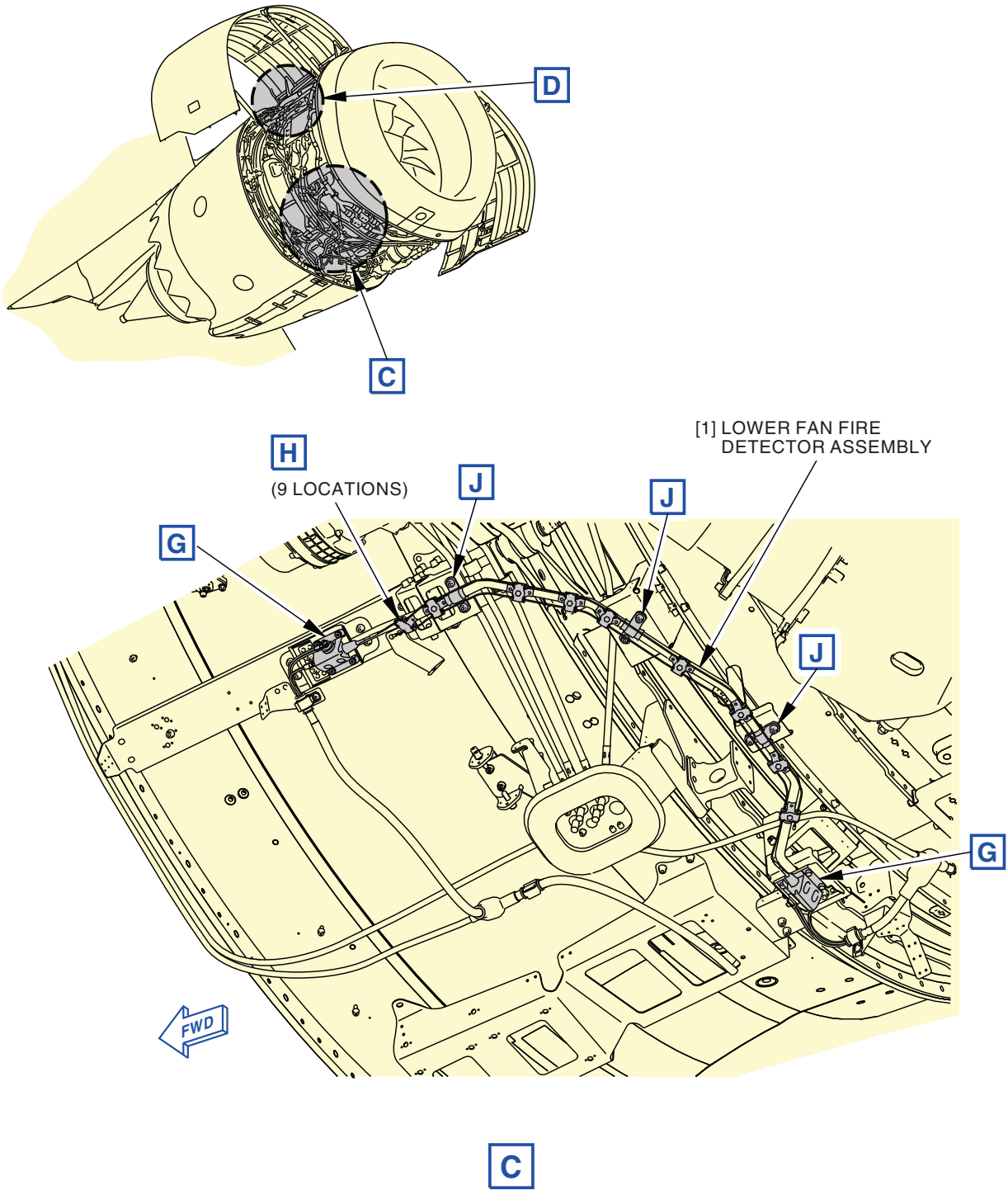
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 2 of 8)

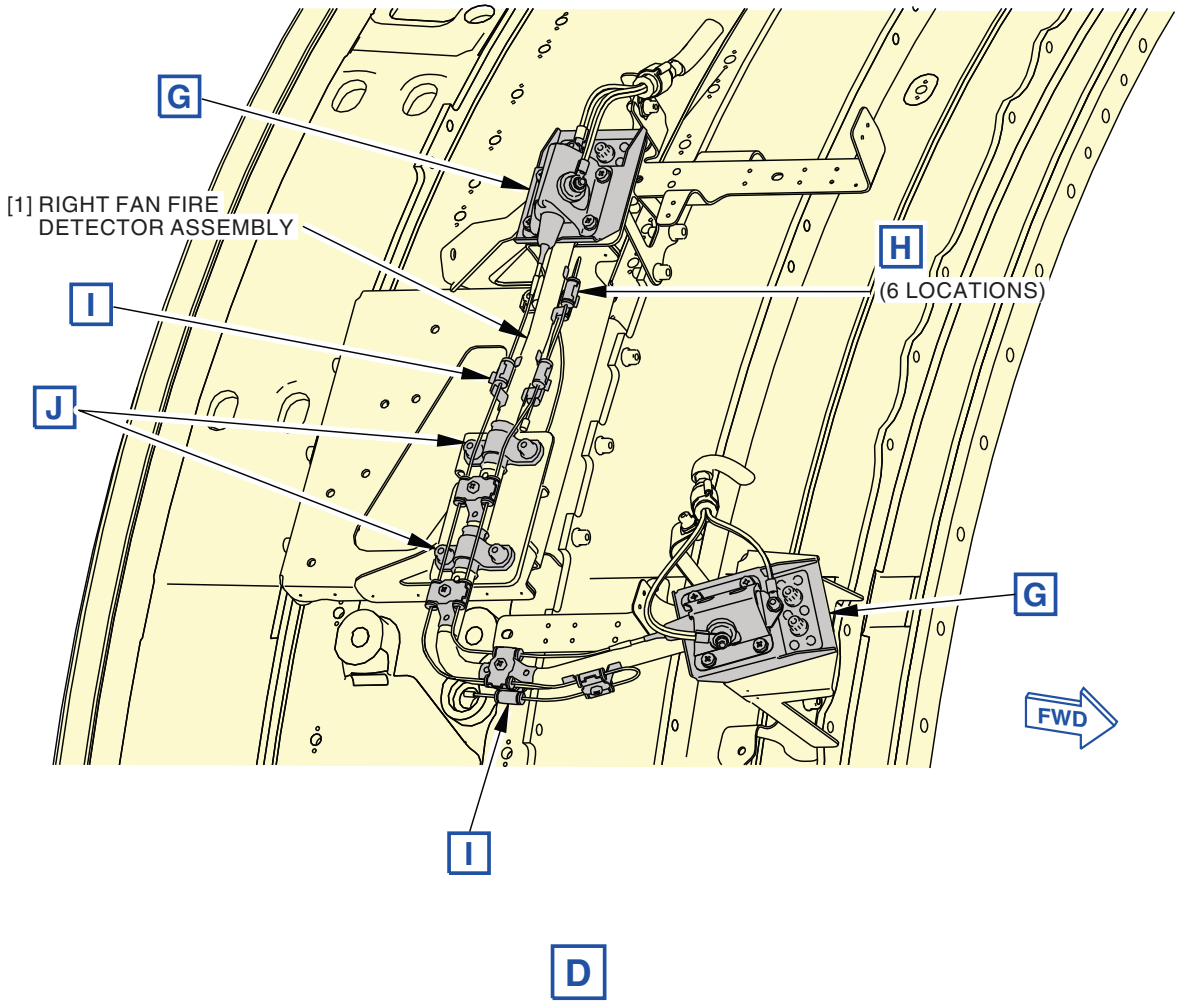
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 3 of 8)

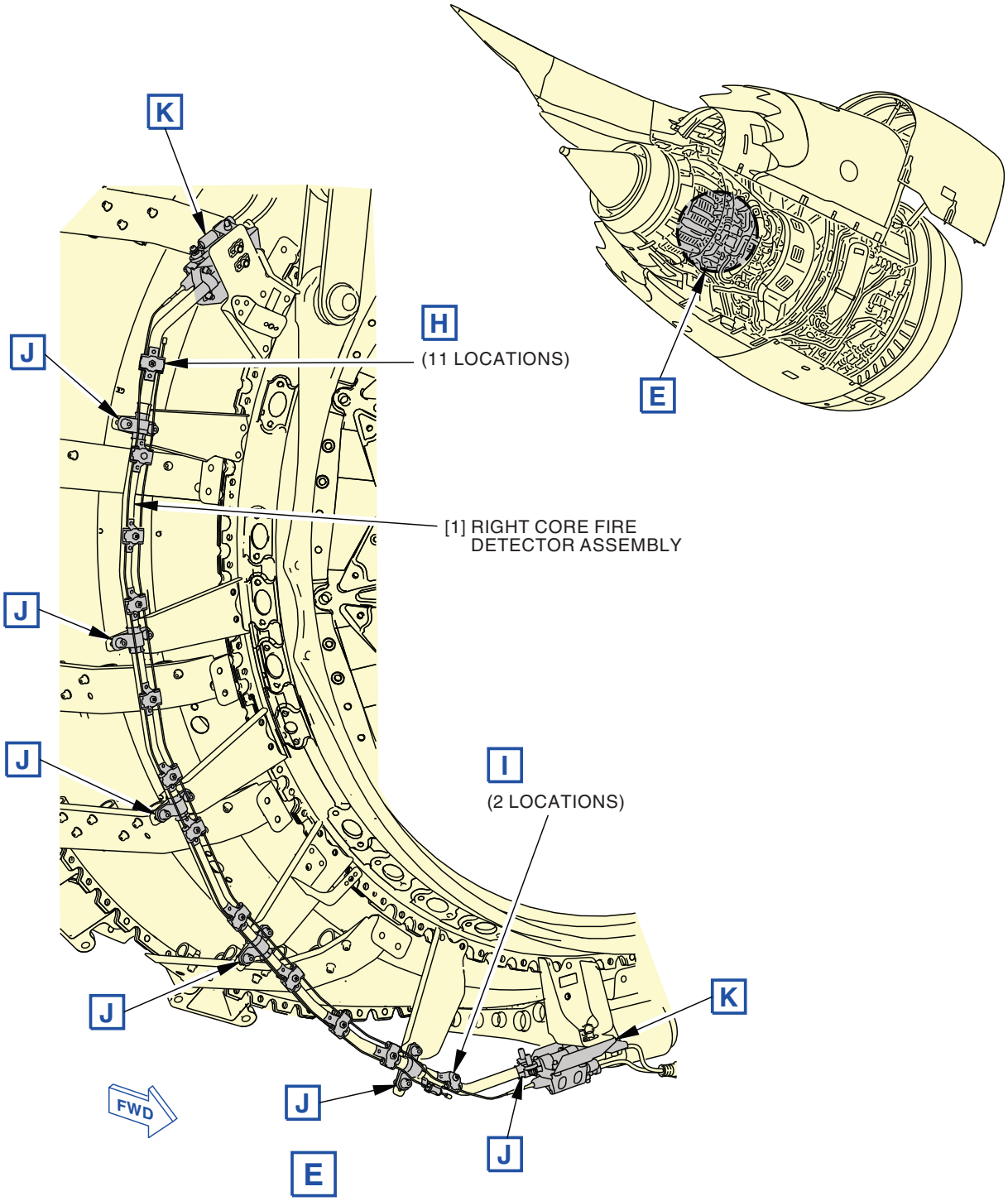
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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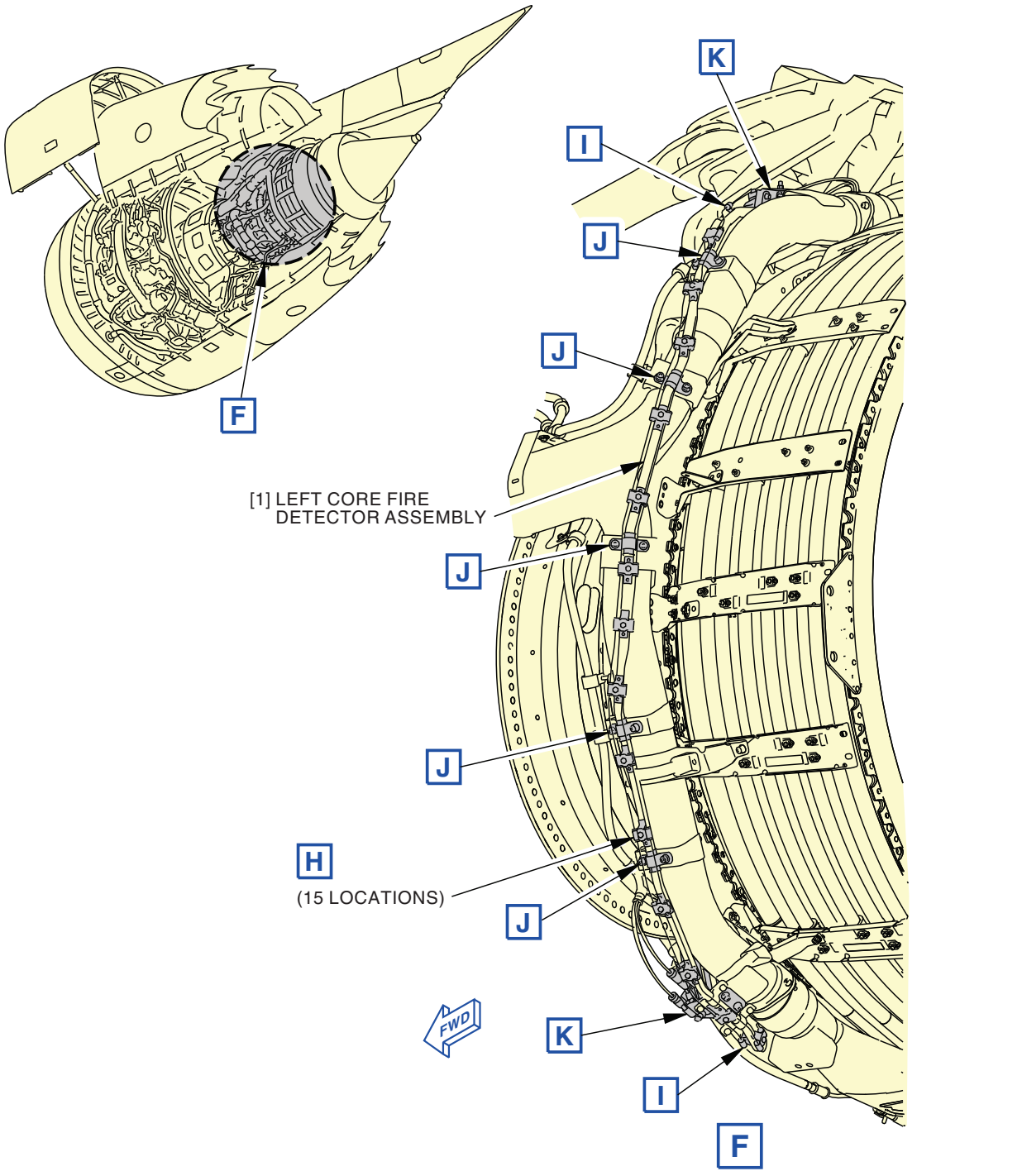
Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 4 of 8)

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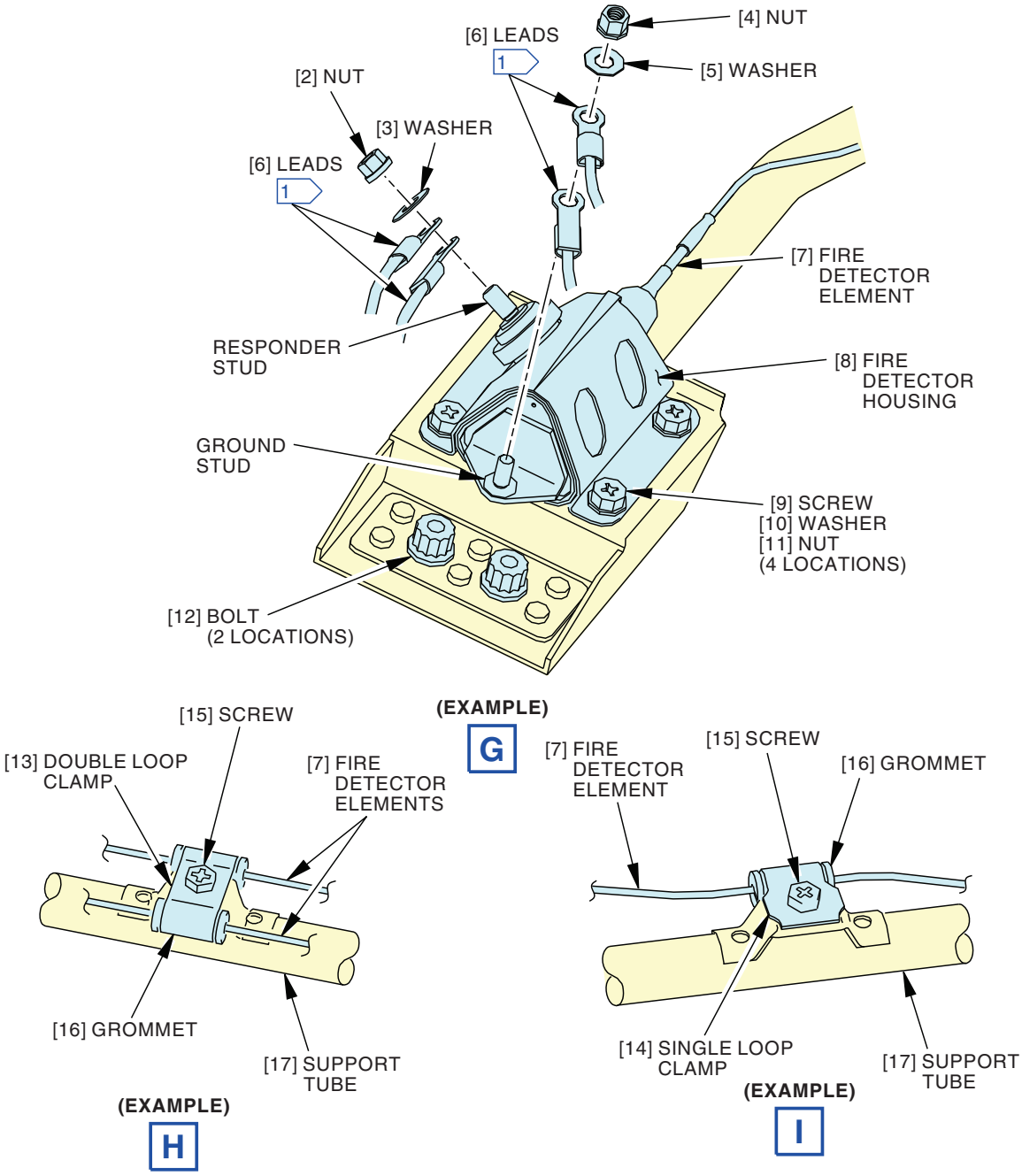
Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 5 of 8)

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Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

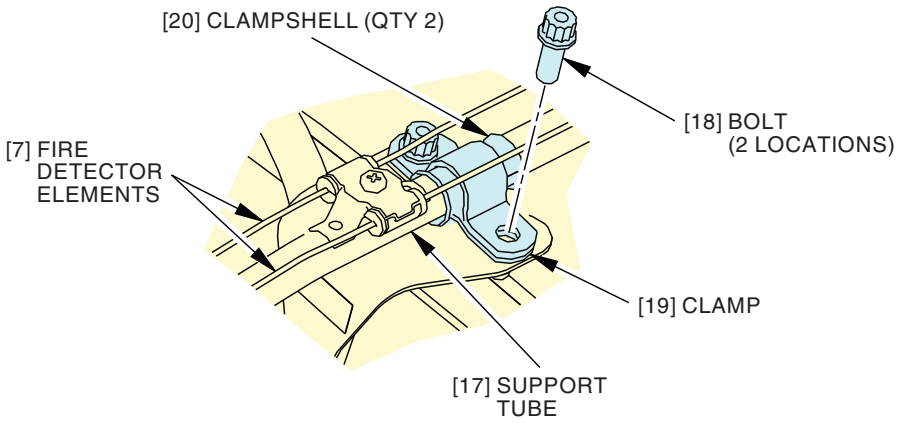


1 A BREAKOUT WIRE TIE MAY BE ADDED TO PREVENT RIDING CONDITION. (LOWER FAN CASE FIRE DETECTOR ASSEMBLY ONLY)

2412074 S00061529873_V3

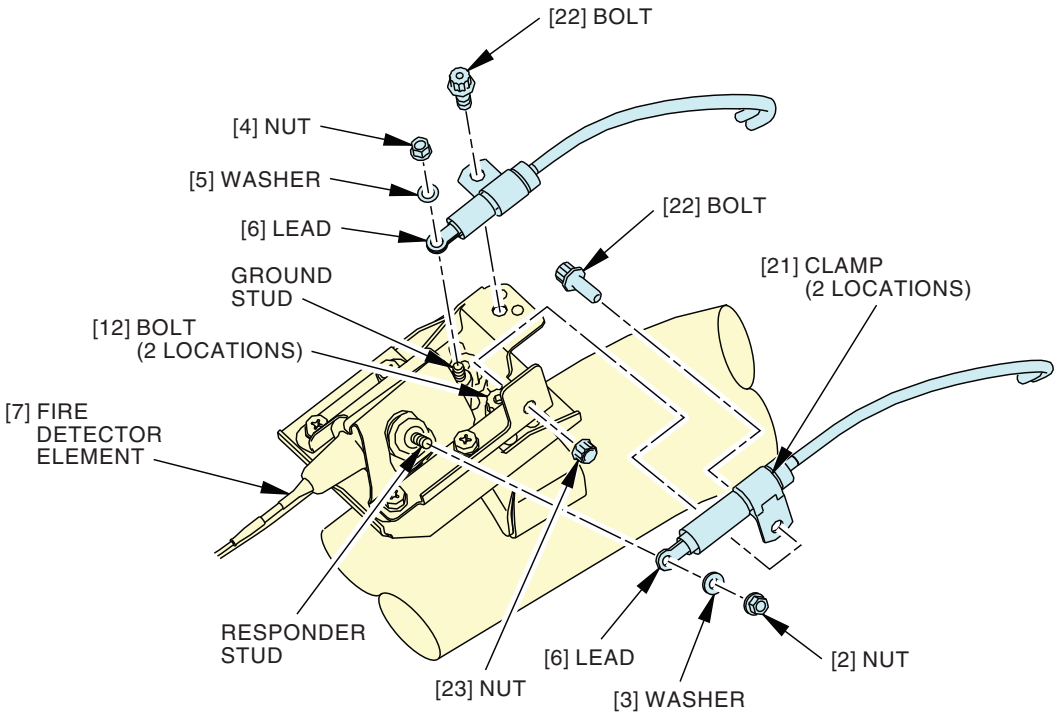
Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

J



(EXAMPLE)

K

2412073 S00061529872_V3

Engine Fire Detector Element Installation
Figure 401/26-11-01-990-801 (Sheet 8 of 8)

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TASK 26-11-01-400-801

3. Engine Fire Detector Element Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the engine fire detector element.

B. References

Reference	Title
26-11-00-710-801	Engine Fire Detection - Operational Test (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Fire detector element	26-11-01-05-045	PTF ALL
		26-11-01-10-055	PTF ALL
		26-11-01-15-035	PTF ALL
		26-11-01-20-055	PTF ALL
		26-11-01-26-060	PTF ALL
		26-11-01-31-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Engine Fire Detector Element Installation

SUBTASK 26-11-01-420-001

(1) Do these steps to install a fire detector element [7]:

(a) Position the fire detector element [7] along the support tube [17].

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(b) Put the fire detector housing [8] in the correct position on the mounting bracket.

(c) Install the four screws [9], washers [10], and nuts [11] that attach the fire detector housing [8] to the mounting bracket.

1) Tighten the nuts [11] to 30 in-lb (3.39 N·m) - 35 in-lb (3.95 N·m).

(d) Put the grommets [16] on the fire detector element [7] at each double loop clamp [13] and single loop clamp [14] location.

(e) Put the fire detector element [7] into each double loop clamp [13] and single loop clamp [14] along the fire detector assembly [1].

(f) Make sure that the end of the detector element extends less than 1 in. (25 mm) past the last grommet [16].

1) If the fire detector element [7] extends more than 1 in. (25 mm), adjust the length of the fire detector element [7] between each double loop clamp [13] and single loop clamp [14] so that any slack is evenly distributed.

(g) Make sure that there is at least 0.12 in. (3.05 mm) between the fire detector element [7] and the support tube [17].

(h) Tighten the screw [15] on each applicable double loop clamp [13] and single loop clamp [14] snug against the grommets [16].

1) Make sure that the screw [15] extends a minimum of 1-1/2 threads through the nut.

2) Make sure that the end of the screw [15] does not touch the support tube [17].

(i) Install the leads [6] to the applicable studs.

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(j) Install the nut [2] to the responder stud.

1) Tighten the nut [2] to 50 in-lb (5.65 N·m) - 54 in-lb (6.10 N·m).

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(k) Install the nut [4] to the ground stud.

1) Tighten the nut [4] to 25 in-lb (2.82 N·m) - 29 in-lb (3.28 N·m).

SUBTASK 26-11-01-280-001

(2) Using the intrinsically safe approved bonding meter, COM-1550, do a check of the resistance between the fire detector housing [8] and the airplane structure.

(a) Make sure that the resistance is less than 0.010 ohm.

(b) If the resistance is more than 0.010 ohm, clean the bonding surfaces with the solvent, B00083, and do the check again.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-11-01-410-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-11-01-410-004

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To close the thrust reverser, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 26-11-01-410-005

(3) To close the fan cowl panel, do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

I. Engine Fire Detector Element Installation Test

SUBTASK 26-11-01-710-001

(1) Do this task: Engine Fire Detection - Operational Test, TASK 26-11-00-710-801.

——— END OF TASK ———

TASK 26-11-01-000-802

4. Engine Fire Detector Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the engine fire detector assembly.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 26-11-01-010-005

(1) To access the fan fire detector assemblies, do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-11-01-010-006

WARNING DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To access the core fire detector assemblies, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-11-01-860-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

E. Fire Detector Assembly Removal

SUBTASK 26-11-01-020-002

(1) Do these steps to remove one of the fire detector assemblies [1]:

NOTE: The fire detector assembly includes the support tube with the fire detector elements installed and the mounting hardware.

CAUTION WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Remove the nut [2] and washer [3] from each responder stud.

CAUTION WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Remove the nut [4] and washer [5] from each ground stud.

(c) Remove the bolts [22], clamps [21], and nuts [23] that hold the leads [6] to the core fire detector bracket.

(d) Remove the leads [6] from the studs.

(e) Remove the bolts [18], clamps [19], and clampshells [20] that hold the support tube [17] in its position.

(f) Remove the bolts [12] that hold the detector in its position.

(g) Remove the fire detector assembly [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-01-400-802

5. Engine Fire Detector Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the engine fire detector assembly.

B. References

Reference	Title
26-11-00-710-801	Engine Fire Detection - Operational Test (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fire detector assembly	26-11-01-01-050	PTF ALL
		26-11-01-02-050	PTF ALL
		26-11-01-05-045	PTF ALL
		26-11-01-10-055	PTF ALL
		26-11-01-15-035	PTF ALL
		26-11-01-20-055	PTF ALL
		26-11-01-26-060	PTF ALL
		26-11-01-31-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Engine Fire Detector Assembly Installation

SUBTASK 26-11-01-420-002

(1) Do these steps to install one of the fire detector assemblies [1]:

(a) Make sure that the ends of the fire detector assembly [1] and brackets are clean.

(b) Position the fire detector assembly [1] on the engine.

(c) Make sure that the preload between the fire detector assembly [1] and attach points is not more than 10 ft-lb (13.6 N·m).

(d) Apply Pure Nickel Special compound, D00006, to the threads of the bolts [12] and bolts [18].

(e) Install the bolts [12] that hold the fire detector assembly [1] to the engine.

1) Tighten the bolts [12] to 78 in-lb (8.8 N·m) - 82 in-lb (9.3 N·m).

(f) Install the bolts [18], clamp [19], and clampshells [20] that hold the support tube [17] in its position.

1) Tighten the bolts [18] to 78 in-lb (8.8 N·m) - 82 in-lb (9.3 N·m).

(g) Make sure that there is at least 0.10 in. (2.54 mm) between the fire detector elements [7] and support tube [17].

(h) Install the leads [6] to the applicable stud.

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(i) Install the nut [4] and washer [5] to each ground stud.

1) Tighten the nut [4] to 25 in-lb (3 N·m) - 29 in-lb (3 N·m).

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(j) Install the nut [2] and washer [3] to each responder stud.

1) Tighten the nuts [2] to 50 in-lb (6 N·m) - 54 in-lb (6 N·m).

CAUTION

DO NOT LET WIRES GET IN THE BOBBIN SPACE OR BETWEEN THE BOBBIN AND THE CLAMP, IF YOU HAVE A SPLIT BOBBIN HARNESS. IF YOU DO NOT OBEY, WIRES CAN RUB AGAINST THE CLAMPS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(k) Install the bolts [22], clamps [21], and nuts [23] that hold the leads [6] to the core fire detector bracket.

SUBTASK 26-11-01-280-002

(2) Measure the resistance between the fire detector housing [8] and titanium fan shroud.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is less than 0.010 ohm (10 milliohm).

(c) If the resistance is more than 0.010 ohm, clean the bonding surfaces with the solvent, B00083, and do the check again.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-11-01-860-004

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-11-01-410-006

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To close the thrust reverser, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 26-11-01-410-007

(3) To close the fan cowl panel, do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

I. Engine Fire Detector Assembly Installation Test

SUBTASK 26-11-01-710-002

(1) Do this task: Engine Fire Detection - Operational Test, TASK 26-11-00-710-801.

——— END OF TASK ———

TASK 26-11-01-000-803

6. Combustor Burn Through Fire Detector Assembly Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the engine combustor burn through fire detector assembly.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 26-11-01-020-006

(1) It is recommended to remove the applicable fan cowl panel, do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

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SUBTASK 26-11-01-020-003

WARNING DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To access the combustor burn through fire detector, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-11-01-020-004

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

E. Engine Combustor Burn Through Fire Detector Assembly Removal

SUBTASK 26-11-01-020-005

(1) Do these steps to remove the engine combustor burn through fire detector assembly [1].

NOTE: The fire detector assembly includes the support tube with the detector elements installed and the mounting hardware.

(a) Remove the nuts [2] and washers [3] from the two responder studs.

(b) Remove the nuts [4] and washers [5] from the two ground studs.

(c) Disconnect the leads [6] from the responder and ground studs.

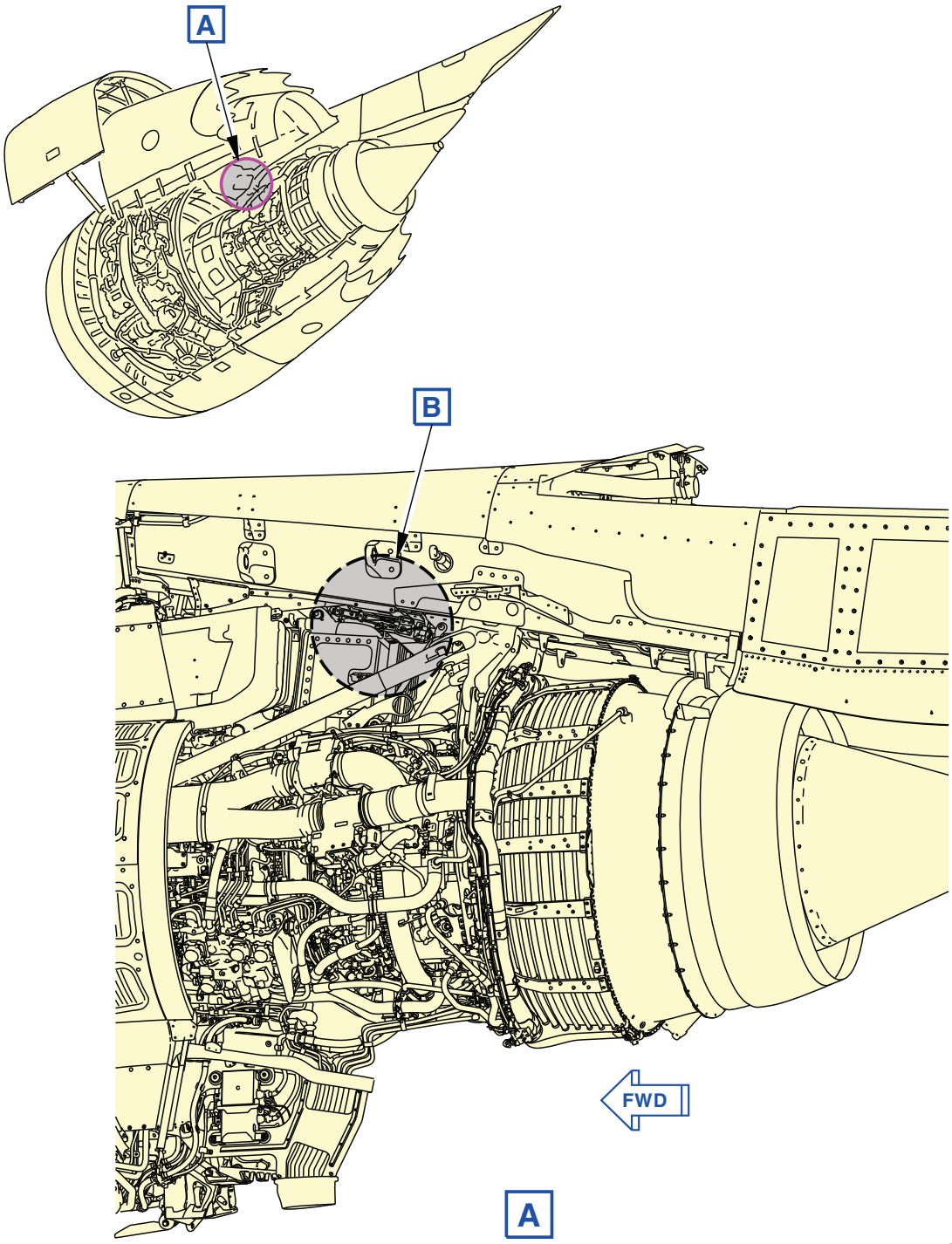
(d) Remove the bolt [26] from the support tube [17].

(e) Remove the bolts [12] from the fire detector assembly [1].

(f) Remove the fire detector assembly [1] from the engine pylon strut.

——— END OF TASK ———

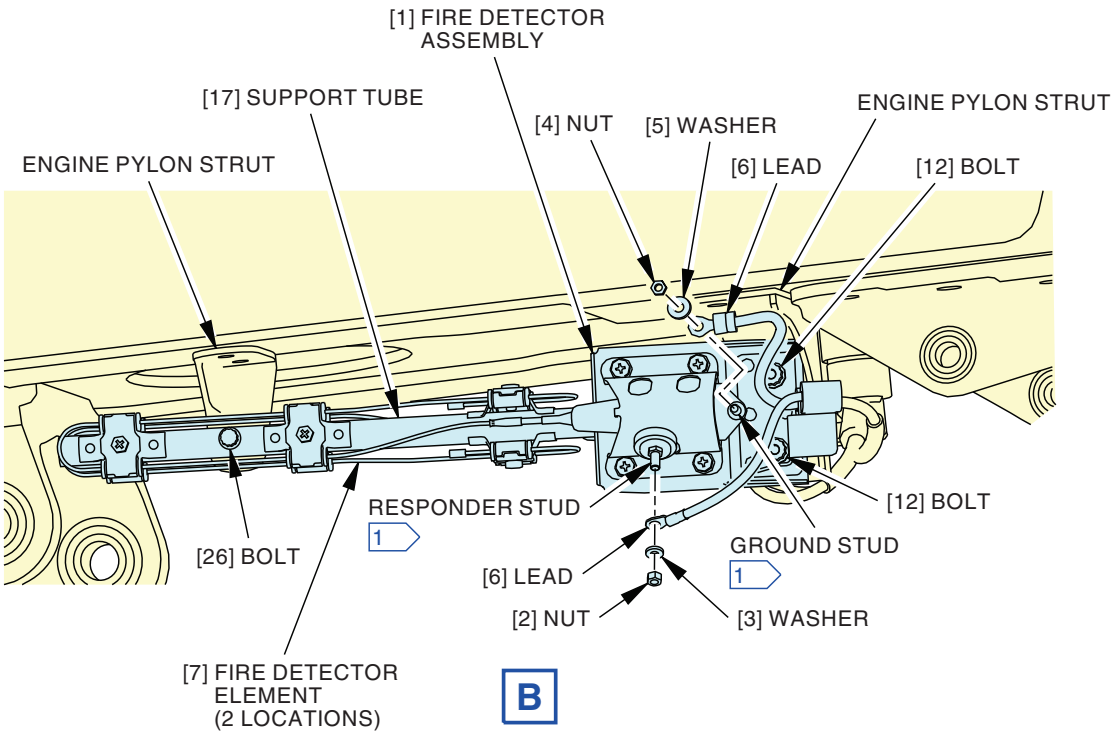
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Combustor Burn Through Fire Detector Assembly
Figure 402/26-11-01-990-802 (Sheet 1 of 2)

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1 SECOND LOCATION OF GROUND AND RESPONDER STUD IS ON THE REVERSE SIDE.

2899708 S0000690949_V2

Combustor Burn Through Fire Detector Assembly
Figure 402/26-11-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-01-400-803

7. Combustor Burn Through Fire Detector Assembly Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the engine combustor burn through fire detector assembly.

B. References

Reference	Title
26-11-00-710-801	Engine Fire Detection - Operational Test (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00130	Alcohol - Isopropyl	TT-I-735
B50073	Alcohol - Isopropyl	ASTM D 770
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D50237	Compound - Antiseize, Pure Nickel Special - Never Seez NSBT-8N/-16N	BAC5008
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008
G50262	Wiper - Cleaning	BMS15-5

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fire detector assembly	26-11-01-01-050	PTF ALL
		26-11-01-02-050	PTF ALL
		26-11-01-05-045	PTF ALL
		26-11-01-10-055	PTF ALL
		26-11-01-15-035	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
1 (cont.)		26-11-01-20-055	PTF ALL
		26-11-01-26-060	PTF ALL
		26-11-01-31-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Engine Combustor Burn Through Fire Detector Assembly Installation

SUBTASK 26-11-01-420-003

(1) Do these steps to install the engine combustor burn through fire detector assembly [1]:

NOTE: The fire detector assembly includes the support tube with the detector elements installed and the mounting hardware.

(a) Use a wiper, G50262, together with alcohol, B50073, or alcohol, B00130, to clean the contact surfaces of the engine pylon strut and fire detector assembly [1].

(b) Install the two bolts [12] to the fire detector assembly [1]:

1) Apply a layer of Pure Nickel Special compound, D50237, Pure Nickel Special compound, D00006, or Never Seez, D50294, to the threads of the bolts [12].

2) Tighten the bolts [12] to 75.0 ±2.0 in-lb (8.5 ±0.2 N·m).

(c) Install the bolt [26] to the support tube [17]:

1) Apply a layer of Pure Nickel Special compound, D50237, Pure Nickel Special compound, D00006, or Never Seez, D50294, to the thread of the bolt [26].

2) Tighten the bolt [26] to 25.0 ±2.0 in-lb (2.8 ±0.2 N·m).

(d) Connect the leads [6] to the two ground studs and two responder studs.

(e) Install the nuts [4] and washers [5] to the two ground studs:

1) Use a wiper, G50262, together with alcohol, B50073, or alcohol, B00130, to clean the ground studs and nuts [4].

2) Tighten the nuts [4] to 27.0 ±2.0 in-lb (3.1 ±0.2 N·m).

(f) Install the nuts [2] and washers [3] to the two responder studs:

1) Use a wiper, G50262, together with alcohol, B50073, or alcohol, B00130, to clean the responder studs and nuts [2].

2) Tighten the nuts [2] to 50 in-lb (6 N·m) - 54 in-lb (6 N·m).

(g) Make sure that there is at least 0.12 in. (3.05 mm) between the fire detector element [7] and the support tube [17].

SUBTASK 26-11-01-420-004

(2) Use the intrinsically safe approved bonding meter, COM-1550, to do a check of the resistance between the fire detector housing [8] and the airplane structure.

(a) Make sure that the resistance is less than 0.010 ohm.

(b) If the resistance is more than 0.010 ohm, clean the bonding surfaces with the solvent, B00083, and do the check again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-11-01-420-005

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

SUBTASK 26-11-01-410-001

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) To close the thrust reverser, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 26-11-01-410-002

(3) To close the fan cowl panel, do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

I. Engine Fire Detector Assembly Installation Test

SUBTASK 26-11-01-420-006

(1) Do this task: Engine Fire Detection - Operational Test, TASK 26-11-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE DETECTORS - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inspection of the wires to the forward transponder of the upper fan case engine fire overheat detector for contact with the strut insulation blanket.

TASK 26-11-01-211-801

2. Engine Fire Detector Inspection

(Figure 601)

A. Consumable Materials

Reference	Description	Specification
G50541	Tubing - Heat Shrinkable, Teflon Sleeve (TFE-R-5/8-X)	AS23053/12
G50544	Tubing - Heat Shrinkable, Teflon Sleeve (TFE-R-1/2-X)	AS23053/12

B. Engine Fire Detector Inspection

SUBTASK 26-11-01-212-001

(1) Make sure that there is no contact between the forward transponder of the upper fan case engine fire/overheat detector and the strut insulation blanket.

SUBTASK 26-11-01-300-001

(2) If there is contact and the wire insulation is damaged, repair the wire as described below and change the routing of the wire.

(a) If no contact is found and the wires are routed, no further action is required.

SUBTASK 26-11-01-030-001

(3) Disconnect the lug from the fire/overheat transponder.

SUBTASK 26-11-01-300-002

(4) Cut a sufficient length of TFE-R 5/8-X tubing, G50541, or TFE-R-1/2-X tubing, G50544, to cover the worn area.

(a) The sleeve must be long enough to exceed the length of the damage by approximately 0.4 in. (10.2 mm).

SUBTASK 26-11-01-300-003

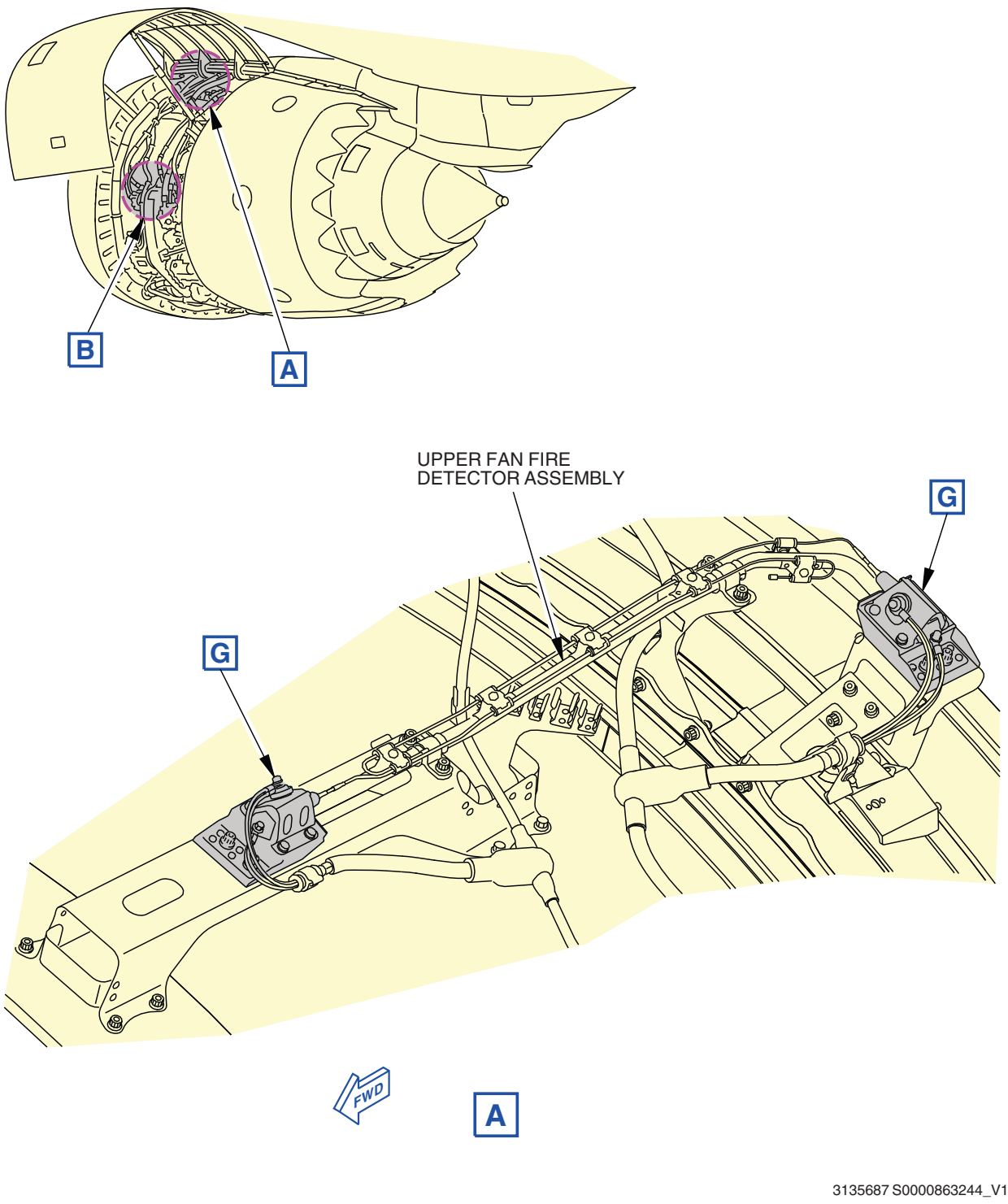
(5) Heat the sleeve at 700°F (371°C) maximum until shrinking operation is completed.

SUBTASK 26-11-01-430-001

(6) Reconnect the lug on the fire detection transponder.

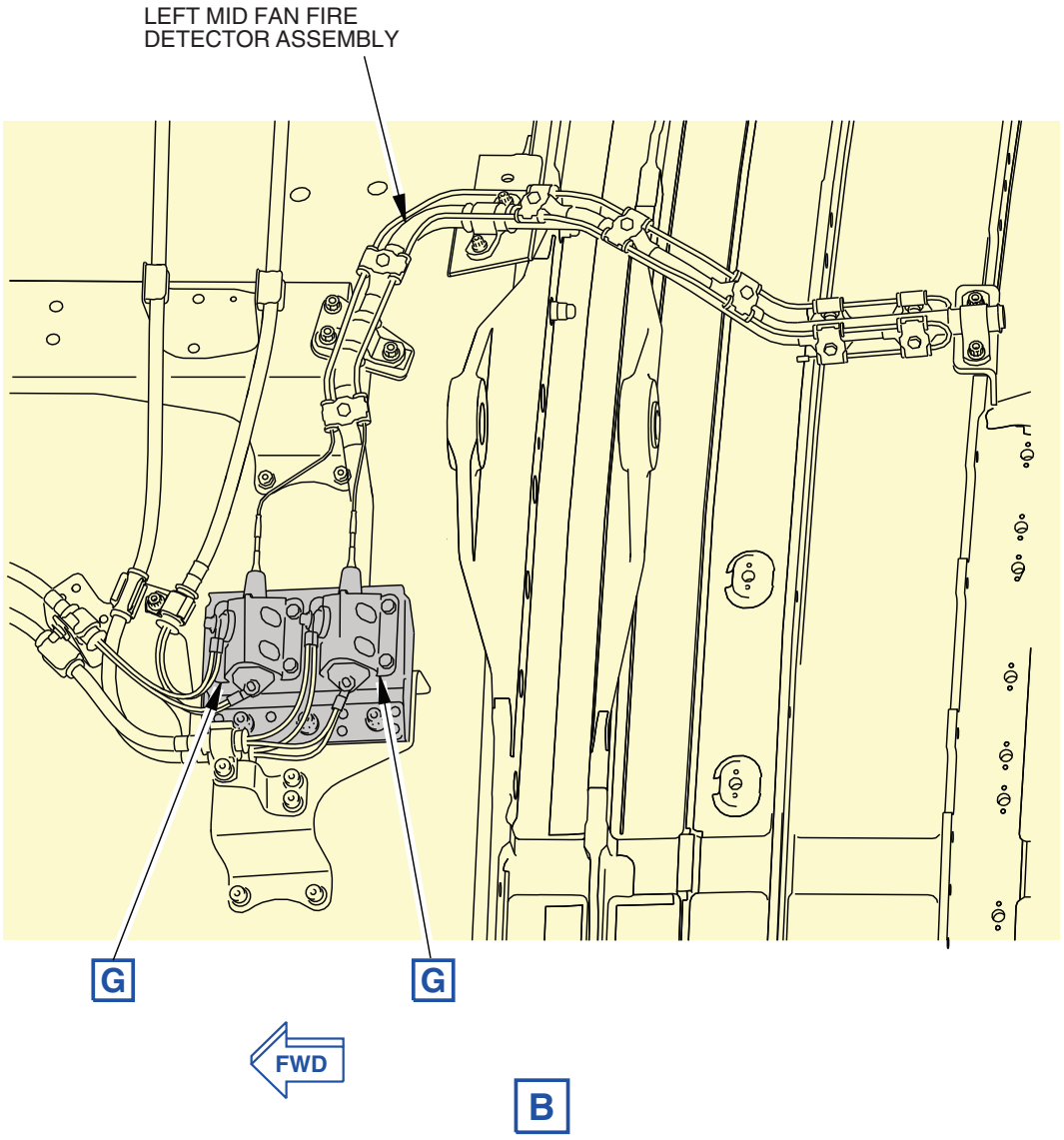
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 1 of 8)

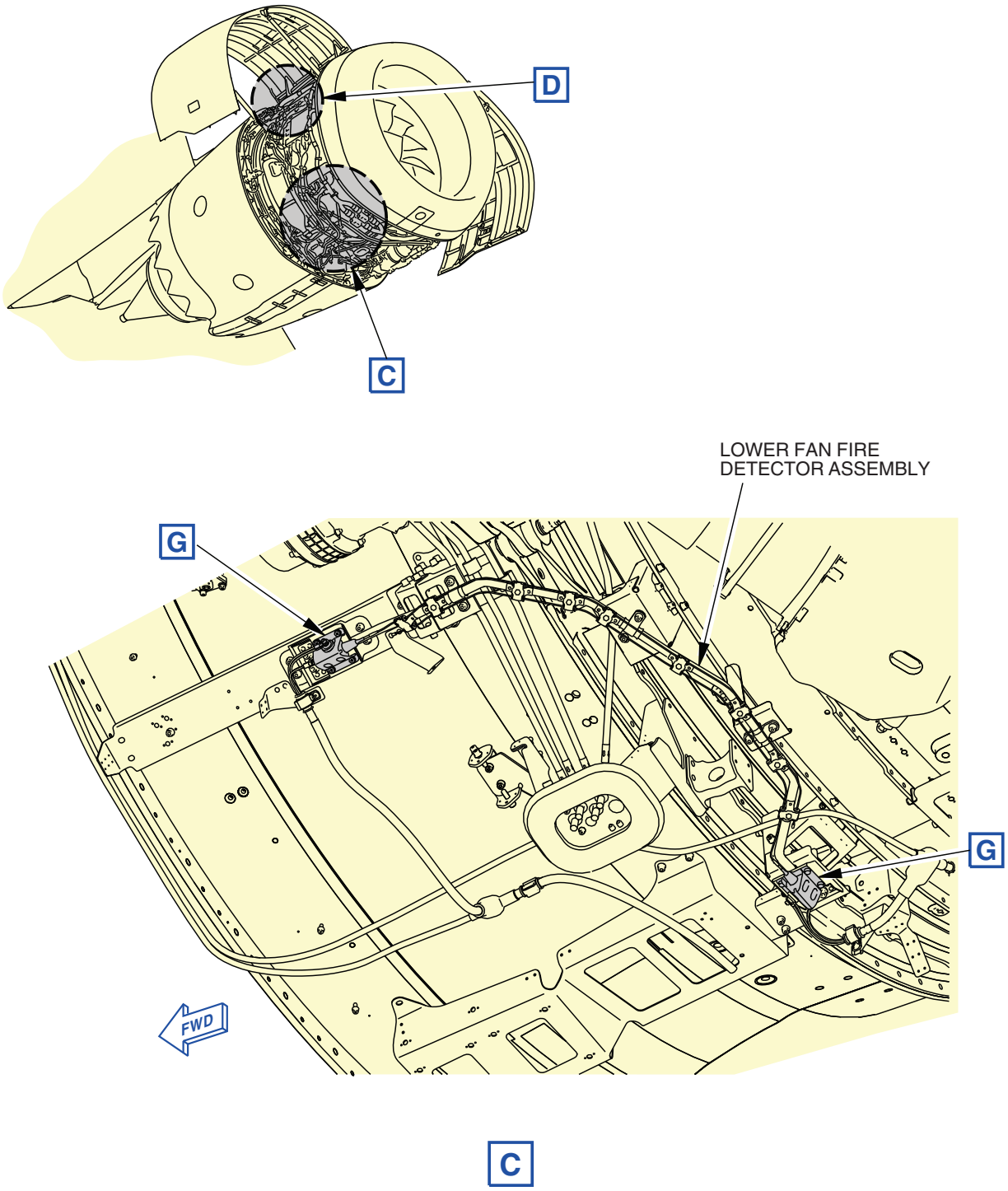
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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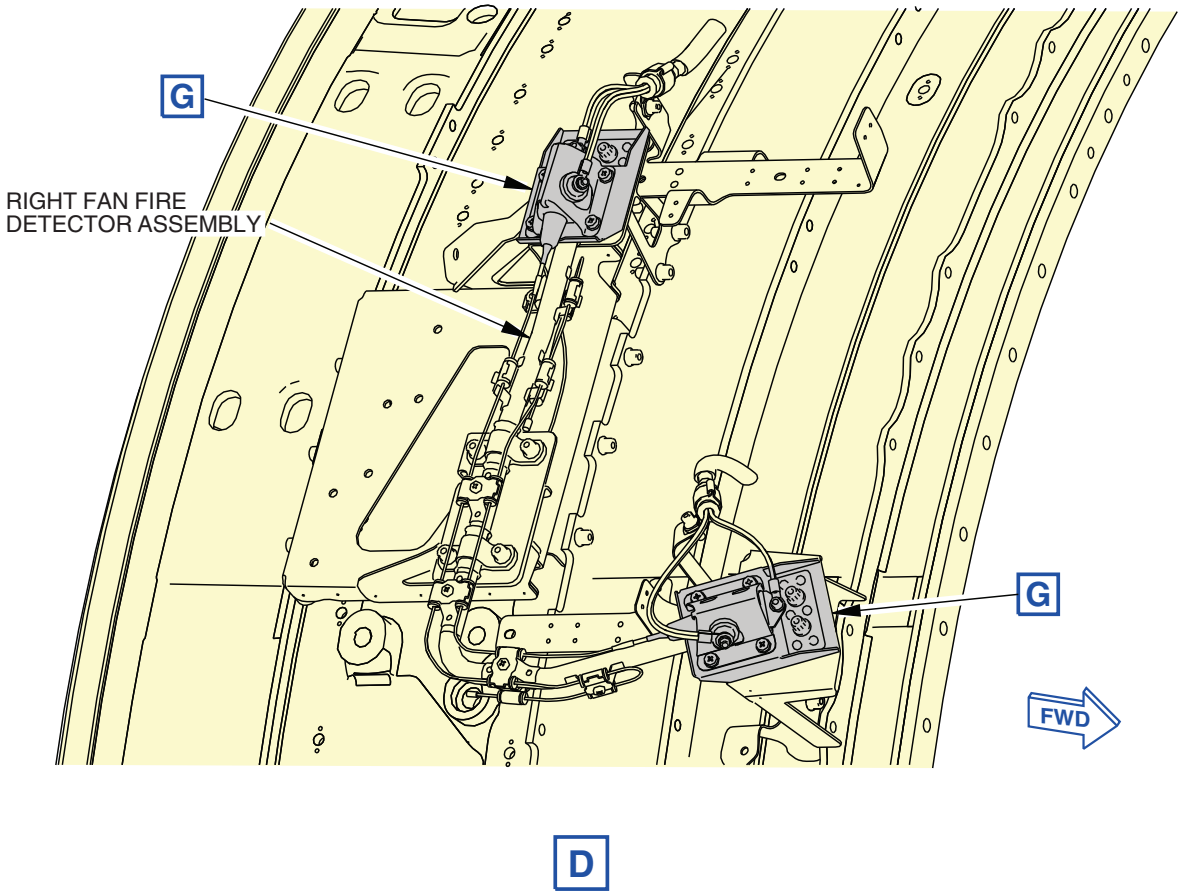
Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 3 of 8)

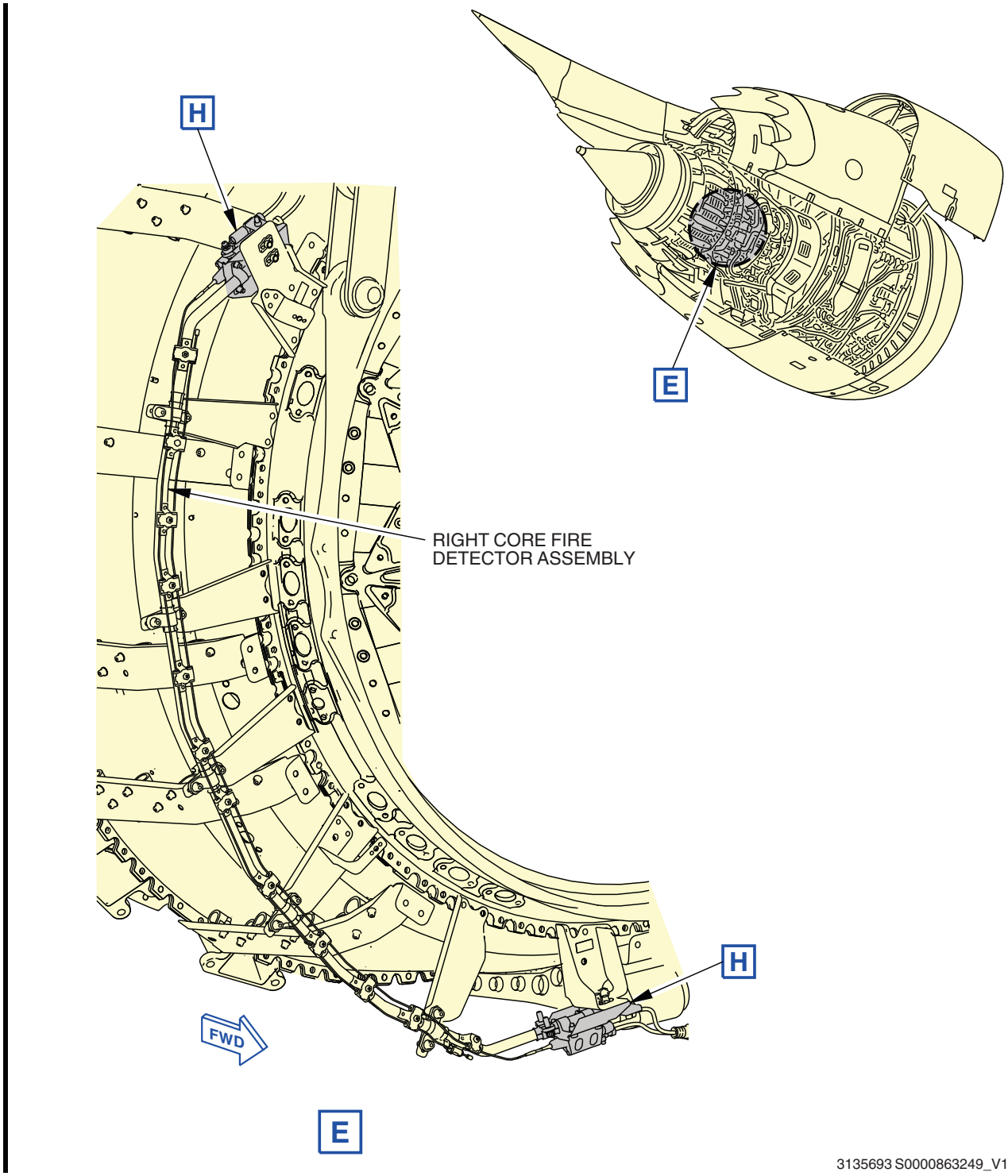
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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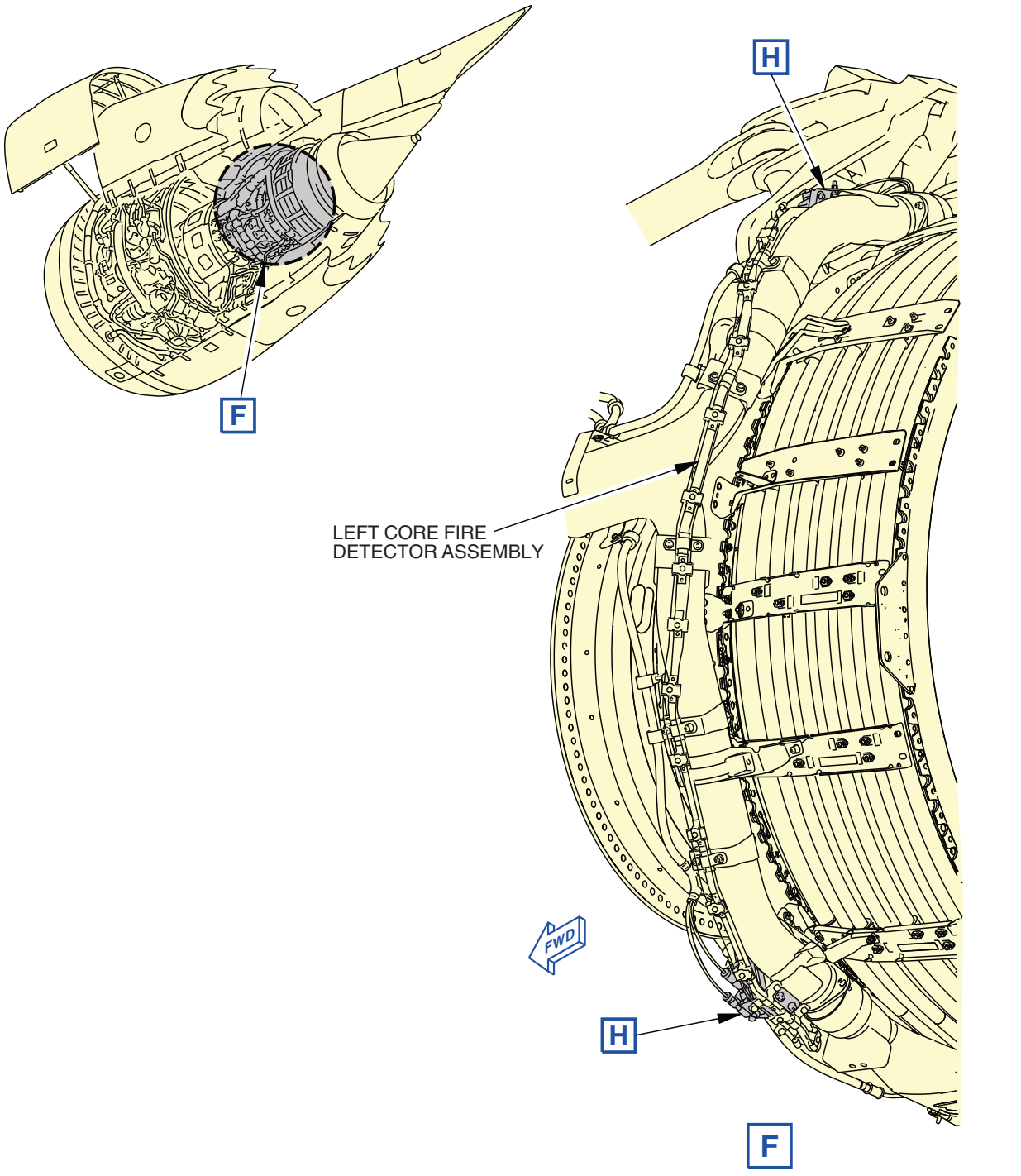
Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 4 of 8)

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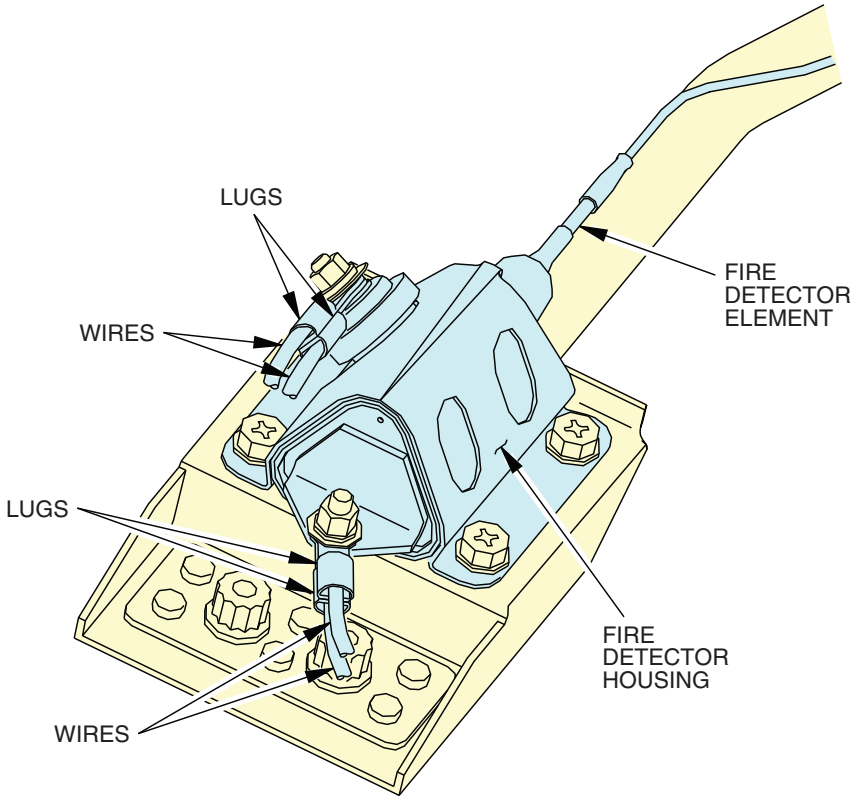
Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 6 of 8)

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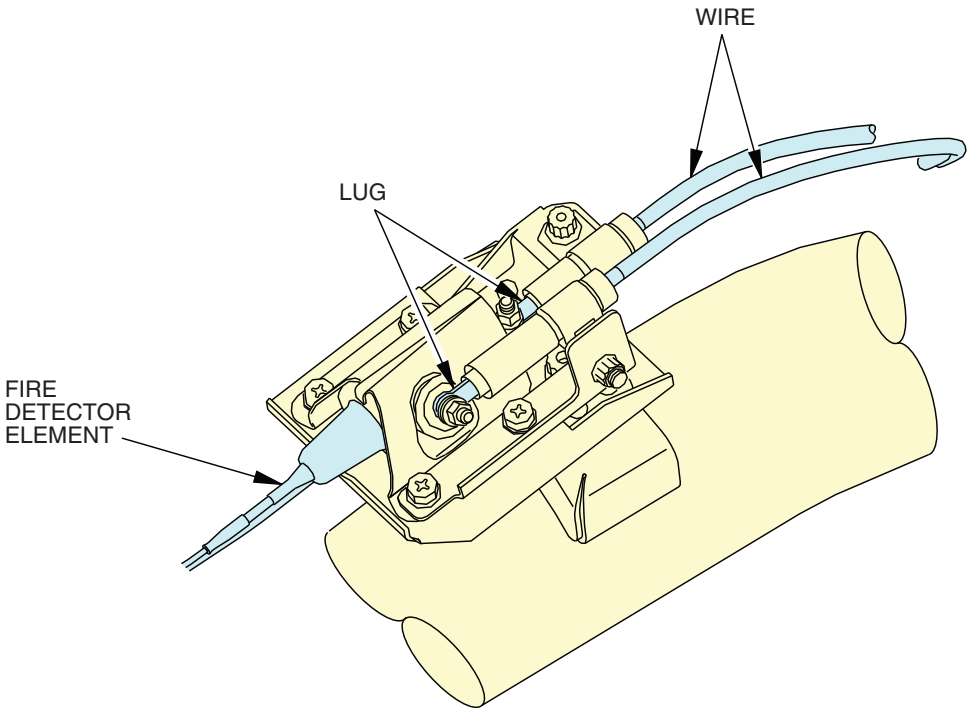
(EXAMPLE)

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Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

H

3135744 S0000863252_V1

Engine Fire Detectors Inspection
Figure 601/26-11-01-990-803 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE DETECTOR HARNESS - REMOVAL/INSTALLATION

1. General

A. The engine fire detection harnesses are W200 and W300. Both of the harnesses connect to one loop of the two engine fire detectors.

B. This procedure has these tasks:

(1) A removal of the engine fire detection harnesses (W200 and W300)

(2) An installation of the engine fire detection harnesses (W200 and W300).

TASK 26-11-02-000-801

2. Engine Fire Detection Harness Removal

(Figure 401, Figure 402, and Figure 403)

A. General

(1) The removal procedure for harness W200 and W300 is the same.

(2) W200 is a component of loop A of the engine fire detection circuit.

(3) W300 is a component of loop B of the engine fire detection circuit.

B. References

Reference	Title
30-21-11-000-801	Engine Anti-Ice Valve Removal (P/B 401)
30-21-21-000-801	Engine Anti-Ice Pressure Switch Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-05-000-801	Intermediate Manifold Pressure (PI) Sensor Removal (P/B 401)
36-11-07-000-801	HPSOV Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
72-23-06-000-801-G00	Fan Frame Strut Panels Removal (P/B 401)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)
77-12-01-000-801-G00	N1 Speed Sensor Removal (P/B 401)
77-12-02-000-801-G00	N2 Speed Sensor Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
79-31-01-000-801-G00	Oil Level Sensor Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 26-11-02-040-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	6	C00148	ANTI-ICE & RAIN ENG 1 & WING CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	7	C01001	ANTI-ICE & RAIN ENG 1 COWL AI VALVE
B	7	C01002	ANTI-ICE & RAIN ENG 2 COWL AI VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	11	C04082	ANTI-ICE & RAIN ENGINE 2 CONTROL
F	6	C04065	IASC EAI INHIBITS OUT L
F	7	C04066	IASC EAI INHIBITS OUT R

SUBTASK 26-11-02-010-001

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-11-02-010-008

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(3) Open the access to the fan and core firewalls.

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SUBTASK 26-11-02-020-010

(4) Remove the blanket [101] from the strut 5 outer fitting [109] (View B, Figure 402).

SUBTASK 26-11-02-000-001

(5) Remove the harness (harness W200 [1] and/or harness W300 [2]) from the harness strut (6:00 o'clock) [104] as follows (Figure 402 and Figure 403):

(a) Remove the Fan Frame Strut 5 Panel (left) (TASK 72-23-06-000-801-G00).

(b) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from the strut 5 outer fitting [109] as follows:

1) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

2) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and strut 5 outer fitting [109].

3) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.

4) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

5) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and around the harnesses.

6) Remove the harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

7) Remove any remaining sealant residue.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

(c) Remove the harness support block [504], harness support block [505], and harness support block [506] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

3) Move the assembly of the harness support block [504], harness support block [505], and harness support block [506] upward and away from the harness strut (6:00 o'clock) [104].

4) Remove the two nuts [112] and two bolts [113] to disassemble the harness support block [504], harness support block [505], and harness support block [506].

5) Cut and remove the sealant on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

6) Remove the harnesses from the harness support block [504], harness support block [505], and harness support block [506].

7) Remove any remaining sealant residue.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

(d) Remove the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [124] that attach the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

3) Move the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] upward and away from the harness strut (6:00 o'clock) [104].

4) Remove the four nuts [126] and four bolts [125] to disassemble the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

5) Cut and remove the sealant on the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and around the harnesses.

6) Remove the harnesses from the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

7) Remove any remaining sealant residue.

PTF ALL

F. Engine Fire Detection Harness Removal

SUBTASK 26-11-02-020-001

(1) Remove the harness W200 [1] and/or harness W300 [2] (Figure 401):

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS ARE CLEAN BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(a) Disconnect the harness connector from the pylon receptacle.

1) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connectors (TASK 70-00-01-910-803-G00).

(b) For the harness W200 [1], do the following steps:

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1) Disconnect the harness connector from the oil level sensor (TASK 79-31-01-000-801-G00).

2) Disconnect the harness connectors from the N1 speed sensor (TASK 77-12-01-000-801-G00).

3) Disconnect the harness connector from the Pressure Regulating and Shutoff Valve (PRSOV) (TASK 36-11-04-000-801).

4) Disconnect the harness connector from the High Pressure Shutoff Valve (HPSOV) (TASK 36-11-07-000-801).

5) Disconnect the harness connector from the intermediate manifold pressure (PI) sensor (TASK 36-11-05-000-801).

(c) For the harness W300 [2], do the following steps:

1) Disconnect the harness connector from the N2 speed sensor (TASK 77-12-02-000-801-G00).

PTF ALL; AIRPLANES WITH UNDER PRESSURE SWITCH (PRE SB 737-30-1074)

2) Disconnect the harness connectors from the Engine Anti-Ice (EAI) over pressure, under pressure, and minimum pressure switches (TASK 30-21-21-000-801).

PTF ALL; AIRPLANES WITHOUT UNDER PRESSURE SWITCH (POST SB 737-30-1074)

3) Disconnect the harness connectors from the EAI over pressure and minimum pressure switches (TASK 30-21-21-000-801).

PTF ALL; AIRPLANES WITH TWO OVER PRESSURE SWITCHES

4) Disconnect the harness connectors from the EAI minimum pressure switch and the over pressure switch jumper bundle (TASK 30-21-21-000-801).

PTF ALL

5) Disconnect the harness connector from the EAI valve (TASK 30-21-11-000-801).

6) Disconnect the harness connector from the Fuel Metering Unit (FMU) (TASK 73-21-04-000-801-G00).

(d) Remove the harness W200 [1] and/or harness W300 [2] from the fan engine brackets (View A, B, C, D, and E, Figure 401).

1) Disengage the harness W200 [1] and/or harness W300 [2] from the engine clips [11].

2) Remove the screws [12] and open the hinged clamps [13] that attach harness W200 [1] and/or harness W300 [2] to the fan engine brackets.

3) Remove the nuts [14] and washers [15] that attach the terminal lugs and ground lugs to the fan fire detectors.

(e) Remove the harness W200 [1] and/or harness W300 [2] from the fan engine support brackets (View A, F, G, H, and J, Figure 401).

1) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the oil tube [26] and engine support bracket [24].

2) Remove the screw [17] and clamp [18] that attach the harness W200 [1] to the engine support bracket [24].

3) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the engine support bracket [27], engine support bracket [28], and engine support bracket [29].

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4) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the engine support bracket [30].

5) Remove the screw [32] and clamp [18] that attach the harness W300 [2] to the engine support bracket [30].

6) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the engine support bracket [31].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

7) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the engine support bracket [33] and engine support bracket [34].

PTF ALL

(f) Remove the harness W200 [1] and/or harness W300 [2] from the core engine brackets (View A, K, L, M, N, and O, Figure 401).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

1) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the core engine bracket [35].

PTF ALL

2) Disengage the harness W200 [1] and/or harness W300 [2] from the clips [25] on the core engine bracket [36].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

3) Remove the screws [17] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [38].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

4) Remove the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [38].

PTF ALL

5) Remove the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [39].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

6) Remove the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [40] and core engine bracket [41].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

7) Remove the screws [53] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [40] and core engine bracket [41].

PTF ALL

8) Remove the screws [17] and clamps [37] that attach the harness W200 [1] to the LPTACC air duct [42] at the location A, location B, and location C (View N, Figure 401).

9) Remove the screws [17] and clamps [37] that attach the harness W300 [2] to the LPTACC air duct [42] at the location D, location E, and location F (View N, Figure 401).

10) Remove the screws [17] and clamps [37] that attach the harness W200 [1] to the core engine bracket [43] and core engine bracket [44].

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11) Remove the screw [17] and clamp [45] that attach the harness W200 [1] to the lower assembly manifold [46].

12) Remove the screws [17] and clamps [37] that attach the harness W300 [2] to the core engine bracket [47], core engine bracket [48], core engine bracket [49], and core engine bracket [50].

(g) Remove the harness W200 [1] and/or harness W300 [2] from the engine brackets (View P and R, Figure 401).

1) Remove the screws [16] that attach the harness W200 [1] to the engine bracket [51] (View R, Figure 401).

2) Remove the screws [16] that attach the harness W300 [2] to the engine bracket [52] (View P, Figure 401).

(h) Remove the bolts [19], nuts [20], and clamps [21] that attach the terminal and ground lug wires to the core fire detector brackets.

CAUTION

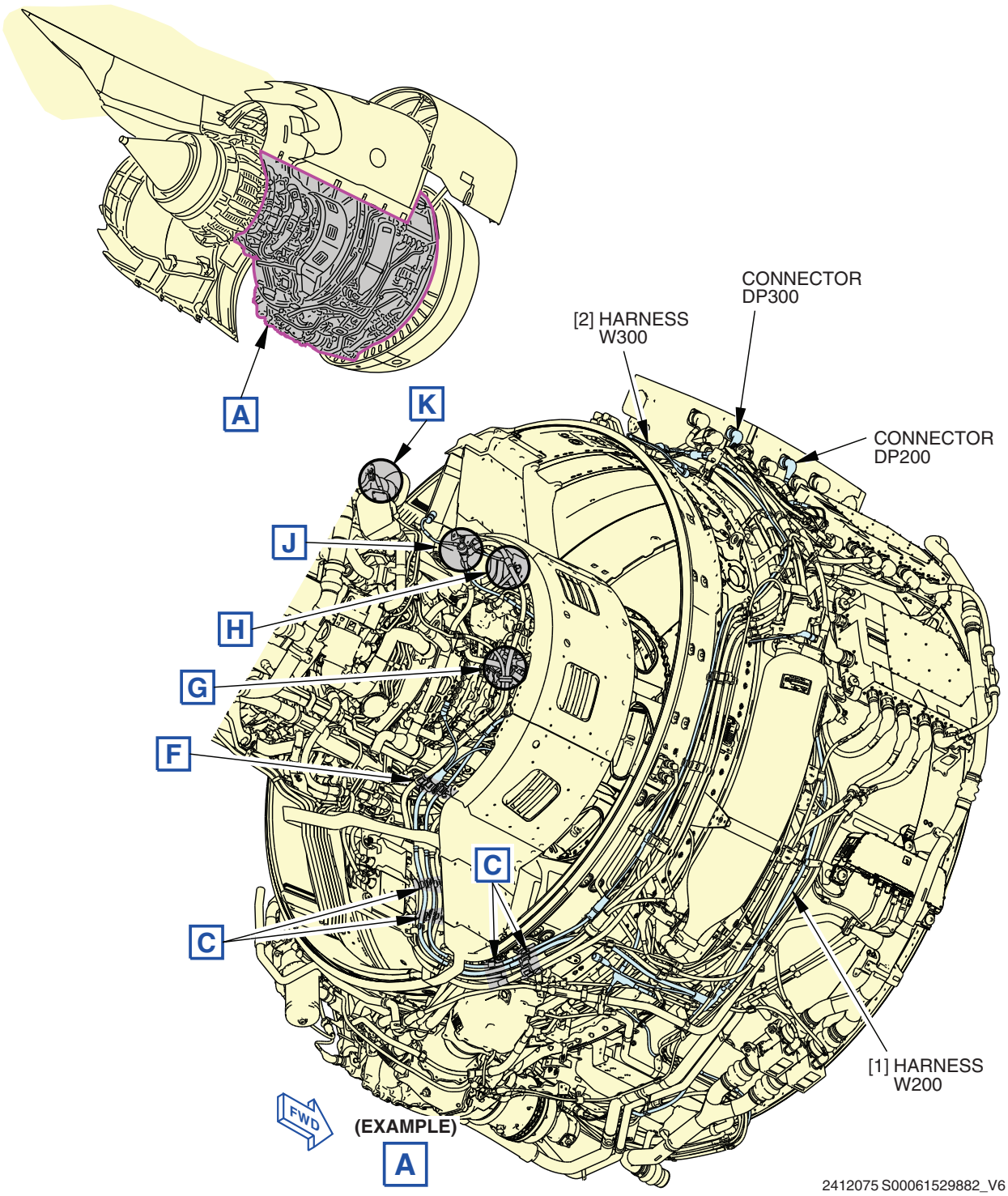
WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(i) Remove the nuts [22] and washers [23] that attach the terminal and ground lugs to the core fire detectors.

(j) Remove the harness W200 [1] and/or harness W300 [2] from the engine.

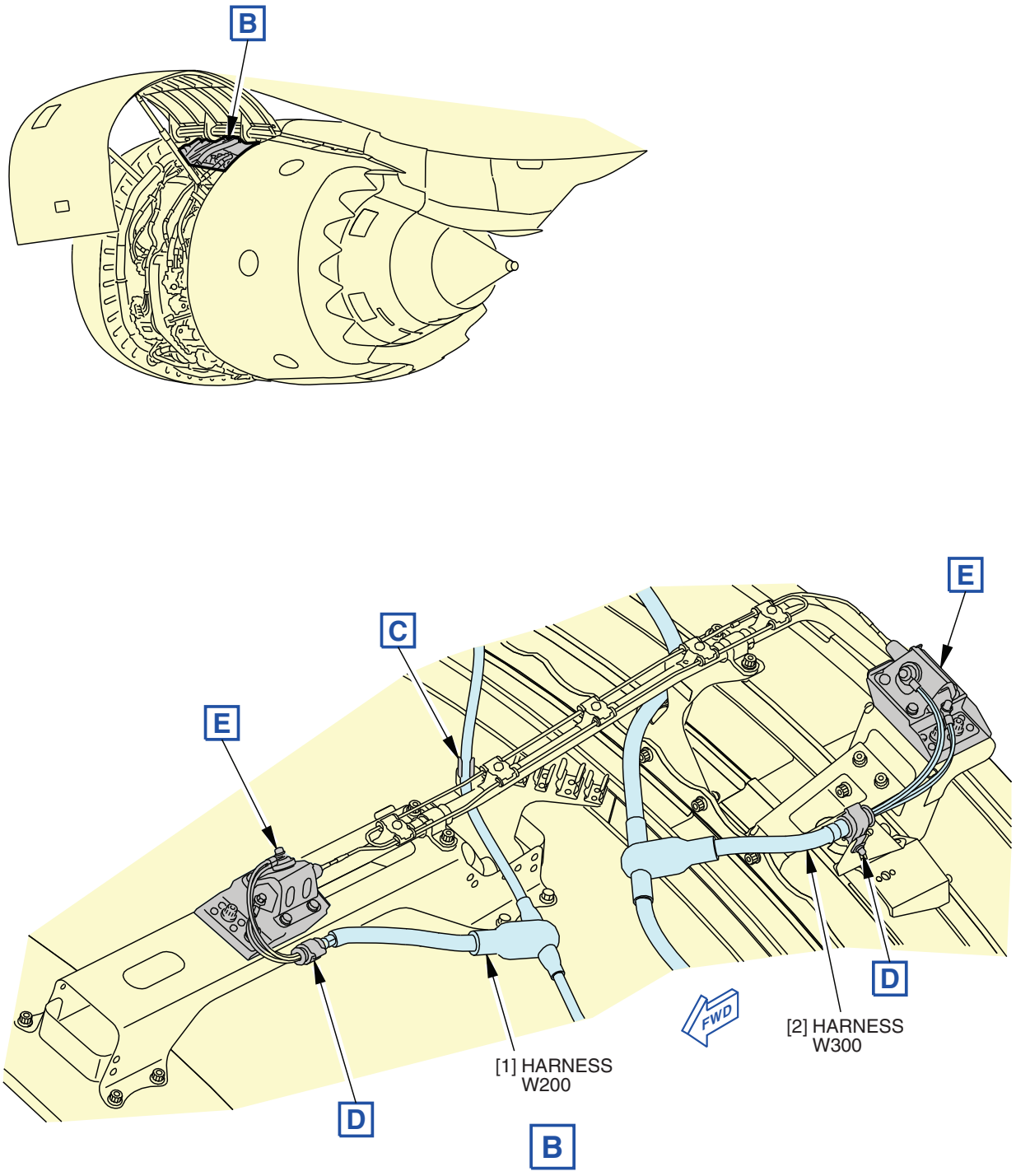
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 1 of 14)

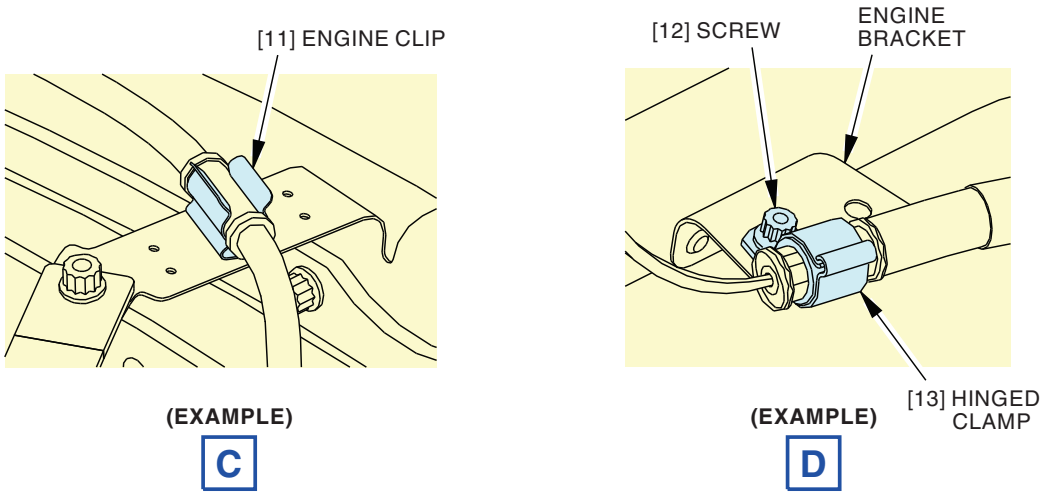
ECCN 9E991 BOEING PROPRIETARY - See title page for details

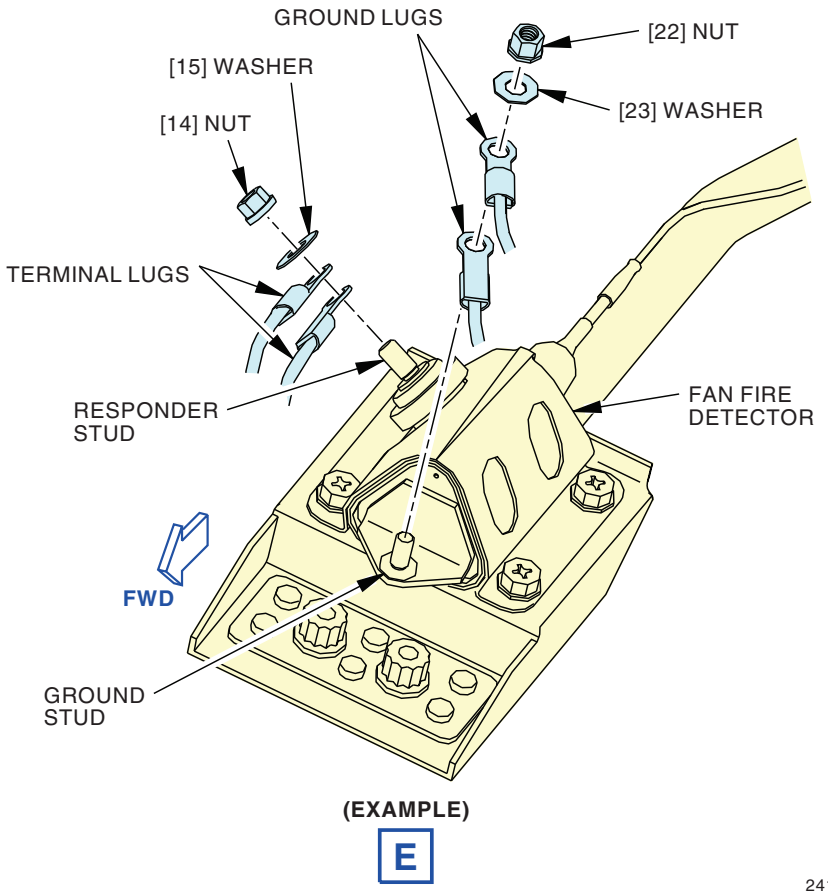


Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 2 of 14)

I

ECCN 9E991 BOEING PROPRIETARY - See title page for details

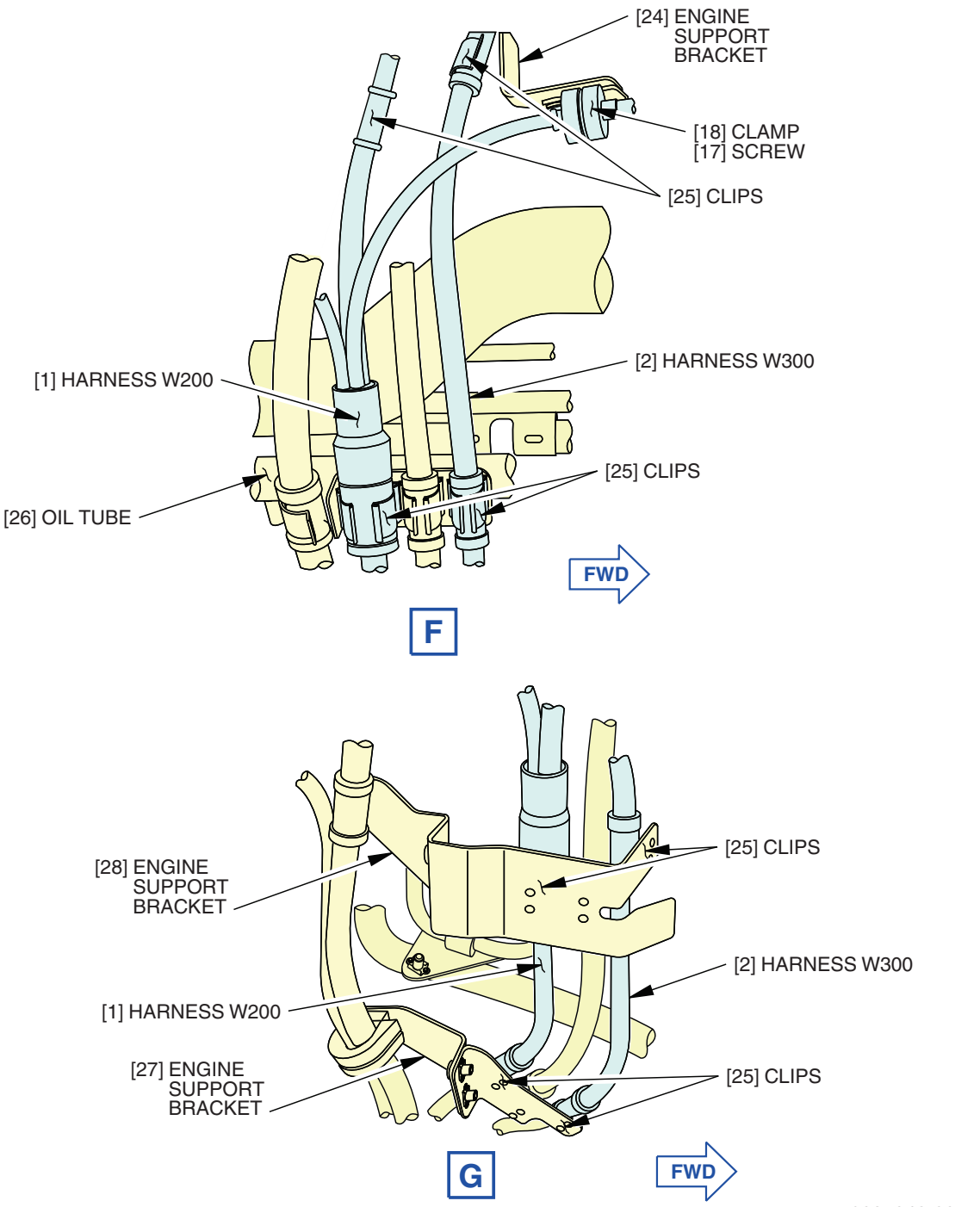




2412077 S00061529884_V4

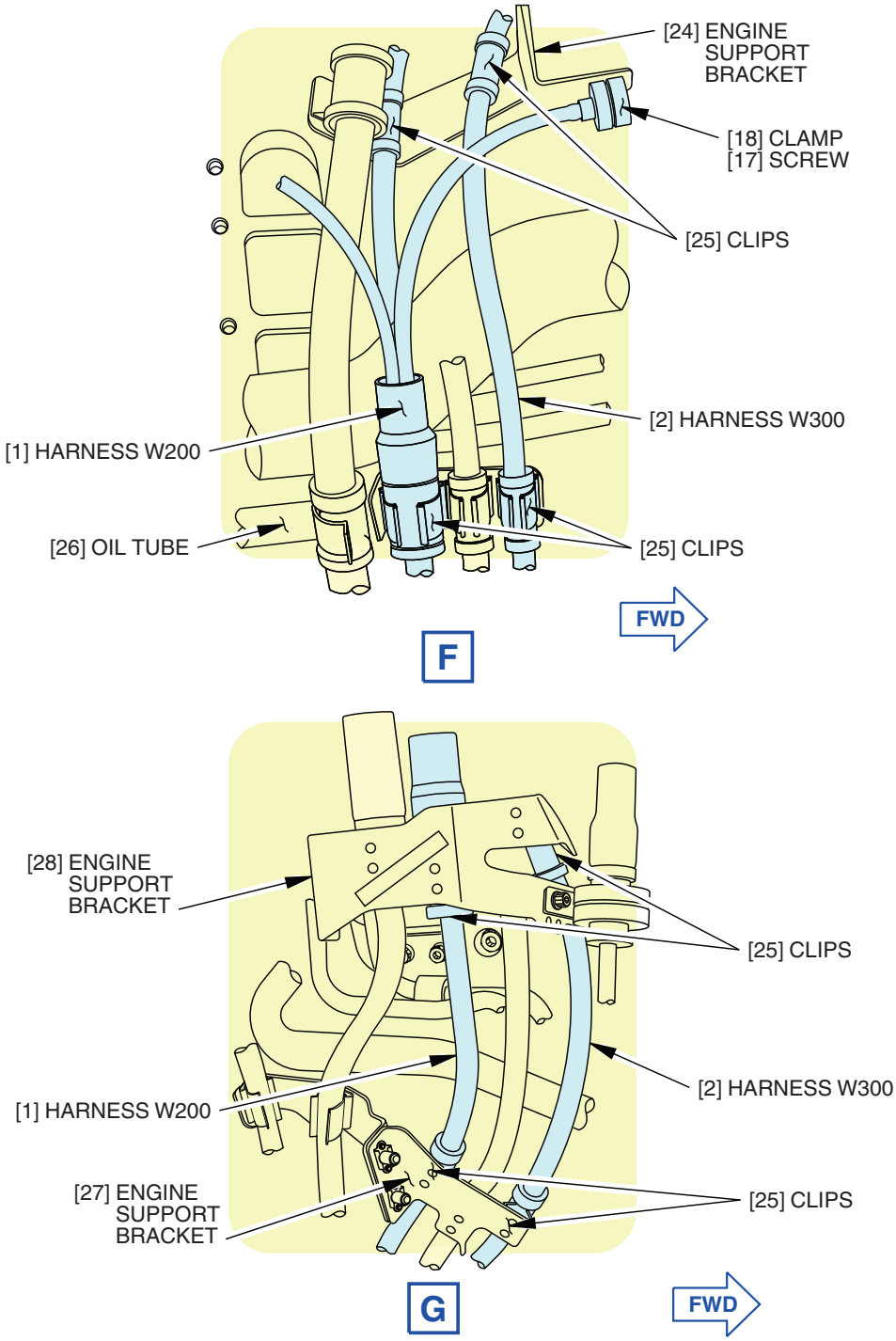
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 3 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



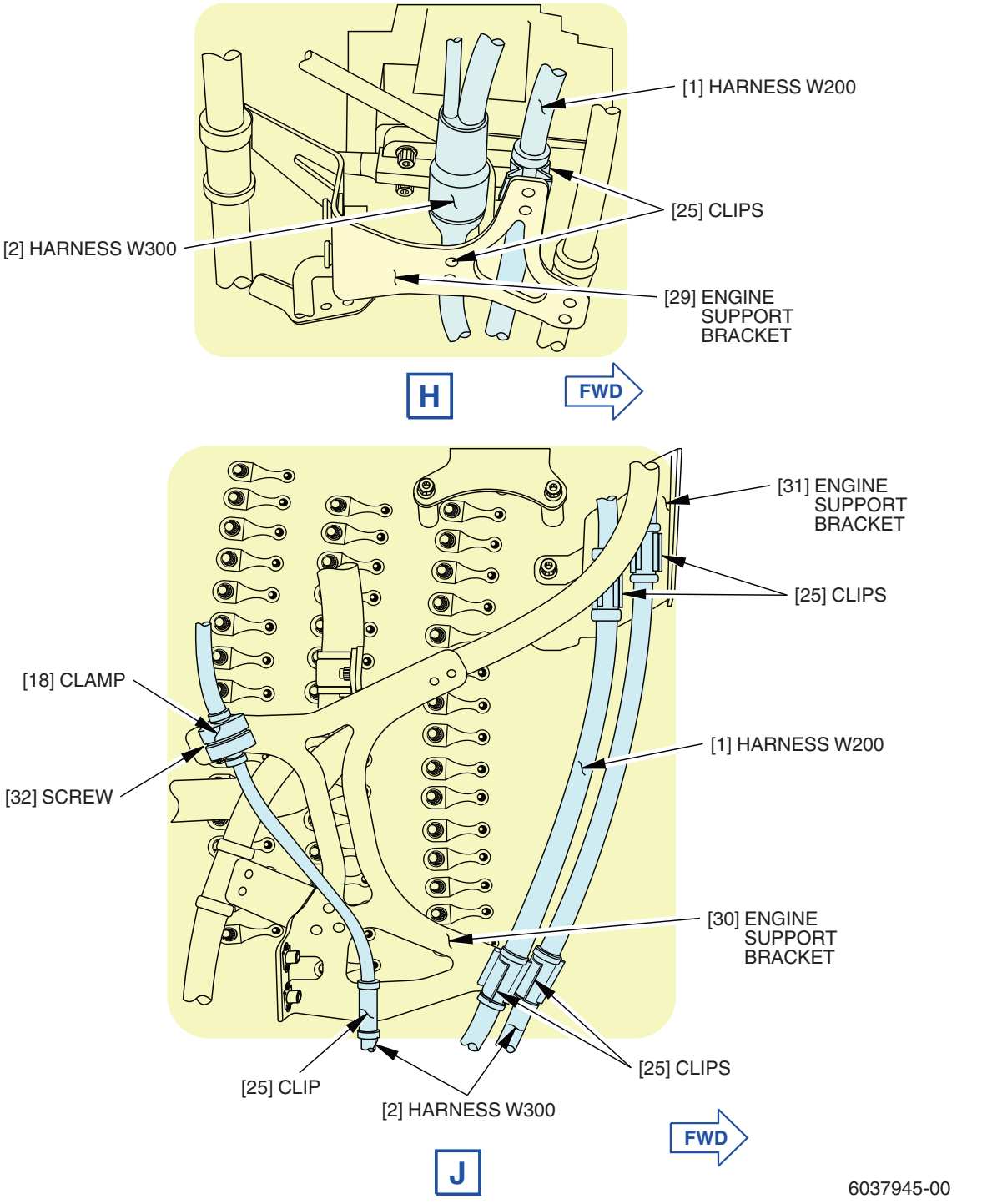
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 4 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



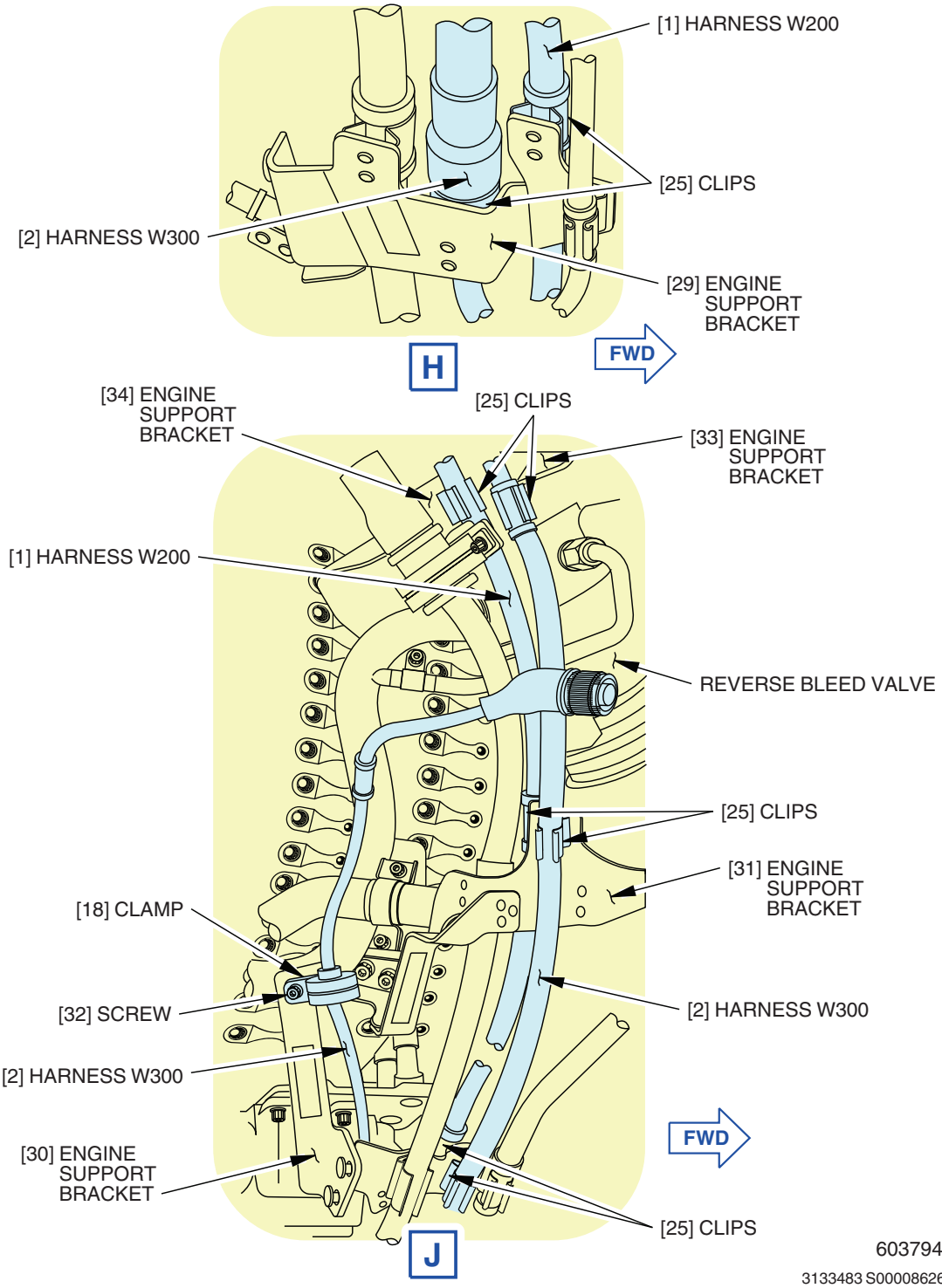
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 5 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



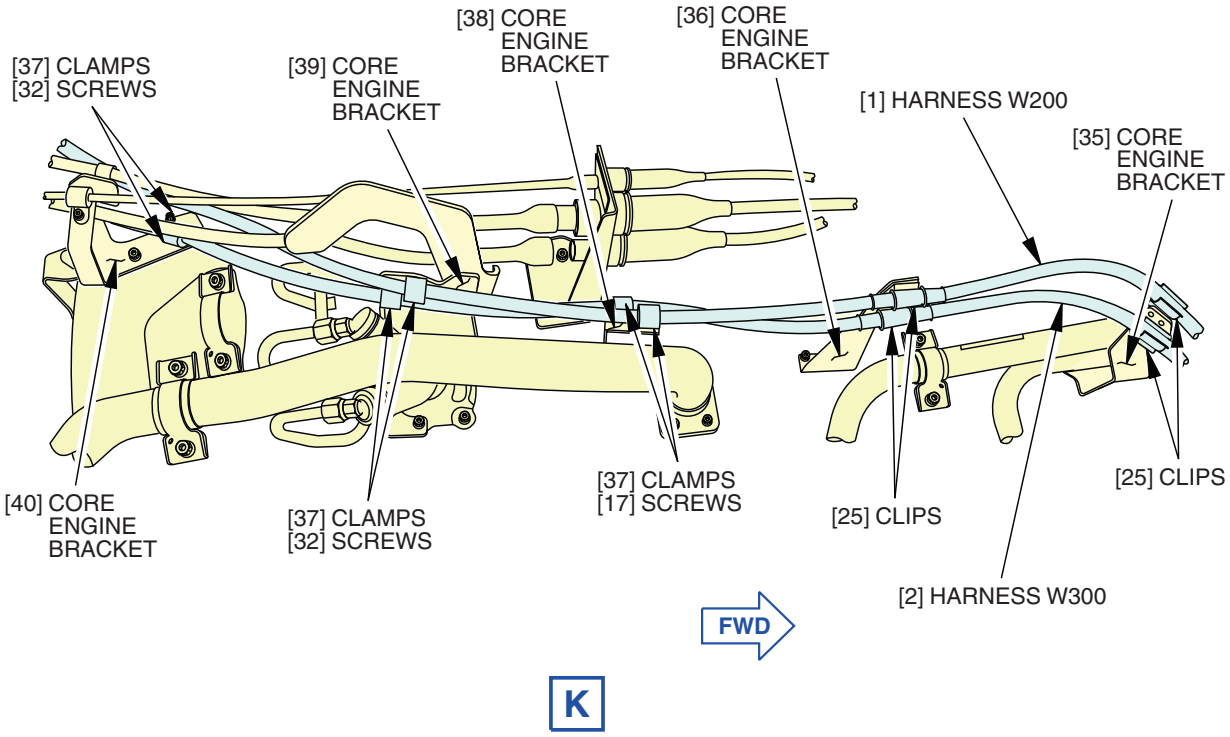
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 6 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 7 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

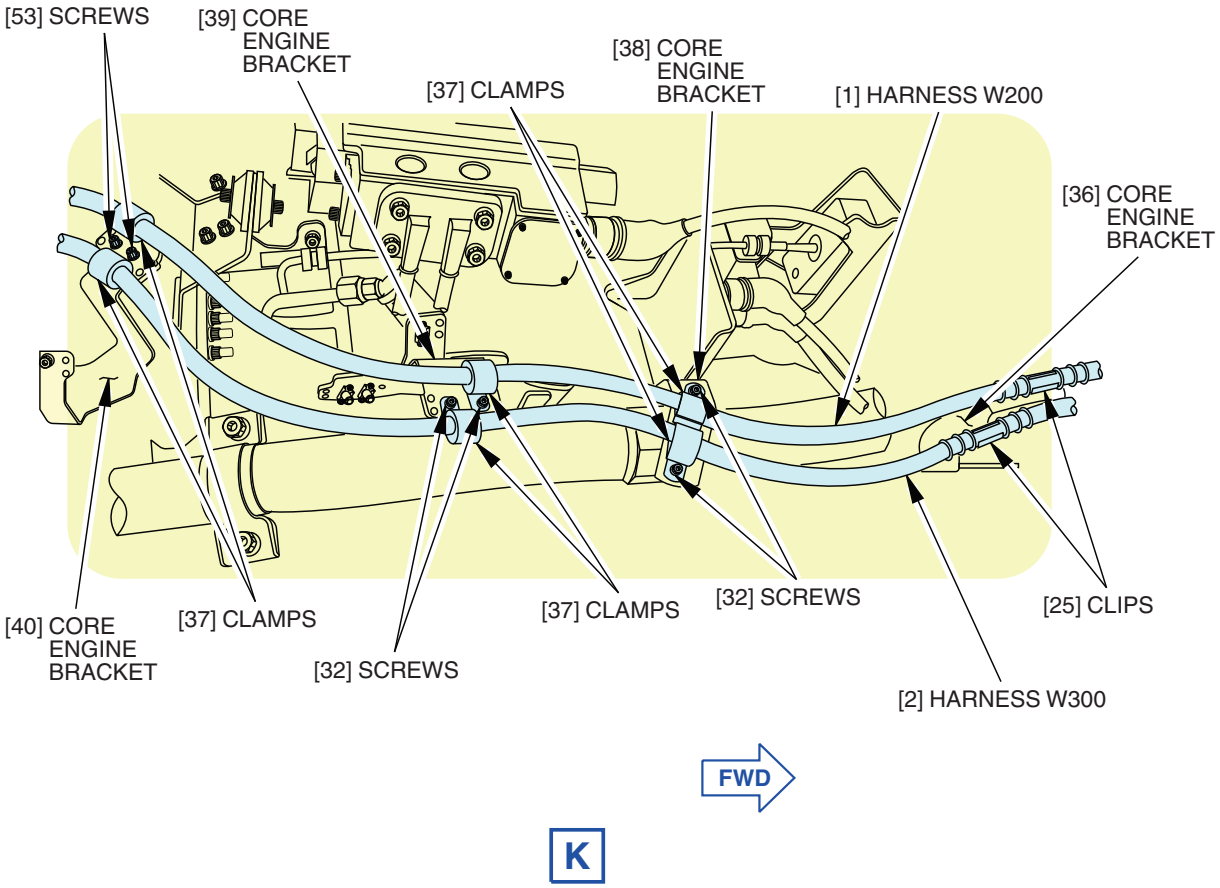


6037951-00

3133487 S0000862601_V1

Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 8 of 14)

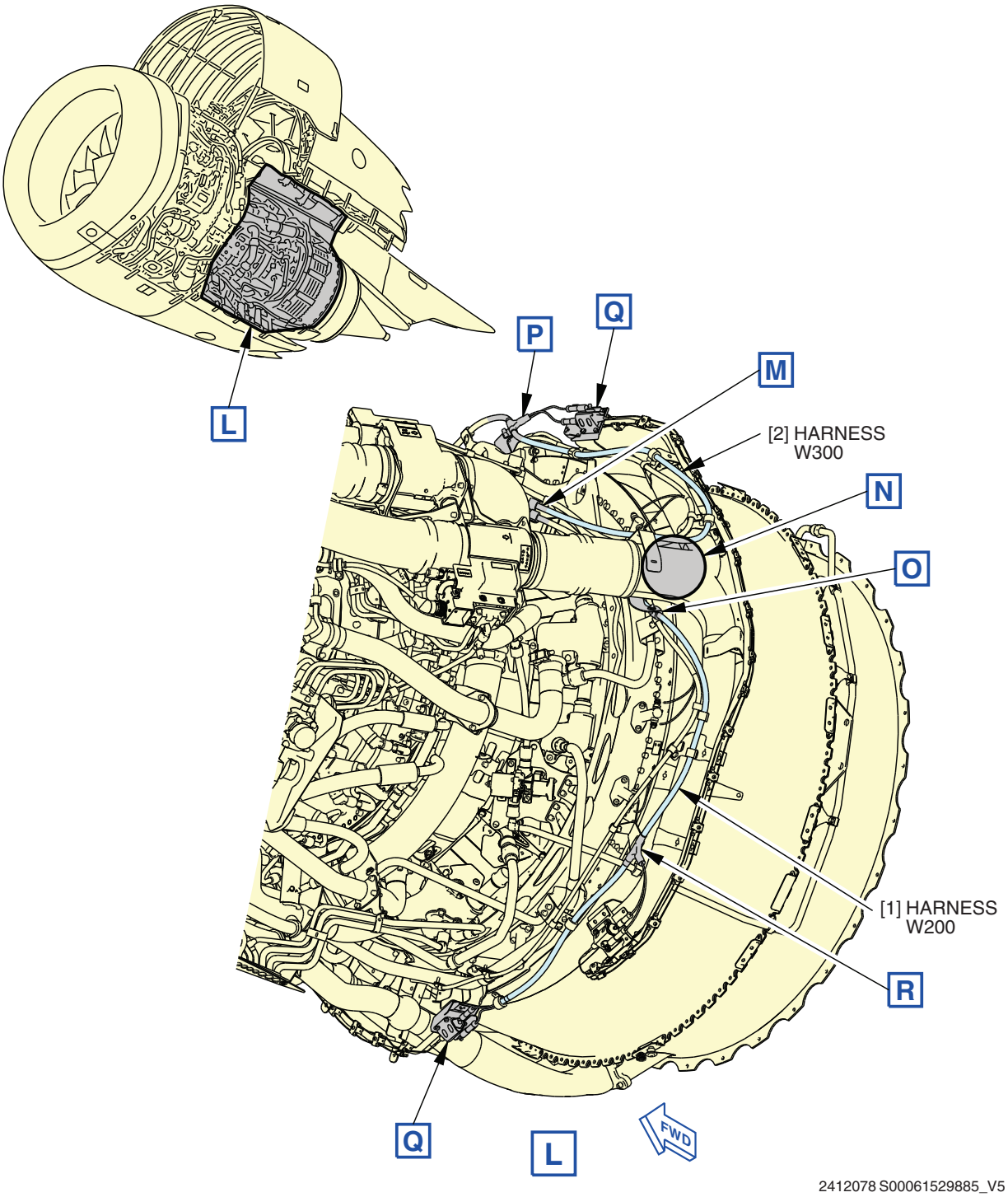
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6037950-00
3133498 S0000862602_V1

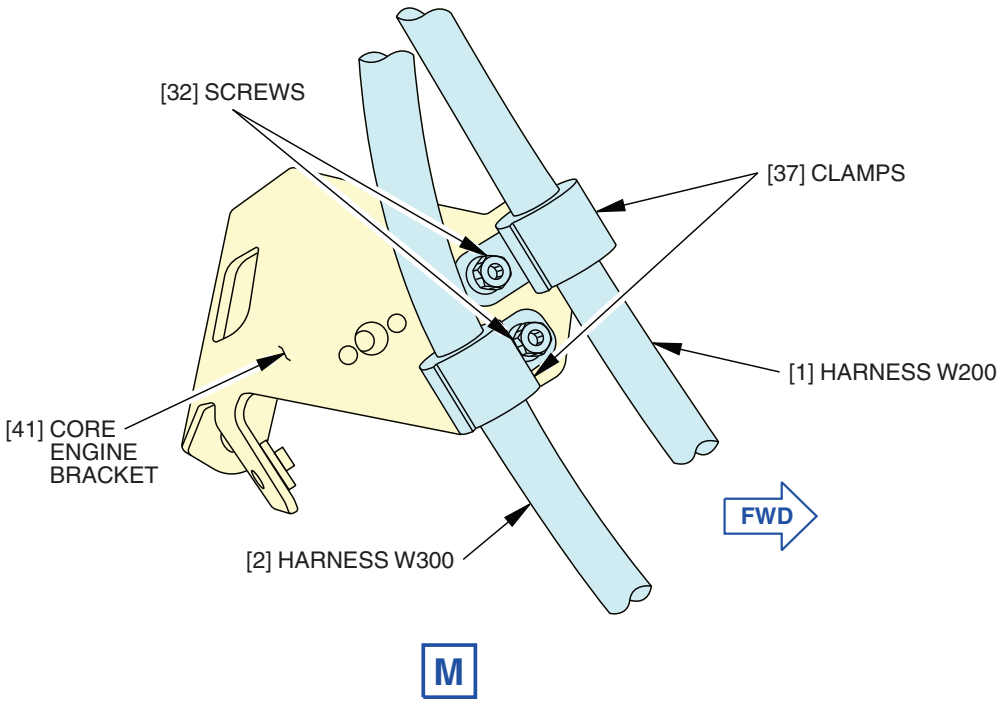
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 9 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 10 of 14)

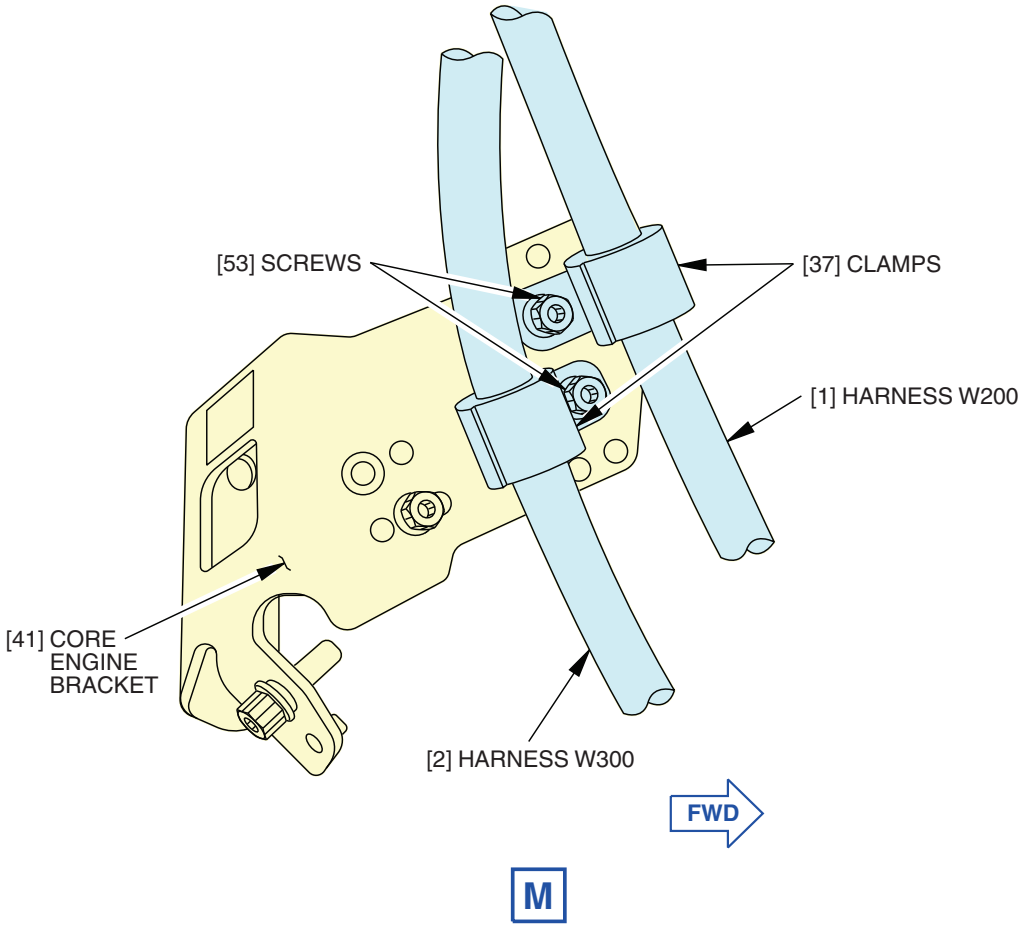
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6037953-00
3133502 S0000862603_V1

Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 11 of 14)

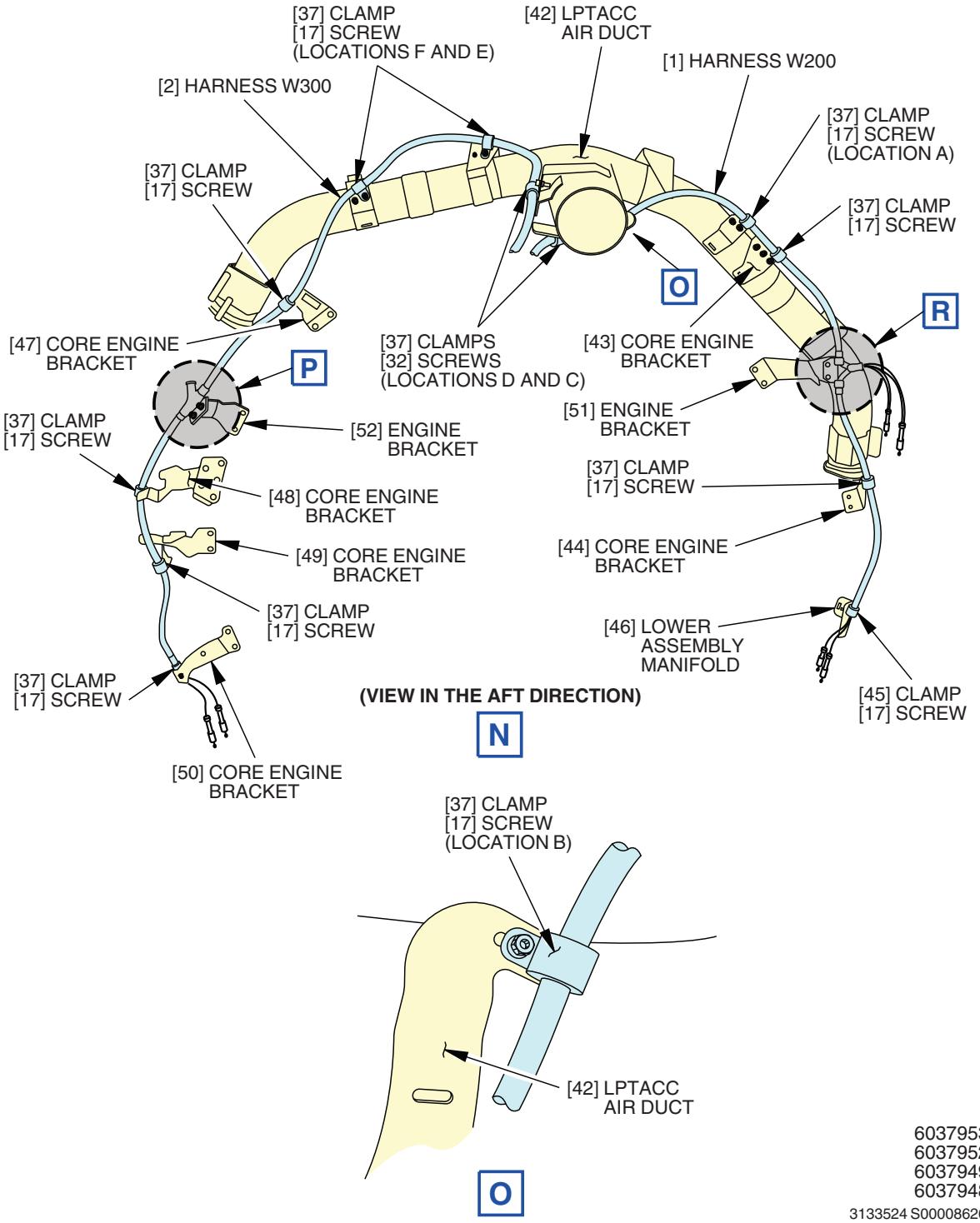
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6037952-00
3133505 S0000862604_V1

Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 12 of 14)

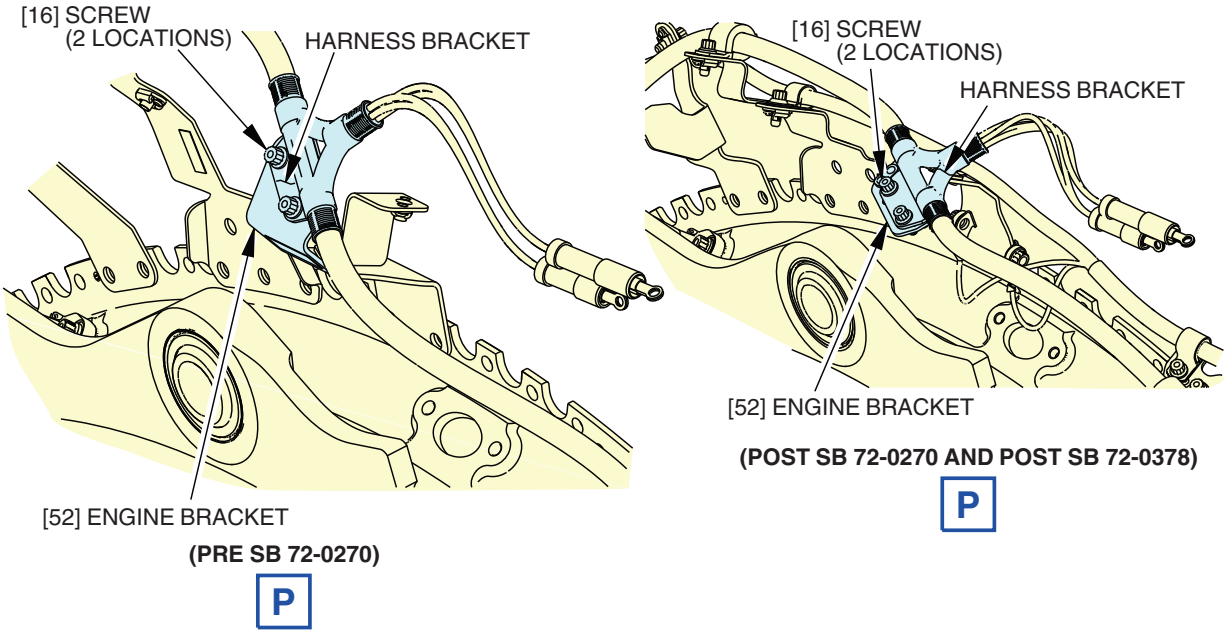
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6037953-00
6037952-00
6037949-00
6037948-00
3133524 S0000862605_V1

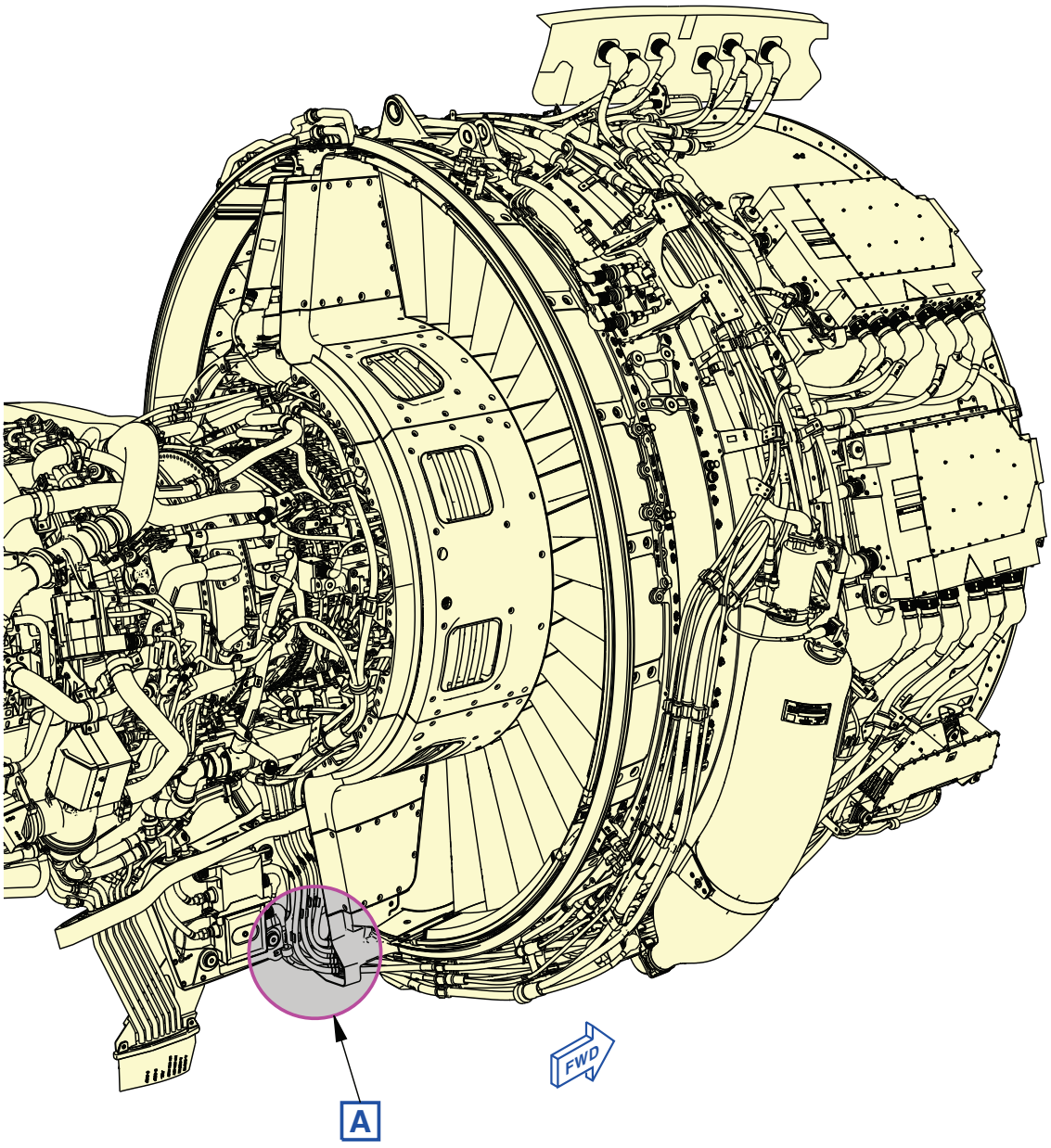
Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 13 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Detection Harnesses Installation
Figure 401/26-11-02-990-801 (Sheet 14 of 14)

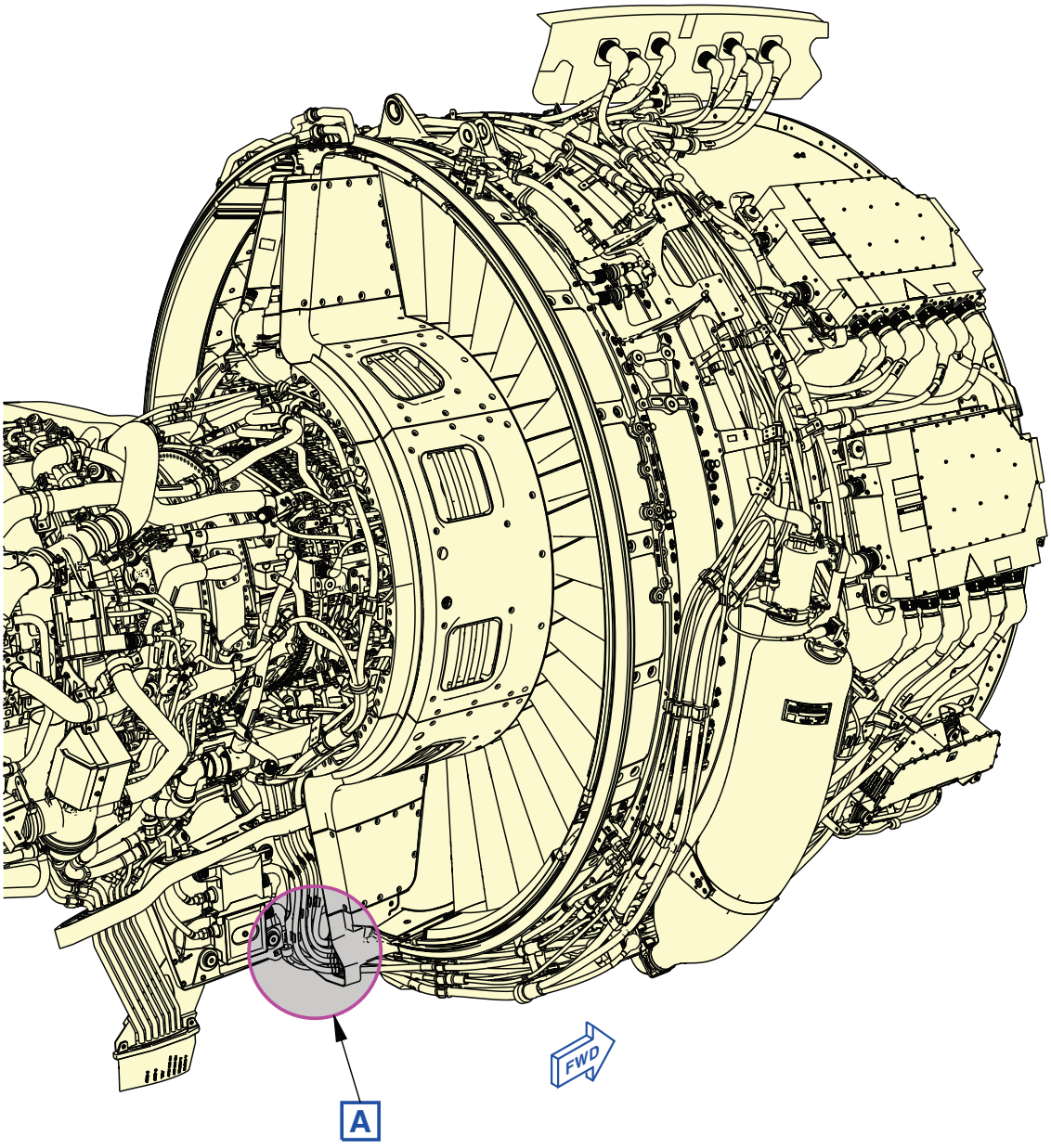
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937794 S0000713383_V3

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 1 of 11)

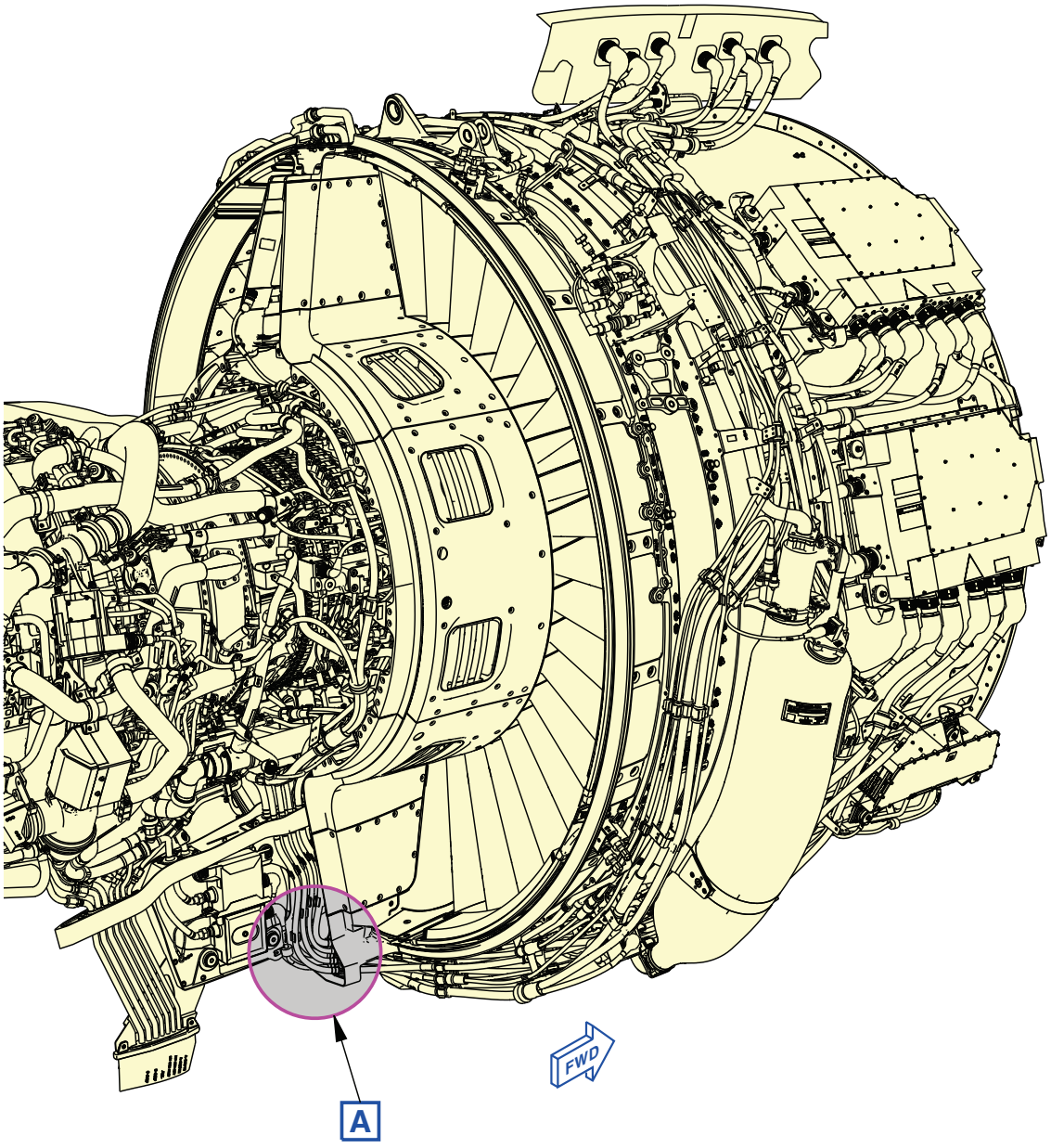
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
3124476 S0000858152_V1

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 2 of 11)

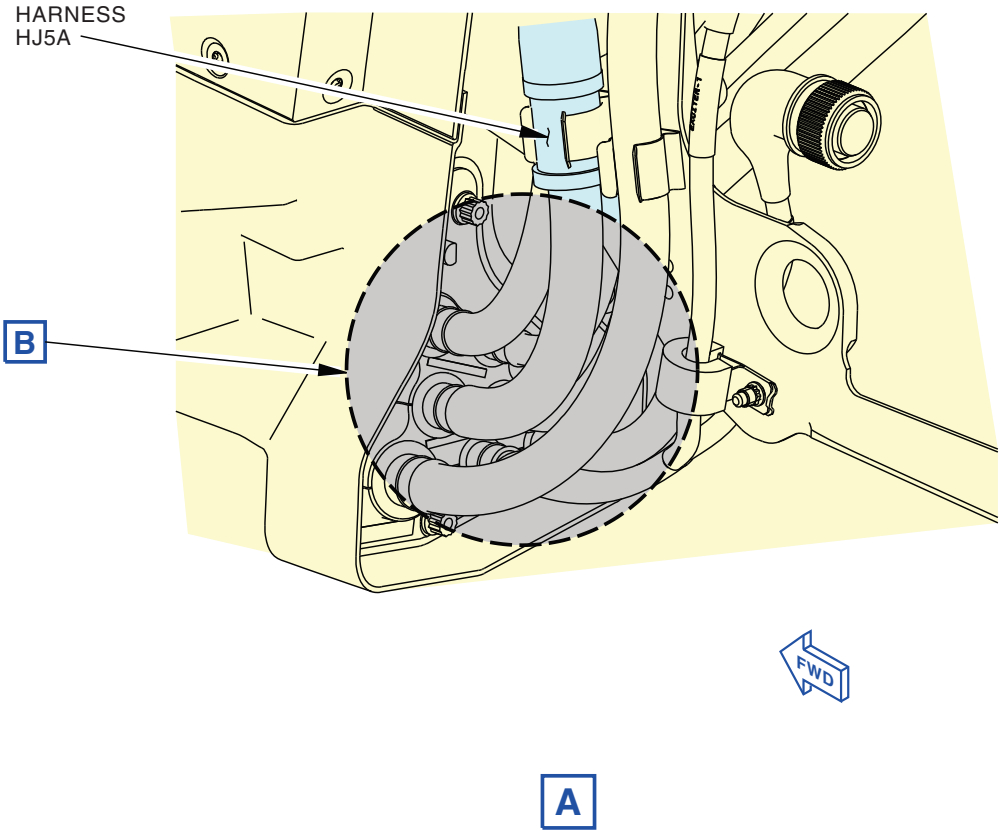
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
3124477 S0000858153_V1

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 3 of 11)

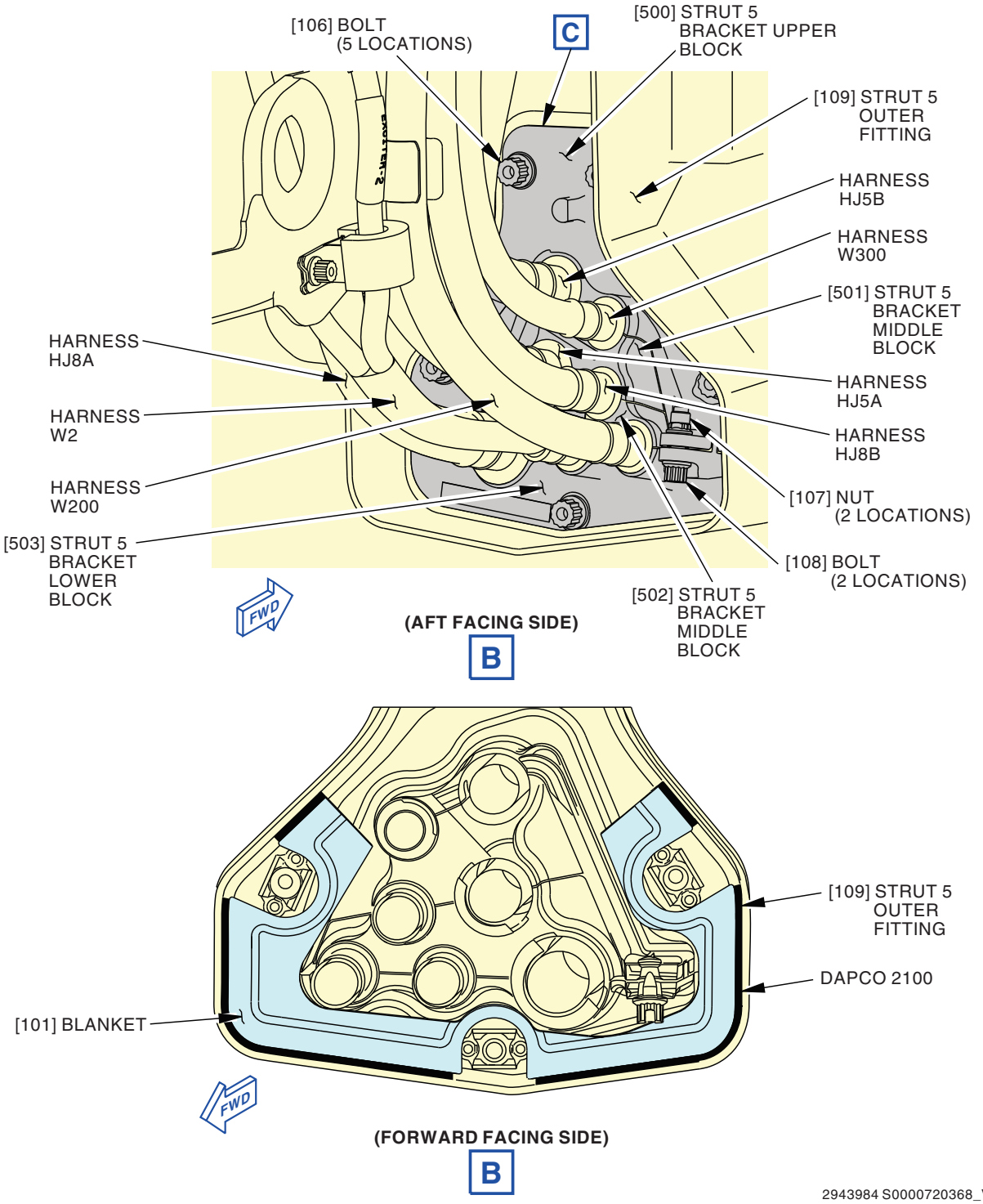
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937795 S0000713384_V2

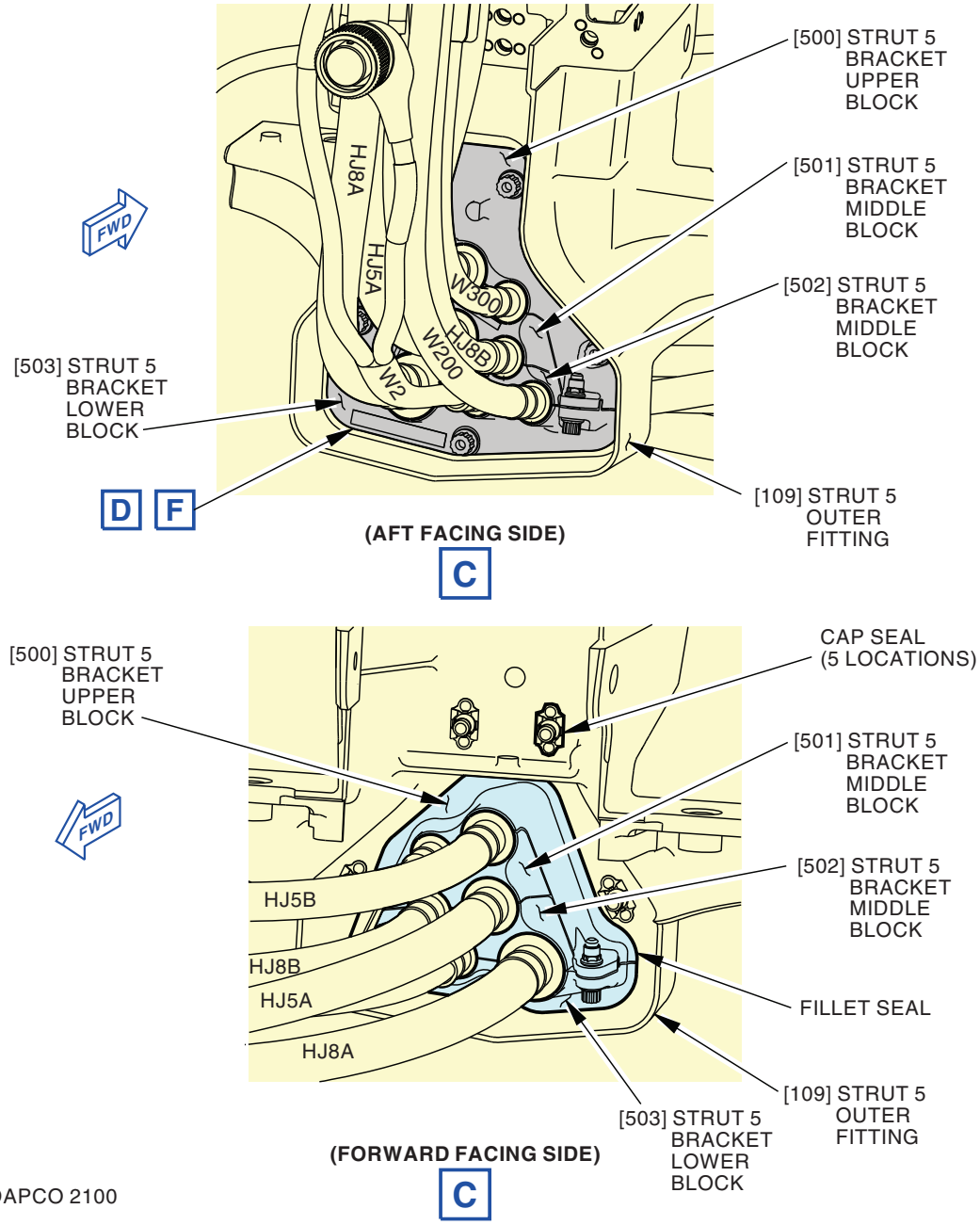
Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

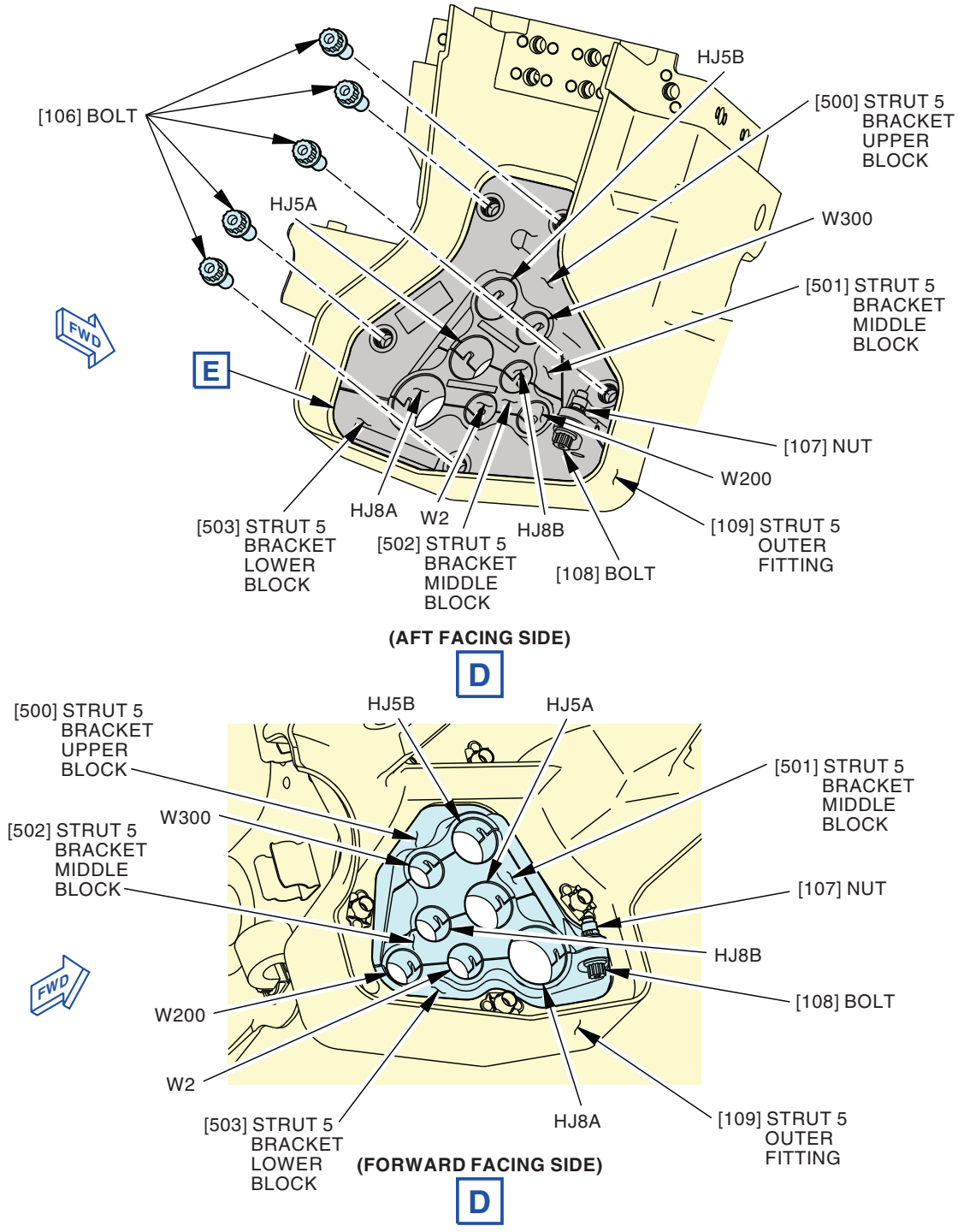
FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 6 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



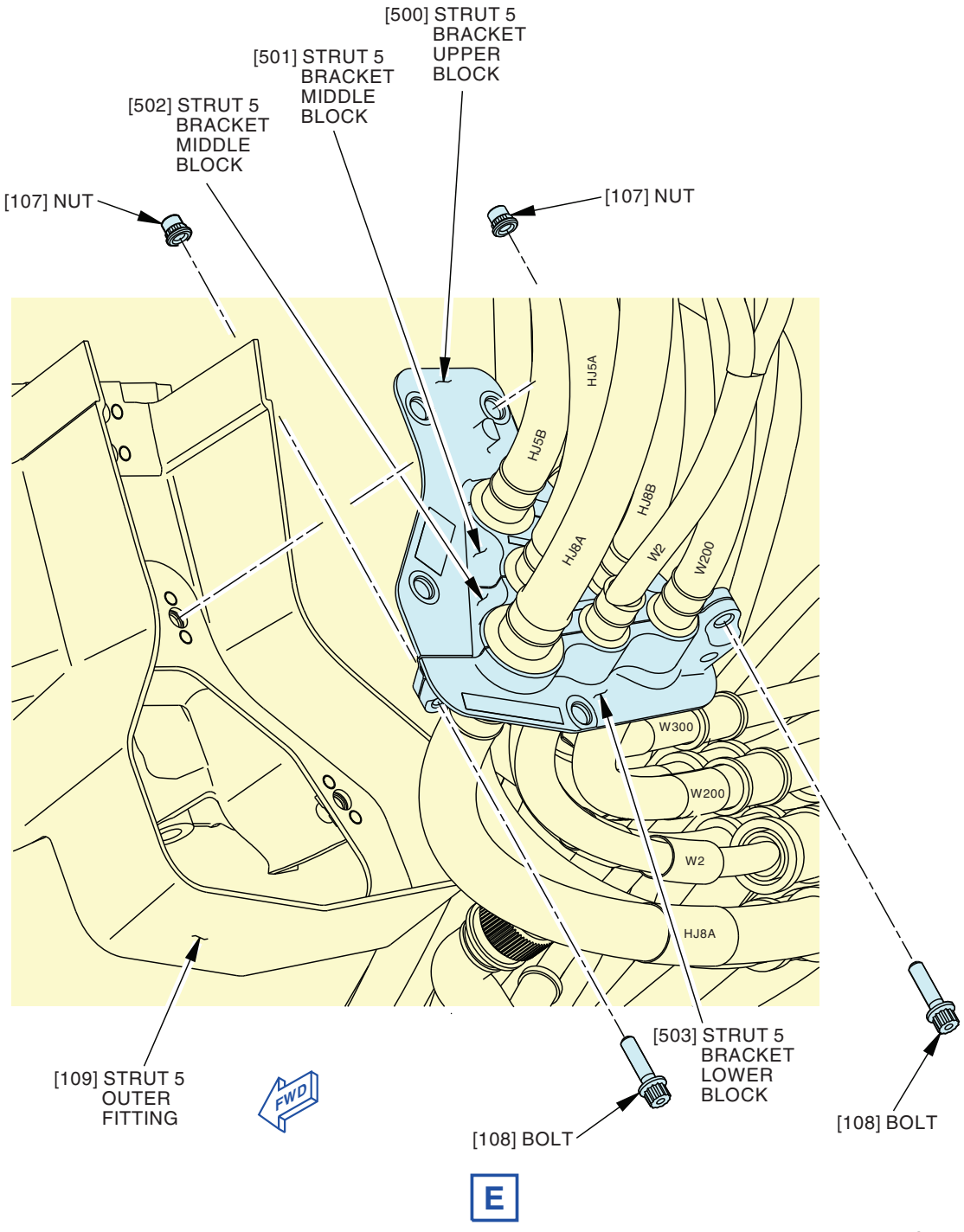
NOTE:

HARNESSES NOT SHOWN

2943980 S0000720369_V2

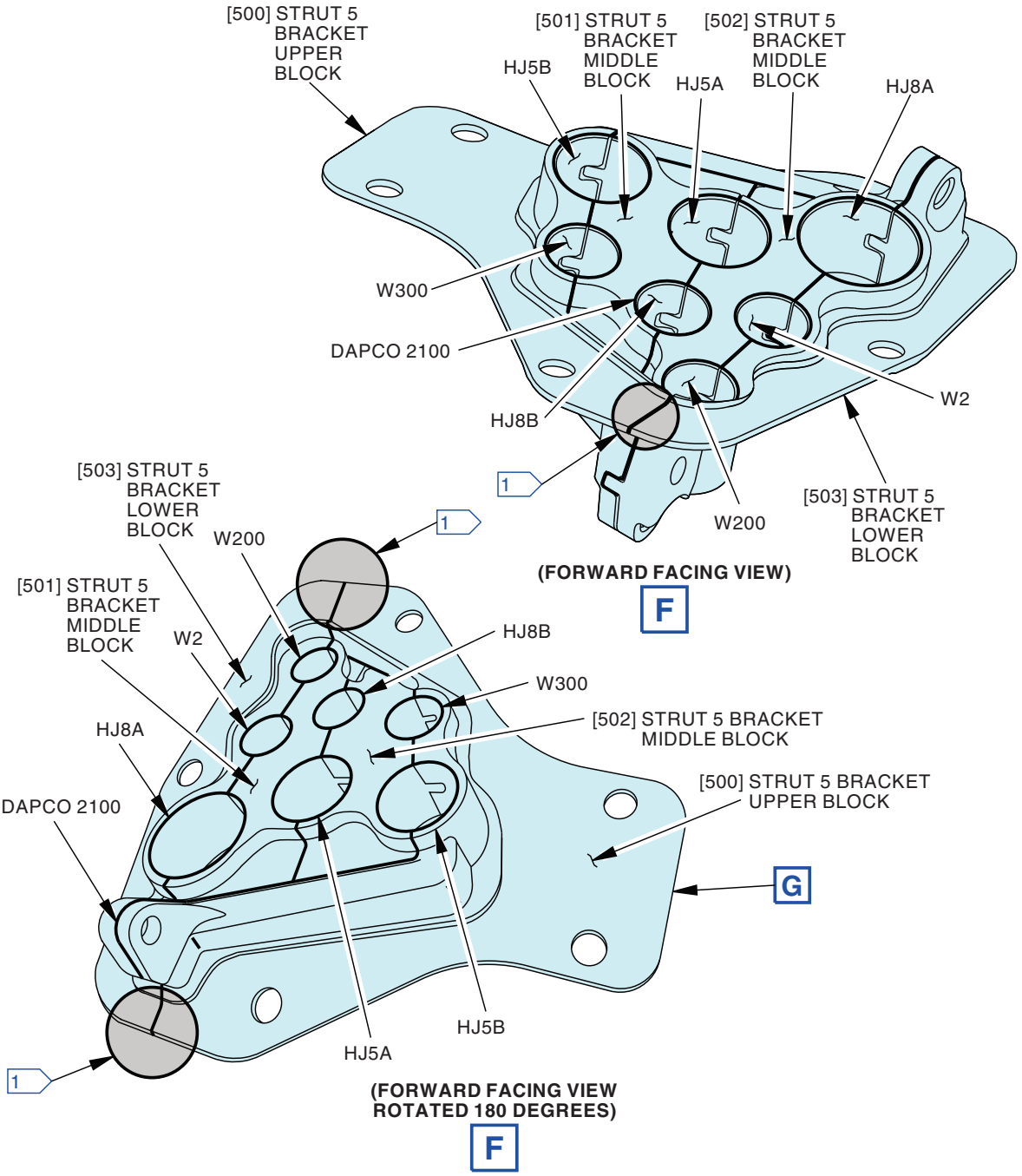
Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

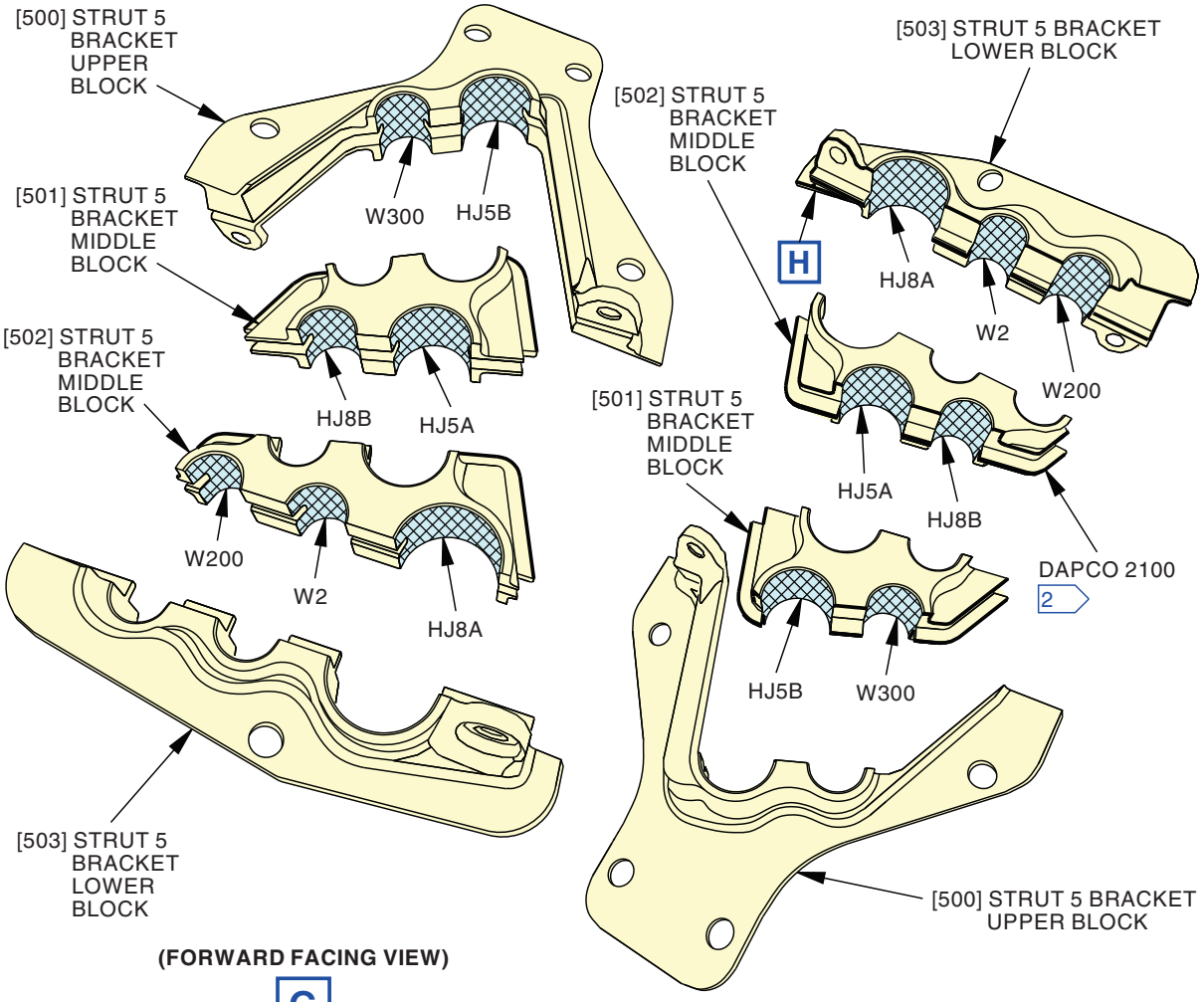
HARNESSES NOT SHOWN.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

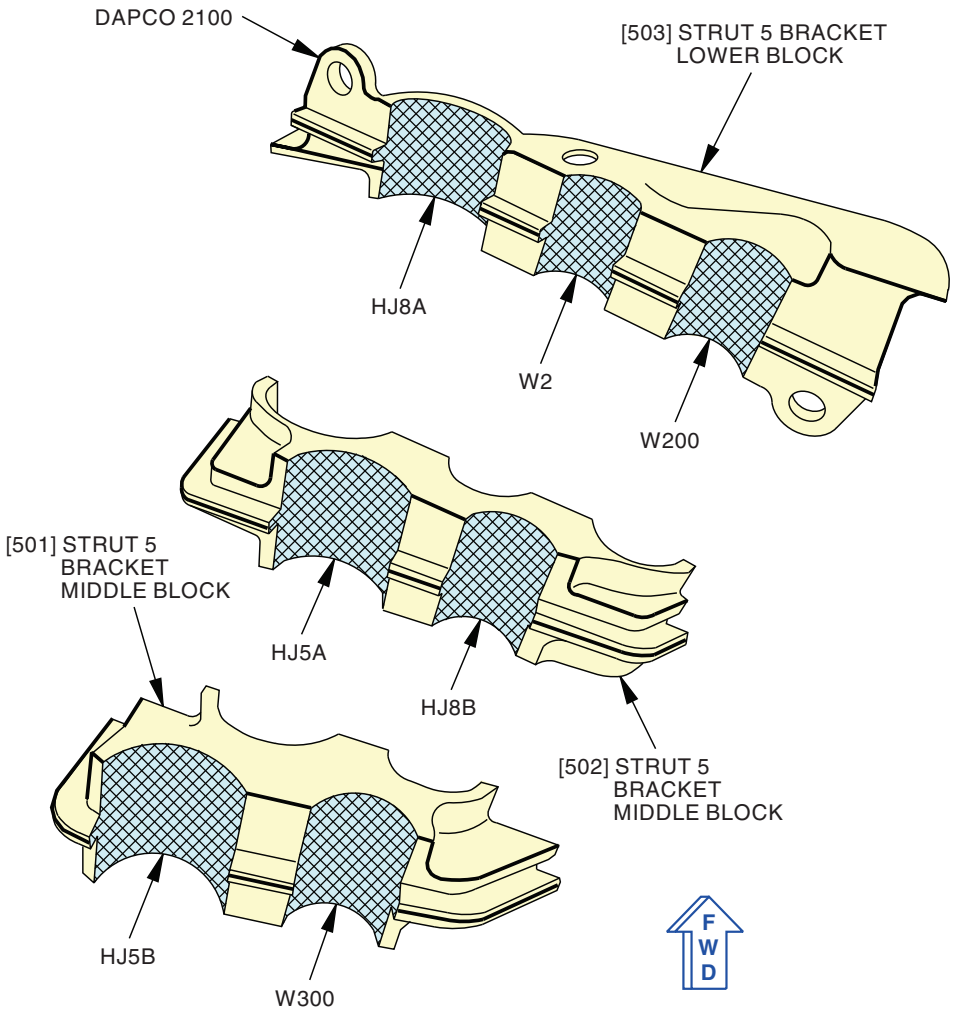
SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 10 of 11)

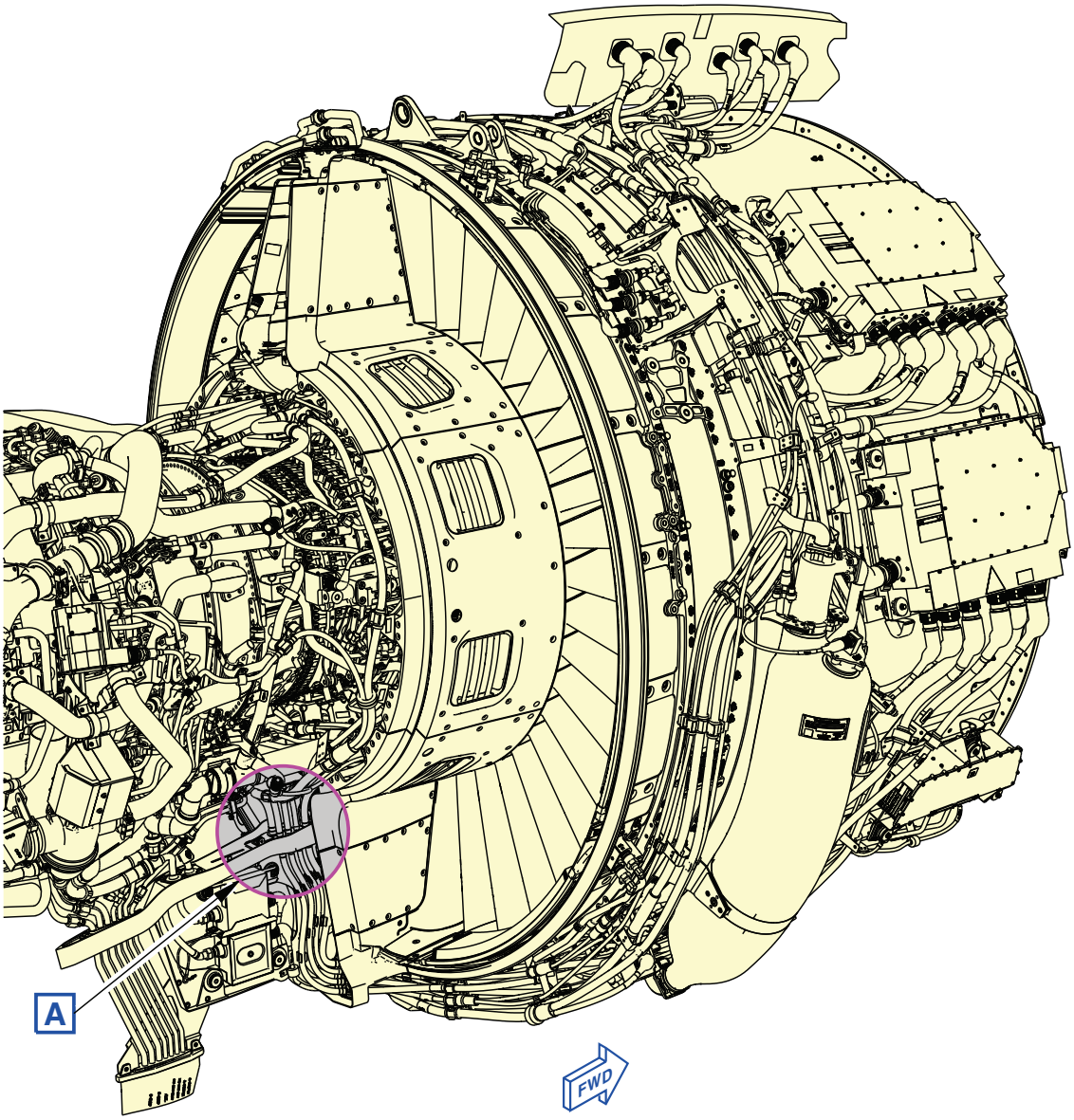
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2949941 S0000726813_V1

Strut Bracket Block Installation
Figure 402/26-11-02-990-824 (Sheet 11 of 11)

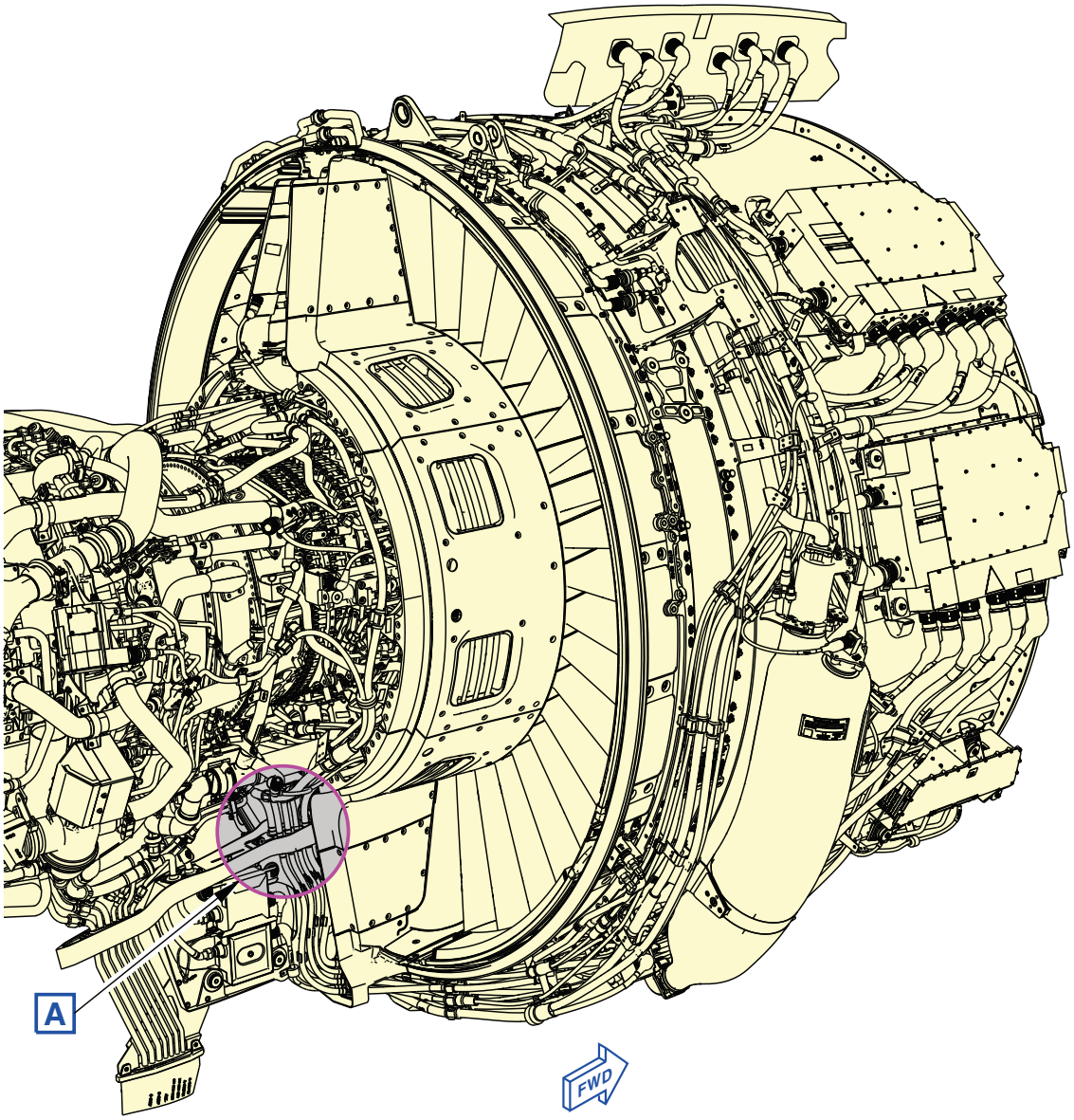
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
2938818 S0000714836_V3

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 1 of 13)

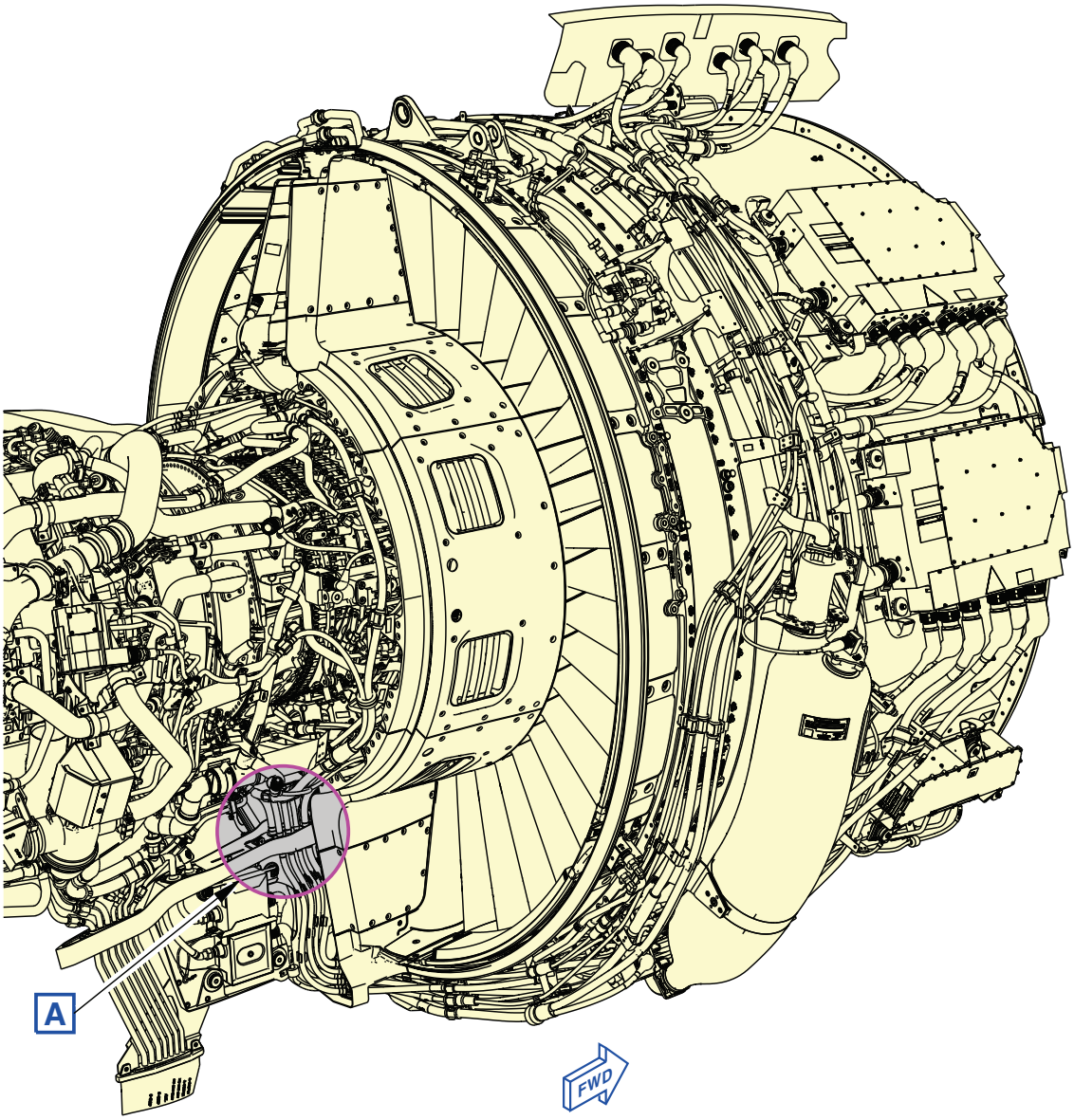
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
3121855 S0000856417_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 2 of 13)

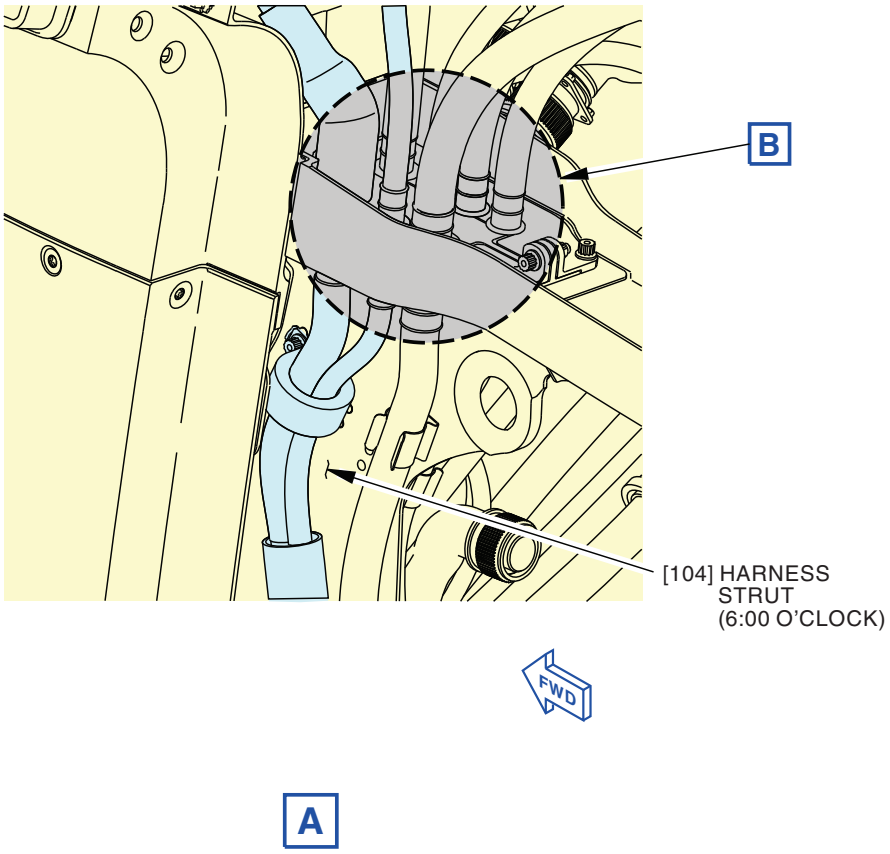
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
3121857 S0000856418_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 3 of 13)

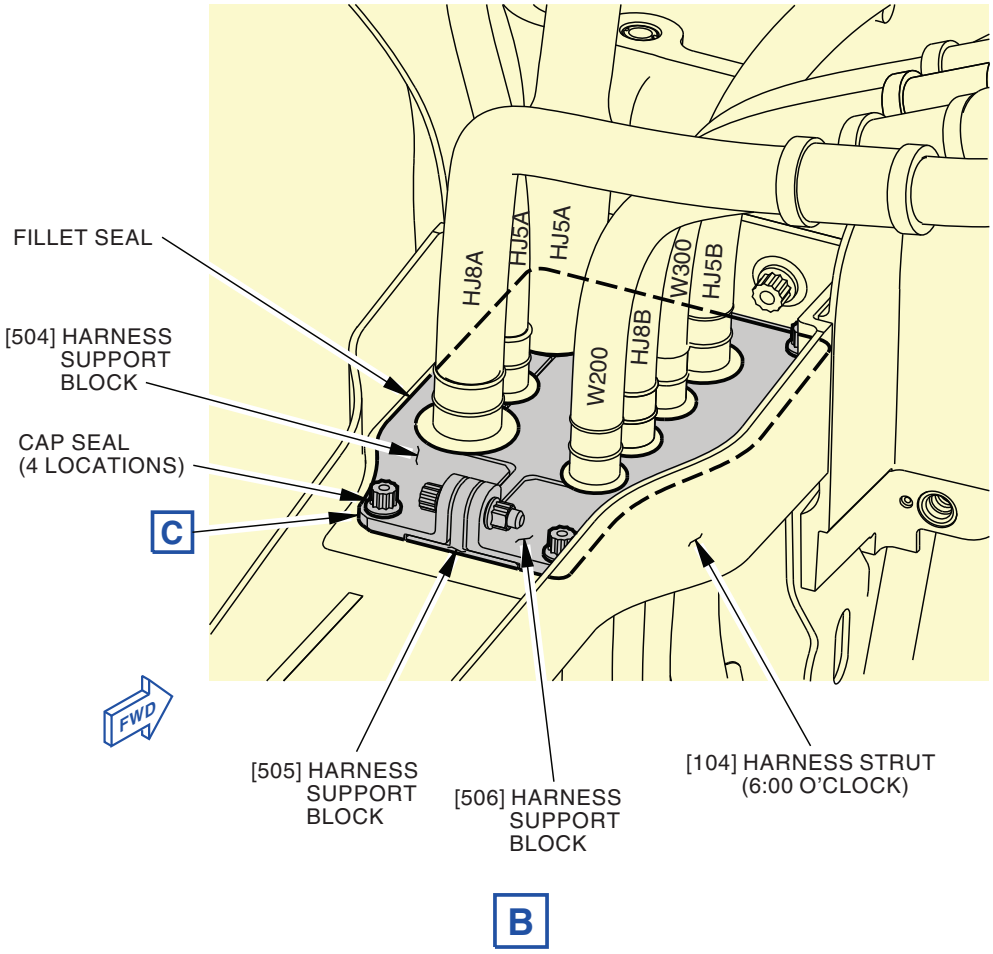
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2952925 S0000730238_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 4 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

——— DAPCO 2100

NOTE:

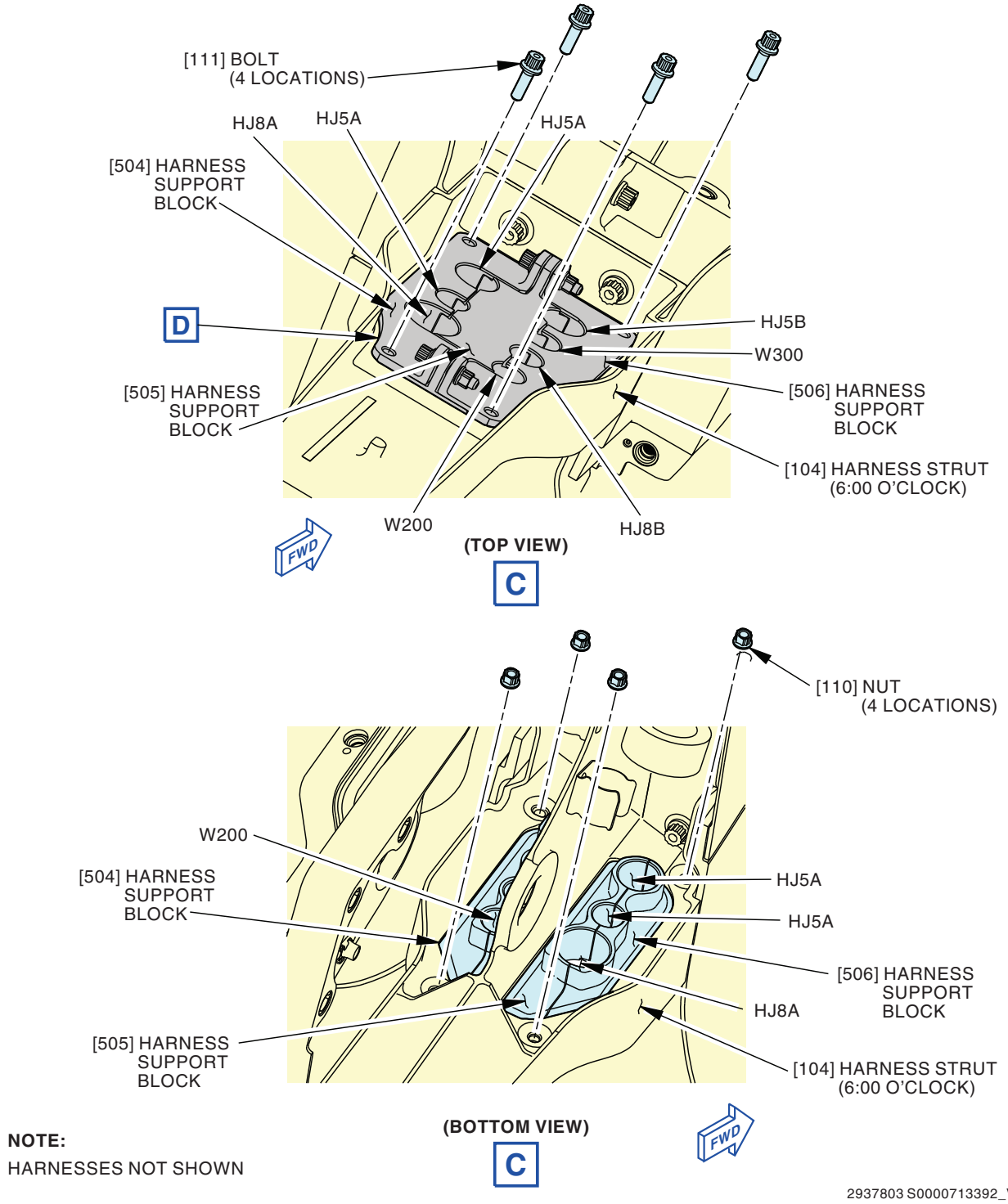
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE HARNESS SUPPORT BLOCK. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937807 S0000713396_V4

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 5 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

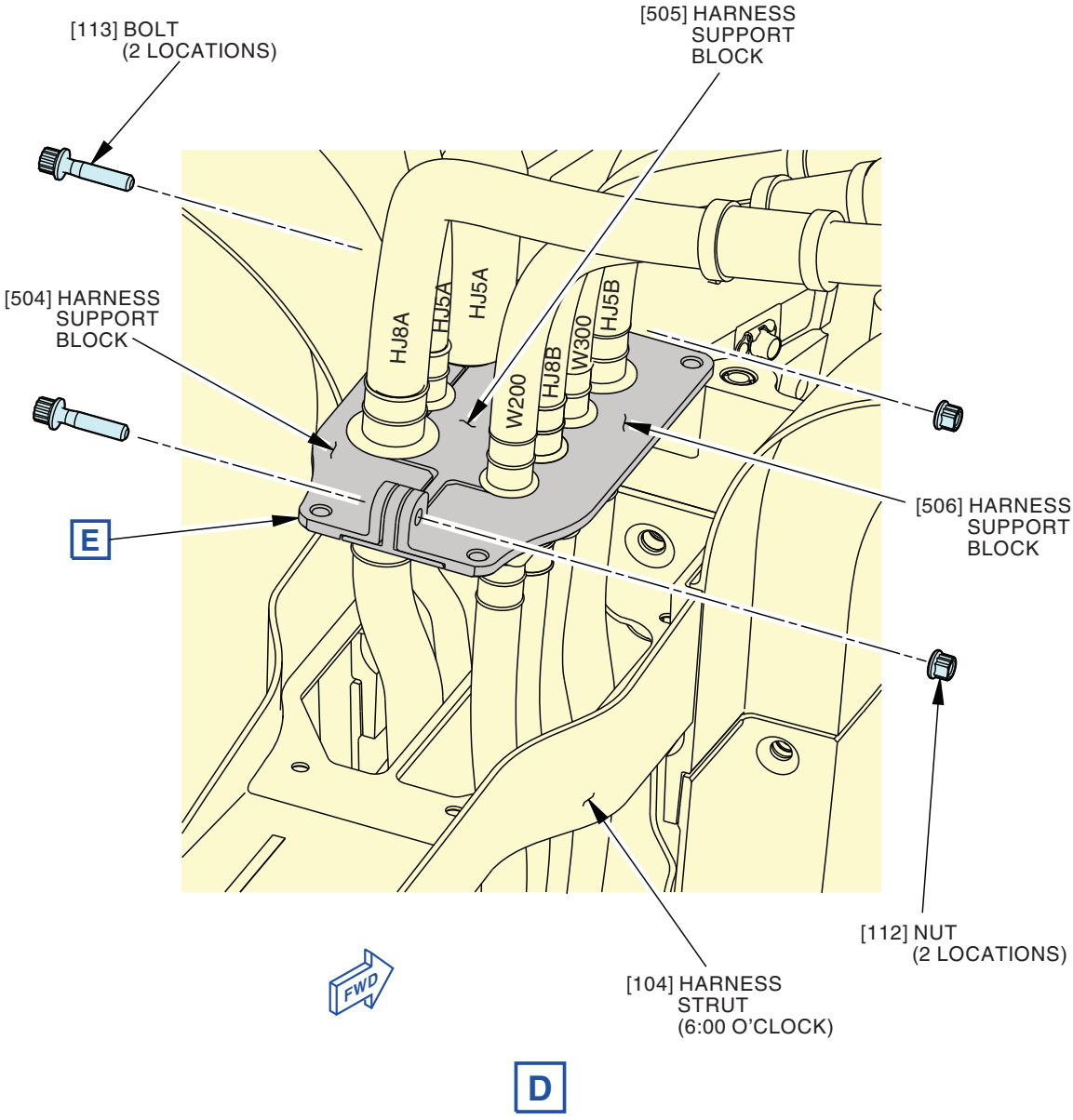


NOTE:
HARNESSES NOT SHOWN

2937803 S0000713392_V2

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 6 of 13)

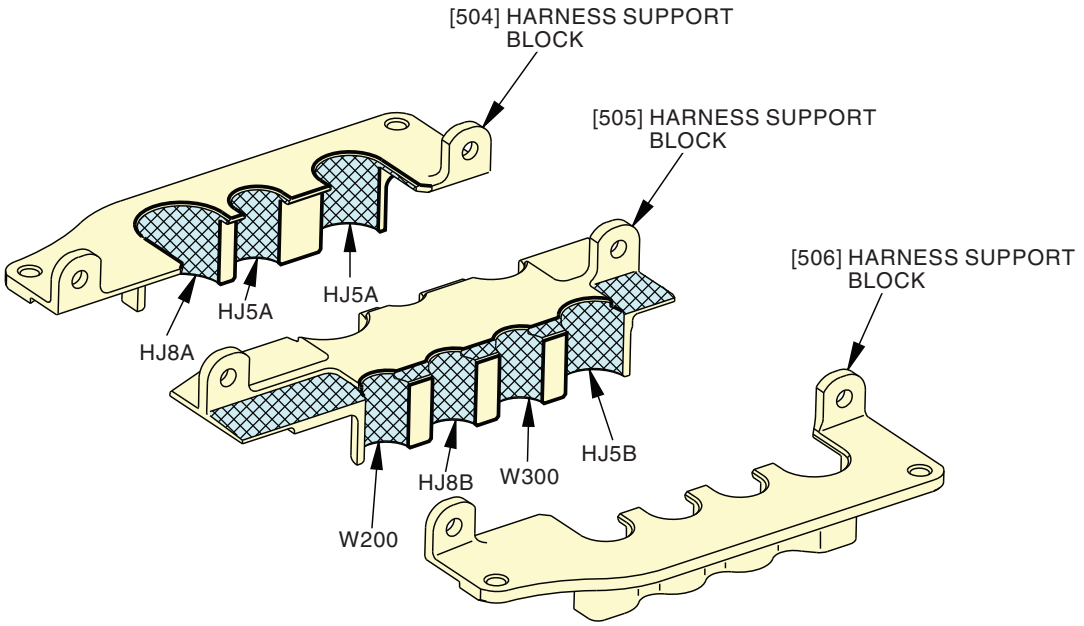
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937804 S0000713393_V2

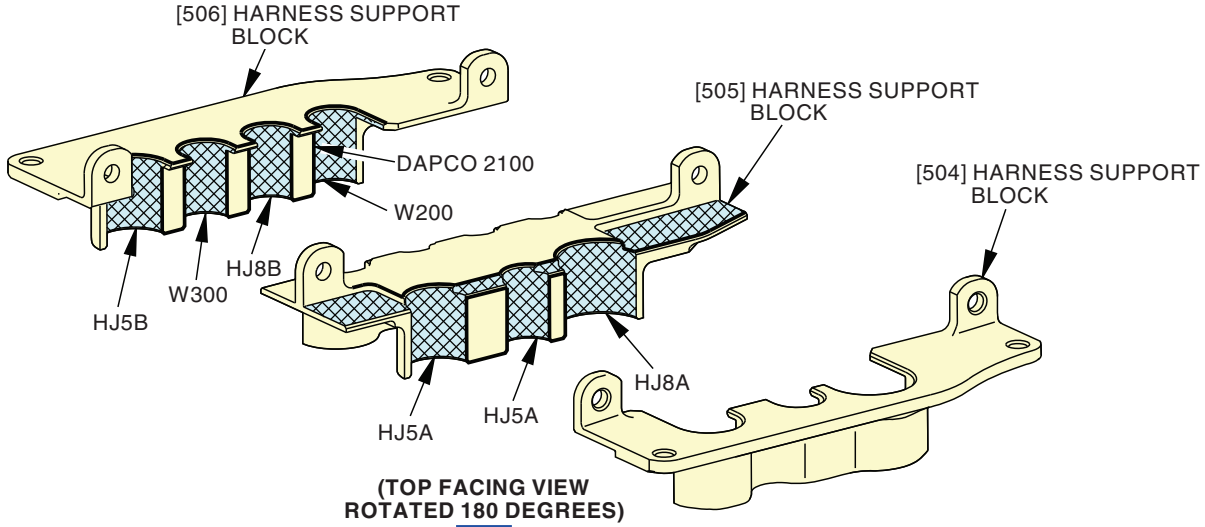
Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 7 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

E



(TOP FACING VIEW ROTATED 180 DEGREES)

E

LEGEND:

DAPCO 2100

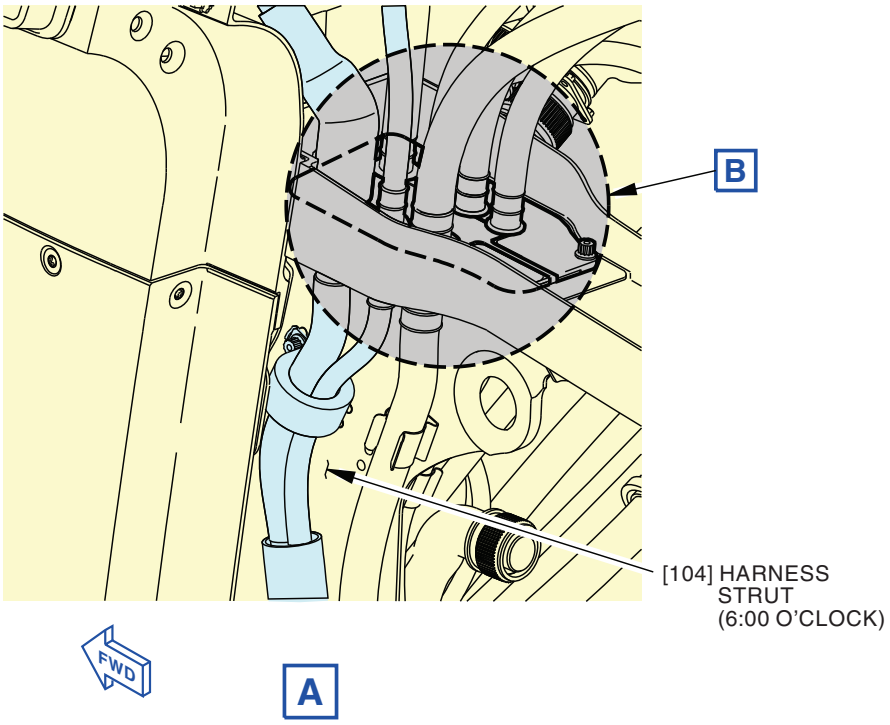
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937805 S0000713394_V4

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 8 of 13)

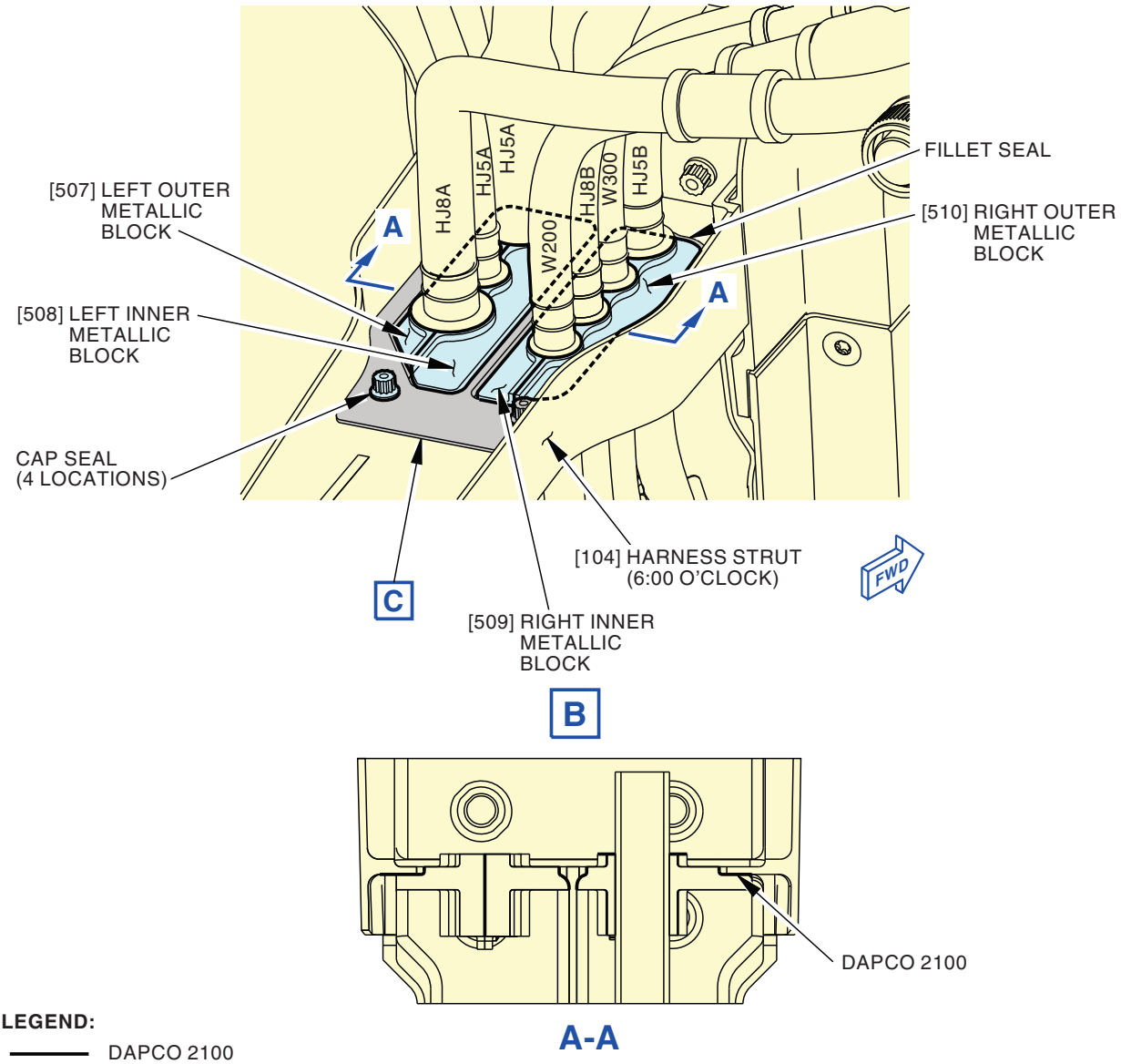
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2962032 S0000739201_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 9 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

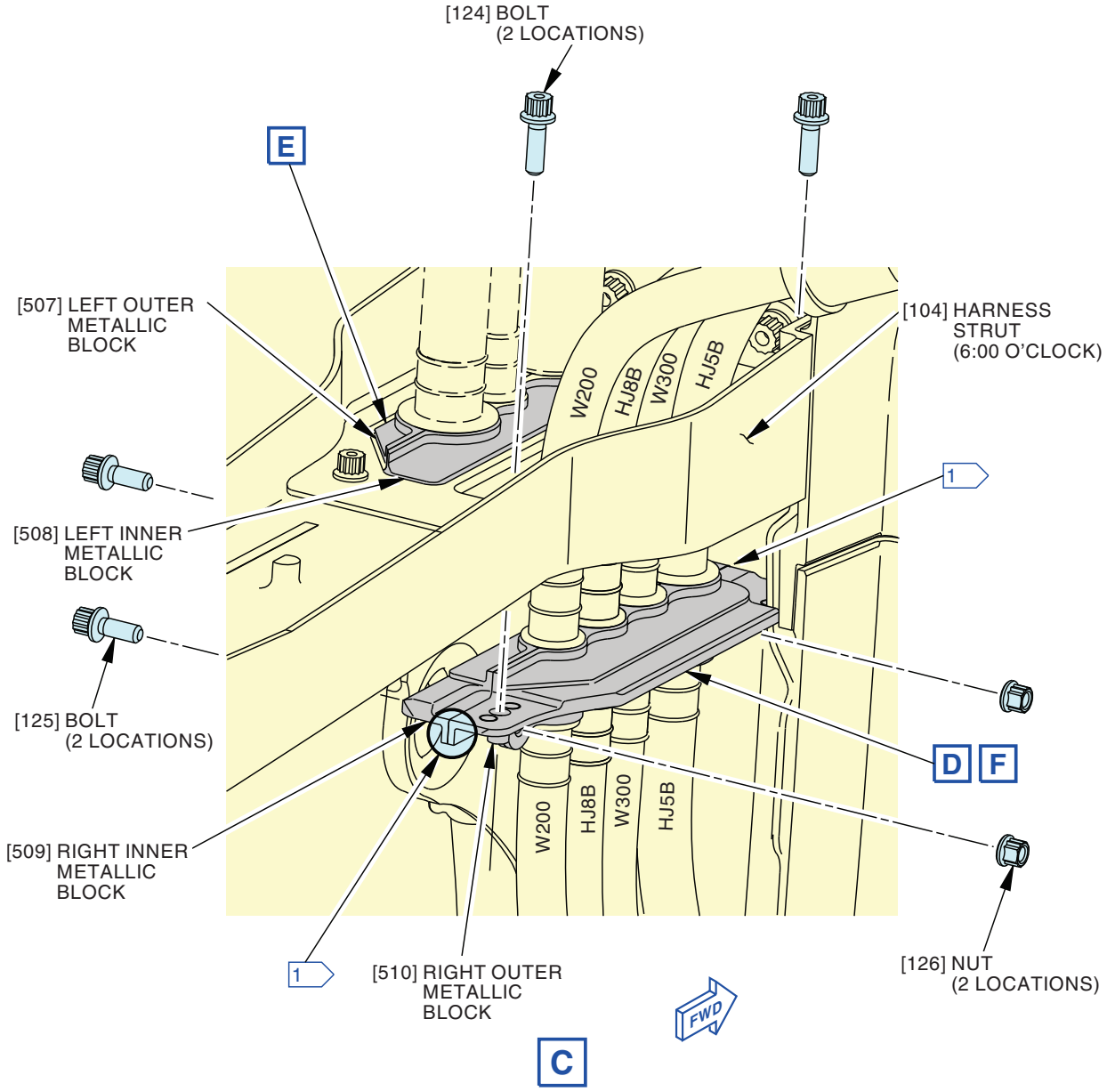
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE METALLIC BLOCKS. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN BOTH RIGHT AND LEFT METALLIC BLOCKS HAVE BEEN REMOVED.

ICNLEAP1B732201D5882800059A00101
2962033 S0000739202_V2

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 10 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

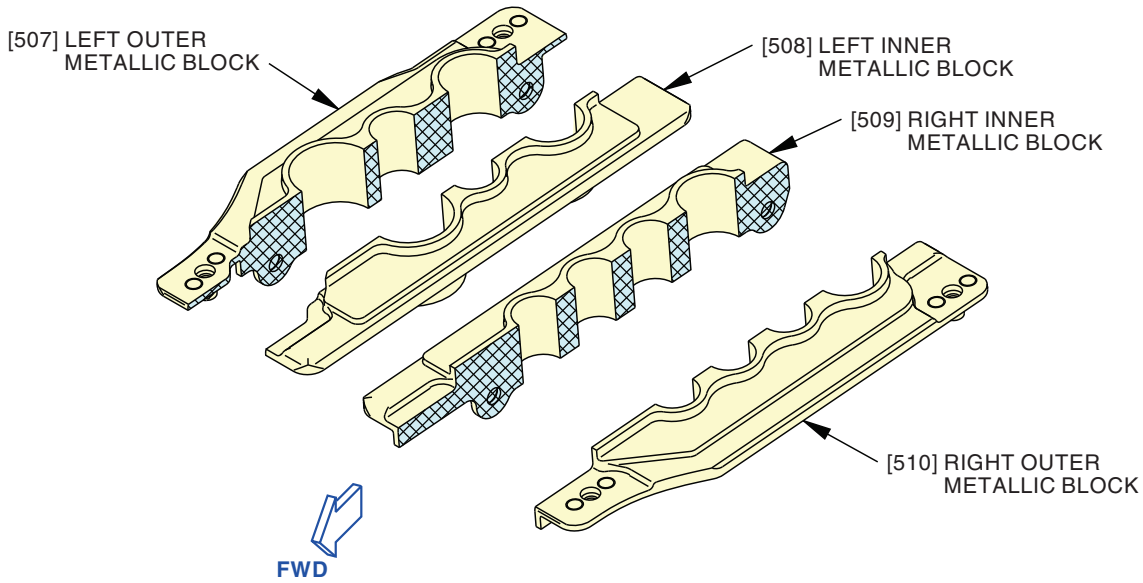
THIS VIEW IS SPECIFIC FOR RIGHT INNER AND OUTER METALLIC BLOCKS. SAME LOGIC IS APPLIED FOR LEFT INNER AND OUTER METALLIC BLOCKS.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

ICNLEAP1B732201D5882800063A00101
2962034 S0000739203_V2

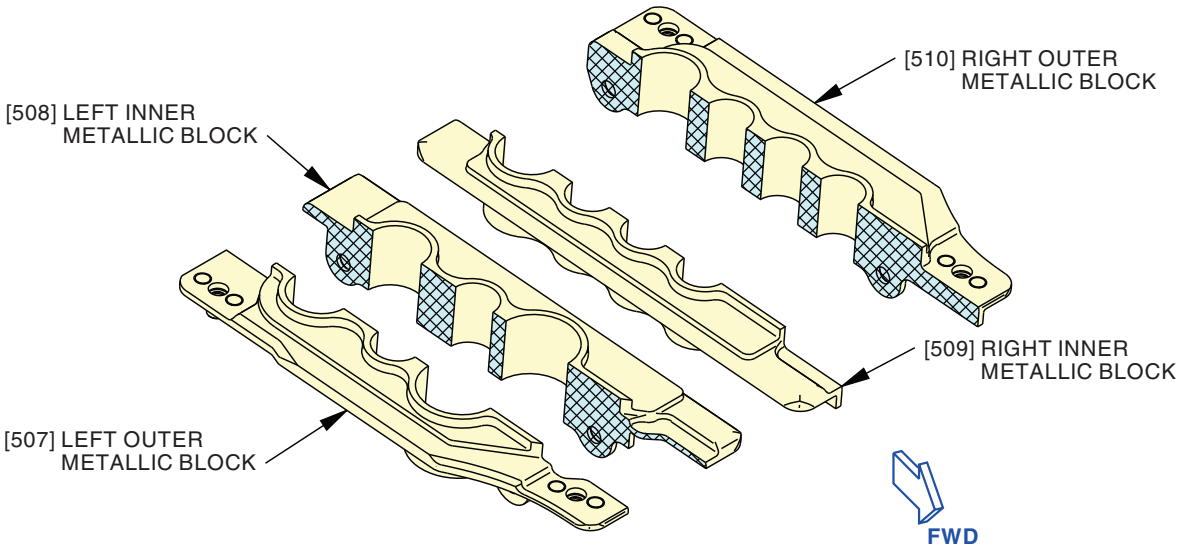
Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 11 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

D



(TOP FACING VIEW ROTATED 90 DEGREES)

D

LEGEND:

DAPCO 2100

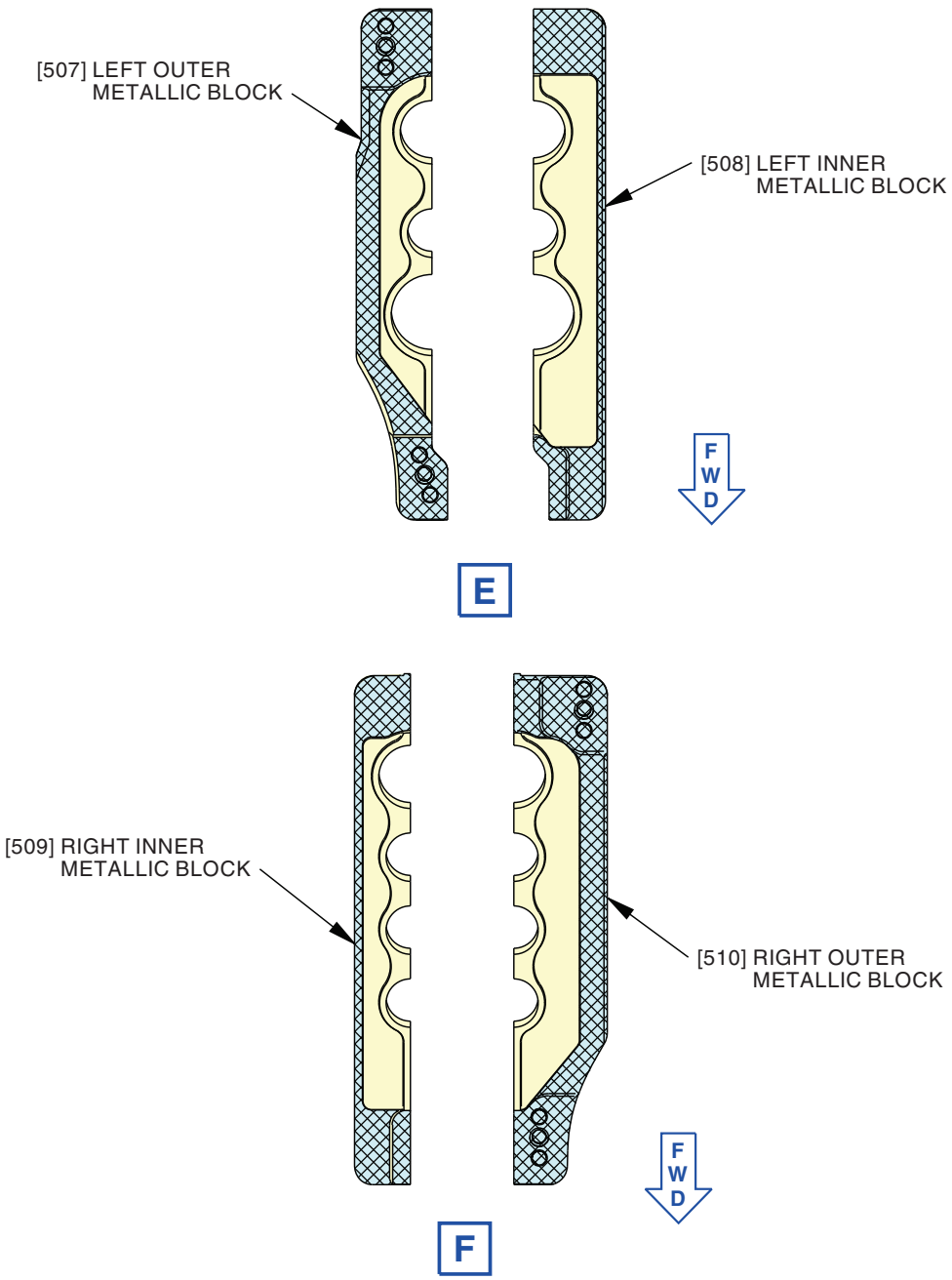
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994643 S0000770273_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 12 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994658 S0000770274_V1

Harness Support Block Installation
Figure 403/26-11-02-990-825 (Sheet 13 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-11-02-400-801

3. Engine Fire Detection Harness Installation

(Figure 401, Figure 402, and Figure 403)

A. General

(1) The installation procedure for harness W200 and W300 is the same.

(2) W200 is a component of loop A of the engine fire detection circuit.

(3) W300 is a component of loop B of the engine fire detection circuit.

B. References

Reference	Title
26-11-00-710-801	Engine Fire Detection - Operational Test (P/B 501)
30-21-11-400-801	Engine Anti-Ice Valve Installation (P/B 401)
30-21-21-400-801	Engine Anti-Ice Pressure Switch Installation (P/B 401)
36-00-00-710-801	Electrical LRU - Replacement Test (P/B 501)
36-11-04-400-801	PRSOV Installation (P/B 401)
36-11-05-400-801	Intermediate Manifold Pressure (PI) Sensor Installation (P/B 401)
36-11-07-400-801	HPSOV Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-23-06-400-801-G00	Fan Frame Strut Panels Installation (P/B 401)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)
77-12-01-400-801-G00	N1 Speed Sensor Installation (P/B 401)
77-12-02-400-801-G00	N2 Speed Sensor Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
79-31-01-400-801-G00	Oil Level Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00418	Oil - Aircraft Turbine Engine Synthetic Base	MIL-PRF-7808
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00637	Oil - Engine - Eastman Turbo Oil 2389	MIL-PRF-7808L grade 3
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008
D50338	Oil - Engine - Hatcol 3211	MIL-PRF-23699 Class STD (Standard)
D50377	Oil - Lubricating, Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699

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E. Engine Fire Detection Harness Installation

SUBTASK 26-11-02-210-002

(1) During the installation, make sure that the harness W200 [1] and harness W300 [2] are not stressed or damaged.

SUBTASK 26-11-02-420-007

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

SUBTASK 26-11-02-420-001

(3) Install the harness W200 [1] and/or harness W300 [2]:

(a) Put the harness W200 [1] and/or harness W300 [2] in position on the engine through the fan and core firewall openings (Figure 401).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(b) Use the teflon-jawed pliers, STD-664, to connect the harness W200 [1] and/or harness W300 [2] to the pylon (TASK 70-00-01-910-803-G00).

(c) Distribute the slack all along the attachment points on the engine.

(d) For the harness W200 [1], do the following steps:

1) Connect the harness connector to the oil level sensor (TASK 79-31-01-400-801-G00).

2) Connect the harness connectors to the N1 speed sensor (TASK 77-12-01-400-801-G00).

3) Connect the harness connector to the Pressure Regulating and Shutoff Valve (PRSOV) (TASK 36-11-04-400-801).

4) Connect the harness connector to the High Pressure Shutoff Valve (HPSOV) (TASK 36-11-07-400-801).

5) Connect the harness connector to the intermediate manifold pressure (PI) sensor (TASK 36-11-05-400-801).

(e) For the harness W300 [2], do the following steps:

1) Connect the harness connector to the N2 speed sensor (TASK 77-12-02-400-801-G00).

PTF ALL; AIRPLANES WITH UNDER PRESSURE SWITCH (PRE SB 737-30-1074)

2) Connect the harness connectors to the Engine Anti-Ice (EAI) over pressure, under pressure, and minimum pressure switches (TASK 30-21-21-400-801).

PTF ALL; AIRPLANES WITHOUT UNDER PRESSURE SWITCH (POST SB 737-30-1074)

3) Connect the harness connectors to the EAI over pressure and minimum pressure switches (TASK 30-21-21-400-801).

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PTF ALL; AIRPLANES WITH TWO OVER PRESSURE SWITCHES

4) Connect the harness connectors to the EAI minimum pressure and the over pressure switch jumper bundle (TASK 30-21-21-400-801).

PTF ALL

5) Connect the harness connector to the EAI valve (TASK 30-21-11-400-801).

6) Connect the harness connector to the Fuel Metering Unit (FMU) (TASK 73-21-04-400-801-G00).

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(f) Attach the terminal lugs to the responder studs on the core fire detectors with the washers [15] and nuts [14] (View Q, Figure 401).

1) Tighten the nuts [14] to 50 in-lb (5.6 N·m) - 54 in-lb (6.1 N·m).

CAUTION

WITH THE SPLIT BOBBIN HARNESS, HOLD THE LUGS IN ITS POSITION WHEN YOU INSTALL OR REMOVE THE NUTS. IF YOU DO NOT OBEY, LUGS CAN TURN AND DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Attach the ground lugs to the ground studs on the core fire detectors with the washers [23] and nuts [22].

1) Tighten the nuts [22] to 25 in-lb (2.82 N·m) - 29 in-lb (3.28 N·m).

CAUTION

DO NOT LET WIRES GET IN THE BOBBIN SPACE OR BETWEEN THE BOBBIN AND THE CLAMP, IF YOU HAVE A SPLIT BOBBIN HARNESS. IF YOU DO NOT OBEY, WIRES CAN RUB AGAINST THE CLAMPS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(h) Attach the terminal and ground lug wires to the core fire detector brackets with the clamps [21], nuts [20], and bolts [19].

1) Lubricate the bolts [19] with Never Seez, D50294.

2) Tighten the nuts [20] to 23 in-lb (2.60 N·m) - 27 in-lb (3.05 N·m).

(i) Install the harness W200 [1] and/or harness W300 [2] to the engine brackets (View P and R, Figure 401):

1) Lubricate the screws [16] with aircraft turbine engine oil, D00418, or Oil - lubricating, D50377.

2) Install the screws [16] that attach the harness W300 [2] to the engine bracket [52] (View P, Figure 401).

3) Install the screws [16] that attach the harness W200 [1] to the engine bracket [51] (View R, Figure 401).

4) Tighten the screws [16] to 53 in-lb (6.0 N·m) - 57 in-lb (6.4 N·m).

(j) Install the harness W200 [1] and/or harness W300 [2] to the core engine brackets (View A, K, L, M, N, and O, Figure 401).

1) Lubricate the screws [17], screws [32], and screws [53] with aircraft turbine engine oil, D00418 (preferred), or Eastman Turbo Oil 2389, D00637 (alternative), or Oil - lubricating, D50377 (preferred), or Hatcol 3211, D50338 (alternative).

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2) Install the screws [17] and clamps [37] that attach the harness W300 [2] to the core engine bracket [47], core engine bracket [48], core engine bracket [49], and core engine bracket [50].

3) Install the screw [17] and clamp [45] that attach the harness W200 [1] to the lower assembly manifold [46].

4) Install the screws [17] and clamps [37] that attach harness W200 [1] to the core engine bracket [43] and core engine bracket [44].

5) Install the screws [17] and clamps [37] that attach the harness W300 [2] to the LPTACC air duct [42] at the location D, location E, and location F (View N, Figure 401).

6) Install the screws [17] and clamps [37] that attach the harness W200 [1] to the LPTACC air duct [42] at the location A, location B, and location C (View N, Figure 401).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

7) Install the screws [53] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [41] and core engine bracket [40].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

8) Install the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [41] and core engine bracket [40].

PTF ALL

9) Install the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [39].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

10) Install the screws [32] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [38].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

11) Install the screws [17] and clamps [37] that attach the harness W200 [1] and/or harness W300 [2] to the core engine bracket [38].

PTF ALL

12) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the core engine bracket [36].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

13) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the core engine bracket [35].

PTF ALL

14) Tighten the screws [17] and screws [32] to 33.5 in-lb (3.8 N·m) - 36.5 in-lb (4.1 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

15) Tighten the screws [53] to 33.5 in-lb (3.8 N·m) - 36.5 in-lb (4.1 N·m).

PTF ALL

(k) Install the harness W200 [1] and/or harness W300 [2] to the engine support brackets (View A, F, G, H, and J, Figure 401).

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1) Lubricate the screw [17] and screw [32] with aircraft turbine engine oil, D00418, (preferred), or Eastman Turbo Oil 2389, D00637 (alternative), or Oil - lubricating, D50377 (preferred), or Hatcol 3211, D50338 (alternative).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

2) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the engine support bracket [33] and engine support bracket [34].

PTF ALL

3) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the engine support bracket [31].
4) Install the screw [32] and clamp [18] that attach the harness W300 [2] to the engine support bracket [30].
5) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the engine support bracket [30].
6) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the engine support bracket [27], engine support bracket [28], and engine support bracket [29].
7) Install the screw [17] and clamp [18] that attach the harness W200 [1] to the engine support bracket [24].
8) Install the harness W200 [1] and/or harness W300 [2] to the clips [25] on the engine support bracket [24] and oil tube [26].
9) Tighten the screw [17] and screw [32] to 33.5 in-lb (3.8 N·m) - 36.5 in-lb (4.1 N·m).

(l) Attach the terminal lugs to the responder studs on the fan fire detectors with the washers [15] and nuts [14] (View E, Figure 401).

1) If the leads are different sizes, install the lead with the small hole under the lead with the large hole.
2) Tighten the nuts [14] to 50 in-lb (5.6 N·m) - 54 in-lb (6.1 N·m).

(m) Install the harness W200 [1] and/or harness W300 [2] on the harness strut (6:00 o'clock) [104] as follows (Figure 402 and Figure 403):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

1) Install the harness support block [504], harness support block [505], and harness support block [506] on the harness strut (6:00 o'clock) [104] as follows:
   a) Lubricate the four bolts [111] with the graphite mineral vaseline, D00601 [CP2101].
   b) Apply Dapco 2100 sealant, A50483 [CP2927], on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses (View E, Figure 403).
   c) Make sure that the harness W200 [1] and/or harness W300 [2] and harnesses are in the correct position on the harness support block [504], harness support block [505], and harness support block [506] (View D, Figure 403).
   d) Install the two bolts [113] and two nuts [112] to assemble the harness support block [504], harness support block [505], and harness support block [506] in the correct position with the harnesses.
   e) Tighten the nuts [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

f) Make sure that there is minimum amount of the squeeze-out on the lower side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

g) Install the harness support block [504], harness support block [505], and harness support block [506] in the correct position on the harness strut (6:00 o'clock) [104].

h) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104] (View B, Figure 403).

i) Install the four bolts [111] and four nuts [110] that attach the assembly of the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

j) Tighten the nuts [110] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

k) Make sure that there is minimum amount of the squeeze-out on the lower side.

l) Apply more sealant to form a correct fillet seal around the contour of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

m) Clean the four nuts [110] and four bolts [111] that will have sealant applied to them.

n) Apply Dapco 2100 sealant, A50483 [CP2927], to the heads of the four bolts [111].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

2) Remove any remaining of old sealant residue left on the harnesses or harness strut (6:00 o'clock) [104].

3) Install the left outer metallic block [507] and left inner metallic block [508] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the two bolts [124] and two bolts [125] with graphite mineral vaseline, D00601 [CP2101].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507] and left inner metallic block [508].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507], left inner metallic block [508], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the left outer metallic block [507] and left inner metallic block [508] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

f) Install the left outer metallic block [507] and left inner metallic block [508] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the left outer metallic block [507] and left inner metallic block [508] to the harness strut (6:00 o'clock) [104].

h) Tighten the bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of left outer metallic block [507] and left inner metallic block [508].

4) Install the new right inner metallic block [509] and right outer metallic block [510] on the bracket assembly as follows:

a) Lubricate the bolts [124] and bolts [125] with graphite mineral vaseline, D00601 [CP2101].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509] and right outer metallic block [510].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the right inner metallic block [509] and right outer metallic block [510] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

f) Install the right inner metallic block [509] and right outer metallic block [510] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the right inner metallic block [509] and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

h) Tighten the bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of right inner metallic block [509] and right outer metallic block [510].

PTF ALL

5) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109] as follows:

a) Lubricate the five bolts [106] and two nuts [107] with the graphite mineral vaseline, D00601 [CP2101].

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b) Apply Dapco 2100 sealant, A50483 [CP2927], to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

c) Make sure that the harness W200 [1] and/or harness W300 [2] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] (View E, Figure 402).

d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Make sure that the fillet seal of the squeeze-out is on the forward facing side.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

g) Make the sealant squeeze-out flush with the surface on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 402).

h) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

i) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and strut 5 outer fitting [109] (View C, Figure 402).

j) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

k) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 402).

m) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

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CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

n) Clean the nut plate side of the bolts [106] and nut plates.

o) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

(n) Attach the ground lugs to the ground studs on the fan fire detectors with washers [23] and nuts [22] (View E, Figure 401).

1) Tighten the nuts [22] to 25 in-lb (2.82 N·m) - 29 in-lb (3.28 N·m).

(o) Attach the harness W200 [1] and/or harness W300 [2] to the fan engine brackets with the hinged clamps [13] and screws [12] (View D, Figure 401).

1) Tighten the screws [12] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

(p) Engage the harness W200 [1] and/or harness W300 [2] in the engine clips [11] (View C, Figure 401).

(q) Install the Fan Frame Strut 5 Panel (left) (TASK 72-23-06-400-801-G00).

SUBTASK 26-11-02-400-001

(4) Attach the blanket [101] to the strut 5 outer fitting [109].

(a) Apply Dapco 2100 sealant, A50483 [CP2927], between blanket [101] and strut 5 outer fitting [109] (View B, Figure 402).

NOTE: Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

SUBTASK 26-11-02-410-001

(5) Close the access to the fan and core firewalls.

(a) Make sure that the harnesses are in the correct position in the fire wall passage holes.

F. Fan Fire Detection Harness Installation Test

SUBTASK 26-11-02-440-001

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	6	C00148	ANTI-ICE & RAIN ENG 1 & WING CONT
A	7	C01001	ANTI-ICE & RAIN ENG 1 COWL AI VALVE
B	7	C01002	ANTI-ICE & RAIN ENG 2 COWL AI VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	11	C04082	ANTI-ICE & RAIN ENGINE 2 CONTROL
F	6	C04065	IASC EAI INHIBITS OUT L
F	7	C04066	IASC EAI INHIBITS OUT R

SUBTASK 26-11-02-710-001

(2) Do this task: Engine Fire Detection - Operational Test, TASK 26-11-00-710-801.

SUBTASK 26-11-02-710-006

(3) Do the following steps for the Line Replaceable Unit (LRU)s that connectors were reconnected:

(a) For the harness W200 [1], do the following steps:

1) For the oil level sensor or N1 speed sensor, do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

2) For the intermediate manifold pressure (PI) sensor, PRSOV, or HPSOV, do this task: Electrical LRU - Replacement Test, TASK 36-00-00-710-801.

(b) For the harness W300 [2], do the following steps:

1) For the N2 speed sensor, do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

2) For the EAI over pressure, under pressure, or minimum pressure switches, do the installation test (TASK 30-21-21-400-801).

3) For the EAI valve, do the installation test (TASK 30-21-11-400-801).

4) For the FMU, do the installation test (TASK 73-21-04-400-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 26-11-02-010-003

WARNING OBEY THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE PROCEDURE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY SMOKE DETECTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Lavatory Smoke Detection - Self Test

(2) Lavatory Smoke Detection - Smoke Test.

B. This procedure does a test of the lavatory smoke detectors.

TASK 26-14-00-710-801

2. Lavatory Smoke Detection - Self Test

(Figure 501)

A. Lavatory Smoke Detection Self Test

SUBTASK 26-14-00-710-001

CAUTION DO NOT USE TOO MUCH FORCE TO PUSH THE SWITCH. TOO MUCH FORCE CAN DAMAGE THE SWITCH.

(1) Push and hold the SELF TEST switch on the smoke detector with a small screwdriver, or other appropriate tool.

SUBTASK 26-14-00-710-002

(2) Make sure that these alarm indications occur:

(a) Lavatories:

1) The LED status indicator turns to a solid red light on the smoke detector.

2) You can hear the smoke detector alarm horn in the lavatory.

SUBTASK 26-14-00-710-003

(3) Release the SELF TEST switch.

(a) Make sure that all external alarm indications stop.

——— END OF TASK ———

TASK 26-14-00-730-801

3. Lavatory Smoke Detection - Smoke Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11202	Smoke Generator - Lavatory/Crew Rest Smoke Detector (or 737 Cargo Area Type Smoke Detectors) Part #: HO-25S Supplier: 61908 Part #: SOLOA10 Supplier: 1WU32 Opt Part #: SOLOA3 Supplier: 1WU32 Opt Part #: SOLOA4 Supplier: 1WU32

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Smoke Test

SUBTASK 26-14-00-860-005

(1) Make sure that the Light Emitting Diode (LED) status indicator on the lavatory smoke detector is a solid green light.

SUBTASK 26-14-00-730-001

(2) Use a smoke generator, COM-11202, to make smoke adjacent to the lavatory smoke detector.

SUBTASK 26-14-00-750-001

(3) Make sure that these alarm indications occur:

(a) Lavatories:

1) The LED status indicator on the smoke detector is a solid red light.

2) You can hear the alarm horn from the smoke detector.

SUBTASK 26-14-00-730-002

(4) Push the horn cancel switch on the smoke detector with an applicable tool.

(a) Make sure that all of the alarm indications stop.

NOTE: The LED status indicator will stay red if smoke is still sensed. The LED status indicator will come on green when the smoke sensor no longer detects smoke.

SUBTASK 26-14-00-860-006

(5) Clear the smoke from the lavatory.

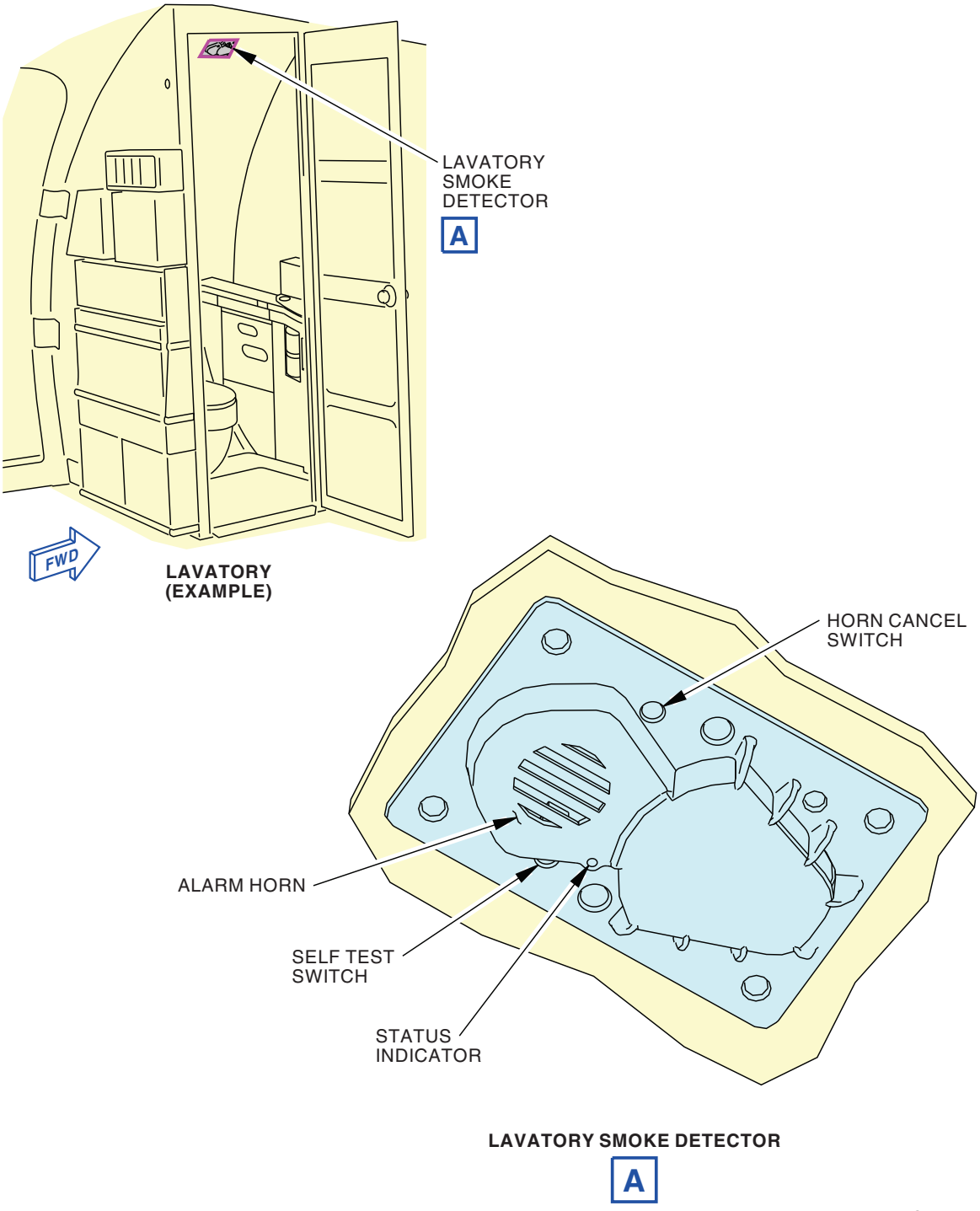
NOTE: If the smoke is not cleared from the lavatory within 30 seconds, the smoke indications will occur again.

SUBTASK 26-14-00-730-003

(6) Do the smoke test again for each of the other lavatory smoke detectors.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Smoke Detection Adjustment
Figure 501/26-14-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY SMOKE DETECTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the lavatory smoke detector.

(2) An installation of the lavatory smoke detector.

TASK 26-14-01-000-801

2. Lavatory Smoke Detector Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Lavatory Smoke Detector.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Removal

SUBTASK 26-14-01-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	12	C01125	LAVATORY SMOKE

D. Lavatory Smoke Detector Removal

SUBTASK 26-14-01-020-002

(1) Remove the screws [14] and spacers [15] that attach the smoke detector [11] to the mounting bracket [13].

SUBTASK 26-14-01-020-001

(2) Disconnect the electrical connector [12].

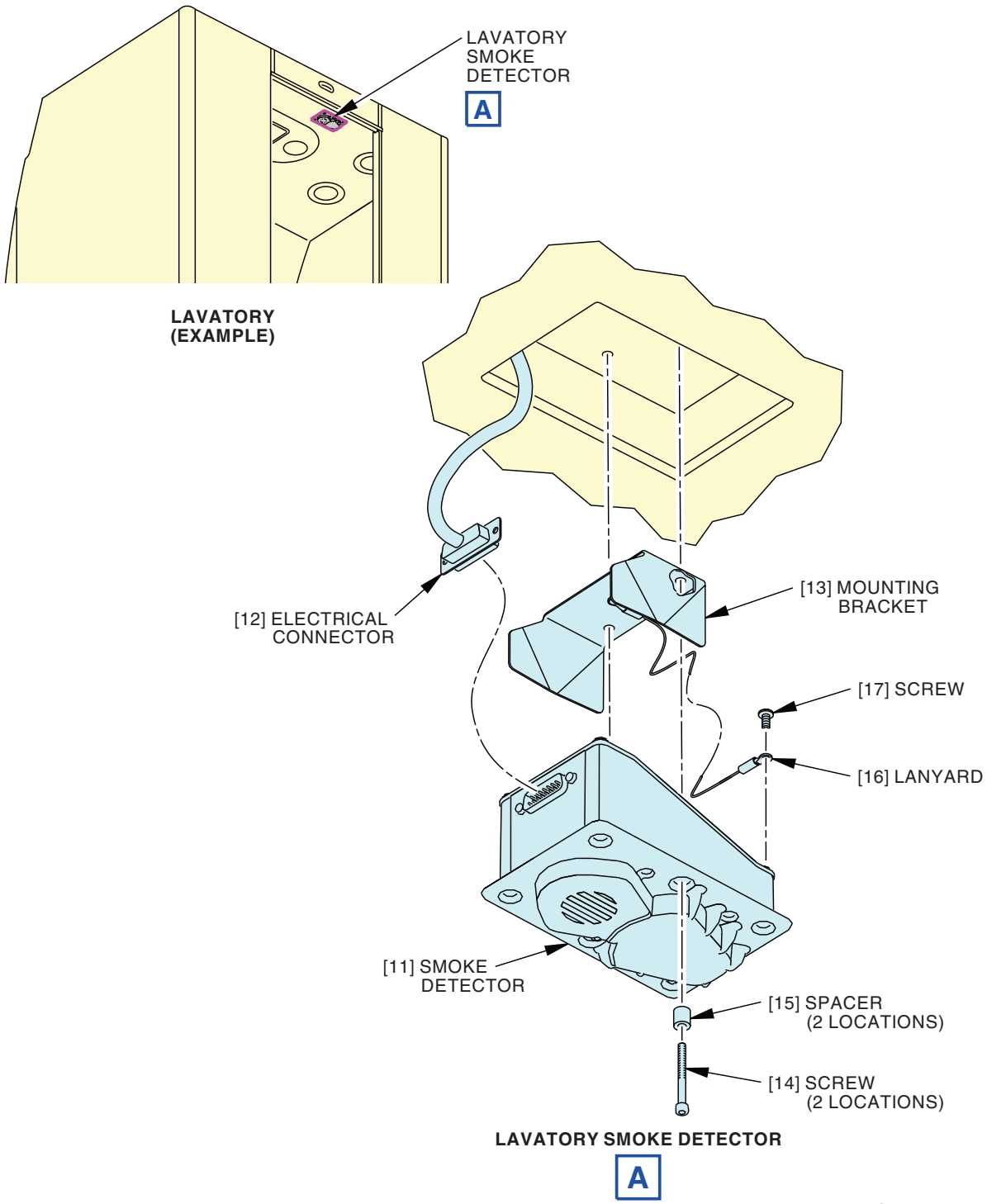
SUBTASK 26-14-01-020-003

(3) Remove the smoke detector [11] and mounting bracket [13] from the ceiling.

NOTE: You do not have to remove the mounting bracket from the smoke detector. The new smoke detector will usually have a mounting bracket attached to it.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Smoke Detector Installation
Figure 401/26-14-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-14-01-400-801

3. Lavatory Smoke Detector Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Lavatory Smoke Detector.

B. References

Reference	Title
26-14-00-710-801	Lavatory Smoke Detection - Self Test (P/B 501)
26-14-00-730-801	Lavatory Smoke Detection - Smoke Test (P/B 501)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Installation

SUBTASK 26-14-01-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	12	C01125	LAVATORY SMOKE

E. Lavatory Smoke Detector Installation

SUBTASK 26-14-01-420-005

(1) Do these steps before you put the smoke detector [11] in its position in the ceiling:

(a) If the lanyard [16] is not connected to the smoke detector [11], attach the lanyard [16] from the mounting bracket [13] to the smoke detector [11] with the screw [17].

1) Tighten the screw [17] to 5 in-lb (0.56 N·m) - 6 in-lb (0.68 N·m).

(b) If the mounting bracket [13] is not connected to the smoke detector [11], attach the mounting bracket [13] to the smoke detector [11] with the spacers [15] and screws [14].

1) Do not tighten the screws [14] at this time.

NOTE: The mounting bracket must be attached loosely to the smoke detector.

(c) Connect the electrical connector [12] to the smoke detector [11].

SUBTASK 26-14-01-420-001

(2) Put and hold the smoke detector [11] and mounting bracket [13] in the correct position in the ceiling.

SUBTASK 26-14-01-420-004

(3) Install the screws [14] to connect the smoke detector [11] to the mounting bracket [13].

SUBTASK 26-14-01-420-002

(4) Tighten the two screws [14] to 9 in-lb (1.02 N·m) - 10 in-lb (1.13 N·m).

SUBTASK 26-14-01-860-003

(5) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	12	C01125	LAVATORY SMOKE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Lavatory Smoke Detector Installation Test

SUBTASK 26-14-01-710-001

(1) Do this task: Lavatory Smoke Detection - Self Test, TASK 26-14-00-710-801.

SUBTASK 26-14-01-730-001

(2) Do this task: Lavatory Smoke Detection - Smoke Test, TASK 26-14-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY SMOKE DETECTOR - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) A cleaning of the lavatory smoke detector.

TASK 26-14-01-100-801

2. Lavatory Smoke Detector Cleaning

(Figure 701)

A. General

(1) This task gives the instructions to clean the lavatory smoke detector.

(2) Each lavatory has a smoke detector installed in the ceiling.

B. References

Reference	Title
26-14-01-000-801	Lavatory Smoke Detector Removal (P/B 401)
26-14-01-400-801	Lavatory Smoke Detector Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle
STD-4107	Source - Air, Clean, Dry, 0 to 60 psi
STD-6402	Swab - Cotton

D. Consumable Materials

Reference	Description	Specification
B50026	Soap - Clear Liquid Dishwashing	
B50073	Alcohol - Isopropyl	ASTM D 770
G01043	Cloth - Lint-free	
G50527	Water	

E. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Prepare for the Cleaning

SUBTASK 26-14-01-000-002

(1) Do this task: Lavatory Smoke Detector Removal, TASK 26-14-01-000-801.

SUBTASK 26-14-01-000-003

CAUTION

BE CAREFUL WHEN YOU REMOVE OR INSTALL THE SCREEN OF THE COVER ASSEMBLY AND LABYRINTH. THIS WILL HELP PREVENT SCRATCHES ON THE LABYRINTH. SCRATCHES ON THE LABYRINTH CAN CAUSE STRAY LIGHT REFLECTIONS AND INCORRECT CALIBRATIONS TO OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

ESDS COMPONENTS AND UNITS MUST BE PROTECTED FROM STATIC DISCHARGE DAMAGE AT ALL TIMES.

(2) Disassemble the lavatory smoke detector as follows:

(a) Remove the screws [3] and washers [4] that attach the cover/Electro-Magnetic Interference (EMI) gasket assembly to the housing/screen assembly [1].

(b) Remove the slide-lock-post [6] and washer [7] that attach the bonding jumper [9] to the connector [8].

NOTE: The slide-lock-post can include one washer, one hex nut, and one slide-lock-post.

(c) Remove the remaining slide-lock-post [6] and washer [7] that attach the connector [8] to the housing/screen assembly [1].

NOTE: The slide-lock-post can include one hex nut, one lock washer, one flat washer, and one slide-lock-post.

(d) Pull the connector [8] back inside the housing/screen assembly [1].

(e) Remove the gasket [10].

(f) Remove the EMI gasket [18] from the cover [5].

1) If it is necessary, remove the nut [21], lock washer [20], and flat washer [19] to disconnect the bonding jumper [22] from the cover/EMI gasket assembly.

(g) Remove the screws [11] and washers [12] that connect the CAA and labyrinth assembly [2] to the housing/screen assembly [1].

(h) Remove the CAA and labyrinth assembly [2].

(i) Disconnect the connector [17] from the CAA and labyrinth assembly [2].

(j) Remove the fasteners [13], washers [14], and nuts [15] that attach the labyrinth top [16] to the CAA and labyrinth assembly [2].

G. Lavatory Smoke Detector Cleaning

SUBTASK 26-14-01-100-005

CAUTION

ESDS COMPONENTS AND UNITS MUST BE PROTECTED FROM STATIC DISCHARGE DAMAGE AT ALL TIMES.

(1) Clean the labyrinth assembly.

(a) Make a solution of soap and water with one part soap, B50026, and twenty parts water, G50527.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION DO NOT PUT THE LABYRINTH FULLY IN THE SOAP AND WATER SOLUTION WHEN YOU CLEAN THE LABYRINTH. LARGE QUANTITIES OF THE SOAP AND WATER SOLUTION CAN CAUSE DAMAGE TO THE LABYRINTH.

(b) Clean the unwanted material from the inner area of the labyrinth with the soft bristle brush, STD-123, and dry, clean air source 0 to 60 psi, STD-4107.

NOTE: You can also use the soap and water solution. Remove the soap and water solution with clean water.

(c) Clean the inner area of the labyrinth with a clean cotton swab, STD-6402, moist with the soap and water solution.

(d) Dry the lenses with a clean and dry cotton swab, STD-6402.

(e) Remove the remaining soap or contamination from the lenses.

(f) Make sure that the labyrinth assembly is fully dry before you assemble the smoke detector.

SUBTASK 26-14-01-100-006

(2) If it is necessary, clean the screen of the housing/screen assembly [1].

WARNING WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

(a) Use a soft bristle brush, STD-123, or dry, clean air source 0 to 60 psi, STD-4107, at maximum pressure of 30 psig (207 kPa) to remove unwanted materials from the screen.

SUBTASK 26-14-01-100-007

(3) Do these steps to clean all non-electrical parts:

WARNING ISOPROPYL ALCOHOL IS FLAMMABLE. AVOID PROLONGED BREATHING OF VAPOR AND PROLONGED OR REPEATED CONTACT WITH SKIN. AVOID CONTACT WITH EYES. CLEAN PARTS IN WELL-VENTILATED AREA AND USE APPROVED SAFETY EQUIPMENT.

(a) Clean the external surfaces with lint-free cloth, G01043, moist with alcohol, B50073.

(b) Dry the surfaces with lint-free cloth, G01043.

SUBTASK 26-14-01-100-008

CAUTION ESDS COMPONENTS AND UNITS MUST BE PROTECTED FROM STATIC DISCHARGE DAMAGE AT ALL TIMES.

(4) Clean the connectors and other electrical components as follows:

WARNING ISOPROPYL ALCOHOL IS FLAMMABLE. AVOID PROLONGED BREATHING OF VAPOR AND PROLONGED OR REPEATED CONTACT WITH SKIN. AVOID CONTACT WITH EYES. CLEAN PARTS IN WELL-VENTILATED AREA AND USE APPROVED SAFETY EQUIPMENT.

(a) Clean the circuit card and connectors with the soft bristle brush, STD-123, and alcohol, B50073.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

(b) Dry the parts with the dry, clean air source 0 to 60 psi, STD-4107, at a maximum pressure of 30 psig (207 kPa).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-14-01-420-006

CAUTION BE CAREFUL WHEN YOU REMOVE OR INSTALL THE SCREEN OF THE COVER ASSEMBLY AND LABYRINTH. THIS WILL HELP PREVENT SCRATCHES ON THE LABYRINTH. SCRATCHES ON THE LABYRINTH CAN CAUSE STRAY LIGHT REFLECTIONS AND INCORRECT CALIBRATIONS TO OCCUR.

CAUTION ESDS COMPONENTS AND UNITS MUST BE PROTECTED FROM STATIC DISCHARGE DAMAGE AT ALL TIMES.

(1) Assemble the lavatory smoke detector as follows:

(a) Install the fasteners [13], washers [14], and nuts [15] to attach the labyrinth top [16] to the CAA and labyrinth assembly [2].

1) Tighten the nuts [15] to 4 in-lb (0.45 N·m) - 5 in-lb (0.56 N·m).

(b) Connect the connector [17] to the CAA and labyrinth assembly [2].

(c) Install the screws [11] and washers [12] to attach the CAA and labyrinth assembly [2] to the housing/screen assembly [1].

1) Tighten the screws [11] to 8 in-lb (0.90 N·m) - 9 in-lb (1.02 N·m).

(d) Put together the cover [5] and EMI gasket [18].

1) If disconnected, connect the bonding jumper [22] to the cover/EMI gasket assembly with the flat washer [19], lock washer [20], and nut [21].

a) Tighten the nut [21] to 8 in-lb (0.90 N·m) - 9 in-lb (1.02 N·m).

(e) Install the gasket [10] on the connector [8].

(f) Put the connector [8] into the opening in the housing/screen assembly [1].

(g) Install the slide-lock-post [6] and washer [7] to attach the connector [8] to the housing/screen assembly [1].

NOTE: The slide-lock-post can include one hex nut, one lock washer, one flat washer, and one slide-lock-post.

1) Make sure that the gasket [10] is between the connector [8] and inner side of the housing/screen assembly [1].

(h) Install the slide-lock-post [6], washer [7], and bonding jumper [9] to attach the connector [8] to the housing/screen assembly [1].

NOTE: The slide-lock-post can include one washer, one hex nut, and one slide-lock-post.

(i) Tighten the slide-lock-posts [6] to 5 in-lb (0.56 N·m) - 6 in-lb (0.68 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Install the screws [3] and washers [4] to attach the cover/EMI gasket assembly to the housing/screen assembly [1].

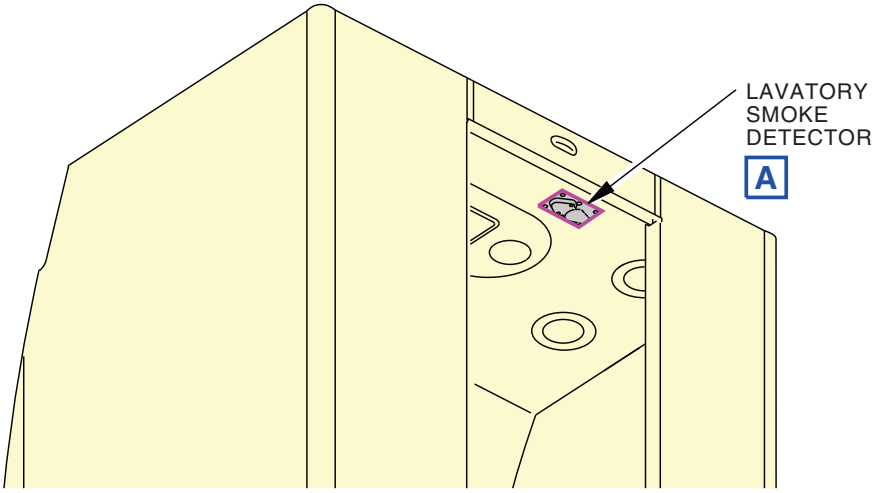
1) Tighten the screws [3] to the 5 in-lb (0.56 N·m) - 6 in-lb (0.68 N·m).

SUBTASK 26-14-01-400-001

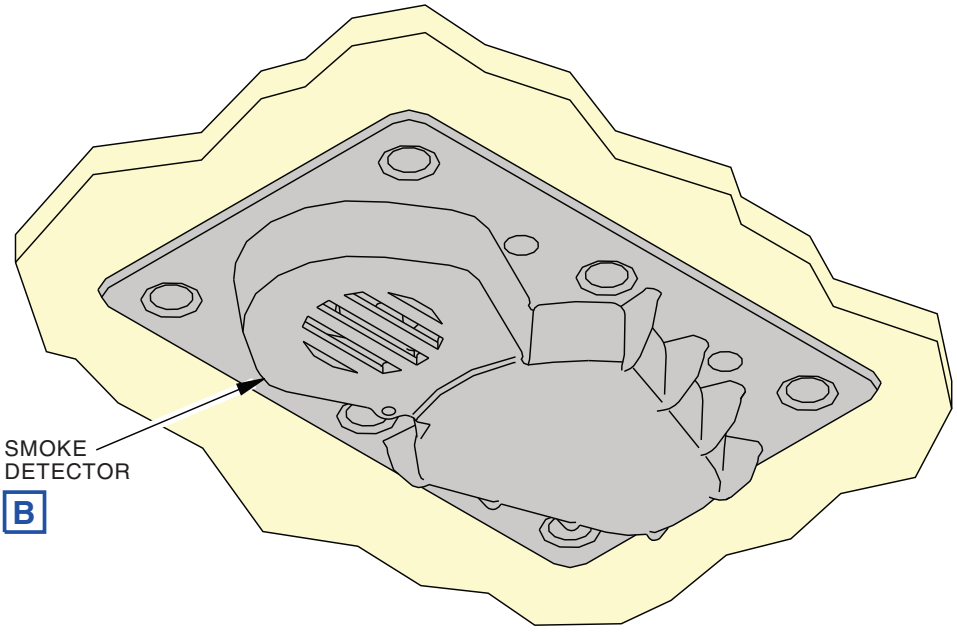
(2) Do this task: Lavatory Smoke Detector Installation, TASK 26-14-01-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**LAVATORY
(EXAMPLE)**



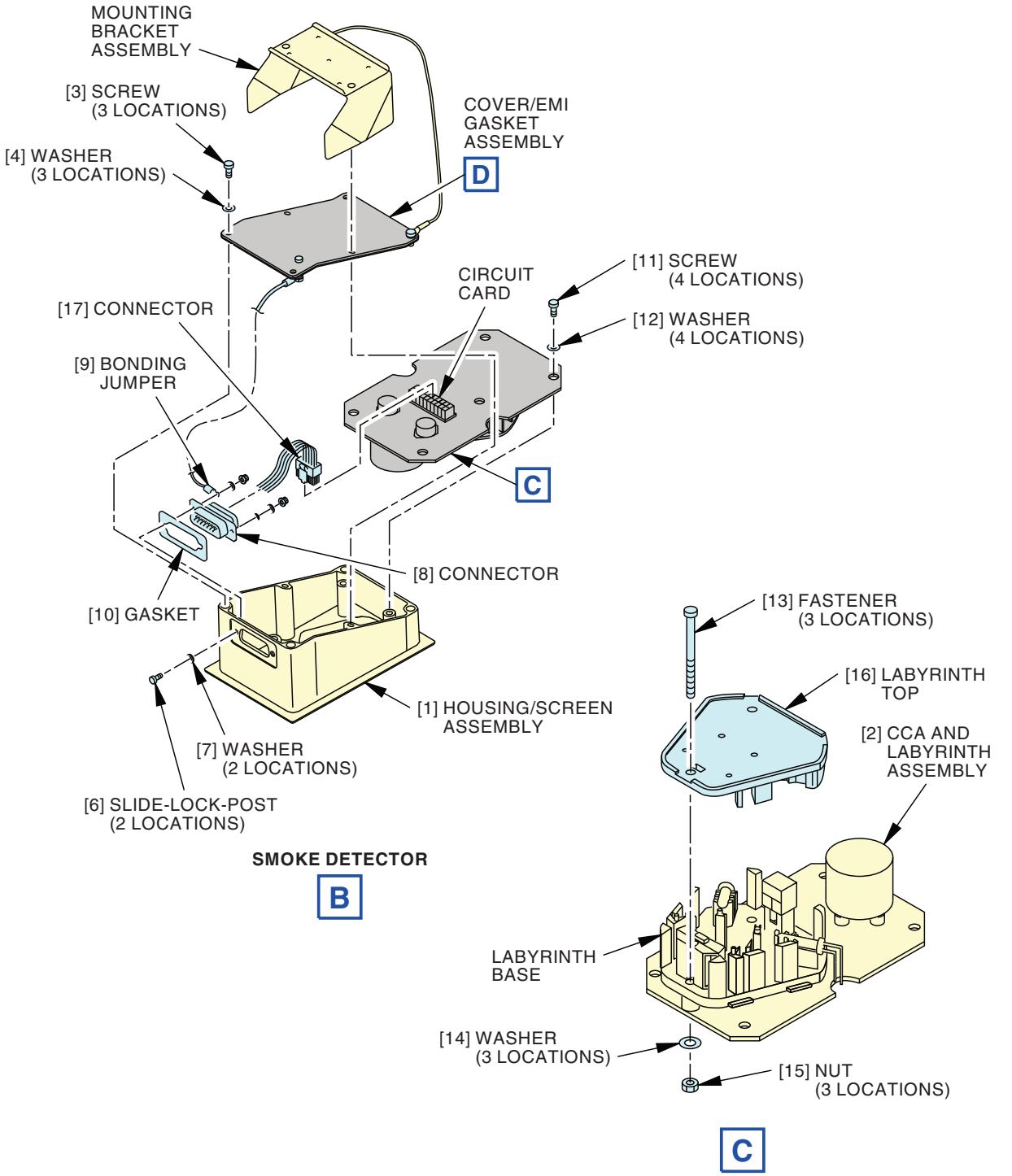
**LAVATORY SMOKE DETECTOR
(EXAMPLE)**

A

2412117 S00061529942_V3

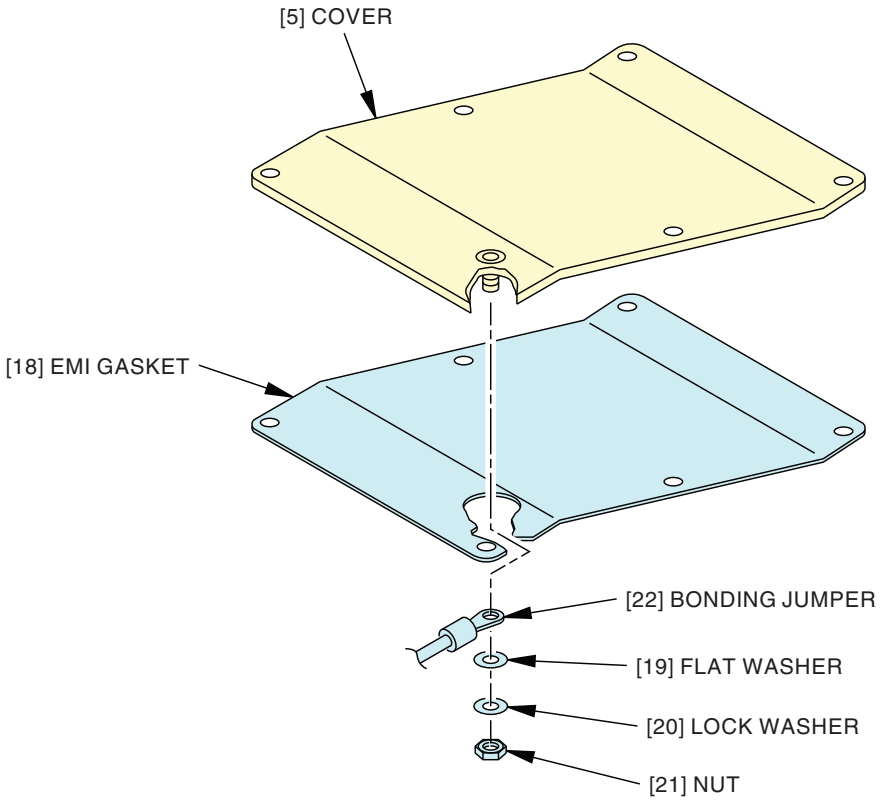
**Lavatory Smoke Detector Cleaning
Figure 701/26-14-01-990-802 (Sheet 1 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Smoke Detector Cleaning
Figure 701/26-14-01-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



COVER/EMI GASKET ASSEMBLY

D

3052307 S0000817443_V1

Lavatory Smoke Detector Cleaning
Figure 701/26-14-01-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE DETECTION - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) APU fire detection - deactivation.

(2) APU fire detection - activation.

TASK 26-15-00-040-801

2. APU Fire Detection - Deactivation

A. General

(1) This task gives instructions to deactivate the APU fire detection system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-15-00-020-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

D. APU Fire Detection - Tryout

NOTE: This tryout is to make sure that the APU fire detection system is in a zero energy state.

SUBTASK 26-15-00-210-001

(1) Move and hold the TEST switch on the Fire Control Panel, P8-1, to the OVHT/FIRE position.

(a) On the Fire Control Panel, make sure that none of these lights come on:

1) APU Fire Switch light.

(b) On the Pilot's Glareshield, P7, make sure that none of these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights.

2) The Captain's and First Officer's FIRE WARN lights.

3) The OVHT/DET light.

(c) Make sure the flight compartment fire bell in the Aural Warning Module, M315, does not come on.

(d) In the right main wheel well, make sure these conditions occur:

1) The APU remote fire light, on the P28 panel, does not come on.

2) The APU fire warning horn does not come on.

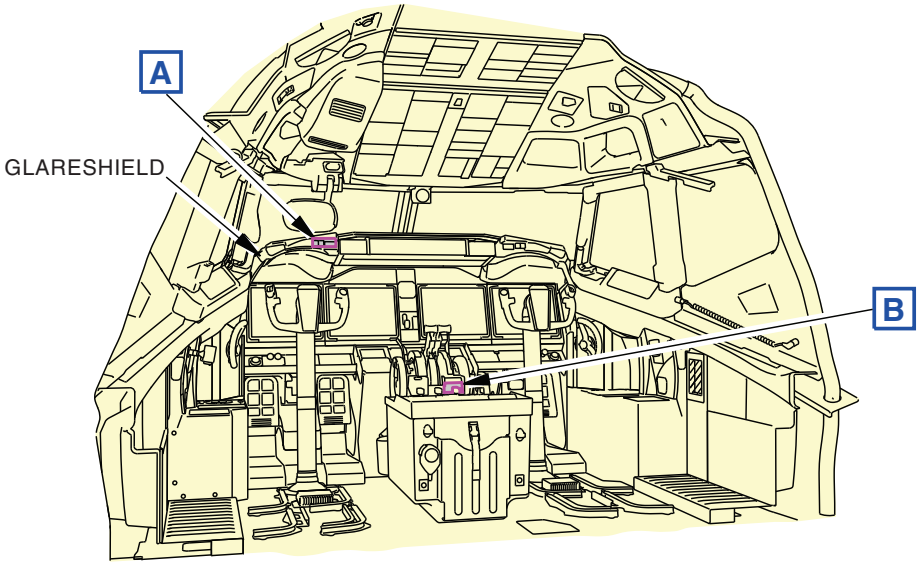
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-00-210-002

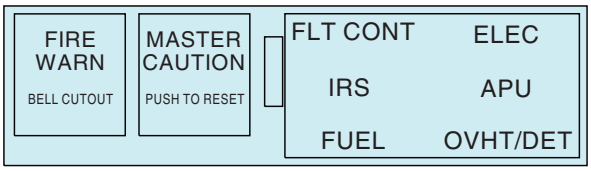
(2) Release the TEST switch on the Engine and APU Fire Control Panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

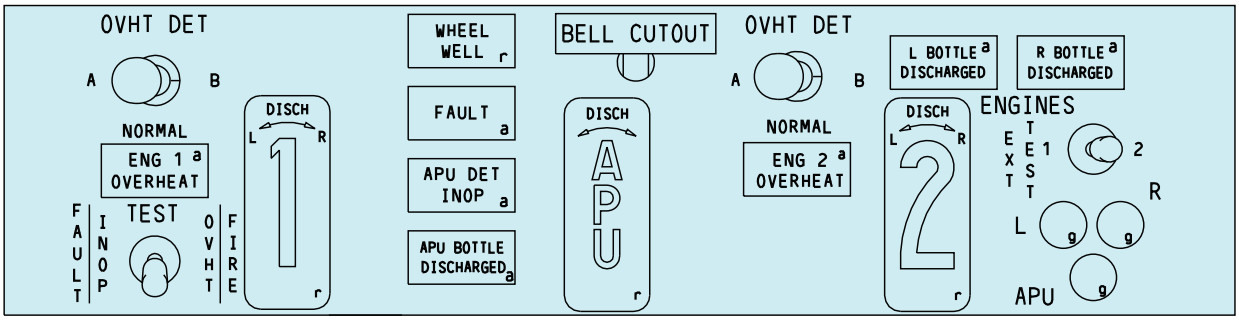


FLIGHT COMPARTMENT



ANNUNCIATOR LAYOUT, CAPTAIN'S

A



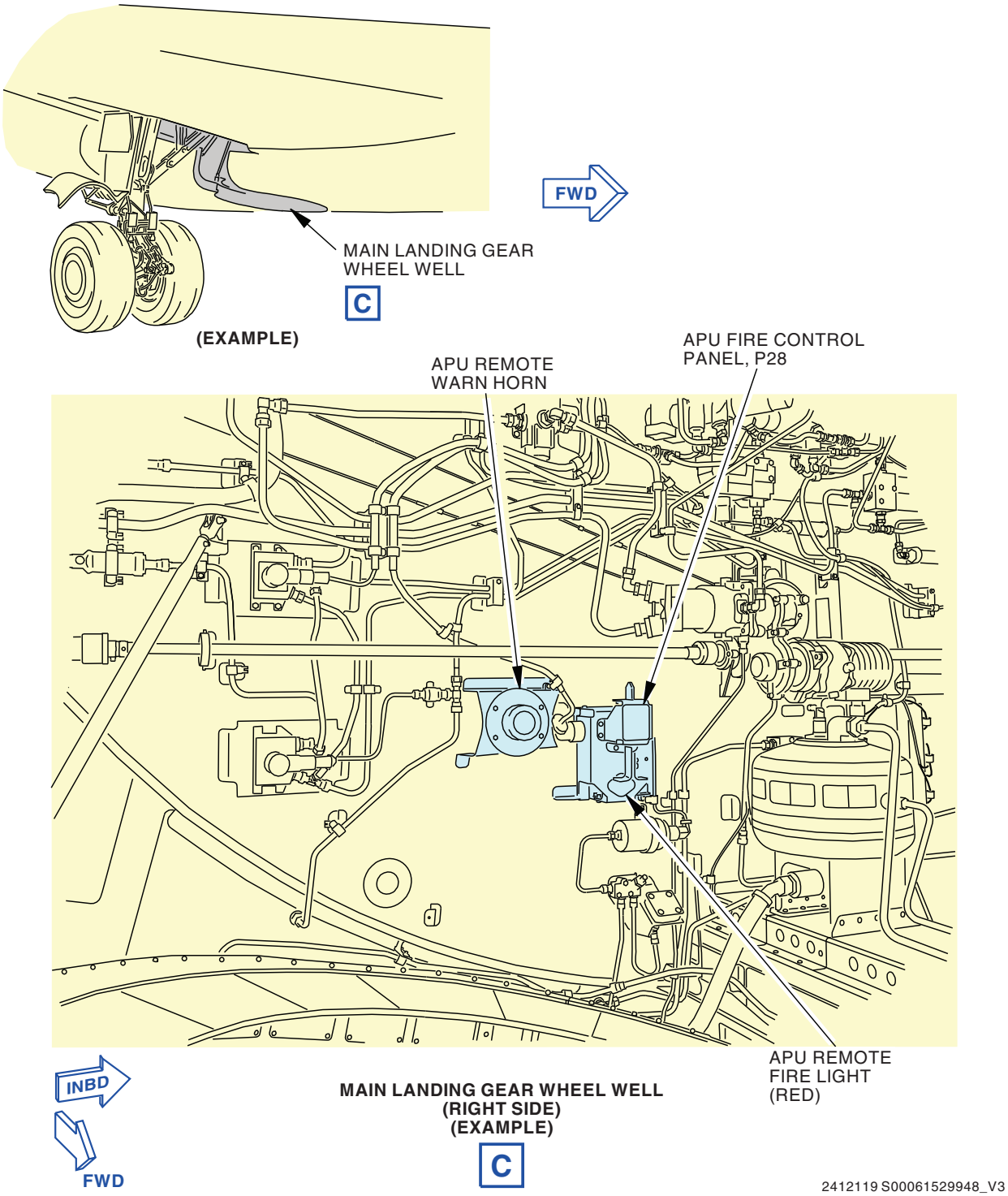
FIRE CONTROL PANEL (P8-1)

B

2412118 S00061529947_V1

APU Fire Detection System
Figure 201/26-15-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fire Detection System
Figure 201/26-15-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-00-440-801

3. APU Fire Detection - Activation

(Figure 201)

A. General

(1) This task gives instructions to activate the APU fire detection system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-15-00-420-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE DETECTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) APU Fire Detection - Operational Test

(2) APU Fire Detection Loop Resistance - System Test

(3) APU Fire Detection Circuit - System Test.

TASK 26-15-00-710-801

2. APU Fire Detection - Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test does a check of the Auxiliary Power Unit (APU) Fire Detection System without disturbing the detection system and it only uses the equipment that is on the airplane. This test will verify the circuit from the Engine and APU Fire Detection Control Module, M279 to the Fire Control Panel, P8-1, and to the APU Fire Control Panel, P28. The test will verify that the Engine and APU Fire Detection Control Module, that continuously monitors the detection loops, operates correctly.

NOTE: The FIRE WARN lights and the ALARM BELL are not activated if the APU fire detection system is inoperative. Only if the engines and APU are functioning properly will the FIRE WARN lights and ALARM BELL be activated. The circuit design precludes latent failure modes due to the parallel path of the test circuit and the actuation circuit.

NOTE: If AC power is removed (no power on main buses) and if there is a faulty APU LOOP, the FIRE BELL, the FIRE WARNING light, the MASTER CAUTION and the OVHT/DET lights are not activated.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14687	Cover - Warning Horn, APU Fire Detection Part #: C26010-1 Supplier: 81205

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Operational Test

SUBTASK 26-15-00-860-002

(1) Open these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

E. FIRE/OVHT Test

SUBTASK 26-15-00-860-004

(1) Put the warning horn cover, SPL-14687, on the APU remote warning horn (located in the main landing gear wheel).

SUBTASK 26-15-00-710-001

(2) Move and hold the TEST switch on the Fire Control Panel, P8-1, to the OVHT/FIRE position to activate the audible signals and the warning lights.

NOTE: When performing the Fire/OVHT Test with a faulted APU fire detector loop, the MASTER CAUTION lights are off, the FIRE WARN and OVHT/DET lights are off, the Fire Bell does not sound and the APU DET INOP light will be on.

NOTE: Ignore the indication lights for the engine and wheel well detection system and the indications on the Engine and APU Fire Detection Control Module, M279, in the Electronic Equipment Bay.

(a) On the Engine and APU Fire Control Panel, P8–1, make sure that this light come on:

1) APU Fire Switch light (Red).

(b) On the Pilot's Glareshield Panel, P7, make sure that these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
2) The Captain's and First Officer's FIRE WARN lights (Red).
3) The OVHT/DET light (Amber).

(c) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, comes on.

(d) In the right main wheel well, make sure that these conditions occur:

1) The APU remote fire light (red), on the APU Fire Control Panel, P28 panel, flashes on and off continuously.
2) The APU fire remote warning horn comes on.

SUBTASK 26-15-00-710-002

(3) Release the TEST switch on the Engine and APU Fire Control Panel, P8-1.

(a) On the Engine and APU Fire Control Panel, P8-1, make sure that this light go off:

1) APU Fire Switch light (Red).

(b) On the Pilot's Glareshield, make sure that these lights go off:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).
2) The Captain's and First Officer's FIRE WARN lights (Red).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) The OVHT/DET light (Amber).

(c) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, stops.

(d) In the right main wheel well, make sure that these conditions occur:

1) The APU remote fire light (red), on the APU Fire Control Panel, P28, goes off.

2) The APU fire remote warning horn stops.

F. FAULT/INOP Test from the Engine and APU Fire Control Panel, P8-1

SUBTASK 26-15-00-710-003

(1) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the FAULT/INOP position.

NOTE: Ignore the indication lights for the engine detection system and the indications on the Engine and APU Fire Detection Control Module, M279, in the Electronic Equipment Bay.

(a) On the Engine and APU Fire Control Panel, make sure that these lights come on:

1) APU DET INOP light (Amber).

NOTE: When performing the FAULT/INOP Test with a faulted APU detector loop, the APU DET INOP light will stay on, the MASTER CAUTION lights are off and the FAULT light will come on during the test.

2) FAULT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure that these lights come on:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

SUBTASK 26-15-00-710-004

(2) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) On the Engine and APU Fire Control Panel, make sure that these lights are off:

1) APU DET INOP light (Amber).

2) FAULT light (Amber).

(b) On the Pilot's Glareshield, P7, make sure that these lights are off:

1) The Captain's and First Officer's MASTER CAUTION lights (Amber).

2) The OVHT/DET light (Amber).

G. FAULT/INOP Test from the Engine and APU Fire Detection Control Module, M279.

(Figure 502)

SUBTASK 26-15-00-710-005

(1) Push and hold the FAULT/INOP TEST switch on the Engine and APU Fire Detection Control Module, in the Electrical and Electronics Compartment.

NOTE: Ignore the indications for the engine detection system and the indications that will occur in the flight compartment on the Engine and APU Fire Control Panel, P8-1, and the Pilot's Glareshield, P7.

(a) On the Engine and APU Fire Detection Control Module, make sure that these lights come on:

1) APU (Amber).

2) The four FAULT CODE lights are ON-ON-ON-ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-00-710-006

(2) Release the FAULT/INOP TEST switch on the Engine and APU Fire Detection Control Module.

(a) On the Engine and APU Fire Detection Control Module, make sure that these lights are off:

1) APU (Amber).

2) The four FAULT CODE lights are OFF-OFF-OFF-OFF.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-15-00-940-001

(1) Remove the warning horn cover, SPL-14687, from the APU remote warning horn (located in the main landing gear wheel).

SUBTASK 26-15-00-860-005

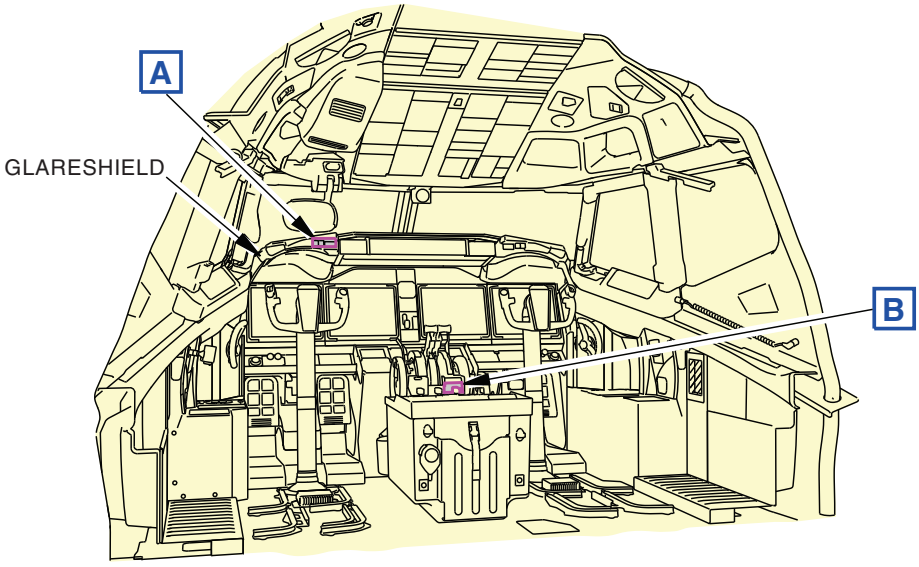
(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

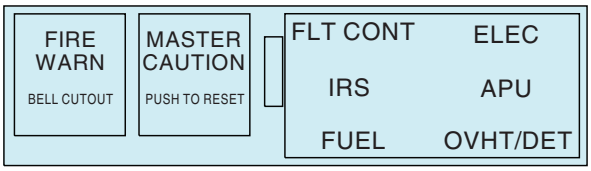
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B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

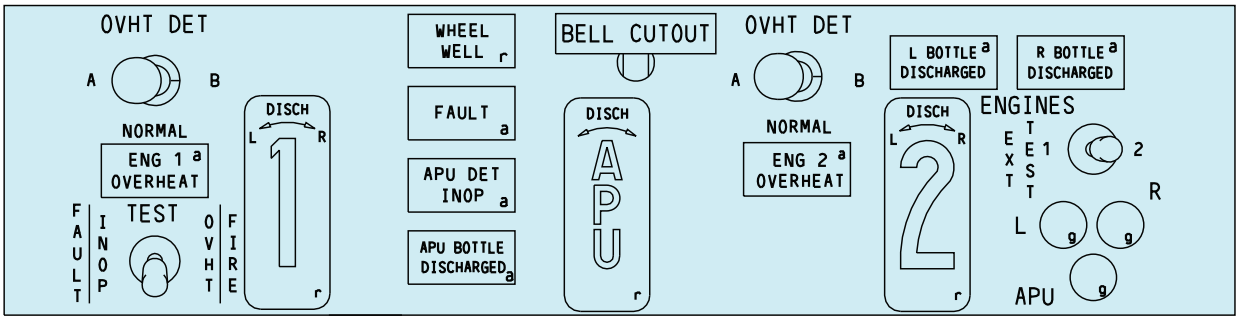


FLIGHT COMPARTMENT



ANNUNCIATOR LAYOUT, CAPTAIN'S

A



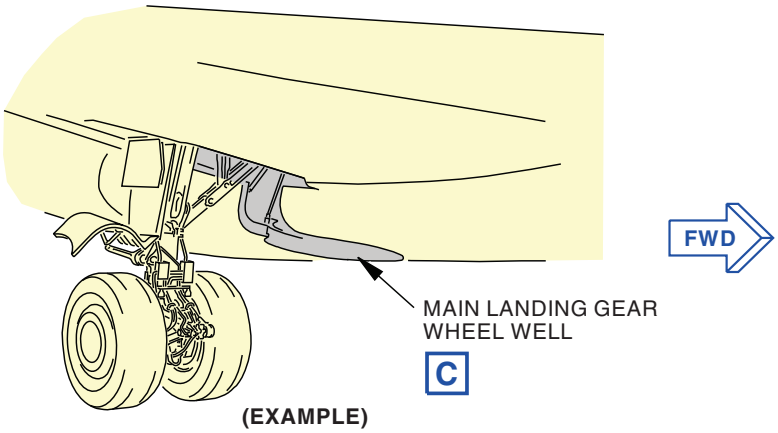
FIRE CONTROL PANEL (P8-1)

B

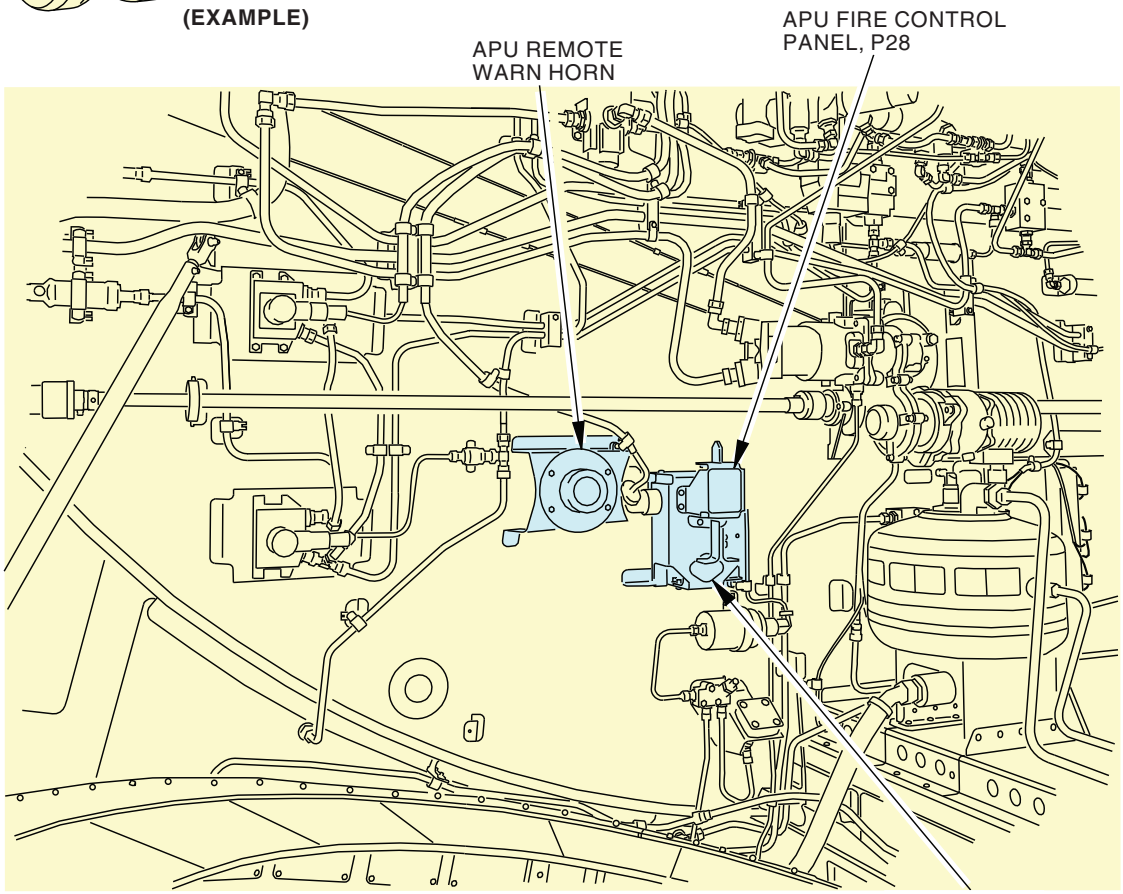
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Auxiliary Power Unit (APU) Fire Detection Operational Test
Figure 501/26-15-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)

C

2412119 S00061529948_V3

Auxiliary Power Unit (APU) Fire Detection Operational Test
Figure 501/26-15-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-00-730-801

3. APU Fire Detection Loop Resistance - System Test

(Figure 502)

A. General

(1) This test does a check of the APU fire detection loop circuit. It disturbs the engine fire detection system by removal of the Engine and APU Fire Detection Control Module, M279.

B. References

Reference	Title
26-10-01-000-801	Engine and APU Fire Detection Module Removal (P/B 401)
26-10-01-400-801	Engine and APU Fire Detection Module Installation (P/B 401)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the System Test

SUBTASK 26-15-00-860-007

(1) Make sure the airplane is in ground mode (TASK 32-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-00-860-008

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-15-00-010-001

(4) To get access to the Electronic Equipment Bay, do this step.

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

G. APU Fire Detection Loop Resistance System Test

SUBTASK 26-15-00-020-002

(1) Do this task: Engine and APU Fire Detection Module Removal, TASK 26-10-01-000-801.

SUBTASK 26-15-00-730-001

(2) Use a digital/analog multimeter, COM-1793, to measure the resistance across pin 25 of connector D1000 and airplane ground.

(a) Make sure the resistance is 1009 ±45 ohms.

SUBTASK 26-15-00-420-002

(3) Do this task: Engine and APU Fire Detection Module Installation, TASK 26-10-01-400-801.

SUBTASK 26-15-00-860-009

(4) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-15-00-710-007

(6) Make sure the APU FAULT AREA light on the Engine and APU Fire Detection Control Module, M279, is off.

SUBTASK 26-15-00-710-008

(7) Make sure the APU DET INOP light on the Engine and APU Fire Control Panel, P8-1, is off.

SUBTASK 26-15-00-710-009

(8) Push the Captain's or First Officer's MASTER CAUTION light, on the Pilot's Glareshield Panel, P7.

SUBTASK 26-15-00-710-010

(9) Make sure the Captain's and First Officer's MASTER CAUTION light, on the Pilot's Glareshield Panel, P7, is off.

SUBTASK 26-15-00-710-011

(10) Make sure the OVHT/DET light, on the Pilot's Glareshield Panel, P7, is off.

H. Put the Airplane Back to Its Usual Condition

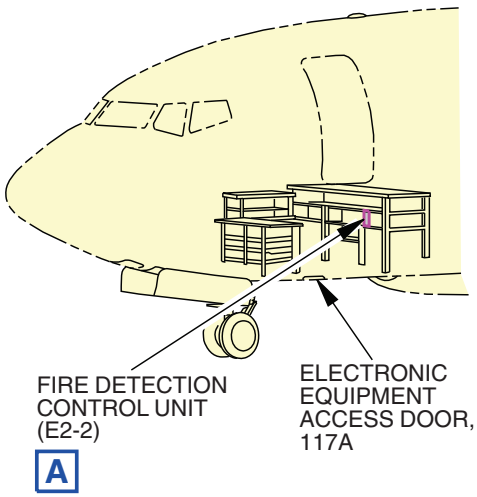
SUBTASK 26-15-00-410-001

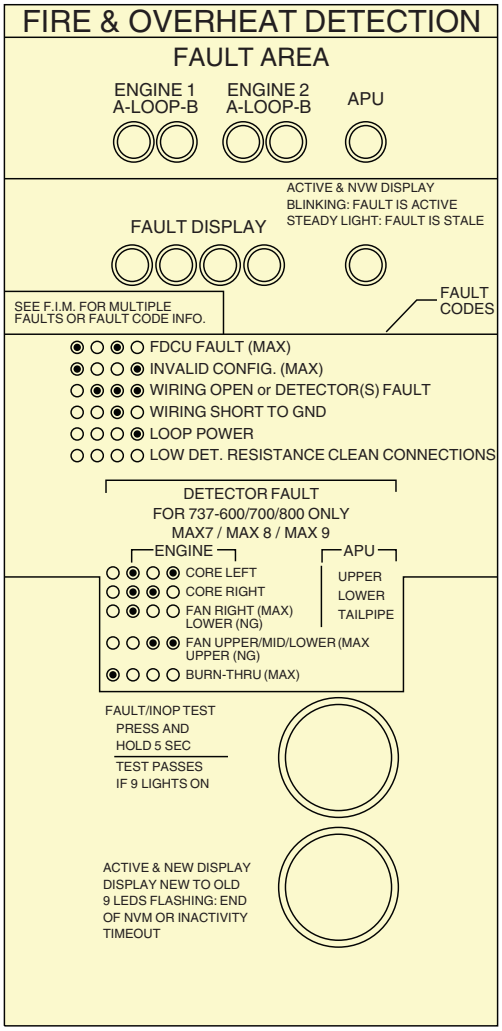
(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





FIRE DETECTION CONTROL UNIT

A

2412120 S00061529953_V3

Auxiliary Power Unit (APU) Fire Detection Resistance Test
Figure 502/26-15-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-00-730-802

4. APU Fire Detection Circuit - System Test

(Figure 501, Figure 503)

A. General

(1) This test does a check of the entire APU fire detection system. It disturbs the APU fire detection system and requires additional test equipment.

(2) The APU Fire Detection - Operational Test, TASK 26-15-00-710-801, does a test to make sure that the Engine and APU Fire Detection Control Module, which continuously monitors the detection loops, operates correctly. This operational test is done without a breakout box.

(3) The task: APU Fire Detection Loop Resistance - System Test, TASK 26-15-00-730-801, does a test of the resistance (only) of the APU fire detection loop circuit. This resistance test is done without a breakout box.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
26-10-01-000-801	Engine and APU Fire Detection Module Removal (P/B 401)
26-10-01-400-801	Engine and APU Fire Detection Module Installation (P/B 401)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
49-11-00-860-802	APU Usual Shutdown (P/B 201)
49-61-12-000-801	Electronic Control Unit Removal (P/B 401)
49-61-12-400-801	Electronic Control Unit Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1655	Test Equipment - APU/Engine Fire Detection System Part #: C26005-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-14687	Cover - Warning Horn, APU Fire Detection Part #: C26010-1 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
315A	APU Cowl Door

F. Prepare for the System Test

SUBTASK 26-15-00-860-010

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 26-15-00-860-011

(2) Make sure the APU is not operating (TASK 49-11-00-860-802).

SUBTASK 26-15-00-860-012

(3) Make sure the airplane is in ground mode (TASK 32-09-00-860-802).

SUBTASK 26-15-00-860-013

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-15-00-010-002

(6) Open these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-00-480-001

(7) Connect the test box equipment, SPL-1655, to the terminal lug of the APU LOWER Fire Detector, M1756, and airplane ground (Figure 503 (Sheet 2)).

(a) Make sure all the toggle switches on the test box are in the NORMAL positions.

(b) Connect one alligator clip to the responder bracket or to a grounding lug.

(c) Connect the other alligator clip to the responder lug.

SUBTASK 26-15-00-480-002

(8) Put the warning horn cover, SPL-14687, on the APU remote warning horn (located in the main landing gear wheel well).

SUBTASK 26-15-00-860-014

(9) Do this task: Engine and APU Fire Detection Module Removal, TASK 26-10-01-000-801.

SUBTASK 26-15-00-760-001

(10) Use a digital/analog multimeter, COM-1793, to measure the resistance across pin 25 of connector D1000 and airplane ground.

(a) Make sure the resistance is 1009 ±45 ohms.

SUBTASK 26-15-00-860-015

(11) Do this task: Engine and APU Fire Detection Module Installation, TASK 26-10-01-400-801.

SUBTASK 26-15-00-860-016

(12) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1
B	19	C01344	APU FIRE SW POWER

(13) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-00-860-017

(14) Make sure the APU FAULT AREA light on the Engine and APU Fire Detection Control Module, M279, is off.

SUBTASK 26-15-00-860-018

(15) Make sure the APU DET INOP light on the Engine and APU Fire Control Panel, P8-1, is off.

G. Fault Indication Circuit Test

SUBTASK 26-15-00-860-019

(1) Push the Captain's or First Officer's MASTER CAUTION light, on the Pilot's Glareshield, P7.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-00-860-020

(2) Make sure the Captain's and First Officer's MASTER CAUTION light, on the Pilot's Glareshield is off.

SUBTASK 26-15-00-860-021

(3) Make sure the OVHT/DET light, on the Pilot's Glareshield is off.

SUBTASK 26-15-00-730-002

(4) On the test box, move the APU LOOP GND FAULT switch to the GND FAULT position.

(a) Make sure the APU DET INOP light (amber) on the Fire Control Panel, P8-1, comes on.

(b) On the Engine and APU Fire Detection Control Module, M279, make sure these conditions occur:

1) The APU LOOP FAULT AREA light comes on.

2) Fault Codes “OFF-OFF-OFF-OFF” (red) for WIRING SHORT TO GND.

SUBTASK 26-15-00-730-003

(5) On the test box, move the APU LOOP GND FAULT switch to the NORMAL position.

(a) Make sure the APU DET INOP light on the Fire Control Panel, P8-1, is off.

(b) On the Engine and APU Fire Detection Control Module, M279, make sure these conditions occur:

1) The APU LOOP FAULT AREA light comes on.

2) Fault Codes “OFF-OFF-OFF-OFF” (red) for WIRING SHORT TO GND.

H. APU Fire Indication Circuit Test

SUBTASK 26-15-00-730-004

(1) On the test box, move the APU LOOP FIRE switch to the FIRE position.

(a) Make sure the "APU" Fire Handle light (red), on the Fire Control Panel, P8-1, comes on.

(b) On the Pilot's Glareshield, P7, make sure these conditions occur:

1) Both the Captain and First Officer's MASTER WARN lights (red) come on.

(c) Make sure the flight compartment fire bell, on the Warning Module, M315, comes on (rings).

SUBTASK 26-15-00-730-005

(2) On the test box, move the APU LOOP FIRE switches to the NORMAL position.

(a) Make sure the "APU" Fire Handle light, on the Engine and APU Fire Control Panel, P8-1, is off.

(b) On the Pilot's Glareshield, P7, make sure these conditions occur:

1) Both he Captain and First Officer's FIRE WARN lights are off.

(c) Make sure the flight compartment fire bell, on the Warning Module, M315, stops (silenced).

SUBTASK 26-15-00-860-022

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-00-860-023

(4) On the test box, move the APU LOOP GND FAULT switch to the GND FAULT position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure these Fire Detection Control Unit (M279) lights come on:

NOTE: See the decal on the front panel of M279 for the fault codes.

1) 'APU LOOP FAULT AREA' light (amber).

2) Fault Codes “OFF-OFF-ON-OFF” (red) for WIRING SHORT TO GND.

SUBTASK 26-15-00-860-024

(5) On the test box, move the APU LOOP GND FAULT switch to the NORMAL position.

(a) Make sure these Fire Detection Control Unit (M279) lights are off:

1) 'APU LOOP FAULT AREA' light (amber).

2) Fault Codes “OFF-OFF-OFF-OFF” (red) for WIRING SHORT TO GND.

SUBTASK 26-15-00-860-025

(6) On the test box, move the APU LOOP GND FAULT switch to the FIRE position.

(a) Make sure these conditions occur in the main wheel well:

1) The APU Remote Fire light (red) on the P28 APU Fire Control panel is flashing on and off at approximately one cycle each second.

2) The APU Fire Warning horn (M277) sounds.

SUBTASK 26-15-00-860-026

(7) On the test box, move the APU LOOP GND FAULT switch to the NORMAL position.

(a) Make sure these conditions occur in the main wheel well:

1) The APU Remote Fire light (red) on the P28 APU Fire Control panel is off.

2) The APU Fire Warning horn (M277) does not sound.

SUBTASK 26-15-00-860-027

(8) Do this task: Electronic Control Unit Removal, TASK 49-61-12-000-801.

SUBTASK 26-15-00-860-028

(9) Connect a digital/analog multimeter, COM-1793, with a 4K Ohm resistor assembly, to measure the voltage across pin B2 of connector D3599B and airplane ground.

NOTE: Connector D3599B is where the APU Electronic Control Unit M1709 was installed on the CVR/ECU rack.

SUBTASK 26-15-00-800-001

(10) On the test box, move the APU LOOP FIRE switches to the FIRE position.

(a) Make sure the voltage (across pin B2) is more than or equal to 24 VDC.

SUBTASK 26-15-00-860-029

(11) On the test box, move the APU LOOP FIRE switches to the NORMAL position.

(a) Make sure the voltage (across pin B2) is 0 ± 2 VDC.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-15-00-020-003

(1) Remove the warning horn cover, SPL-14687, on the APU remote warning horn (located in the main landing gear wheel well).

SUBTASK 26-15-00-860-030

(2) Remove the digital/analog multimeter, COM-1793, with the 4K Ohm resistor assembly from pin B2 of connector D3599B and airplane ground.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-00-480-003

(3) Remove the equipment, SPL-1655, from the terminal lug of the APU LOWER Fire Detector, M1756, and airplane ground.

SUBTASK 26-15-00-420-003

(4) Do this task: Electronic Control Unit Installation, TASK 49-61-12-400-801.

SUBTASK 26-15-00-860-031

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-00-860-032

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

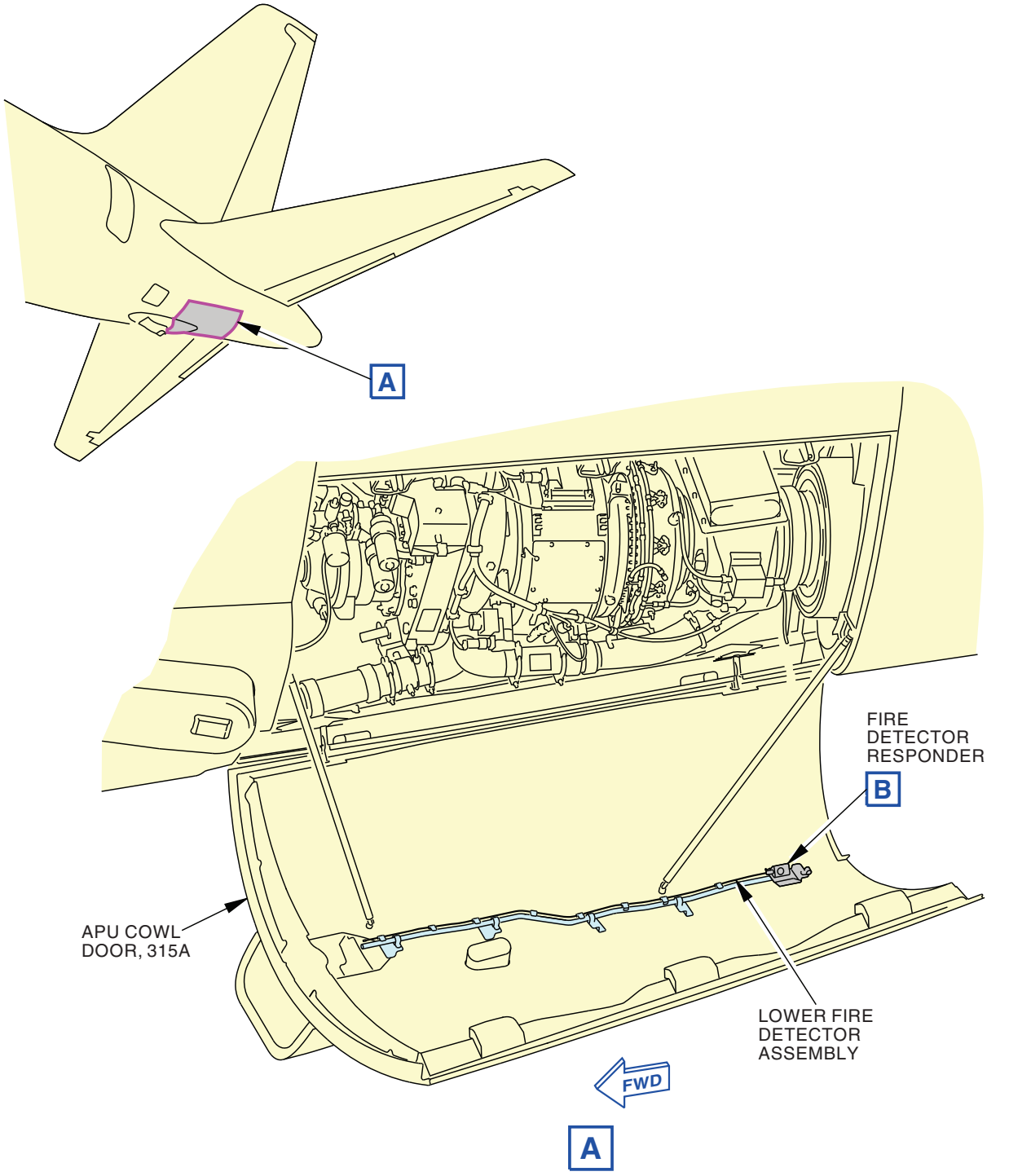
SUBTASK 26-15-00-410-002

(7) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
315A	APU Cowl Door

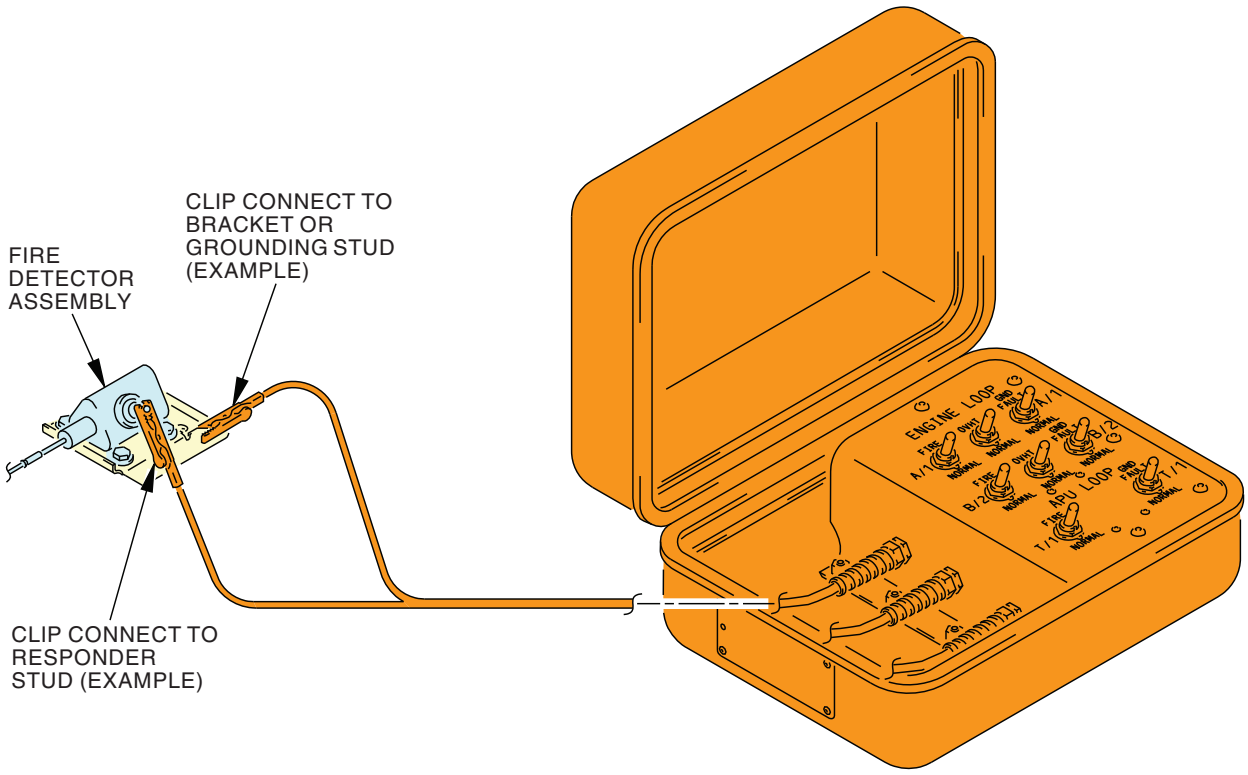
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Fire Detection Circuit Test
Figure 503/26-15-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

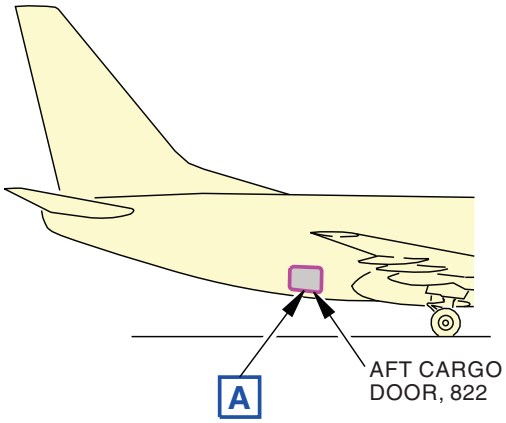


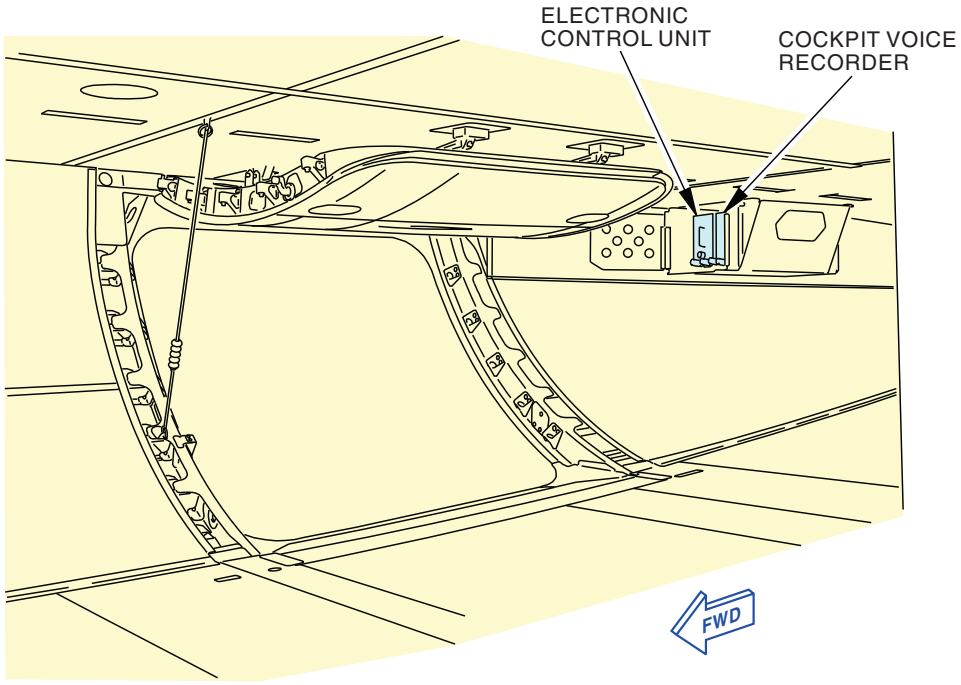
B

2412122 S00061529956_V1

Auxiliary Power Unit (APU) Fire Detection Circuit Test
Figure 503/26-15-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2412123 S00061529957_V1

Electronic Control Unit Test
Figure 504/26-15-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU OVERHEAT DETECTORS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Auxiliary Power Unit (APU) overheat detector elements

(2) An installation of the APU overheat detector elements

(3) A removal of the APU overheat detector assemblies

(4) An installation of the APU overheat detector assemblies.

B. There are three overheat detectors in the APU compartment. One overheat detector is installed on the APU door. The second is installed on the upper firewall panel in the APU compartment. The third detector is installed above the APU exhaust duct. The overheat detectors are single loop detectors.

TASK 26-15-01-000-801

2. APU Overheat Detector Element Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Auxiliary Power Unit (APU) overheat detector element.

B. Location Zones

Zone	Area
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Access Panels

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

D. Prepare for the Removal

SUBTASK 26-15-01-840-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-15-01-010-001

(2) To get access to the upper or lower APU detector, open this access panel:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-002

(3) To get access to the APU tailpipe overheat detector, open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

E. APU Overheat Detector Element Removal

SUBTASK 26-15-01-000-001

(1) Do these steps to remove the detector element [7] from the detector housing:

(a) Loosen the screw [10] on each clamp [9].

(b) Remove the nut [5] and washer [4] from the terminal lug.

(c) Disconnect the lead from the terminal lug.

(d) Remove the screws [2], washers [16], and nuts [17] from the overheat detector housing.

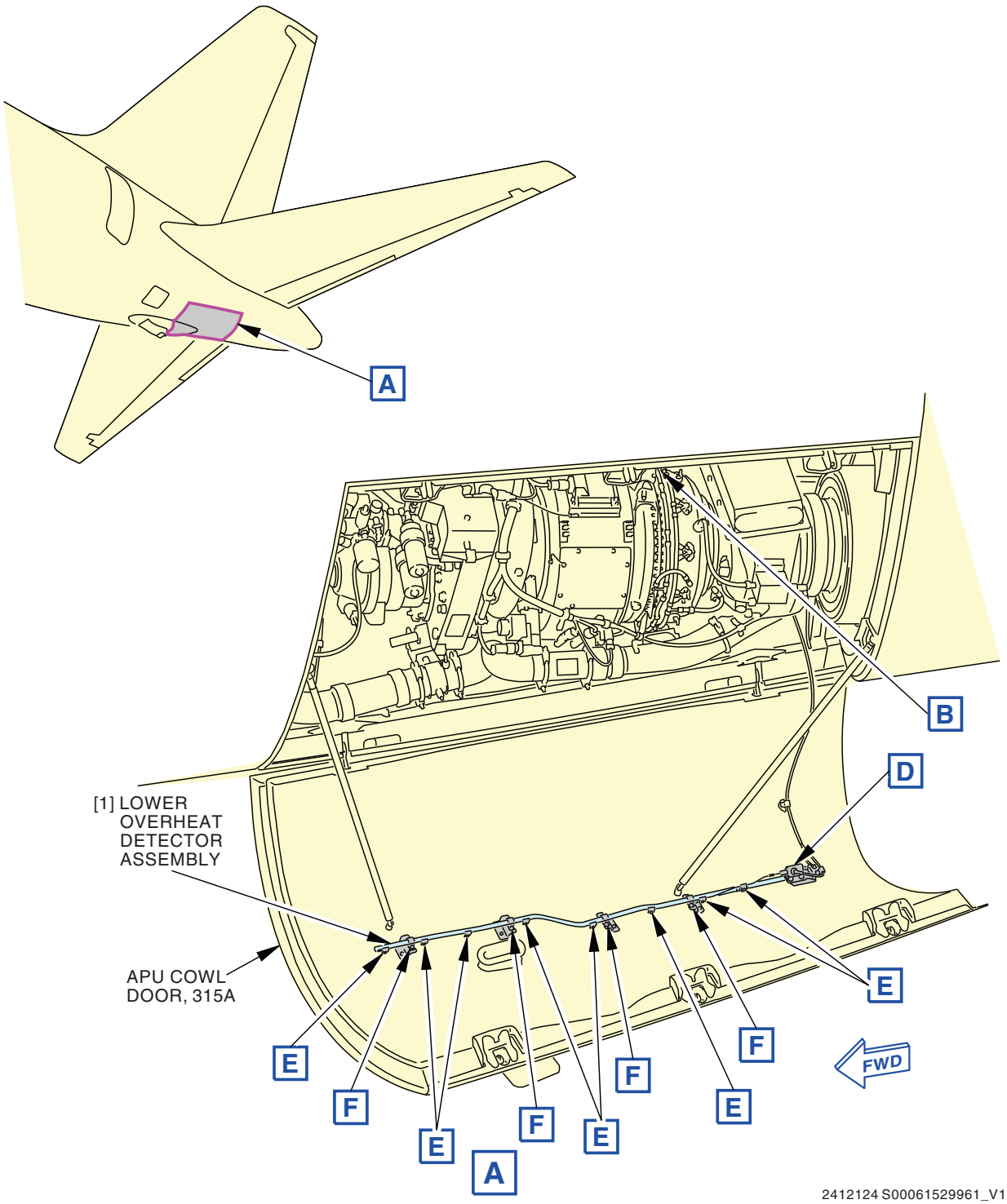
(e) Remove the element [7] from the clamp [9].

(f) Remove the grommets [8] from the element [7] to use on the new element.

(g) Remove the element [7].

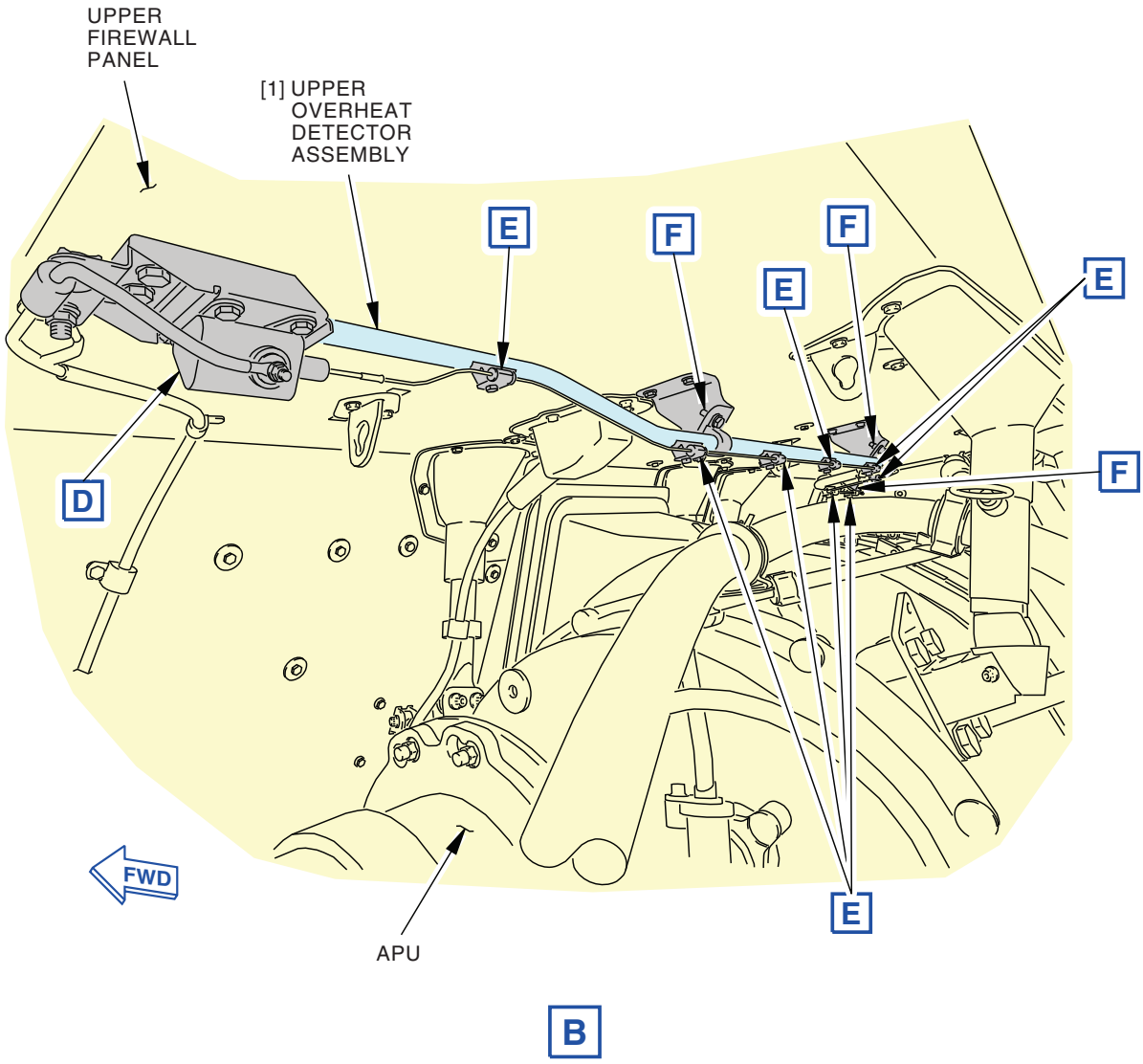
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Overheat Detector Installation
Figure 401/26-15-01-990-801 (Sheet 1 of 4)

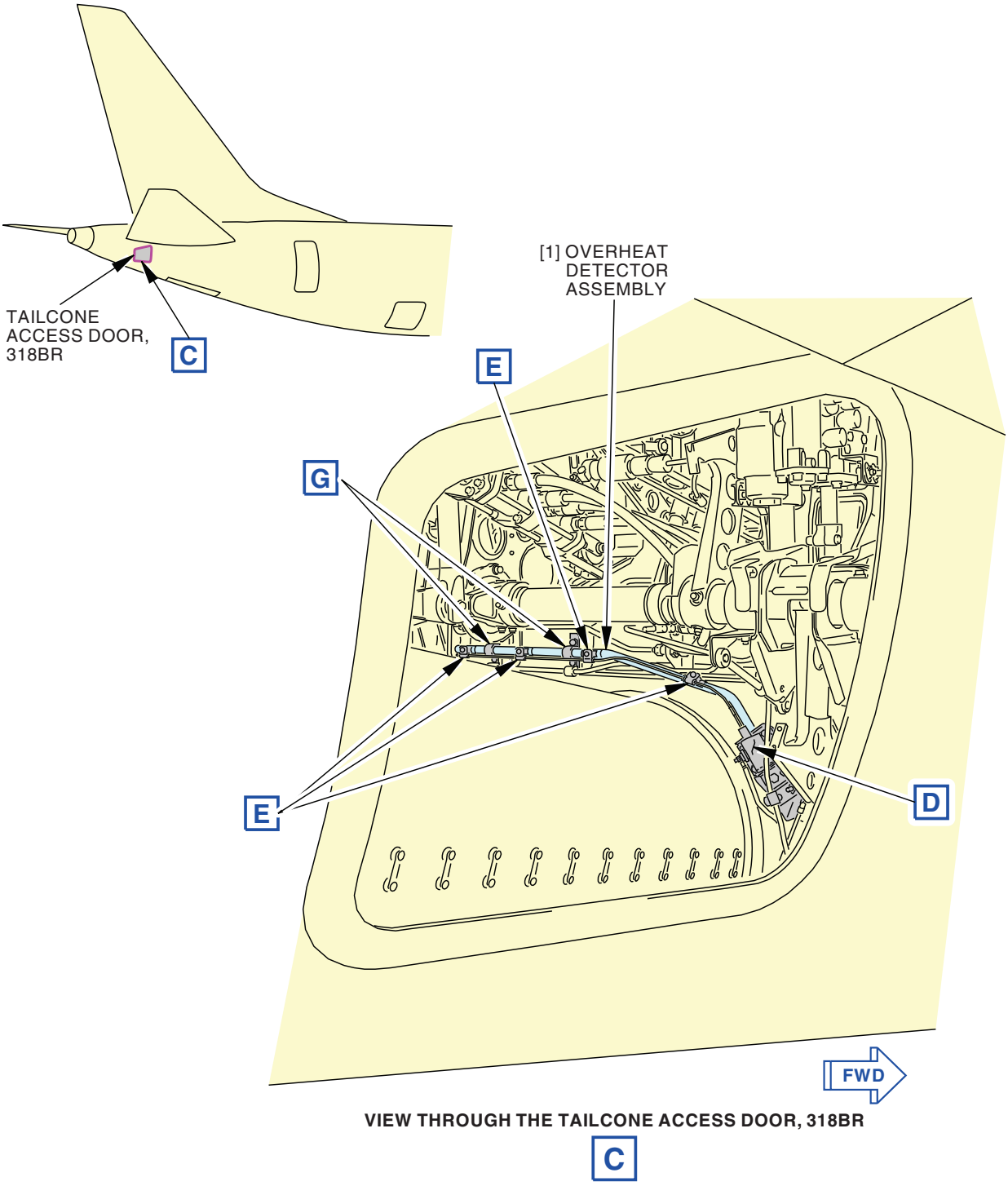
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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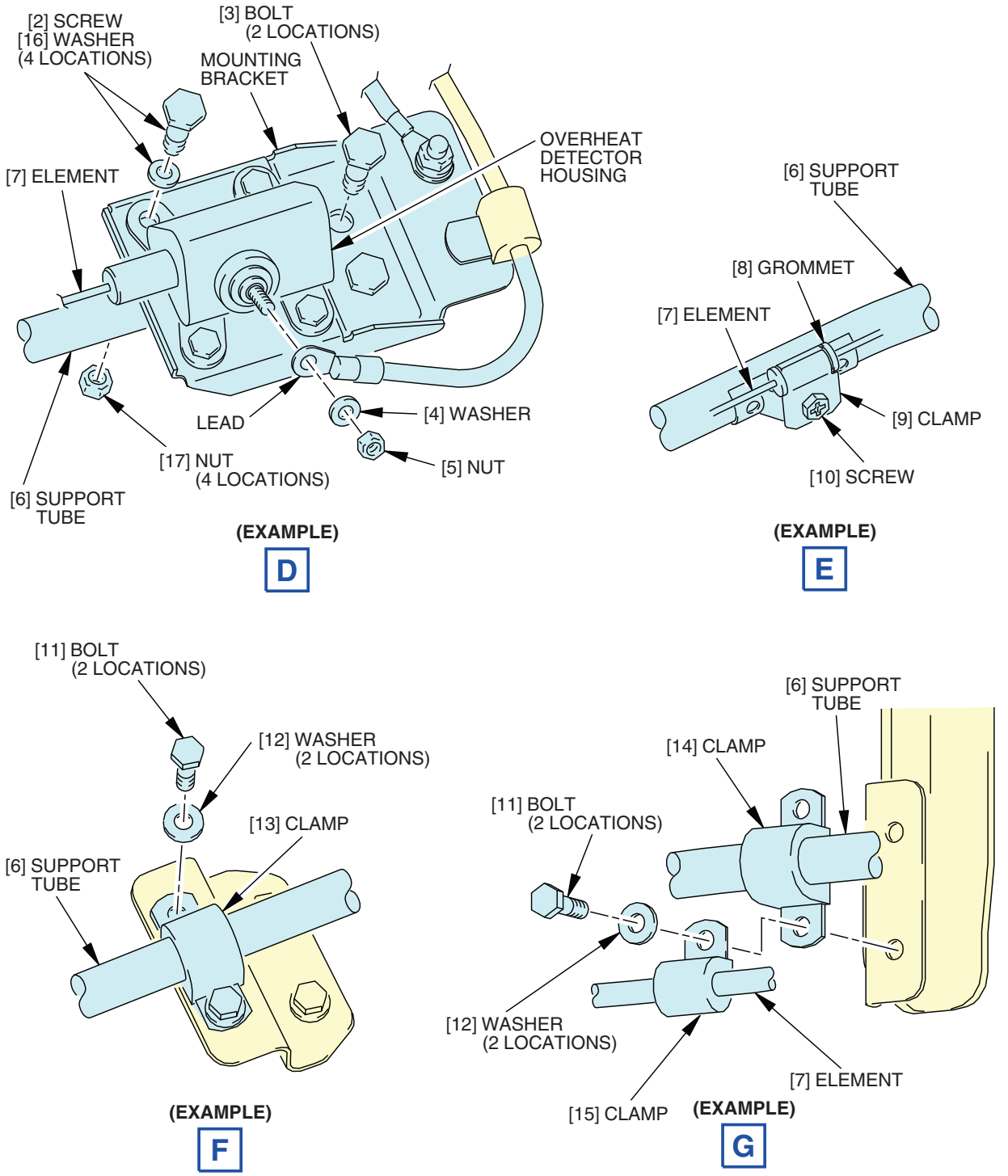
Auxiliary Power Unit (APU) Overheat Detector Installation
Figure 401/26-15-01-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Overheat Detector Installation
Figure 401/26-15-01-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Overheat Detector Installation
Figure 401/26-15-01-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-01-400-801

3. APU Overheat Detector Element Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Auxiliary Power Unit (APU) overheat detector element.

B. References

Reference	Title
26-15-00-710-801	APU Fire Detection - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50154	Silcoset 152 - cold cure, white, silicone compound	
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
D00004	Compound - Antiseize, Graphite-Petrolatum	SAE AMS 2518

E. Location Zones

Zone	Area
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Installation

SUBTASK 26-15-01-840-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-01-010-003

(2) To get access to the upper or lower APU detector, open this access panel:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-004

(3) To get access to the APU tailpipe overheat detector, open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

H. APU Overheat Detector Element Installation

SUBTASK 26-15-01-420-001

(1) Do these steps to install an overheat detector element [7]:

(a) Make sure that the detector housing and mounting bracket are clean.

(b) Put the overheat detector element [7] in position along the support tube [6].

(c) Put the overheat detector housing in the correct position on the mounting bracket.

(d) Install the screws [2], washers [16], and nuts [17] that attach the overheat detector housing to the mounting bracket.

1) Tighten the nuts [17] to 25 in-lb (2.8 N·m) - 30 in-lb (3.4 N·m).

(e) Install the grommets [8] on the detector element [7] at each clamp [9] location.

(f) Install the element [7] in the clamp [9].

(g) Make sure that the end of the detector element [7] ends less than 1 in. (25 mm) past the last grommet [8].

1) If the detector element [7] extends more than 1 in. (25 mm), adjust the length of the element [7] between each clamp [9] so that all slack is evenly distributed.

(h) Make sure that there is a minimum of 0.12 in. (3.0 mm) between the overheat detector element [7] and support tube [6].

CAUTION

FOR THE LAST ASSEMBLY, USE LUBRICATED SCREWS. IF SCREWS ARE REUSED, OR IF NON-LUBRICATED SCREWS ARE USED THEY CAN FREEZE BECAUSE OF HIGH TEMPERATURES. THE THREADS ON THESE SCREWS MUST HAVE A LAYER OF COMPOUND, D00004, BEFORE INSTALLATION.

(i) Apply compound, D00004, to the screw [10] threads.

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(j) Tighten the screw [10] on each clamp [9] until 1-1/2 to 3 threads extend through the nut.

1) Make sure that the grommets [8] and clamp [9] are tight.

2) Make sure that the ends of the screws [10] do not touch the support tube [6].

(k) Apply compound, D00004, to the threads of the terminal lug.

(l) Put the lead in position on the terminal lug, and attach with a washer [4] and nut [5].

(m) Tighten the nut [5] to 32 in-lb (3.6 N·m) - 38 in-lb (4.3 N·m).

(n) Apply Silcoset 152 adhesive, A50154, to the terminal lugs.

SUBTASK 26-15-01-280-001

(2) Do a check of the resistance between the element stud and the airplane structure.

(a) Use an intrinsically safe approved bonding meter, COM-1550, or equivalent, that can measure 0.0001 ohm (0.1 milliohm).

(b) Make sure that the resistance is less than 0.0025 ohm (2.5 milliohms).

(c) If the resistance is 0.0025 ohm (2.5 milliohms) or more, clean the bonding surfaces with the solvent, B00083, and do the check again.

I. APU Overheat Detector Assembly Installation Test

SUBTASK 26-15-01-840-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-01-710-001

(2) Do this task: APU Fire Detection - Operational Test, TASK 26-15-00-710-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 26-15-01-010-005

(1) Close this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 26-15-01-010-006

(2) Close this access panel:

Number	Name/Location
315A	APU Cowl Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-01-000-802

4. APU Overheat Detector Assembly Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU Overheat Detector Assembly.

B. Location Zones

Zone	Area
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Access Panels

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

D. Prepare for the Removal

SUBTASK 26-15-01-840-004

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-01-010-007

(2) To get access to the upper or lower APU detector, open this access door:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-008

(3) To get access to the APU tailpipe overheat detector, open this access door:

Number	Name/Location
318BR	Tailcone Access Door

E. APU Overheat Detector Assembly Removal

SUBTASK 26-15-01-000-002

(1) Do these steps to remove the detector assembly [1] from the APU compartment.

(a) Remove the nut [5] and washer [4] from the terminal lug.

(b) Disconnect the lead from the terminal lug.

(c) Remove the bolts [3] from the overheat detector housing.

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(d) Remove the bolts [11] and washers [12] that attach each clamp [13] to the APU compartment.

1) Remove the clamp [13] from the support tube [6].

(e) Remove the bolts [11] and washer [12] that attach each clamp [15] and clamp [14] to the APU compartment.

(f) Remove the detector assembly [1] from the APU compartment.

——— END OF TASK ———

TASK 26-15-01-400-802

5. APU Overheat Detector Assembly Installation

(Figure 401)

A. General

(1) This task gives instructions to install the APU Overheat Detector Assembly.

B. References

Reference	Title
26-15-00-710-801	APU Fire Detection - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50154	Silcoset 152 - cold cure, white, silicone compound	
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

G. Prepare for the Installation

SUBTASK 26-15-01-840-005

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-01-010-009

(2) To get access to the upper or lower APU detector, open this access door:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-010

(3) To get access to the APU tailpipe overheat detector, open this access door:

Number	Name/Location
318BR	Tailcone Access Door

H. APU Overheat Detector Assembly Installation

SUBTASK 26-15-01-420-002

(1) Do these steps to install a overheat detector assembly [1]:

(a) Put the detector assembly [1] in position in the APU compartment and install the clamps [13] on the support tube [6].

1) Attach each clamp [13] with two bolts [11] and washers [12].

2) Tighten the bolts [11] to 35.0 ±3.0 in-lb (4.0 ±0.4 N·m).

(b) Put the detector assembly [1] in position in the APU compartment and install the clamp [14] and clamp [15] on the support tube [6].

1) Attach each clamp [14] and clamp [15] with the bolts [11] and washers [12].

2) Tighten the bolts [11] to 35.0 ±3.0 in-lb (4.0 ±0.4 N·m).

(c) Install the two bolts [3] that attach the overheat detector housing to the APU compartment.

(d) Tighten the bolts [3] to 80.0 ±8.0 in-lb (9 ±1 N·m).

(e) Put the lead in position on the terminal lug, and attach it with a washer [4] and nut [5].

(f) Tighten the nut [5] to 35.0 ±3.0 in-lb (4.0 ±0.4 N·m).

(g) Apply Silcoset 152 adhesive, A50154, to the terminal lugs.

SUBTASK 26-15-01-280-002

(2) Do a check of the resistance between the stud and the airplane structure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use an intrinsically safe approved bonding meter, COM-1550, or equivalent, which can measure 0.0001 ohm (0.1 milliohm).

(b) Make sure that the resistance is less than 0.0025 ohm (2.5 milliohms).

(c) If the resistance is more than 0.0025 ohm (2.5 milliohms) or more, clean the bonding surfaces with the solvent, B00083, and do the check again.

I. APU Overheat Detector Assembly Installation Test

SUBTASK 26-15-01-840-006

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-15-01-710-002

(2) Do this task: APU Fire Detection - Operational Test, TASK 26-15-00-710-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 26-15-01-010-011

(1) Close this access door:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-012

(2) Close this access door:

Number	Name/Location
318BR	Tailcone Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU OVERHEAT DETECTORS - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Inspection of the APU Overheat Detectors.

TASK 26-15-01-210-801

2. APU Overheat Detectors Inspection

A. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

B. Access Panels

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

C. APU Overheat Detectors Inspection

SUBTASK 26-15-01-010-013

(1) To get access to the upper or lower APU detector, open this access door:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 26-15-01-010-014

(2) To get access to the APU tailpipe overheat detector, open this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 26-15-01-210-001

(3) Do a general visual inspection of the APU overheat detectors.

(a) Make sure the APU overheat detectors are in good condition.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 26-15-01-010-015

(1) Close these access doors:

Number	Name/Location
315A	APU Cowl Door
318BR	Tailcone Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE WARNING HORN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) APU Fire Warning Horn Removal

(2) APU Fire Warning Horn Installation.

B. The APU Fire Warning Horn M277 is located on the aft bulkhead of the right main wheel well.

TASK 26-15-02-000-801

2. APU Fire Warning Horn Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU Fire Warning Horn.

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Prepare for the Removal

SUBTASK 26-15-02-010-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU

D. APU Fire Warning Horn Removal

SUBTASK 26-15-02-020-008

(1) Remove the screws [2] and washers [3] that connect the APU fire warning horn [1] to the fire horn bracket.

SUBTASK 26-15-02-020-009

(2) Carefully move the APU fire warning horn [1] away from the fire horn bracket to get sufficient access to disconnect the electrical wire terminals [4].

SUBTASK 26-15-02-020-001

(3) Tag each electrical wire terminal [4] with its location.

SUBTASK 26-15-02-020-002

(4) Loosen the electrical wire terminals [4].

SUBTASK 26-15-02-020-003

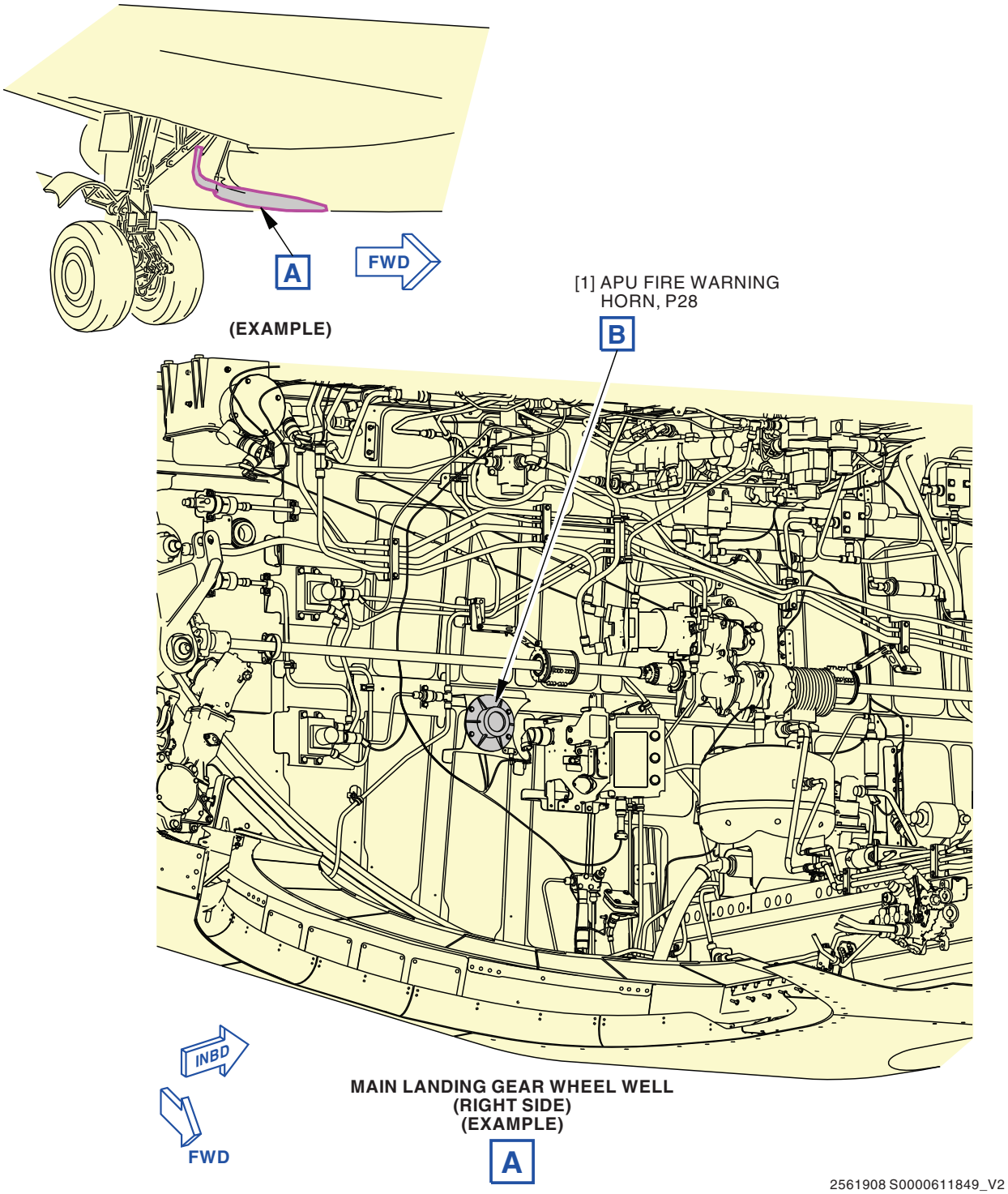
(5) Disconnect the electrical wire terminals [4] from the APU fire warning horn [1].

SUBTASK 26-15-02-020-005

(6) Remove the APU fire warning horn [1].

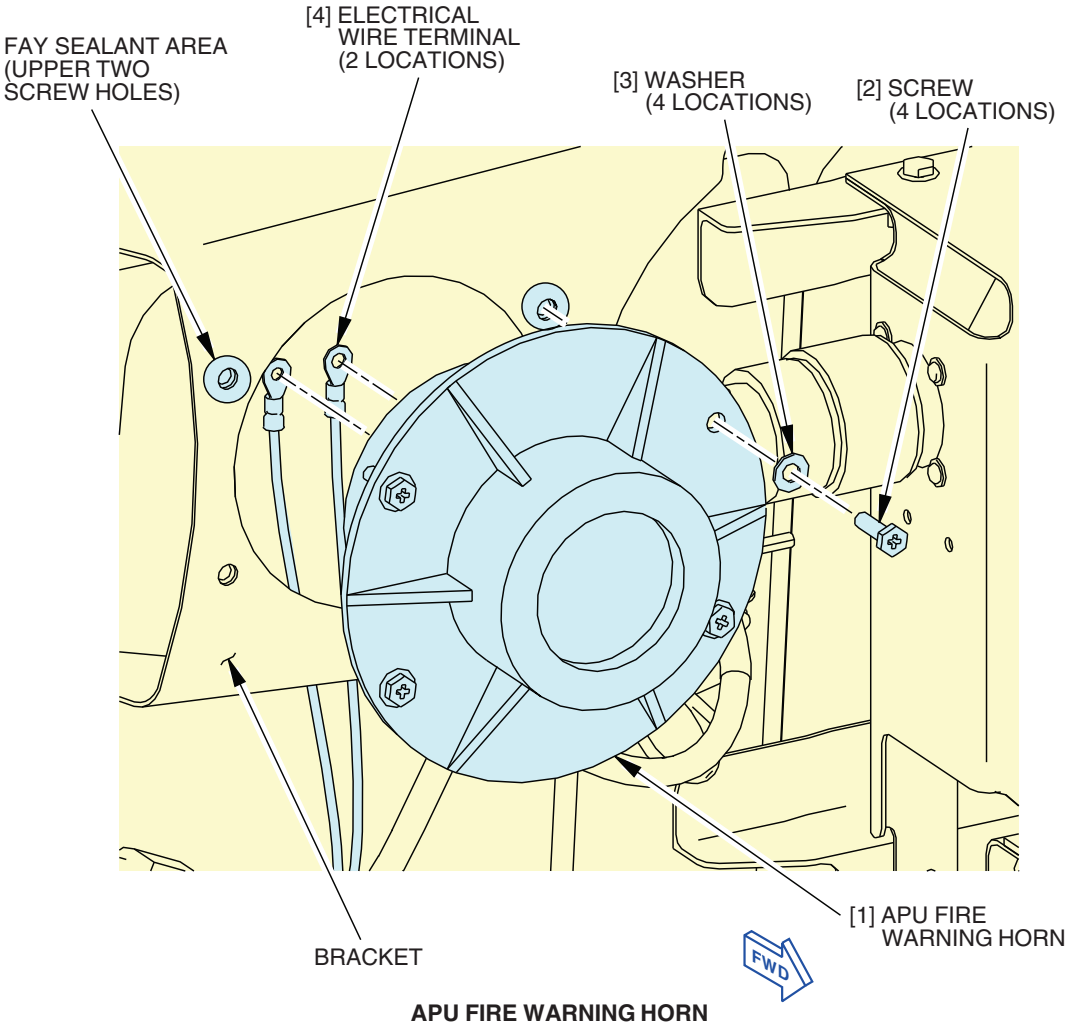
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fire Warning Horn Installation
Figure 401/26-15-02-990-801 (Sheet 1 of 2)

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**APU FIRE WARNING HORN
(EXAMPLE)**

B

2561909 S0000611850_V2

**APU Fire Warning Horn Installation
Figure 401/26-15-02-990-801 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-15-02-400-801

3. APU Fire Warning Horn Installation

(Figure 401)

A. General

(1) This task gives instructions to install the APU Fire Warning Horn.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
26-15-00-710-801	APU Fire Detection - Operational Test (P/B 501)
SOPM SWPM 20-20-00	Standard Overhaul Practices Manual
SWPM 20-20-00 Paragraph 14.D	Category 2 Fay Sealed Fay Surface Bonds

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-765	Scraper - Plastic

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50115 [P08-050]	Adhesive/Sealant - Flame Resistant Curring Agent - Dapco #2030	PWA 36751-2
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	APU fire warning horn	26-15-02-01-005	PTF ALL

F. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

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G. Prepare for the Installation

SUBTASK 26-15-02-010-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU

H. APU Fire Warning Horn Installation

SUBTASK 26-15-02-010-003

(1) Get access to the fire horn bracket.

SUBTASK 26-15-02-020-007

(2) Remove any existing sealant with plastic scraper, STD-765.

SUBTASK 26-15-02-100-001

(3) Prepare the upper two screw hole mating surfaces between the fire horn bracket and the APU fire warning horn [1].

(a) Do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

SUBTASK 26-15-02-420-001

(4) Apply fay seal with sealant, A50110, or flame resistant adhesive/sealant, A50115 [P08-050], or sealant, A50231, or sealant, A50296, to the mating surfaces of the APU fire warning horn [1] and to the fire horn bracket, where the upper screws [2] are installed (SWPM 20-20-00 Paragraph 14.D).

SUBTASK 26-15-02-420-003

(5) Connect the electrical wire terminals [4] to the APU fire warning horn [1].

SUBTASK 26-15-02-020-006

(6) Remove the location tags from the electrical wire terminals [4].

SUBTASK 26-15-02-420-002

(7) Install the screws [2] and washers [3] that attach the APU fire warning horn [1] to the fire horn bracket.

(a) Make sure that the squeeze out is continuous and visible along edges of bonding interfaces.

SUBTASK 26-15-02-700-001

(8) Measure the bonding resistance between the fire horn bracket and APU fire warning horn [1] (SOPM SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the measured resistance is not more than 0.50 milliohms (0.00050 ohms).

SUBTASK 26-15-02-410-001

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	23	C00403	FIRE PROTECTION DETECTION APU

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. APU Fire Warning Horn Installation Test

SUBTASK 26-15-02-480-001

(1) Do this task: APU Fire Detection - Operational Test, TASK 26-15-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO BAY SMOKE DETECTION - ADJUSTMENT/TEST

1. General

A. Both cargo bays have smoke detectors installed in the ceiling. When the concentration of smoke at the sensor gets to a preset threshold, the smoke detector alarm occurs.

B. This procedure consists of these tasks:

(1) Cargo Bay Smoke Detection - Operational Test.

(2) Cargo Bay Smoke Detection - Smoke Test.

(3) Cargo Electronic Unit (CEU) - BITE Test

TASK 26-16-00-710-801

2. Cargo Bay Smoke Detection - Operational Test

(Figure 501)

A. Prepare for the Operational Test

SUBTASK 26-16-00-860-020

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

B. Cargo Bay Smoke Detection Operational Test

SUBTASK 26-16-00-860-021

(1) Make sure FWD and AFT DET SELECT switches on the Cargo Fire Control Panel are set to NORM.

SUBTASK 26-16-00-710-001

(2) Push and hold the TEST switch on the Cargo Fire Control Panel.

(a) On the Cargo Fire Control panel, look for these indications:

1) The ARM FWD and AFT red lights comes on.

2) The DISCH light comes on.

3) The EXT FWD and EXT AFT light comes on.

4) The DETECTOR FAULT light is off.

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's FIRE WARN lights (Red).

(c) Make sure you can hear the flight compartment fire bell.

SUBTASK 26-16-00-730-001

(3) Push the Captain's or First Officer's FIRE WARN switch.

NOTE: If you push the either of the FIRE WARN lights on the P7 panel, or the BELL CUTOUT switch on the Engine and APU Fire Control panel, the same conditions will occur. The FIRE WARN and BELL CUTOUT switches cause the same resets but from different locations.

(a) On the Pilot's Glareshield, P7, make sure these lights go off:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) The Captain's and First Officer's FIRE WARN lights (Red).

(b) Make sure you can not hear the fire bell.

SUBTASK 26-16-00-860-003

(4) Release the TEST switch.

(a) On the Cargo Bay Fire Control panel, look for these indications:

1) The ARM FWD and AFT red lights goes off.

2) The DISCH light goes off.

3) The EXT FWD and AFT lights go off.

SUBTASK 26-16-00-860-022

(5) Set the FWD and AFT DET SELECT switches to “A”.

SUBTASK 26-16-00-710-002

(6) Push and hold the TEST switch.

(a) On the Cargo Fire Control panel, look for these indications:

1) The ARM FWD and AFT red light comes on.

2) The DISCH light comes on.

3) The EXT FWD and EXT AFT light comes on.

4) The DETECTOR FAULT light is off.

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's FIRE WARN lights (Red).

(c) Make sure you hear the flight compartment fire bell.

SUBTASK 26-16-00-730-026

(7) Push the Captain's or First Officer's FIRE WARN switch.

NOTE: If you push the either of the FIRE WARN lights on the P7 panel, or the BELL CUTOUT switch on the Engine and APU Fire Control panel, the same conditions will occur. The FIRE WARN and BELL CUTOUT switches cause the same resets but from different locations.

(a) On the Pilot's Glareshield, P7, make sure these lights go off:

1) The Captain's and First Officer's FIRE WARN lights (Red).

(b) Make sure you do not hear the fire bell.

SUBTASK 26-16-00-860-023

(8) Release the TEST switch.

(a) On the Cargo Bay Fire Control panel, look for these indications:

1) The ARM FWD and AFT red lights goes off.

2) The DISCH light goes off.

3) The EXT FWD and AFT lights go off.

SUBTASK 26-16-00-860-024

(9) Set the FWD and AFT DET SELECT switches to “B”.

SUBTASK 26-16-00-710-003

(10) Push and hold the TEST switch.

(a) On the Cargo Fire Control panel, look for these indications:

1) The ARM FWD and AFT red lights comes on.

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2) The DISCH light comes on.

3) The EXT FWD and EXT AFT light comes on.

4) The DETECTOR FAULT light is off.

(b) On the Pilot's Glareshield, P7, make sure these lights come on:

1) The Captain's and First Officer's FIRE WARN lights (Red).

(c) Make sure you hear the flight compartment fire bell.

SUBTASK 26-16-00-730-027

(11) Push the Captain's or First Officer's FIRE WARN switch.

NOTE: If you push the either of the FIRE WARN lights on the P7 panel, or the BELL CUTOUT switch on the Engine and APU Fire Control panel, the same conditions will occur. The FIRE WARN and BELL CUTOUT switches cause the same resets but from different locations.

(a) On the Pilot's Glareshield, P7, make sure these lights go off:

1) The Captain's and First Officer's FIRE WARN lights (Red).

(b) Make sure that you do not hear the fire bell.

SUBTASK 26-16-00-860-025

(12) Release the TEST switch.

(a) On the Cargo Bay Fire Control panel, look for this indication:

1) The ARM FWD and AFT red light goes off.

2) The DISCH light goes off.

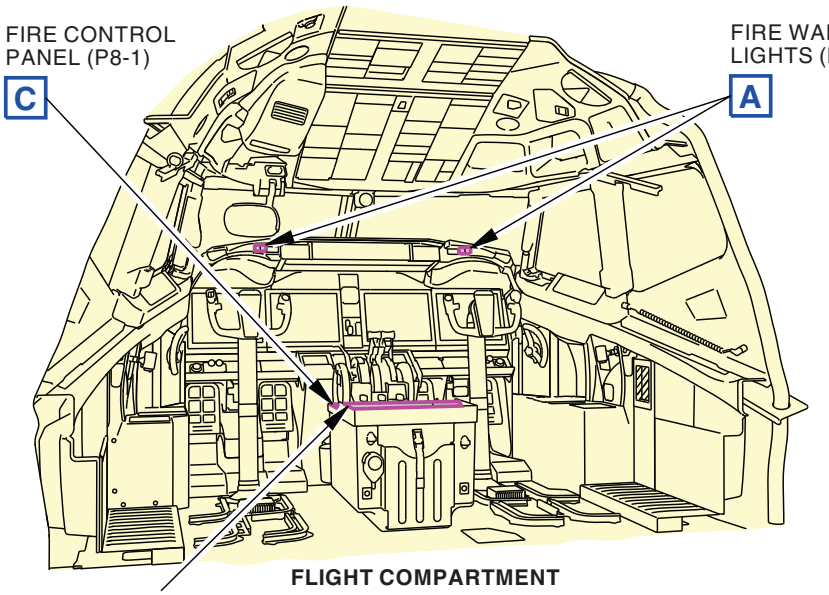
3) The EXT FWD and EXT AFT light goes off.

SUBTASK 26-16-00-860-026

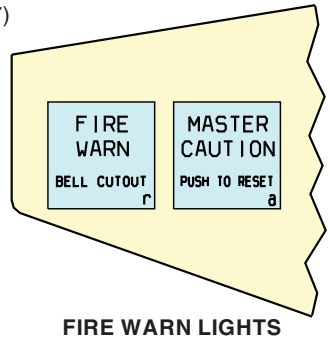
(13) Set the FWD and AFT DET SELECT switches to back to NORM.

——— END OF TASK ———

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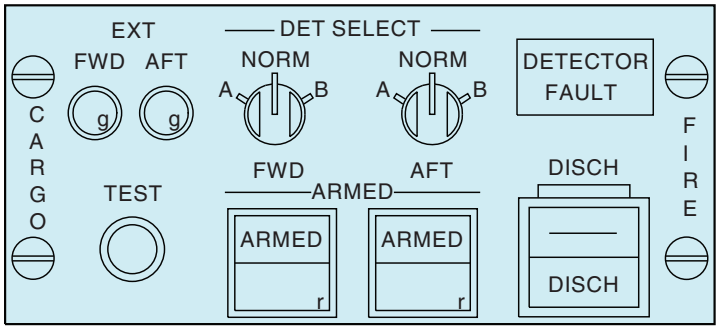


FLIGHT COMPARTMENT



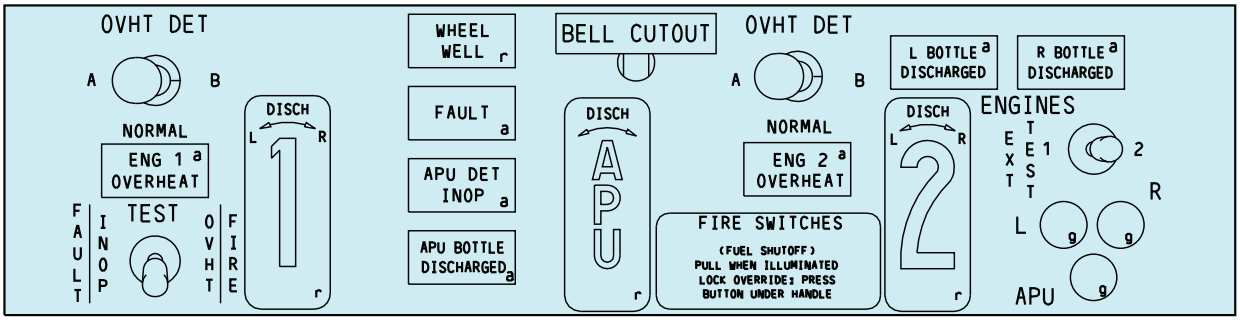
FIRE WARN LIGHTS (P7) (EXAMPLE)

A



CARGO FIRE CONTROL PANEL (P8)

B



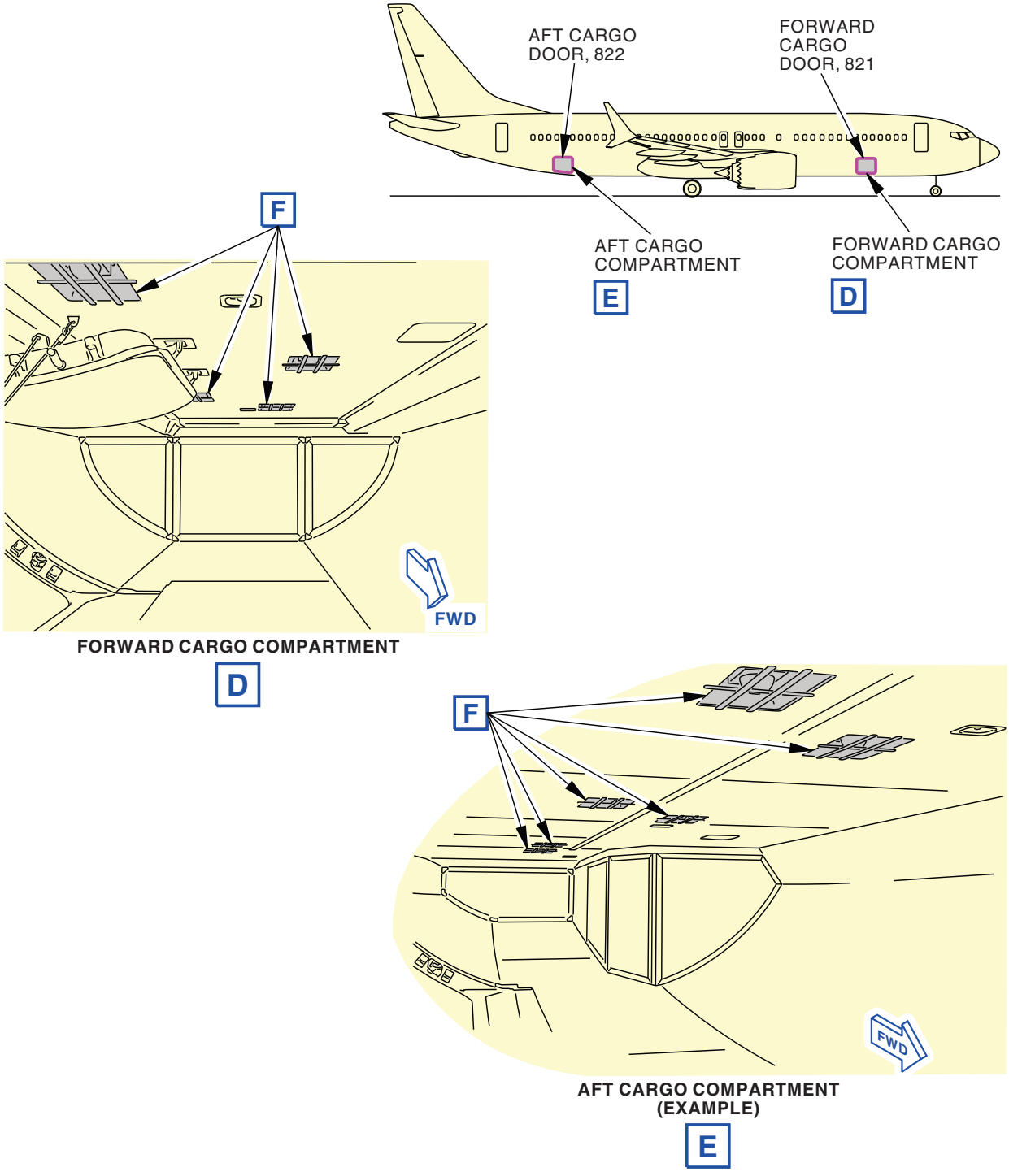
FIRE CONTROL PANEL (P8-1)

C

2412128 S00061529978_V1

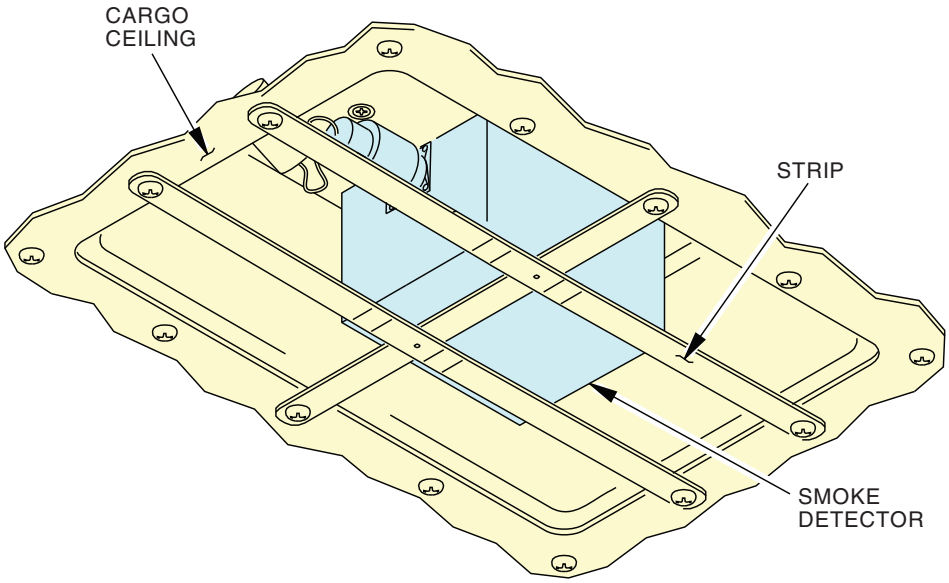
Cargo Bay Smoke Detection Test
Figure 501/26-16-00-990-802 (Sheet 1 of 3)

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Cargo Bay Smoke Detection Test
Figure 501/26-16-00-990-802 (Sheet 2 of 3)

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(EXAMPLE)

F

2412130 S00061529983_V2

Cargo Bay Smoke Detection Test
Figure 501/26-16-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-16-00-730-801

3. Cargo Bay Smoke Detection - Smoke Test

(Figure 501)

A. General

(1) When the DET SEL switch is set to A, and any smoke detector in loop A detects smoke, an alarm will occur. Similarly, when the DET SEL switch is set to B, and the B loop detects smoke, the alarm will occur. When the DET SEL switch is set to NORM, smoke must be detected by a smoke detector in loop A and loop B (any smoke detector in both loops, they do not have to be a pair) before the alarm will occur.

NOTE: If you apply too much smoke to a smoke detector, the excess smoke could affect adjacent smoke detectors. Try to limit the quantity of smoke applied to each smoke detector.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11202	Smoke Generator - Lavatory/Crew Rest Smoke Detector (or 737 Cargo Area Type Smoke Detectors) Part #: HO-25S Supplier: 61908 Part #: SOLOA10 Supplier: 1WU32 Opt Part #: SOLOA3 Supplier: 1WU32 Opt Part #: SOLOA4 Supplier: 1WU32

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 26-16-00-860-005

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-16-00-860-019

(2) Wait ten seconds for the smoke detectors to run internal self-testing.

E. Forward Cargo Bay Smoke Detection Smoke Test

NOTE: The subsequent test procedures are for each smoke detector pair in the FWD cargo compartment.

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SUBTASK 26-16-00-860-006

(1) On the P8 cargo fire control panel, set the FWD DET SELECT switch to A position.

SUBTASK 26-16-00-730-002

(2) Use a smoke generator, COM-11202, to apply smoke to the A loop smoke detector for the pair in the test (Table 501).

SUBTASK 26-16-00-730-003

(3) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the FWD CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

SUBTASK 26-16-00-730-004

(4) On the P8 cargo fire control panel, set the FWD DET SELECT switch to NORM position.

(a) Make sure that the FWD CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns off within 10 seconds.

SUBTASK 26-16-00-860-008

(5) Clear the smoke from the A loop smoke detector.

SUBTASK 26-16-00-860-011

(6) On the P8 cargo fire control panel, set the FWD DET SELECT switch to B position.

SUBTASK 26-16-00-730-006

(7) Use a smoke generator, COM-11202, to apply smoke to the B loop smoke detector for the pair in the test (Table 501).

SUBTASK 26-16-00-730-007

(8) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the FWD CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

SUBTASK 26-16-00-730-008

(9) On the P8 cargo fire control panel, set the FWD DET SELECT switch to NORM.

(a) Make sure that the FWD CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns off within 10 seconds.

SUBTASK 26-16-00-860-012

(10) Clear the smoke from the B loop smoke detector.

SUBTASK 26-16-00-730-009

(11) Use a smoke generator, COM-11202, to apply smoke to both A and B loop smoke detectors for the pair in the test.

SUBTASK 26-16-00-730-010

(12) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the FWD CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

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SUBTASK 26-16-00-860-013

(13) Clear the smoke from the A and B loop smoke detectors.

SUBTASK 26-16-00-730-011

(14) Make sure that the FWD CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns when the smoke is gone.

SUBTASK 26-16-00-730-012

(15) On the P8 cargo fire control panel, set the FWD DET SELECT switch to A position.

(a) Make sure that the FWD CARGO FIRE light and FIRE WARN lights stay off, and the fire bell does not sound.

SUBTASK 26-16-00-730-013

(16) On the P8 cargo fire control panel, set the FWD DET SELECT switch to B position.

(a) Make sure that the FWD CARGO FIRE light and FIRE WARN lights stay off, and the fire bell does not sound.

SUBTASK 26-16-00-730-005

(17) Do the Smoke Test for each smoke detector pair in the FWD cargo compartment (Table 501).

Table 501/26-16-00-993-801

SMOKE DETECTOR PAIR	FWD DET SWITCH POSITION	AFT DET SWITCH POSITION	CARGO BAY	LOCATION	EQUIPMENT NUMBER
A1	A	NORM	FWD	FWD LEFT	M2238
B1	B	NORM	FWD	FWD RIGHT	M2240
A2	A	NORM	FWD	AFT LEFT	M2239
B2	B	NORM	FWD	AFT RIGHT	M2241

F. Aft Cargo Bay Smoke Detection Smoke Test

NOTE: The subsequent test procedures are for each smoke detector pair in the AFT cargo compartment.

SUBTASK 26-16-00-860-014

(1) On the P8 cargo fire control panel, set the AFT DET SELECT switch to A position.

SUBTASK 26-16-00-730-014

(2) Use a smoke generator, COM-11202, to apply smoke to the A loop smoke detector for the pair in the test (Table 502).

SUBTASK 26-16-00-730-015

(3) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the AFT CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

SUBTASK 26-16-00-730-016

(4) On the P8 cargo fire control panel, set the AFT DET SELECT switch to NORM position.

(a) Make sure that the AFT CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns off within 10 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-16-00-860-015

(5) Clear the smoke from the A loop smoke detector.

SUBTASK 26-16-00-860-016

(6) On the P8 cargo fire control panel, set the AFT DET SELECT switch to B position (Table 502).

SUBTASK 26-16-00-730-017

(7) Use a smoke generator, COM-11202, to apply smoke to the B loop smoke detector for the pair in the test.

SUBTASK 26-16-00-730-018

(8) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the AFT CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

SUBTASK 26-16-00-730-019

(9) On the P8 cargo fire control panel, set the AFT DET SELECT switch to NORM position.

(a) Make sure that the AFT CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns off within 10 seconds.

SUBTASK 26-16-00-860-017

(10) Clear the smoke from the B loop smoke detector.

SUBTASK 26-16-00-730-020

(11) Use a smoke generator, COM-11202, to apply smoke to both A and B loop smoke detectors for the pair in the test (Table 502).

SUBTASK 26-16-00-730-021

(12) Make sure that these indications occur:

(a) On the P8 cargo fire control panel, the AFT CARGO FIRE light (red) comes on.

(b) On the P7 pilot's glareshield, the Captain's and First Officer's FIRE WARN lights (red) come on.

(c) In the flight compartment, the fire bell comes on.

SUBTASK 26-16-00-860-018

(13) Clear the smoke from the A and B loop smoke detectors.

SUBTASK 26-16-00-730-022

(14) Make sure that the AFT CARGO FIRE light and FIRE WARN lights go off, and the fire bell turns off when the smoke is cleared.

SUBTASK 26-16-00-730-023

(15) On the P8 cargo fire control panel, set the AFT DET SELECT switch to A position.

(a) Make sure that the AFT CARGO FIRE light and FIRE WARN lights stay off, and the fire bell does not sound.

SUBTASK 26-16-00-730-024

(16) On the P8 cargo fire control panel, set the AFT DET SELECT switch to B position.

(a) Make sure that the AFT CARGO FIRE light and FIRE WARN lights stay off, and the fire bell does not sound.

SUBTASK 26-16-00-730-025

(17) Do the Smoke Test for each smoke detector pair in the AFT cargo compartment (Table 502).

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